

pro_gan_cifar10

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#

BÀI TẬP CUỐI KỲ

##

Môn: Thị giác máy tính chuyên sâu

1 Second-level requirement:

1.0.1 Train and generate images for one class of the dataset using the mechanism of ProGAN.

Ghi chú: Phần code dưới đây đã tham khảo từ nguồn: <https://machinelearningmastery.com/how-to-train-a-progressive-growing-gan-in-keras-for-synthesizing-faces/>.

```
[1]: !pip install tensorflow_datasets

# Import libraries
from google.colab import files
import numpy as np
import tensorflow as tf
import tensorflow_datasets as tfds
import keras

from sklearn.model_selection import train_test_split
from skimage.transform import resize
from matplotlib import pyplot as plt
from math import sqrt, ceil
from keras import Model
from keras import utils
from keras import callbacks
```

```

from keras import optimizers
from keras import backend
from keras.initializers import RandomNormal
from keras.constraints import max_norm
from keras.datasets import cifar10
from keras.optimizers.legacy import Adam, SGD
from keras.models import Sequential
from keras.layers import Add, Layer, Input, Activation, Conv2D, MaxPooling2D,
↳Conv2DTranspose, UpSampling2D, LeakyReLU, Reshape, Dropout,
↳BatchNormalization, GlobalMaxPooling2D, AveragePooling2D, Flatten, Dense

```

Requirement already satisfied: tensorflow_datasets in
/usr/local/lib/python3.10/dist-packages (4.9.4)

Requirement already satisfied: absl-py in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (1.4.0)

Requirement already satisfied: click in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (8.1.7)

Requirement already satisfied: dm-tree in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (0.1.8)

Requirement already satisfied: etils[enp,epath,etree]>=0.9.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (1.7.0)

Requirement already satisfied: numpy in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (1.25.2)

Requirement already satisfied: promise in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (2.3)

Requirement already satisfied: protobuf>=3.20 in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (3.20.3)

Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (5.9.5)

Requirement already satisfied: requests>=2.19.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (2.31.0)

Requirement already satisfied: tensorflow-metadata in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (1.15.0)

Requirement already satisfied: termcolor in /usr/local/lib/python3.10/dist-
packages (from tensorflow_datasets) (2.4.0)

Requirement already satisfied: toml in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (0.10.2)

Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (4.66.2)

Requirement already satisfied: wrapt in /usr/local/lib/python3.10/dist-packages
(from tensorflow_datasets) (1.14.1)

Requirement already satisfied: array-record>=0.5.0 in
/usr/local/lib/python3.10/dist-packages (from tensorflow_datasets) (0.5.1)

Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages
(from etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (2023.6.0)

Requirement already satisfied: importlib_resources in
/usr/local/lib/python3.10/dist-packages (from
etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (6.4.0)

Requirement already satisfied: typing_extensions in
 /usr/local/lib/python3.10/dist-packages (from
 etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (4.11.0)
 Requirement already satisfied: zipp in /usr/local/lib/python3.10/dist-packages
 (from etils[enp,epath,etree]>=0.9.0->tensorflow_datasets) (3.18.1)
 Requirement already satisfied: charset-normalizer<4,>=2 in
 /usr/local/lib/python3.10/dist-packages (from
 requests>=2.19.0->tensorflow_datasets) (3.3.2)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-
 packages (from requests>=2.19.0->tensorflow_datasets) (3.7)
 Requirement already satisfied: urllib3<3,>=1.21.1 in
 /usr/local/lib/python3.10/dist-packages (from
 requests>=2.19.0->tensorflow_datasets) (2.0.7)
 Requirement already satisfied: certifi>=2017.4.17 in
 /usr/local/lib/python3.10/dist-packages (from
 requests>=2.19.0->tensorflow_datasets) (2024.2.2)
 Requirement already satisfied: six in /usr/local/lib/python3.10/dist-packages
 (from promise->tensorflow_datasets) (1.16.0)

```
[2]: # Get the full dataset (batch_size=-1) in NumPy arrays from the returned tf.
      ↪Tensor object
cifar10_train = tfds.load(name="cifar10", split=tfds.Split.TRAIN, batch_size=-1,
      ↪)
cifar10_test = tfds.load(name="cifar10", split=tfds.Split.TEST, batch_size=-1)

# Convert tfds dataset to numpy array records
cifar10_train = tfds.as_numpy(cifar10_train)
cifar10_test = tfds.as_numpy(cifar10_test)

# Seperate feature X and label Y
X_train, y_train = cifar10_train["image"], cifar10_train["label"]
X_test, y_test = cifar10_test["image"], cifar10_test["label"]

# Print shapes of the entire training and test set of CIFAR 10
print("X_train shape: " + str(X_train.shape))
print("X_test shape: " + str(X_test.shape))
print("y_train shape: " + str(y_train.shape))
print("y_test shape: " + str(y_test.shape))
```

Downloading and preparing dataset 162.17 MiB (download: 162.17 MiB, generated:
 132.40 MiB, total: 294.58 MiB) to /root/tensorflow_datasets/cifar10/3.0.2...

Dl Completed...: 0 url [00:00, ? url/s]

Dl Size...: 0 MiB [00:00, ? MiB/s]

Extraction completed...: 0 file [00:00, ? file/s]

Generating splits...: 0%| | 0/2 [00:00<?, ? splits/s]

```

Generating train examples...: 0%|          | 0/50000 [00:00<?, ? examples/s]
Shuffling /root/tensorflow_datasets/cifar10/3.0.2.incompleteTBL1WV/cifar10-train.
↳tfrecord*...: 0%|          ...
Generating test examples...: 0%|          | 0/10000 [00:00<?, ? examples/s]
Shuffling /root/tensorflow_datasets/cifar10/3.0.2.incompleteTBL1WV/cifar10-test.
↳tfrecord*...: 0%|          |...

Dataset cifar10 downloaded and prepared to
/root/tensorflow_datasets/cifar10/3.0.2. Subsequent calls will reuse this data.
X_train shape: (50000, 32, 32, 3)
X_test shape: (10000, 32, 32, 3)
y_train shape: (50000,)
y_test shape: (10000,)

```

```

[3]: # Get samples of a certain label. You can choose from one of the classes:
# 0 - airplane; 1 - automobile; 2 - bird; 3 - cat; 4 - deer; 5 - dog; 6 - frog;
↳7 - horse; 8 - ship; 9 - truck

label = 2                                # Class '2' is bird
train_indices = np.where(y_train == label) # Get indices of the label
train_set = X_train[train_indices]         # Get train samples of the label
test_indices = np.where(y_test == label)   # Get indices of the label
test_set = X_test[test_indices]            # Get test samples of the label

# Print shapes of datasets of the selected class
print("Train_set shape of the selected class: " + str(train_set.shape))
print("Test_set shape of the selected class: " + str(test_set.shape))

# NOTE: Training a GAN doesn't require to have train/test split. You should
↳merge the train_set and test_set into one dataset, which can be used as real
↳data samples
data = np.vstack((train_set, test_set))
print("Dataset shape of the selected class: " + str(data.shape))

```

```

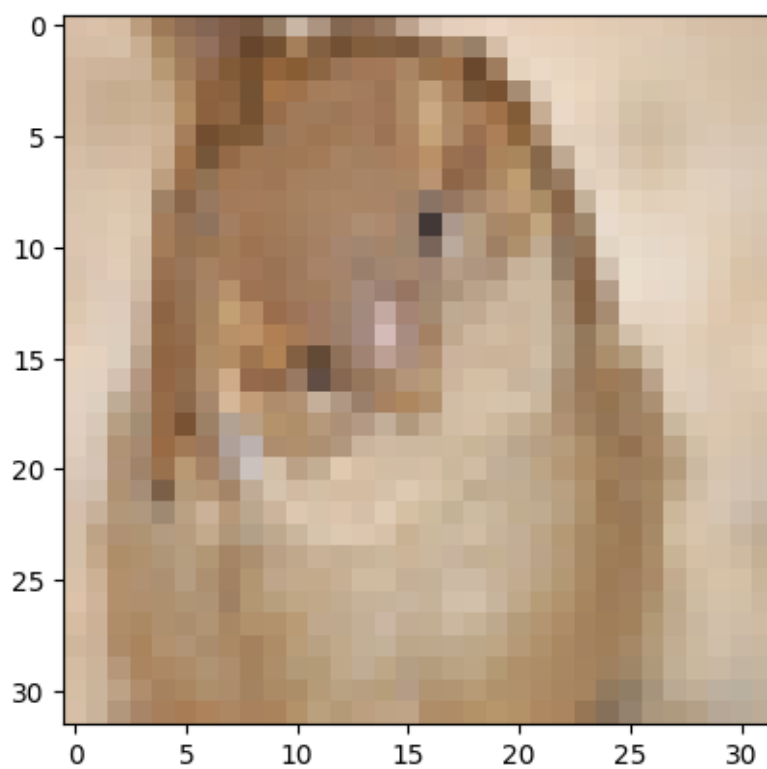
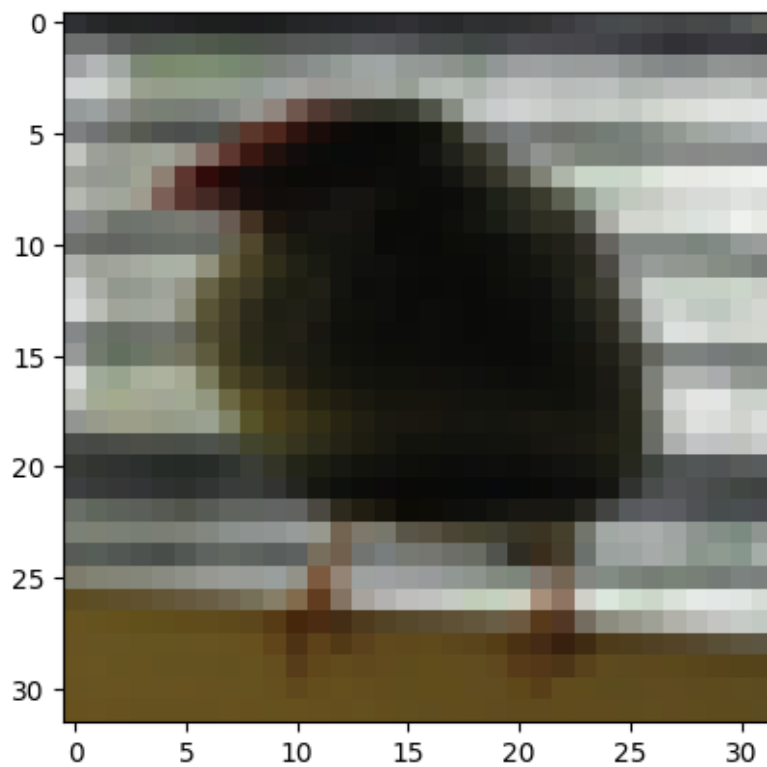
Train_set shape of the selected class: (5000, 32, 32, 3)
Test_set shape of the selected class: (1000, 32, 32, 3)
Dataset shape of the selected class: (6000, 32, 32, 3)

```

```

[4]: # Visualize some data samples of class '2' (bird)
import matplotlib.pyplot as plt
imgplot = plt.imshow(data[50])
plt.show()
imgplot = plt.imshow(data[5500])
plt.show()

```



```
[5]: # Weighted sum layer
class WeightedSum(Add):
    # Init with the default value
    def __init__(self, alpha=0.0, **kwargs):
        super(WeightedSum, self).__init__(**kwargs)
        self.alpha = backend.variable(alpha, name='ws_alpha')

    # Outputs weighted sum of inputs
    def _merge_function(self, inputs):
        # Supports weighted sum of only two inputs
        assert (len(inputs) == 2)

        #  $((1-a) * input1) + (a * input2)$ 
        output = ((1.0 - self.alpha) * inputs[0]) + (self.alpha * inputs[1])
        return output
```

2 Build the Generator

```
[6]: # Add a generator block
def add_generator_block(old_model):
    # Getting the end of the last block
    block_end = old_model.layers[-2].output

    # Upsampling, and defining new block
    upsampling = UpSampling2D()(block_end)
    g = Conv2D(128, (3,3), padding='same')(upsampling)
    g = LeakyReLU(alpha=0.2)(g)

    g = Conv2D(64, (3,3), padding='same')(g)
    g = LeakyReLU(alpha=0.2)(g)

    # Adding new output layer
    out_image = Conv2D(3, (1,1), padding='same')(g)

    # Defining model
    model1 = Model(old_model.input, out_image)

    # Getting the output layer from old model
    out_old = old_model.layers[-1]

    # Connecting the upsample to the old output layer
    out_image2 = out_old(upsampling)

    # Defining new output image as the weighted sum of the old and new models
```

```
merged = WeightedSum()([out_image2, out_image])

# Defining model
model2 = Model(old_model.input, merged)

return [model1, model2]
```

```
[7]: # Defining generator models
def define_generator(latent_dim, n_blocks, in_dim=4):
    model_list = list()

    # Base model latent input
    in_latent = Input(shape=(latent_dim,))

    # Linearly scaling up to activation maps
    g = Dense(256 * in_dim * in_dim)(in_latent)
    g = Reshape((in_dim, in_dim, 256))(g)

    # Conv 3x3, input block
    g = Conv2D(128, (3,3), padding='same')(g)
    g = LeakyReLU(alpha=0.2)(g)

    # Conv 3x3
    g = Conv2D(64, (3,3), padding='same')(g)
    g = LeakyReLU(alpha=0.2)(g)

    # Conv 1x1, output block
    out_image = Conv2D(3, (1,1), activation='tanh', padding='same')(g)

    # Defining model
    model = Model(in_latent, out_image)

    # Storing model
    model_list.append([model, model])

    # Creating submodels
    for i in range(1, n_blocks):
        # Getting prior model without the fade-on
        old_model = model_list[i - 1][0]

        # Creating new model for next resolution
        models = add_generator_block(old_model)

        # Storing model
        model_list.append(models)

    return model_list
```

```
# Show the fade-in version of the fourth Generator model, which has output_
↳ resolution of 32x32
generators = define_generator(256, 4)
generators[3][1].summary()
```

Model: "model_6"

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 256)]	0	[]
dense (Dense) ['input_1[0][0]']	(None, 4096)	1052672	
reshape (Reshape) ['dense[0][0]']	(None, 4, 4, 256)	0	
conv2d (Conv2D) ['reshape[0][0]']	(None, 4, 4, 128)	295040	
leaky_re_lu (LeakyReLU) ['conv2d[0][0]']	(None, 4, 4, 128)	0	
conv2d_1 (Conv2D) ['leaky_re_lu[0][0]']	(None, 4, 4, 64)	73792	
leaky_re_lu_1 (LeakyReLU) ['conv2d_1[0][0]']	(None, 4, 4, 64)	0	
up_sampling2d (UpSampling2D) ['leaky_re_lu_1[0][0]']	(None, 8, 8, 64)	0	
conv2d_3 (Conv2D) ['up_sampling2d[0][0]']	(None, 8, 8, 128)	73856	
leaky_re_lu_2 (LeakyReLU) ['conv2d_3[0][0]']	(None, 8, 8, 128)	0	
conv2d_4 (Conv2D) ['leaky_re_lu_2[0][0]']	(None, 8, 8, 64)	73792	
leaky_re_lu_3 (LeakyReLU) ['conv2d_4[0][0]']	(None, 8, 8, 64)	0	

up_sampling2d_1 (UpSampling2D)	(None, 16, 16, 64)	0
['leaky_re_lu_3[0][0]']		
conv2d_6 (Conv2D)	(None, 16, 16, 128)	73856
['up_sampling2d_1[0][0]']		
leaky_re_lu_4 (LeakyReLU)	(None, 16, 16, 128)	0
['conv2d_6[0][0]']		
conv2d_7 (Conv2D)	(None, 16, 16, 64)	73792
['leaky_re_lu_4[0][0]']		
leaky_re_lu_5 (LeakyReLU)	(None, 16, 16, 64)	0
['conv2d_7[0][0]']		
up_sampling2d_2 (UpSampling2D)	(None, 32, 32, 64)	0
['leaky_re_lu_5[0][0]']		
conv2d_9 (Conv2D)	(None, 32, 32, 128)	73856
['up_sampling2d_2[0][0]']		
leaky_re_lu_6 (LeakyReLU)	(None, 32, 32, 128)	0
['conv2d_9[0][0]']		
conv2d_10 (Conv2D)	(None, 32, 32, 64)	73792
['leaky_re_lu_6[0][0]']		
leaky_re_lu_7 (LeakyReLU)	(None, 32, 32, 64)	0
['conv2d_10[0][0]']		
conv2d_8 (Conv2D)	multiple	195
['up_sampling2d_2[0][0]']		
conv2d_11 (Conv2D)	(None, 32, 32, 3)	195
['leaky_re_lu_7[0][0]']		
weighted_sum_2 (WeightedSum)	(None, 32, 32, 3)	1
['conv2d_8[1][0]', m) 'conv2d_11[0][0]']		

=====

Total params: 1864839 (7.11 MB)
Trainable params: 1864839 (7.11 MB)
Non-trainable params: 0 (0.00 Byte)

3 Build the Discriminator

```
[8]: # Add a discriminator block
def add_discriminator_block(old_model, n_input_layers=3):
    # Getting shape of existing model
    in_shape = list(old_model.input.shape)

    # Defining new input shape as double the size
    i_s = (int(in_shape[-2])*2, int(in_shape[-2])*2, int(in_shape[-1]))
    in_img = Input(shape=i_s)

    # Defining new input processing layer
    d = Conv2D(64, (1,1), padding='same')(in_img)
    d = LeakyReLU(alpha=0.2)(d)

    # Defining new block
    d = Conv2D(128, (3,3), padding='same')(d)
    d = LeakyReLU(alpha=0.2)(d)

    d = Conv2D(64, (3,3), padding='same')(d)
    d = LeakyReLU(alpha=0.2)(d)

    d = AveragePooling2D()(d)
    block_new = d

    # Skipping the input
    for i in range(n_input_layers, len(old_model.layers)):
        d = old_model.layers[i](d)

    # Defining straight-through model
    m1 = Model(in_img, d)

    # Compiling model
    opt = Adam(learning_rate=0.0001, beta_1=0.5)
    m1.compile(loss='binary_crossentropy', optimizer=opt, metrics=['accuracy'])

    # Downsampling the new larger image
    downsample = AveragePooling2D()(in_img)

    # Connecting old input
    block_old = old_model.layers[1](downsample)
    block_old = old_model.layers[2](block_old)

    # Fade in output of old model input layer with new input
```

```

d = WeightedSum()([block_old, block_new])

# Skipping the input
for i in range(n_input_layers, len(old_model.layers)):
    d = old_model.layers[i](d)

# Defining straight-through model
m2 = Model(in_img, d)

# Compiling model
m2.compile(loss='binary_crossentropy', optimizer=opt, metrics=['accuracy'])

return [m1, m2]

```

```

[9]: # Define the discriminator models for each image resolution
def define_discriminator(n_blocks, input_shape=(4,4,3)):
    model_list = list()

    # Base model input
    in_image = Input(shape=input_shape)

    # Conv 1x1
    d = Conv2D(64, (1,1), padding='same')(in_image)
    d = LeakyReLU(alpha=0.2)(d)

    # Conv 3x3 (output block)
    d = Conv2D(128, (3,3), padding='same')(d)
    d = LeakyReLU(alpha=0.2)(d)

    # Conv 3x3
    d = Conv2D(256, (3,3), padding='same')(d)
    d = LeakyReLU(alpha=0.2)(d)

    # Dense output layer
    d = Flatten()(d)
    d = Dropout(0.5)(d)
    out_class = Dense(1, activation='sigmoid')(d)

    # Define model
    model = Model(in_image, out_class)

    # Compile model
    opt = Adam(learning_rate=0.0001, beta_1=0.5)
    model.compile(loss='binary_crossentropy', optimizer=opt, metrics=['accuracy'])

    # Store model
    model_list.append([model, model])

```

```

# Create submodels
for i in range(1, n_blocks):
    # Get prior model without the fade-on
    old_model = model_list[i - 1][0]

    # Create new model for next resolution
    models = add_discriminator_block(old_model)

    # Store model
    model_list.append(models)

return model_list

# Show the fade-in version of the fourth Discriminator model, which has input_
↪ resolution of 32x32
discriminators = define_discriminator(4)
discriminators[3][1].summary()

```

Model: "model_13"

Layer (type)	Output Shape	Param #	Connected to
input_5 (InputLayer)	[(None, 32, 32, 3)]	0	[]
conv2d_21 (Conv2D)	(None, 32, 32, 64)	256	['input_5[0][0]']
leaky_re_lu_17 (LeakyReLU)	(None, 32, 32, 64)	0	['conv2d_21[0][0]']
conv2d_22 (Conv2D)	(None, 32, 32, 128)	73856	['leaky_re_lu_17[0][0]']
leaky_re_lu_18 (LeakyReLU)	(None, 32, 32, 128)	0	['conv2d_22[0][0]']
average_pooling2d_5 (AveragePooling2D)	(None, 16, 16, 3)	0	['input_5[0][0]']
conv2d_23 (Conv2D)	(None, 32, 32, 64)	73792	['leaky_re_lu_18[0][0]']
conv2d_18 (Conv2D)	(None, 16, 16, 64)	256	['average_pooling2d_5[0][0]']

leaky_re_lu_19 (LeakyReLU)	(None, 32, 32, 64)	0
['conv2d_23[0][0]']		
leaky_re_lu_14 (LeakyReLU)	(None, 16, 16, 64)	0
['conv2d_18[1][0]']		
average_pooling2d_4 (AveragePooling2D)	(None, 16, 16, 64)	0
['leaky_re_lu_19[0][0]']		
weighted_sum_5 (WeightedSum)	(None, 16, 16, 64)	1
['leaky_re_lu_14[1][0]', m) 'average_pooling2d_4[0][0]']		
conv2d_19 (Conv2D)	(None, 16, 16, 128)	73856
['weighted_sum_5[0][0]']		
leaky_re_lu_15 (LeakyReLU)	(None, 16, 16, 128)	0
['conv2d_19[2][0]']		
conv2d_20 (Conv2D)	(None, 16, 16, 64)	73792
['leaky_re_lu_15[2][0]']		
leaky_re_lu_16 (LeakyReLU)	(None, 16, 16, 64)	0
['conv2d_20[2][0]']		
average_pooling2d_2 (AveragePooling2D)	(None, 8, 8, 64)	0
['leaky_re_lu_16[2][0]']		
conv2d_16 (Conv2D)	(None, 8, 8, 128)	73856
['average_pooling2d_2[2][0]']		
leaky_re_lu_12 (LeakyReLU)	(None, 8, 8, 128)	0
['conv2d_16[4][0]']		
conv2d_17 (Conv2D)	(None, 8, 8, 64)	73792
['leaky_re_lu_12[4][0]']		
leaky_re_lu_13 (LeakyReLU)	(None, 8, 8, 64)	0
['conv2d_17[4][0]']		
average_pooling2d (AveragePooling2D)	(None, 4, 4, 64)	0
['leaky_re_lu_13[4][0]']		

conv2d_13 (Conv2D)	(None, 4, 4, 128)	73856
['average_pooling2d[4][0]']		
leaky_re_lu_9 (LeakyReLU)	(None, 4, 4, 128)	0
['conv2d_13[6][0]']		
conv2d_14 (Conv2D)	(None, 4, 4, 256)	295168
['leaky_re_lu_9[6][0]']		
leaky_re_lu_10 (LeakyReLU)	(None, 4, 4, 256)	0
['conv2d_14[6][0]']		
flatten (Flatten)	(None, 4096)	0
['leaky_re_lu_10[6][0]']		
dropout (Dropout)	(None, 4096)	0
['flatten[6][0]']		
dense_1 (Dense)	(None, 1)	4097
['dropout[6][0]']		

```
=====
=====
Total params: 816578 (3.11 MB)
Trainable params: 816578 (3.11 MB)
Non-trainable params: 0 (0.00 Byte)
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```

4 Build composite models for training Generators via Discriminators

```
[10]: # Define composite models for training generators via discriminators
def define_composite(discriminators, generators):
    model_list = list()

    # Creating composite models
    for i in range(len(discriminators)):
        g_models, d_models = generators[i], discriminators[i]

        # Straight-through model
        d_models[0].trainable = False
        model1 = Sequential()
        model1.add(g_models[0])
        model1.add(d_models[0])
        opt = Adam(learning_rate=0.0001, beta_1=0.5)
```

```

model1.compile(loss='binary_crossentropy', optimizer=opt)

# Fade-in model
d_models[1].trainable = False
model2 = Sequential()
model2.add(g_models[1])
model2.add(d_models[1])
model2.compile(loss='binary_crossentropy', optimizer=opt)

# Storing
model_list.append([model1, model2])

return model_list

```

```

[11]: # Update the alpha value on each instance of WeightedSum
def update_fadein(models, step, n_steps):
    # Calculate current alpha (linear from 0 to 1)
    alpha = step / float(n_steps - 1)

    # Update the alpha for each model
    for model in models:
        for layer in model.layers:
            if isinstance(layer, WeightedSum):
                backend.set_value(layer.alpha, alpha)

```

```

[12]: # Scale images to preferred size
def scale_dataset(images, new_shape):
    images_list = list()
    for image in images:
        # Resize with nearest neighbor interpolation
        new_image = resize(image, new_shape, 0)

        # Storing
        images_list.append(new_image)

    return np.asarray(images_list)

```

5 Prepare real data samples from the CIFAR 10 dataset

```

[13]: # Load and normalize CIFAR 10 images of the selected class
def load_real_samples():
    # Load dataset
    (X_train, y_train), (X_test, y_test) = cifar10.load_data()
    label = 2 # Class '2' is bird
    train_indices = np.where(y_train == label)[0] # Get indices of the label
    train_set = X_train[train_indices]

```

```

test_indices = np.where(y_test == label)[0]    # Get indices of the label
test_set = X_test[test_indices]
X = np.vstack((train_set, test_set))

# Convert pixel values from unsigned int to float32 for normalization
↪ calculation
X = X.astype('float32')

# Because the generator is using tanh activation, we need to normalize real
↪ images from range [0,255] to range [-1,1]
X = (X - 127.5) / 127.5

return X # Return the set of normalized images

# Select real samples from the real dataset for training purpose
def generate_real_samples(dataset, n_samples):
    # Generate random n_samples sample indices from the dataset
    ix = np.random.randint(0, dataset.shape[0], size = n_samples)

    # Get the images from the list of indices
    X = dataset[ix]

    # Generate labels for the real samples (label=1)
    y = np.ones((n_samples, 1))

    return X, y # Return the real images and labels

```

6 Generate fake data samples by passing a random vector through the Generator

```

[14]: # Generate random latent vectors as input for Generator
def generate_latent_points(latent_dim, n_samples):
    # Generate points in the latent space
    x_input = np.random.randn(latent_dim * n_samples)

    # Reshape to a batch of inputs for the Generator
    x_input = x_input.reshape(n_samples, latent_dim)

    return x_input

# Use Generator to generate n_samples fake samples
def generate_fake_samples(g_model, latent_dim, n_samples):
    # Generate random latent vectors as input for Generator
    x_input = generate_latent_points(latent_dim, n_samples)

```



```

# Generate fake data samples by passing the random latent vectors through the
↪Generator
X = g_model.predict(x_input)

# Generate labels for the fake samples (label=0)
y = np.zeros((n_samples, 1))

return X, y # Return the fake images and labels

```

7 Save the Generator's outputs and the models

```

[15]: # Generating samples, saving as a plot and saving the model
def summarize_performance(status, g_model, d_model, gan_model, latent_dim,
↪n_samples=49):

    # Devising a name
    gs = g_model.output_shape
    nm = '%03dx%03d-%s' % (gs[1], gs[2], status)

    # Generating images
    X, _ = generate_fake_samples(g_model, latent_dim, n_samples)

    # Plotting real images
    square = int(sqrt(n_samples))
    for i in range(n_samples):
        plt.subplot(square, square, 1 + i)
        plt.axis('off')
        plt.imshow(((X[i] * 127.5) + 127.5).astype(np.uint8)) # Convert pixel
↪values from range [-1,1] into range [0,255]

    # Saving a plot to file
    f1 = 'plot_%s.png' % (nm)
    plt.savefig(f1)
    plt.close()

    # Saving the generator model
    f2 = 'g_model_%s.sav' % (nm)
    g_model.save(f2)

    # Saving the discriminator model
    f3 = 'd_model_%s.sav' % (nm)
    d_model.save(f3)

    # Saving the composite model
    f4 = 'gan_model_%s.sav' % (nm)
    gan_model.save(f4)

```

```
print('>Saved: %s, %s, %s and %s' % (f1, f2, f3, f4))
```

8 Training process with the Generator and the Discriminator

```
[16]: # Train a Generator and a Discriminator
def train_epochs(g_model, d_model, gan_model, dataset, latent_dim, n_epochs,
    ↪n_batch, fadein=False):
    bat_per_epo = int(dataset.shape[0] / n_batch) # Number of iterations per
    ↪epoch
    half_batch = int(n_batch / 2) # A batch to the Discriminator
    ↪contains real samples (half-batch) and fake samples (half-batch)
    n_steps = bat_per_epo * n_epochs # Calculate the number of
    ↪training iterations

    g_loss_values = [] # Array containing
    ↪Generator loss over iterations
    d_loss_real_values = [] # Array containing
    ↪Discriminator loss of real samples over iterations
    d_loss_fake_values = [] # Array containing
    ↪Discriminator loss of fake samples over iterations

    # Manually enumerating epochs
    for x in range(n_steps):
        # Updating alpha for all WeightedSum layers when fading in new blocks
        if fadein:
            update_fadein([g_model, d_model, gan_model], x, n_steps)

        # Preparing real and fake samples
        X_real, y_real = generate_real_samples(dataset, half_batch)
        X_fake, y_fake = generate_fake_samples(g_model, latent_dim, half_batch)

        # Updating discriminator model
        d_loss1, _ = d_model.train_on_batch(X_real, y_real)
        d_loss2, _ = d_model.train_on_batch(X_fake, y_fake)

        # Updating the generator via the discriminator's error
        X_gan = generate_latent_points(latent_dim, n_batch)
        y_gan = np.ones((n_batch, 1))
        g_loss = gan_model.train_on_batch(X_gan, y_gan)

        # Summarizing loss on this batch
        i = ceil((x + 1) / bat_per_epo) # Calculate the current epoch
        k = (i - 1) * bat_per_epo # Calculate the iterations in
    ↪the previous epochs
```

```

        j = x + 1 - k                                # Calculate the current
iteration in the current epoch
        print('Epoch %d, Iteration %d/%d, D_loss_real = %.3f, D_loss_fake = %.3f,
G_loss = %.3f' % (i, j, bat_per_epo, d_loss1, d_loss2, g_loss))

        # Add loss values to historical arrays
        g_loss_values.append(g_loss)
        d_loss_real_values.append(d_loss1)
        d_loss_fake_values.append(d_loss2)

    return d_loss_real_values, d_loss_fake_values, g_loss_values    #
Return the loss arrays of Generator and Discriminator

```

```

[17]: # Train the Generators and the Discriminators
def train(g_models, d_models, gan_models, dataset, latent_dim, e_norm,
e_fadein, n_batch):
    # Fitting the baseline model
    gn, dn, gan_n = g_models[0][0], d_models[0][0], gan_models[0][0]

    # Scaling dataset to appropriate size
    gs = gn.output_shape
    sd = scale_dataset(dataset, gs[1:])
    print('Scaled Data', sd.shape)

    # Training normal or straight-through models
    train_epochs(gn, dn, gan_n, sd, latent_dim, e_norm[0], n_batch[0])
    summarize_performance('tuned', gn, dn, gan_n, latent_dim)

    # Processing each level of growth
    for i in range(1, len(g_models)):
        # Retrieving models for this level of growth
        [gn, g_fadein] = g_models[i]
        [dn, d_fadein] = d_models[i]
        [gan_n, gan_fadein] = gan_models[i]

        # Scaling the dataset to appropriate size
        gs = gn.output_shape
        sd = scale_dataset(dataset, gs[1:])
        print('Scaled Data', sd.shape)

        # Training fade-in models for next level of growth
        d_losses_real_fadein, d_losses_fake_fadein, g_losses_fadein =
train_epochs(g_fadein, d_fadein, gan_fadein, sd, latent_dim, e_fadein[i],
n_batch[i], True)
        summarize_performance('faded', g_fadein, d_fadein, gan_fadein, latent_dim)

```

```

    # -> During the fade-in phase, a portion of the generator and discriminator
    ↪ operates at a lower resolution while the newly introduced layers are
    ↪ gradually scaled up.

    # This phase is designed to ease the transition from one resolution to
    ↪ another, by helping to reduce the risk of instability when moving to higher
    ↪ resolutions.

    # Training normal or straight-through models
    d_losses_real_norm, d_losses_fake_norm, g_losses_norm = train_epochs(gn,
    ↪ dn, gan_n, sd, latent_dim, e_norm[i], n_batch[i])
    summarize_performance('tuned', gn, dn, gan_n, latent_dim)

    # Return the losses of the last level of growth (32x32 resolution)
    return d_losses_real_fadein, d_losses_fake_fadein, g_losses_fadein,
    ↪ d_losses_real_norm, d_losses_fake_norm, g_losses_norm

```

9 Main code

```

[18]: # Size of the latent space
latent_dim = 256
n_blocks = 4          # Number of growth phases. For example, 4 == [4, 8, 16, 32]
epochs = [30, 40, 50, 60]
batch_size = [128, 128, 128, 128]

# Create the models
g_models = define_generator(latent_dim, n_blocks)
d_models = define_discriminator(n_blocks)
gan_models = define_composite(d_models, g_models)

# loading image data
dataset = load_real_samples()

# Train the ProGAN
d_losses_real_fadein, d_losses_fake_fadein, g_losses_fadein,
    ↪ d_losses_real_norm, d_losses_fake_norm, g_losses_norm = train(g_models,
    ↪ d_models, gan_models, dataset, latent_dim, epochs, epochs, batch_size)

```

```

Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 [=====] - 4s 0us/step
Scaled Data (6000, 4, 4, 3)
2/2 [=====] - 1s 4ms/step
Epoch 1, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss =
0.695
2/2 [=====] - 0s 3ms/step

```

Epoch 1, Iteration 3/46, D_loss_real = 0.677, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 4/46, D_loss_real = 0.665, D_loss_fake = 0.700, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 5/46, D_loss_real = 0.653, D_loss_fake = 0.711, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 6/46, D_loss_real = 0.637, D_loss_fake = 0.724, G_loss = 0.669
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 7/46, D_loss_real = 0.636, D_loss_fake = 0.731, G_loss = 0.663
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 8/46, D_loss_real = 0.635, D_loss_fake = 0.741, G_loss = 0.660
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 9/46, D_loss_real = 0.626, D_loss_fake = 0.739, G_loss = 0.665
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 10/46, D_loss_real = 0.628, D_loss_fake = 0.731, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 11/46, D_loss_real = 0.620, D_loss_fake = 0.721, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 12/46, D_loss_real = 0.630, D_loss_fake = 0.701, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 13/46, D_loss_real = 0.634, D_loss_fake = 0.692, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 14/46, D_loss_real = 0.635, D_loss_fake = 0.673, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 15/46, D_loss_real = 0.640, D_loss_fake = 0.663, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 16/46, D_loss_real = 0.639, D_loss_fake = 0.658, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 17/46, D_loss_real = 0.643, D_loss_fake = 0.669, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 18/46, D_loss_real = 0.622, D_loss_fake = 0.672, G_loss = 0.736
2/2 [=====] - 0s 3ms/step

Epoch 1, Iteration 19/46, D_loss_real = 0.604, D_loss_fake = 0.669, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 20/46, D_loss_real = 0.583, D_loss_fake = 0.684, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 21/46, D_loss_real = 0.590, D_loss_fake = 0.700, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 22/46, D_loss_real = 0.578, D_loss_fake = 0.747, G_loss = 0.665
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 23/46, D_loss_real = 0.591, D_loss_fake = 0.791, G_loss = 0.640
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 24/46, D_loss_real = 0.597, D_loss_fake = 0.807, G_loss = 0.611
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 25/46, D_loss_real = 0.611, D_loss_fake = 0.832, G_loss = 0.604
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 26/46, D_loss_real = 0.552, D_loss_fake = 0.824, G_loss = 0.610
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 27/46, D_loss_real = 0.607, D_loss_fake = 0.805, G_loss = 0.646
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 28/46, D_loss_real = 0.580, D_loss_fake = 0.774, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 29/46, D_loss_real = 0.612, D_loss_fake = 0.702, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 30/46, D_loss_real = 0.594, D_loss_fake = 0.639, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 31/46, D_loss_real = 0.573, D_loss_fake = 0.580, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 32/46, D_loss_real = 0.584, D_loss_fake = 0.539, G_loss = 0.947
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 33/46, D_loss_real = 0.608, D_loss_fake = 0.529, G_loss = 0.957
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 34/46, D_loss_real = 0.580, D_loss_fake = 0.527, G_loss = 0.935
2/2 [=====] - 0s 3ms/step

Epoch 1, Iteration 35/46, D_loss_real = 0.571, D_loss_fake = 0.557, G_loss = 0.925
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 36/46, D_loss_real = 0.561, D_loss_fake = 0.618, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 37/46, D_loss_real = 0.559, D_loss_fake = 0.708, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 38/46, D_loss_real = 0.573, D_loss_fake = 0.773, G_loss = 0.653
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 39/46, D_loss_real = 0.608, D_loss_fake = 0.841, G_loss = 0.623
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 40/46, D_loss_real = 0.556, D_loss_fake = 0.844, G_loss = 0.617
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 41/46, D_loss_real = 0.587, D_loss_fake = 0.827, G_loss = 0.626
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 42/46, D_loss_real = 0.593, D_loss_fake = 0.785, G_loss = 0.656
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 43/46, D_loss_real = 0.631, D_loss_fake = 0.747, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 44/46, D_loss_real = 0.629, D_loss_fake = 0.718, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 45/46, D_loss_real = 0.635, D_loss_fake = 0.694, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 46/46, D_loss_real = 0.632, D_loss_fake = 0.657, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 1/46, D_loss_real = 0.642, D_loss_fake = 0.639, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 2/46, D_loss_real = 0.665, D_loss_fake = 0.625, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 3/46, D_loss_real = 0.650, D_loss_fake = 0.609, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 4/46, D_loss_real = 0.628, D_loss_fake = 0.608, G_loss = 0.819
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 5/46, D_loss_real = 0.668, D_loss_fake = 0.585, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 6/46, D_loss_real = 0.637, D_loss_fake = 0.587, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 7/46, D_loss_real = 0.647, D_loss_fake = 0.580, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 8/46, D_loss_real = 0.646, D_loss_fake = 0.568, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 9/46, D_loss_real = 0.629, D_loss_fake = 0.560, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 10/46, D_loss_real = 0.642, D_loss_fake = 0.556, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 11/46, D_loss_real = 0.669, D_loss_fake = 0.562, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 12/46, D_loss_real = 0.637, D_loss_fake = 0.564, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 13/46, D_loss_real = 0.614, D_loss_fake = 0.587, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 14/46, D_loss_real = 0.608, D_loss_fake = 0.622, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 15/46, D_loss_real = 0.594, D_loss_fake = 0.644, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 16/46, D_loss_real = 0.573, D_loss_fake = 0.706, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 17/46, D_loss_real = 0.575, D_loss_fake = 0.736, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 18/46, D_loss_real = 0.593, D_loss_fake = 0.742, G_loss = 0.668
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 19/46, D_loss_real = 0.586, D_loss_fake = 0.749, G_loss = 0.667
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 20/46, D_loss_real = 0.566, D_loss_fake = 0.740, G_loss = 0.668
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 21/46, D_loss_real = 0.553, D_loss_fake = 0.728, G_loss = 0.668
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 22/46, D_loss_real = 0.557, D_loss_fake = 0.733, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 23/46, D_loss_real = 0.523, D_loss_fake = 0.735, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 24/46, D_loss_real = 0.521, D_loss_fake = 0.733, G_loss = 0.666
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 25/46, D_loss_real = 0.538, D_loss_fake = 0.750, G_loss = 0.664
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 26/46, D_loss_real = 0.563, D_loss_fake = 0.755, G_loss = 0.652
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 27/46, D_loss_real = 0.531, D_loss_fake = 0.764, G_loss = 0.632
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 28/46, D_loss_real = 0.533, D_loss_fake = 0.768, G_loss = 0.628
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 29/46, D_loss_real = 0.541, D_loss_fake = 0.791, G_loss = 0.628
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 30/46, D_loss_real = 0.513, D_loss_fake = 0.774, G_loss = 0.617
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 31/46, D_loss_real = 0.528, D_loss_fake = 0.793, G_loss = 0.628
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 32/46, D_loss_real = 0.540, D_loss_fake = 0.812, G_loss = 0.610
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 33/46, D_loss_real = 0.554, D_loss_fake = 0.791, G_loss = 0.616
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 34/46, D_loss_real = 0.529, D_loss_fake = 0.822, G_loss = 0.608
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 35/46, D_loss_real = 0.533, D_loss_fake = 0.806, G_loss = 0.603
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 36/46, D_loss_real = 0.590, D_loss_fake = 0.828, G_loss = 0.604
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 37/46, D_loss_real = 0.572, D_loss_fake = 0.803, G_loss = 0.601
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 38/46, D_loss_real = 0.538, D_loss_fake = 0.834, G_loss = 0.596
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 39/46, D_loss_real = 0.557, D_loss_fake = 0.823, G_loss = 0.610
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 40/46, D_loss_real = 0.551, D_loss_fake = 0.830, G_loss = 0.597
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 41/46, D_loss_real = 0.556, D_loss_fake = 0.828, G_loss = 0.608
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 42/46, D_loss_real = 0.593, D_loss_fake = 0.819, G_loss = 0.612
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 43/46, D_loss_real = 0.532, D_loss_fake = 0.846, G_loss = 0.608
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 44/46, D_loss_real = 0.600, D_loss_fake = 0.797, G_loss = 0.603
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 45/46, D_loss_real = 0.578, D_loss_fake = 0.794, G_loss = 0.606
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 46/46, D_loss_real = 0.574, D_loss_fake = 0.803, G_loss = 0.615
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 1/46, D_loss_real = 0.571, D_loss_fake = 0.807, G_loss = 0.622
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 2/46, D_loss_real = 0.570, D_loss_fake = 0.796, G_loss = 0.617
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 3/46, D_loss_real = 0.585, D_loss_fake = 0.800, G_loss = 0.620
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 4/46, D_loss_real = 0.581, D_loss_fake = 0.828, G_loss = 0.626
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 5/46, D_loss_real = 0.578, D_loss_fake = 0.779, G_loss = 0.647
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 6/46, D_loss_real = 0.567, D_loss_fake = 0.759, G_loss = 0.646
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 7/46, D_loss_real = 0.595, D_loss_fake = 0.789, G_loss = 0.647
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 8/46, D_loss_real = 0.576, D_loss_fake = 0.761, G_loss = 0.666
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 9/46, D_loss_real = 0.623, D_loss_fake = 0.766, G_loss = 0.665
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 10/46, D_loss_real = 0.627, D_loss_fake = 0.749, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 11/46, D_loss_real = 0.609, D_loss_fake = 0.759, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 12/46, D_loss_real = 0.639, D_loss_fake = 0.738, G_loss = 0.672
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 13/46, D_loss_real = 0.631, D_loss_fake = 0.739, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 14/46, D_loss_real = 0.662, D_loss_fake = 0.757, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 15/46, D_loss_real = 0.643, D_loss_fake = 0.764, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 16/46, D_loss_real = 0.625, D_loss_fake = 0.767, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.720, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 18/46, D_loss_real = 0.641, D_loss_fake = 0.711, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 19/46, D_loss_real = 0.669, D_loss_fake = 0.733, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 20/46, D_loss_real = 0.652, D_loss_fake = 0.730, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 21/46, D_loss_real = 0.672, D_loss_fake = 0.722, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.732, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 23/46, D_loss_real = 0.683, D_loss_fake = 0.710, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.720, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 26/46, D_loss_real = 0.706, D_loss_fake = 0.705, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 27/46, D_loss_real = 0.722, D_loss_fake = 0.695, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 28/46, D_loss_real = 0.735, D_loss_fake = 0.677, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 29/46, D_loss_real = 0.730, D_loss_fake = 0.677, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 30/46, D_loss_real = 0.765, D_loss_fake = 0.673, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 31/46, D_loss_real = 0.743, D_loss_fake = 0.661, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 32/46, D_loss_real = 0.748, D_loss_fake = 0.658, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 33/46, D_loss_real = 0.761, D_loss_fake = 0.663, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 34/46, D_loss_real = 0.737, D_loss_fake = 0.678, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 35/46, D_loss_real = 0.750, D_loss_fake = 0.669, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 36/46, D_loss_real = 0.737, D_loss_fake = 0.667, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 37/46, D_loss_real = 0.772, D_loss_fake = 0.683, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 38/46, D_loss_real = 0.750, D_loss_fake = 0.658, G_loss = 0.745
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 39/46, D_loss_real = 0.745, D_loss_fake = 0.652, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 40/46, D_loss_real = 0.763, D_loss_fake = 0.660, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 41/46, D_loss_real = 0.754, D_loss_fake = 0.662, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 42/46, D_loss_real = 0.757, D_loss_fake = 0.673, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 43/46, D_loss_real = 0.754, D_loss_fake = 0.660, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 44/46, D_loss_real = 0.763, D_loss_fake = 0.653, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 45/46, D_loss_real = 0.781, D_loss_fake = 0.654, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 46/46, D_loss_real = 0.765, D_loss_fake = 0.659, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 1/46, D_loss_real = 0.740, D_loss_fake = 0.664, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 2/46, D_loss_real = 0.752, D_loss_fake = 0.652, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 3/46, D_loss_real = 0.763, D_loss_fake = 0.652, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 4/46, D_loss_real = 0.765, D_loss_fake = 0.640, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 5/46, D_loss_real = 0.770, D_loss_fake = 0.633, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 6/46, D_loss_real = 0.785, D_loss_fake = 0.634, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 7/46, D_loss_real = 0.779, D_loss_fake = 0.647, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 8/46, D_loss_real = 0.763, D_loss_fake = 0.633, G_loss = 0.763
2/2 [=====] - 0s 3ms/step

Epoch 4, Iteration 9/46, D_loss_real = 0.764, D_loss_fake = 0.642, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 10/46, D_loss_real = 0.761, D_loss_fake = 0.634, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 11/46, D_loss_real = 0.763, D_loss_fake = 0.643, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 12/46, D_loss_real = 0.756, D_loss_fake = 0.645, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 13/46, D_loss_real = 0.759, D_loss_fake = 0.642, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 14/46, D_loss_real = 0.764, D_loss_fake = 0.645, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 15/46, D_loss_real = 0.772, D_loss_fake = 0.651, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 16/46, D_loss_real = 0.765, D_loss_fake = 0.645, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 17/46, D_loss_real = 0.762, D_loss_fake = 0.651, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 18/46, D_loss_real = 0.758, D_loss_fake = 0.651, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 19/46, D_loss_real = 0.759, D_loss_fake = 0.650, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 20/46, D_loss_real = 0.753, D_loss_fake = 0.644, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 21/46, D_loss_real = 0.764, D_loss_fake = 0.646, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 22/46, D_loss_real = 0.777, D_loss_fake = 0.658, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 23/46, D_loss_real = 0.759, D_loss_fake = 0.645, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 24/46, D_loss_real = 0.755, D_loss_fake = 0.638, G_loss = 0.752
2/2 [=====] - 0s 3ms/step

Epoch 4, Iteration 25/46, D_loss_real = 0.771, D_loss_fake = 0.654, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 26/46, D_loss_real = 0.770, D_loss_fake = 0.648, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 27/46, D_loss_real = 0.762, D_loss_fake = 0.634, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 28/46, D_loss_real = 0.776, D_loss_fake = 0.641, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 29/46, D_loss_real = 0.767, D_loss_fake = 0.632, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 30/46, D_loss_real = 0.758, D_loss_fake = 0.631, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 31/46, D_loss_real = 0.771, D_loss_fake = 0.621, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 32/46, D_loss_real = 0.760, D_loss_fake = 0.620, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 33/46, D_loss_real = 0.747, D_loss_fake = 0.630, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 34/46, D_loss_real = 0.764, D_loss_fake = 0.618, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 35/46, D_loss_real = 0.756, D_loss_fake = 0.639, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 36/46, D_loss_real = 0.770, D_loss_fake = 0.654, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 37/46, D_loss_real = 0.756, D_loss_fake = 0.672, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 38/46, D_loss_real = 0.766, D_loss_fake = 0.671, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 39/46, D_loss_real = 0.758, D_loss_fake = 0.658, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 40/46, D_loss_real = 0.738, D_loss_fake = 0.651, G_loss = 0.745
2/2 [=====] - 0s 3ms/step

Epoch 4, Iteration 41/46, D_loss_real = 0.720, D_loss_fake = 0.651, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 42/46, D_loss_real = 0.734, D_loss_fake = 0.643, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 43/46, D_loss_real = 0.732, D_loss_fake = 0.639, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 44/46, D_loss_real = 0.708, D_loss_fake = 0.629, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.632, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 46/46, D_loss_real = 0.709, D_loss_fake = 0.640, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 1/46, D_loss_real = 0.697, D_loss_fake = 0.655, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 2/46, D_loss_real = 0.715, D_loss_fake = 0.677, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 3/46, D_loss_real = 0.676, D_loss_fake = 0.714, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.729, G_loss = 0.655
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 5/46, D_loss_real = 0.707, D_loss_fake = 0.748, G_loss = 0.650
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 6/46, D_loss_real = 0.712, D_loss_fake = 0.772, G_loss = 0.670
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 7/46, D_loss_real = 0.712, D_loss_fake = 0.750, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 8/46, D_loss_real = 0.714, D_loss_fake = 0.723, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 9/46, D_loss_real = 0.704, D_loss_fake = 0.681, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 10/46, D_loss_real = 0.712, D_loss_fake = 0.638, G_loss = 0.791
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 11/46, D_loss_real = 0.716, D_loss_fake = 0.610, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 12/46, D_loss_real = 0.720, D_loss_fake = 0.575, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 13/46, D_loss_real = 0.732, D_loss_fake = 0.578, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 14/46, D_loss_real = 0.747, D_loss_fake = 0.571, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 15/46, D_loss_real = 0.759, D_loss_fake = 0.599, G_loss = 0.834
2/2 [=====] - 0s 13ms/step
Epoch 5, Iteration 16/46, D_loss_real = 0.727, D_loss_fake = 0.627, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 17/46, D_loss_real = 0.717, D_loss_fake = 0.670, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 18/46, D_loss_real = 0.735, D_loss_fake = 0.704, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 19/46, D_loss_real = 0.709, D_loss_fake = 0.746, G_loss = 0.659
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 20/46, D_loss_real = 0.710, D_loss_fake = 0.778, G_loss = 0.640
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 21/46, D_loss_real = 0.715, D_loss_fake = 0.796, G_loss = 0.632
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 22/46, D_loss_real = 0.714, D_loss_fake = 0.787, G_loss = 0.647
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 23/46, D_loss_real = 0.731, D_loss_fake = 0.748, G_loss = 0.669
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 24/46, D_loss_real = 0.708, D_loss_fake = 0.714, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 25/46, D_loss_real = 0.725, D_loss_fake = 0.684, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 26/46, D_loss_real = 0.719, D_loss_fake = 0.670, G_loss = 0.754
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.639, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 28/46, D_loss_real = 0.720, D_loss_fake = 0.617, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.603, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 30/46, D_loss_real = 0.725, D_loss_fake = 0.602, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 31/46, D_loss_real = 0.725, D_loss_fake = 0.620, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 32/46, D_loss_real = 0.710, D_loss_fake = 0.628, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 33/46, D_loss_real = 0.711, D_loss_fake = 0.642, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 34/46, D_loss_real = 0.713, D_loss_fake = 0.696, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 35/46, D_loss_real = 0.722, D_loss_fake = 0.703, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 36/46, D_loss_real = 0.719, D_loss_fake = 0.720, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 37/46, D_loss_real = 0.703, D_loss_fake = 0.738, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 38/46, D_loss_real = 0.731, D_loss_fake = 0.726, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 39/46, D_loss_real = 0.746, D_loss_fake = 0.718, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 40/46, D_loss_real = 0.729, D_loss_fake = 0.706, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 41/46, D_loss_real = 0.751, D_loss_fake = 0.682, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 42/46, D_loss_real = 0.736, D_loss_fake = 0.666, G_loss = 0.750
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 43/46, D_loss_real = 0.739, D_loss_fake = 0.637, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 44/46, D_loss_real = 0.751, D_loss_fake = 0.646, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 45/46, D_loss_real = 0.768, D_loss_fake = 0.627, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 46/46, D_loss_real = 0.752, D_loss_fake = 0.612, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 1/46, D_loss_real = 0.760, D_loss_fake = 0.623, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 2/46, D_loss_real = 0.753, D_loss_fake = 0.615, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 3/46, D_loss_real = 0.757, D_loss_fake = 0.623, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 4/46, D_loss_real = 0.759, D_loss_fake = 0.629, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 5/46, D_loss_real = 0.743, D_loss_fake = 0.634, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 6/46, D_loss_real = 0.755, D_loss_fake = 0.641, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 7/46, D_loss_real = 0.735, D_loss_fake = 0.654, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 8/46, D_loss_real = 0.734, D_loss_fake = 0.646, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 9/46, D_loss_real = 0.741, D_loss_fake = 0.654, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 10/46, D_loss_real = 0.743, D_loss_fake = 0.661, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 11/46, D_loss_real = 0.743, D_loss_fake = 0.657, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 12/46, D_loss_real = 0.738, D_loss_fake = 0.650, G_loss = 0.742
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 13/46, D_loss_real = 0.739, D_loss_fake = 0.668, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 14/46, D_loss_real = 0.729, D_loss_fake = 0.662, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 15/46, D_loss_real = 0.722, D_loss_fake = 0.655, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 16/46, D_loss_real = 0.718, D_loss_fake = 0.655, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 17/46, D_loss_real = 0.725, D_loss_fake = 0.656, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 18/46, D_loss_real = 0.715, D_loss_fake = 0.647, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 19/46, D_loss_real = 0.727, D_loss_fake = 0.657, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 20/46, D_loss_real = 0.720, D_loss_fake = 0.657, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 21/46, D_loss_real = 0.723, D_loss_fake = 0.656, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 22/46, D_loss_real = 0.721, D_loss_fake = 0.663, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 23/46, D_loss_real = 0.718, D_loss_fake = 0.670, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 24/46, D_loss_real = 0.711, D_loss_fake = 0.670, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 25/46, D_loss_real = 0.712, D_loss_fake = 0.675, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.677, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.672, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 28/46, D_loss_real = 0.710, D_loss_fake = 0.674, G_loss = 0.724
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 29/46, D_loss_real = 0.699, D_loss_fake = 0.675, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 30/46, D_loss_real = 0.693, D_loss_fake = 0.675, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.703, D_loss_fake = 0.668, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.671, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 35/46, D_loss_real = 0.717, D_loss_fake = 0.672, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 36/46, D_loss_real = 0.693, D_loss_fake = 0.668, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 37/46, D_loss_real = 0.703, D_loss_fake = 0.675, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.665, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.663, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.691, D_loss_fake = 0.661, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.667, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.712, D_loss_fake = 0.659, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.655, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.705, D_loss_fake = 0.658, G_loss = 0.737
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 45/46, D_loss_real = 0.700, D_loss_fake = 0.666, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.666, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.670, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.687, D_loss_fake = 0.672, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.685, D_loss_fake = 0.684, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.670, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.674, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.673, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.698, D_loss_fake = 0.673, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.689, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.678, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.682, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss = 0.720
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 15/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.669, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.699, D_loss_fake = 0.677, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.708, D_loss_fake = 0.674, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.687, D_loss_fake = 0.677, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.711, D_loss_fake = 0.668, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.675, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.660, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.717, D_loss_fake = 0.661, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.732, D_loss_fake = 0.668, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.724, D_loss_fake = 0.657, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.718, D_loss_fake = 0.662, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.656, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.727, D_loss_fake = 0.649, G_loss = 0.754
2/2 [=====] - 0s 12ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.690, D_loss_fake = 0.653, G_loss = 0.751
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 31/46, D_loss_real = 0.727, D_loss_fake = 0.674, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.740, D_loss_fake = 0.668, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.734, D_loss_fake = 0.678, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.730, D_loss_fake = 0.680, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.727, D_loss_fake = 0.705, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.735, D_loss_fake = 0.698, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.718, D_loss_fake = 0.672, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.723, D_loss_fake = 0.654, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.726, D_loss_fake = 0.654, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.729, D_loss_fake = 0.639, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.726, D_loss_fake = 0.627, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.727, D_loss_fake = 0.621, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.727, D_loss_fake = 0.622, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.628, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.730, D_loss_fake = 0.637, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.716, D_loss_fake = 0.639, G_loss = 0.761
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 1/46, D_loss_real = 0.731, D_loss_fake = 0.665, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.726, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.713, D_loss_fake = 0.722, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.720, D_loss_fake = 0.718, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.720, G_loss = 0.668
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.720, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.694, D_loss_fake = 0.717, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.721, D_loss_fake = 0.715, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.720, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.697, D_loss_fake = 0.699, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.680, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.715, D_loss_fake = 0.669, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.713, D_loss_fake = 0.658, G_loss = 0.743
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 17/46, D_loss_real = 0.704, D_loss_fake = 0.646, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.642, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.724, D_loss_fake = 0.636, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.722, D_loss_fake = 0.632, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.706, D_loss_fake = 0.621, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.720, D_loss_fake = 0.624, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.725, D_loss_fake = 0.616, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.681, D_loss_fake = 0.629, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.719, D_loss_fake = 0.628, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.736, D_loss_fake = 0.640, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.709, D_loss_fake = 0.685, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.749, D_loss_fake = 0.694, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.722, D_loss_fake = 0.706, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.710, D_loss_fake = 0.724, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.713, D_loss_fake = 0.749, G_loss = 0.668
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 33/46, D_loss_real = 0.703, D_loss_fake = 0.742, G_loss = 0.672
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.726, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.718, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.719, D_loss_fake = 0.727, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.667, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.655, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.706, D_loss_fake = 0.650, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.656, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.649, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.702, D_loss_fake = 0.662, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.675, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.673, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.703, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.714, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.683, D_loss_fake = 0.727, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.728, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.718, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.719, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.720, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.724, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.716, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.712, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.704, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.709, D_loss_fake = 0.692, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.703, D_loss_fake = 0.680, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.679, G_loss = 0.719
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.677, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.702, D_loss_fake = 0.687, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.716, D_loss_fake = 0.706, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.714, D_loss_fake = 0.709, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.703, D_loss_fake = 0.725, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.705, D_loss_fake = 0.717, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.711, D_loss_fake = 0.730, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.698, D_loss_fake = 0.709, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.702, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.706, D_loss_fake = 0.704, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.701, D_loss_fake = 0.708, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 35/46, D_loss_real = 0.711, D_loss_fake = 0.707, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.703, D_loss_fake = 0.709, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.704, D_loss_fake = 0.702, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.703, D_loss_fake = 0.701, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.707, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.707, D_loss_fake = 0.706, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.705, D_loss_fake = 0.696, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.704, D_loss_fake = 0.695, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.700, D_loss_fake = 0.686, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.713, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.713, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 5/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.705, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.708, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.714, G_loss = 0.676
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.708, D_loss_fake = 0.710, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.711, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.700, D_loss_fake = 0.715, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.708, D_loss_fake = 0.711, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.697, D_loss_fake = 0.714, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.712, D_loss_fake = 0.707, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.704, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.709, D_loss_fake = 0.699, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.723
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 21/46, D_loss_real = 0.708, D_loss_fake = 0.674, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.714, D_loss_fake = 0.669, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.706, D_loss_fake = 0.670, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.721, D_loss_fake = 0.663, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.714, D_loss_fake = 0.666, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.714, D_loss_fake = 0.664, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.676, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.686, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.711, D_loss_fake = 0.681, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.738, D_loss_fake = 0.684, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.728, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.731, D_loss_fake = 0.697, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.724, D_loss_fake = 0.705, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.723, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.720, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 37/46, D_loss_real = 0.715, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.708, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.704, D_loss_fake = 0.705, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.709, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.706, D_loss_fake = 0.668, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.713, D_loss_fake = 0.677, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.706, D_loss_fake = 0.676, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 7/46, D_loss_real = 0.703, D_loss_fake = 0.705, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.697, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.705, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.704, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.705, D_loss_fake = 0.722, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.694, D_loss_fake = 0.718, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.706, D_loss_fake = 0.706, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.705, D_loss_fake = 0.709, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.702, D_loss_fake = 0.696, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.704, D_loss_fake = 0.681, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.709, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.707, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.707, D_loss_fake = 0.673, G_loss = 0.728
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 23/46, D_loss_real = 0.715, D_loss_fake = 0.670, G_loss = 0.724
2/2 [=====] - 0s 5ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.708, D_loss_fake = 0.658, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.713, D_loss_fake = 0.664, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.716, D_loss_fake = 0.662, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.663, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.711, D_loss_fake = 0.660, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.714, D_loss_fake = 0.665, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.717, D_loss_fake = 0.662, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.716, D_loss_fake = 0.667, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.715, D_loss_fake = 0.668, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.702, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.705, D_loss_fake = 0.676, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.709, D_loss_fake = 0.685, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.709, D_loss_fake = 0.687, G_loss = 0.708
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 39/46, D_loss_real = 0.713, D_loss_fake = 0.685, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.710, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.713, D_loss_fake = 0.684, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.710, D_loss_fake = 0.676, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.712, D_loss_fake = 0.679, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.712, D_loss_fake = 0.676, G_loss = 0.718
2/2 [=====] - 0s 5ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.712, D_loss_fake = 0.672, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.707, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.713, D_loss_fake = 0.671, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.705, D_loss_fake = 0.670, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.671, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.710, D_loss_fake = 0.676, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.711, D_loss_fake = 0.673, G_loss = 0.720
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 9/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.705, D_loss_fake = 0.671, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.712, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.702, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.703, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.703, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.705, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 25/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.681, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.674, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.674, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.679, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.702, D_loss_fake = 0.670, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.712, D_loss_fake = 0.680, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.704, D_loss_fake = 0.678, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.706, D_loss_fake = 0.688, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.707, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.682, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.678, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.700, D_loss_fake = 0.676, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.700, D_loss_fake = 0.676, G_loss = 0.713
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 11/46, D_loss_real = 0.703, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.703, D_loss_fake = 0.679, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.706, D_loss_fake = 0.695, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.709
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 27/46, D_loss_real = 0.702, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.705, D_loss_fake = 0.682, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.704, D_loss_fake = 0.673, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.670, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.675, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.706, D_loss_fake = 0.670, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.711, D_loss_fake = 0.670, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.669, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.712, D_loss_fake = 0.675, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.673, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.674, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.708, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.706, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 43/46, D_loss_real = 0.711, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.705, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.708, D_loss_fake = 0.693, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.703, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.704, D_loss_fake = 0.690, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.680, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.700, D_loss_fake = 0.680, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.703, D_loss_fake = 0.679, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.710
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.674, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.707, D_loss_fake = 0.676, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.676, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.704, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.702, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.708, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.701, D_loss_fake = 0.680, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 45/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.699, D_loss_fake = 0.697, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.702, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.703, D_loss_fake = 0.704, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.704, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.697, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.704, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 31/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.701, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.702, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.689, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.678, G_loss = 0.716
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 1/46, D_loss_real = 0.699, D_loss_fake = 0.673, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.677, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.674, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.699, D_loss_fake = 0.680, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.701, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.681, D_loss_fake = 0.694, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.706, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.682, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.708, G_loss = 0.688
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.705, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.691, D_loss_fake = 0.705, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.679, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.681, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.679, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 35/46, D_loss_real = 0.690, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.709, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.709, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.698, D_loss_fake = 0.712, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.684, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.705, G_loss = 0.686
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 5/46, D_loss_real = 0.683, D_loss_fake = 0.712, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.716, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.708, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.673, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.701, D_loss_fake = 0.678, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.705, D_loss_fake = 0.685, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 21/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.705, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.713, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.704, D_loss_fake = 0.678, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.680, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.677, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.709, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.708
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.681, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.706, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 4ms/step

Epoch 19, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.683, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.682, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.688, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 39/46, D_loss_real = 0.704, D_loss_fake = 0.681, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.713, D_loss_fake = 0.677, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.680, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.700, D_loss_fake = 0.681, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 9/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.701, D_loss_fake = 0.691, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.680, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.679, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.675, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.673, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.679, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.714, D_loss_fake = 0.692, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.680, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.712, D_loss_fake = 0.683, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.672, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.704, D_loss_fake = 0.671, G_loss = 0.726
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 41/46, D_loss_real = 0.712, D_loss_fake = 0.667, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.664, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.707, D_loss_fake = 0.669, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.714, D_loss_fake = 0.665, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.714, D_loss_fake = 0.675, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.669, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.674, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.710, D_loss_fake = 0.677, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 11/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.709, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.710, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.715, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.686, D_loss_fake = 0.715, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.717, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.709, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.690, D_loss_fake = 0.702, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.685, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.692, D_loss_fake = 0.680, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.694, D_loss_fake = 0.677, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.681, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.688, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.682, D_loss_fake = 0.708, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.707, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.716, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.721, G_loss = 0.679
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 13/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.707, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.693, D_loss_fake = 0.706, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.714, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.704, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.679, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.675, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 12ms/step

Epoch 22, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.677, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.686, D_loss_fake = 0.706, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.717, G_loss = 0.683
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.682, D_loss_fake = 0.710, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.712, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.691, D_loss_fake = 0.716, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.713, G_loss = 0.678
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.714, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.714, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.710, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.680, D_loss_fake = 0.696, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.675, D_loss_fake = 0.704, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.706, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.674, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.676, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.700, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.711, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.678, D_loss_fake = 0.711, G_loss = 0.684
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 15/46, D_loss_real = 0.681, D_loss_fake = 0.702, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.674, D_loss_fake = 0.715, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.673, D_loss_fake = 0.711, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.673, D_loss_fake = 0.717, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.670, D_loss_fake = 0.722, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.663, D_loss_fake = 0.720, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.679, D_loss_fake = 0.721, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.677, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.681, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.679, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.676, D_loss_fake = 0.713, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.712, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.685, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.671, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.708, D_loss_fake = 0.666, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.669, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.673, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.703, D_loss_fake = 0.672, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.679, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.695, D_loss_fake = 0.672, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.707
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 1/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.704, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.702, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.704, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.703, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.698, D_loss_fake = 0.700, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 5ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.697, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.693, D_loss_fake = 0.680, G_loss = 0.714
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.706, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.688, D_loss_fake = 0.709, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.691, D_loss_fake = 0.707, G_loss = 0.688
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 33/46, D_loss_real = 0.688, D_loss_fake = 0.702, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.697, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 19/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.708, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.687, D_loss_fake = 0.708, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.682, D_loss_fake = 0.713, G_loss = 0.686
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.707, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.719, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.676, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.712, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.675, D_loss_fake = 0.714, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.677, D_loss_fake = 0.708, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.673, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.670, D_loss_fake = 0.693, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.681, D_loss_fake = 0.702, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.676, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.683, D_loss_fake = 0.705, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.672, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.721, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.701, D_loss_fake = 0.714, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.709, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.685, D_loss_fake = 0.714, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.707, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.710, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.709, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.664, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.673, G_loss = 0.724
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 21/46, D_loss_real = 0.703, D_loss_fake = 0.678, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.669, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.697, D_loss_fake = 0.700, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.698, D_loss_fake = 0.704, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.703, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.707, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.703, D_loss_fake = 0.713, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.712, D_loss_fake = 0.685, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.701, D_loss_fake = 0.678, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.703, D_loss_fake = 0.673, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.705, D_loss_fake = 0.668, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.682, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.671, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.720, D_loss_fake = 0.679, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.721, D_loss_fake = 0.676, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.711, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 7/46, D_loss_real = 0.708, D_loss_fake = 0.690, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.714, D_loss_fake = 0.681, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.707, D_loss_fake = 0.682, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.714, D_loss_fake = 0.689, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.707, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.717, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.677, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.713, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.713, D_loss_fake = 0.672, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.707, D_loss_fake = 0.675, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.710, D_loss_fake = 0.677, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.709, D_loss_fake = 0.677, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.707, D_loss_fake = 0.675, G_loss = 0.715
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.675, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.705, D_loss_fake = 0.678, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.677, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.680, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.691, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.688, D_loss_fake = 0.702, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 9/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.702, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.704, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.702, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.702, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.678, D_loss_fake = 0.704, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 11ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.675, D_loss_fake = 0.682, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.694, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.692, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.681, D_loss_fake = 0.722, G_loss = 0.674
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.718, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss = 0.682
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.722, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.724, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.724, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.693, D_loss_fake = 0.718, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.712, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.674, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.672, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.668, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.673, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.674, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.676, G_loss = 0.727
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 11/46, D_loss_real = 0.704, D_loss_fake = 0.666, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.704, D_loss_fake = 0.674, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.706, D_loss_fake = 0.697, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.707, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.687, D_loss_fake = 0.708, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.709, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.700, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 27/46, D_loss_real = 0.703, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.676, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.695, D_loss_fake = 0.674, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.679, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.675, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.701, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.702, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.705, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.697, D_loss_fake = 0.704, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.675, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.703, D_loss_fake = 0.680, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.685, D_loss_fake = 0.679, G_loss = 0.715
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.680, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.703, D_loss_fake = 0.694, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.699, D_loss_fake = 0.677, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.710, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.678, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.707, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.702, D_loss_fake = 0.689, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

```

Epoch 30, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 77ms/step

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet
to be built. `model.compile_metrics` will be empty until you train or evaluate
the model.

>Saved: plot_004x004-tuned.png, g_model_004x004-tuned.sav,
d_model_004x004-tuned.sav and gan_model_004x004-tuned.sav
Scaled Data (6000, 8, 8, 3)
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 1/46, D_loss_real = 0.700, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 3/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 5/46, D_loss_real = 0.676, D_loss_fake = 0.704, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 6/46, D_loss_real = 0.669, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 7/46, D_loss_real = 0.673, D_loss_fake = 0.705, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 8/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 9/46, D_loss_real = 0.661, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 10/46, D_loss_real = 0.657, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.710, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 12/46, D_loss_real = 0.668, D_loss_fake = 0.712, G_loss =

```

```

0.684
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 13/46, D_loss_real = 0.658, D_loss_fake = 0.729, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 14/46, D_loss_real = 0.667, D_loss_fake = 0.729, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 15/46, D_loss_real = 0.669, D_loss_fake = 0.742, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 16/46, D_loss_real = 0.666, D_loss_fake = 0.733, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 17/46, D_loss_real = 0.680, D_loss_fake = 0.740, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 18/46, D_loss_real = 0.674, D_loss_fake = 0.720, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 19/46, D_loss_real = 0.664, D_loss_fake = 0.708, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 20/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.698, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 22/46, D_loss_real = 0.676, D_loss_fake = 0.686, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 23/46, D_loss_real = 0.677, D_loss_fake = 0.691, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 24/46, D_loss_real = 0.672, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 25/46, D_loss_real = 0.677, D_loss_fake = 0.680, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 26/46, D_loss_real = 0.659, D_loss_fake = 0.689, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 27/46, D_loss_real = 0.664, D_loss_fake = 0.694, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 28/46, D_loss_real = 0.679, D_loss_fake = 0.723, G_loss =

```

```

0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 29/46, D_loss_real = 0.670, D_loss_fake = 0.718, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 30/46, D_loss_real = 0.673, D_loss_fake = 0.727, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.758, G_loss =
0.651
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 32/46, D_loss_real = 0.671, D_loss_fake = 0.744, G_loss =
0.641
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 33/46, D_loss_real = 0.670, D_loss_fake = 0.754, G_loss =
0.646
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.748, G_loss =
0.655
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.740, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 36/46, D_loss_real = 0.677, D_loss_fake = 0.735, G_loss =
0.662
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.727, G_loss =
0.668
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 38/46, D_loss_real = 0.681, D_loss_fake = 0.724, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 39/46, D_loss_real = 0.680, D_loss_fake = 0.715, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 40/46, D_loss_real = 0.686, D_loss_fake = 0.710, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 41/46, D_loss_real = 0.691, D_loss_fake = 0.702, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.686, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 44/46, D_loss_real = 0.682, D_loss_fake = 0.675, G_loss =

```

```

0.723
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 45/46, D_loss_real = 0.686, D_loss_fake = 0.668, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 46/46, D_loss_real = 0.690, D_loss_fake = 0.660, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.653, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.640, G_loss =
0.773
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.641, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 4/46, D_loss_real = 0.721, D_loss_fake = 0.620, G_loss =
0.773
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.653, G_loss =
0.773
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.657, G_loss =
0.763
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 7/46, D_loss_real = 0.719, D_loss_fake = 0.672, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.709, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 10/46, D_loss_real = 0.709, D_loss_fake = 0.685, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 11/46, D_loss_real = 0.709, D_loss_fake = 0.720, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 12/46, D_loss_real = 0.704, D_loss_fake = 0.721, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 13/46, D_loss_real = 0.706, D_loss_fake = 0.717, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 14/46, D_loss_real = 0.706, D_loss_fake = 0.741, G_loss =

```

```

0.684
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 15/46, D_loss_real = 0.707, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 16/46, D_loss_real = 0.705, D_loss_fake = 0.711, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.693, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 18/46, D_loss_real = 0.713, D_loss_fake = 0.690, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 19/46, D_loss_real = 0.703, D_loss_fake = 0.677, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 20/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 21/46, D_loss_real = 0.706, D_loss_fake = 0.661, G_loss =
0.738
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 22/46, D_loss_real = 0.710, D_loss_fake = 0.660, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 23/46, D_loss_real = 0.710, D_loss_fake = 0.665, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.659, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.669, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 26/46, D_loss_real = 0.714, D_loss_fake = 0.679, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 27/46, D_loss_real = 0.686, D_loss_fake = 0.685, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 28/46, D_loss_real = 0.718, D_loss_fake = 0.684, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.700, G_loss =

```

```

0.693
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 31/46, D_loss_real = 0.693, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 33/46, D_loss_real = 0.689, D_loss_fake = 0.713, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.708, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.706, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.717, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.724, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 38/46, D_loss_real = 0.687, D_loss_fake = 0.714, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 39/46, D_loss_real = 0.697, D_loss_fake = 0.706, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.702, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 42/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss =

```



```

0.709
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 1/46, D_loss_real = 0.689, D_loss_fake = 0.682, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.678, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 4/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.683, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 8/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 10/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 11/46, D_loss_real = 0.696, D_loss_fake = 0.715, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.712, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.700, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.693, G_loss =

```

```

0.685
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 18/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 21/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 26/46, D_loss_real = 0.680, D_loss_fake = 0.705, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 27/46, D_loss_real = 0.677, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 5ms/step
Epoch 3, Iteration 28/46, D_loss_real = 0.676, D_loss_fake = 0.704, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.704, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 30/46, D_loss_real = 0.681, D_loss_fake = 0.710, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 31/46, D_loss_real = 0.678, D_loss_fake = 0.702, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.707, G_loss =

```

```

0.684
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 33/46, D_loss_real = 0.680, D_loss_fake = 0.711, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.713, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 35/46, D_loss_real = 0.687, D_loss_fake = 0.708, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.709, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 37/46, D_loss_real = 0.681, D_loss_fake = 0.706, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 38/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.711, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 41/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.697, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 46/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss =

```

```

0.692
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.690, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 9/46, D_loss_real = 0.682, D_loss_fake = 0.685, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.674, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 11/46, D_loss_real = 0.675, D_loss_fake = 0.683, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 12/46, D_loss_real = 0.674, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.684, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 15/46, D_loss_real = 0.665, D_loss_fake = 0.701, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 16/46, D_loss_real = 0.654, D_loss_fake = 0.708, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 17/46, D_loss_real = 0.681, D_loss_fake = 0.714, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 18/46, D_loss_real = 0.671, D_loss_fake = 0.753, G_loss =

```

```

0.665
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.753, G_loss =
0.657
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 20/46, D_loss_real = 0.668, D_loss_fake = 0.748, G_loss =
0.652
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 21/46, D_loss_real = 0.661, D_loss_fake = 0.752, G_loss =
0.654
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.742, G_loss =
0.662
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 23/46, D_loss_real = 0.663, D_loss_fake = 0.731, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 24/46, D_loss_real = 0.662, D_loss_fake = 0.701, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 25/46, D_loss_real = 0.654, D_loss_fake = 0.699, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 26/46, D_loss_real = 0.659, D_loss_fake = 0.687, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.673, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 28/46, D_loss_real = 0.654, D_loss_fake = 0.688, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 29/46, D_loss_real = 0.655, D_loss_fake = 0.679, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 30/46, D_loss_real = 0.641, D_loss_fake = 0.706, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 31/46, D_loss_real = 0.648, D_loss_fake = 0.730, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 32/46, D_loss_real = 0.659, D_loss_fake = 0.751, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 33/46, D_loss_real = 0.675, D_loss_fake = 0.770, G_loss =
0.652
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 34/46, D_loss_real = 0.677, D_loss_fake = 0.779, G_loss =

```

```

0.646
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.766, G_loss =
0.641
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 36/46, D_loss_real = 0.685, D_loss_fake = 0.758, G_loss =
0.653
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.747, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.724, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.710, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 43/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 44/46, D_loss_real = 0.707, D_loss_fake = 0.668, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 45/46, D_loss_real = 0.703, D_loss_fake = 0.657, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.660, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 1/46, D_loss_real = 0.712, D_loss_fake = 0.659, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 2/46, D_loss_real = 0.712, D_loss_fake = 0.654, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 3/46, D_loss_real = 0.717, D_loss_fake = 0.664, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 4/46, D_loss_real = 0.726, D_loss_fake = 0.670, G_loss =

```

```

0.727
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 5/46, D_loss_real = 0.720, D_loss_fake = 0.670, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 6/46, D_loss_real = 0.720, D_loss_fake = 0.673, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 7/46, D_loss_real = 0.727, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 8/46, D_loss_real = 0.717, D_loss_fake = 0.677, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 9/46, D_loss_real = 0.716, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 10/46, D_loss_real = 0.718, D_loss_fake = 0.680, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 11/46, D_loss_real = 0.716, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 12/46, D_loss_real = 0.716, D_loss_fake = 0.671, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 13/46, D_loss_real = 0.713, D_loss_fake = 0.678, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 14/46, D_loss_real = 0.718, D_loss_fake = 0.679, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 15/46, D_loss_real = 0.711, D_loss_fake = 0.679, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 16/46, D_loss_real = 0.718, D_loss_fake = 0.672, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 17/46, D_loss_real = 0.715, D_loss_fake = 0.680, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 18/46, D_loss_real = 0.711, D_loss_fake = 0.678, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 19/46, D_loss_real = 0.710, D_loss_fake = 0.679, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 20/46, D_loss_real = 0.710, D_loss_fake = 0.679, G_loss =

```

```

0.716
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 21/46, D_loss_real = 0.703, D_loss_fake = 0.674, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 23/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 24/46, D_loss_real = 0.703, D_loss_fake = 0.687, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 25/46, D_loss_real = 0.707, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.686, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.697, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 30/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 31/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.705, G_loss =

```



```

0.691
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 38/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 41/46, D_loss_real = 0.679, D_loss_fake = 0.695, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 43/46, D_loss_real = 0.678, D_loss_fake = 0.693, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 44/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 45/46, D_loss_real = 0.676, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 46/46, D_loss_real = 0.672, D_loss_fake = 0.706, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.710, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 2/46, D_loss_real = 0.656, D_loss_fake = 0.706, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 3/46, D_loss_real = 0.665, D_loss_fake = 0.723, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 4/46, D_loss_real = 0.664, D_loss_fake = 0.725, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 5/46, D_loss_real = 0.658, D_loss_fake = 0.728, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 6/46, D_loss_real = 0.655, D_loss_fake = 0.733, G_loss =

```

```

0.664
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 7/46, D_loss_real = 0.667, D_loss_fake = 0.736, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 8/46, D_loss_real = 0.668, D_loss_fake = 0.732, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.729, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 10/46, D_loss_real = 0.674, D_loss_fake = 0.711, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 11/46, D_loss_real = 0.674, D_loss_fake = 0.707, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.705, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 13/46, D_loss_real = 0.679, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 14/46, D_loss_real = 0.678, D_loss_fake = 0.693, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 15/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.680, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.676, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.678, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.675, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 21/46, D_loss_real = 0.674, D_loss_fake = 0.687, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.681, G_loss =

```

```

0.697
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 23/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 25/46, D_loss_real = 0.695, D_loss_fake = 0.727, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.731, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.712, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.717, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.719, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 30/46, D_loss_real = 0.693, D_loss_fake = 0.725, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.723, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.709, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.683, D_loss_fake = 0.714, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 35/46, D_loss_real = 0.700, D_loss_fake = 0.676, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.674, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 37/46, D_loss_real = 0.709, D_loss_fake = 0.668, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.699, D_loss_fake = 0.660, G_loss =

```

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0.750
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.700, D_loss_fake = 0.665, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.658, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.701, D_loss_fake = 0.669, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.704, D_loss_fake = 0.662, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.684, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.702, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.674, D_loss_fake = 0.694, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.674, D_loss_fake = 0.709, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.712, G_loss =

```

```

0.678
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.661, D_loss_fake = 0.720, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.726, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.671, D_loss_fake = 0.708, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.672, D_loss_fake = 0.728, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.661, D_loss_fake = 0.729, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.659, D_loss_fake = 0.731, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.651, D_loss_fake = 0.728, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.655, D_loss_fake = 0.727, G_loss =
0.660
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.656, D_loss_fake = 0.733, G_loss =
0.652
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.652, D_loss_fake = 0.764, G_loss =
0.660
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.646, D_loss_fake = 0.783, G_loss =
0.666
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.665, D_loss_fake = 0.740, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.728, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.679, D_loss_fake = 0.715, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss =

```

```

0.699
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.690, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.675, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.668, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.666, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.717, D_loss_fake = 0.672, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.708, D_loss_fake = 0.682, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.683, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.686, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.708, D_loss_fake = 0.686, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.704, G_loss =

```

```

0.691
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.711, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.710, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.709, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.713, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.674, D_loss_fake = 0.714, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.675, D_loss_fake = 0.719, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.672, D_loss_fake = 0.715, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.669, D_loss_fake = 0.721, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.713, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.670, D_loss_fake = 0.711, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.666, D_loss_fake = 0.712, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.672, D_loss_fake = 0.705, G_loss =

```

```

0.690
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.669, D_loss_fake = 0.712, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.671, D_loss_fake = 0.702, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.669, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.700, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.693, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.676, D_loss_fake = 0.700, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.696, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.700, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.705, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.675, D_loss_fake = 0.704, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.672, D_loss_fake = 0.709, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.674, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.672, D_loss_fake = 0.707, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.673, D_loss_fake = 0.706, G_loss =

```



```

0.691
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.673, D_loss_fake = 0.705, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.703, D_loss_fake = 0.683, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.670, G_loss =

```

```

0.727
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.668, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.674, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.671, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.679, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.671, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.701, D_loss_fake = 0.668, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.716, D_loss_fake = 0.659, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.711, D_loss_fake = 0.653, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.715, D_loss_fake = 0.689, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.704, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.715, D_loss_fake = 0.717, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.700, D_loss_fake = 0.702, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.711, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.703, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.706, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.705, G_loss =

```

```

0.687
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.679, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.684, D_loss_fake = 0.706, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.666, D_loss_fake = 0.701, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.676, D_loss_fake = 0.706, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.662, D_loss_fake = 0.705, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.657, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.649, D_loss_fake = 0.720, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.628, D_loss_fake = 0.729, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.761, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.663, D_loss_fake = 0.765, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.660, D_loss_fake = 0.714, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.700, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.689, G_loss =
0.757
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.676, D_loss_fake = 0.662, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.657, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.666, G_loss =

```

```

0.739
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.676, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.705, D_loss_fake = 0.693, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.698, D_loss_fake = 0.711, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.699, D_loss_fake = 0.722, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.681, D_loss_fake = 0.729, G_loss =
0.667
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.674, D_loss_fake = 0.728, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.667, D_loss_fake = 0.721, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.680, D_loss_fake = 0.717, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.730, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.733, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.682, D_loss_fake = 0.715, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.674, D_loss_fake = 0.693, G_loss =

```

```

0.708
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.682, D_loss_fake = 0.710, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.678, D_loss_fake = 0.713, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.679, D_loss_fake = 0.707, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.688, D_loss_fake = 0.713, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.717, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.673, D_loss_fake = 0.723, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.674, D_loss_fake = 0.721, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.674, D_loss_fake = 0.716, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.715, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.676, D_loss_fake = 0.709, G_loss =

```

```

0.679
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.672, D_loss_fake = 0.706, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.674, D_loss_fake = 0.706, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.674, D_loss_fake = 0.711, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.676, D_loss_fake = 0.707, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.706, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.705, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.705, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.675, D_loss_fake = 0.700, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.712, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.670, G_loss =

```

```

0.734
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.656, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.661, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.713, D_loss_fake = 0.648, G_loss =
0.747
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.714, D_loss_fake = 0.661, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.659, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.669, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.709, D_loss_fake = 0.662, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.720, D_loss_fake = 0.669, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.669, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.707, D_loss_fake = 0.687, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.688, D_loss_fake = 0.685, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.678, D_loss_fake = 0.706, G_loss =

```

```

0.690
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.673, D_loss_fake = 0.705, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.681, D_loss_fake = 0.711, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.667, D_loss_fake = 0.713, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.715, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.716, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.665, D_loss_fake = 0.713, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.660, D_loss_fake = 0.718, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.659, D_loss_fake = 0.723, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.661, D_loss_fake = 0.720, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.655, D_loss_fake = 0.719, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.643, D_loss_fake = 0.715, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.642, D_loss_fake = 0.719, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.651, D_loss_fake = 0.730, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.645, D_loss_fake = 0.725, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.643, D_loss_fake = 0.727, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.647, D_loss_fake = 0.734, G_loss =

```



```

0.667
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.653, D_loss_fake = 0.730, G_loss =
0.662
2/2 [=====] - 0s 5ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.655, D_loss_fake = 0.740, G_loss =
0.661
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.648, D_loss_fake = 0.738, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.645, D_loss_fake = 0.741, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.643, D_loss_fake = 0.730, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.659, D_loss_fake = 0.714, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.662, D_loss_fake = 0.721, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.662, D_loss_fake = 0.709, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.707, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.659, D_loss_fake = 0.697, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.666, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.666, D_loss_fake = 0.693, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.705, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.665, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.665, D_loss_fake = 0.714, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss =

```

```

0.705
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.668, D_loss_fake = 0.701, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.681, D_loss_fake = 0.715, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.730, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.727, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.675, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.657, G_loss =
0.789
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.717, D_loss_fake = 0.632, G_loss =
0.781
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.724, D_loss_fake = 0.655, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.712, D_loss_fake = 0.653, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.717, D_loss_fake = 0.663, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.720, D_loss_fake = 0.669, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.704, D_loss_fake = 0.665, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.727, D_loss_fake = 0.671, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.698, D_loss_fake = 0.678, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.675, G_loss =

```

```

0.729
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.670, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.676, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.704, D_loss_fake = 0.678, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.678, D_loss_fake = 0.696, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.684, D_loss_fake = 0.705, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.672, D_loss_fake = 0.693, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.681, D_loss_fake = 0.713, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.707, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.657, D_loss_fake = 0.711, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.676, D_loss_fake = 0.733, G_loss =

```

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0.667
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.667, D_loss_fake = 0.757, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.672, D_loss_fake = 0.728, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.717, G_loss =
0.668
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.678, D_loss_fake = 0.726, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.679, D_loss_fake = 0.712, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.670, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.698, D_loss_fake = 0.669, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.668, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.683, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.702, D_loss_fake = 0.699, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss =

```

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0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.707, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.678, D_loss_fake = 0.700, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.673, D_loss_fake = 0.704, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.672, D_loss_fake = 0.697, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.673, D_loss_fake = 0.705, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.699, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.672, D_loss_fake = 0.698, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.708, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.723, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.669, D_loss_fake = 0.713, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.675, D_loss_fake = 0.714, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.739, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.684, D_loss_fake = 0.715, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.685, D_loss_fake = 0.728, G_loss =

```

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0.685
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.683, D_loss_fake = 0.717, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.683, D_loss_fake = 0.700, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.688, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.693, D_loss_fake = 0.702, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.699, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.694, D_loss_fake = 0.709, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.721, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.712, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.720, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.722, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.712, G_loss =

```

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0.682
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.707, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.699, D_loss_fake = 0.706, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.701, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.714, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.699, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.699, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.679, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.678, D_loss_fake = 0.683, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =

```

```

0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.691, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.676, D_loss_fake = 0.683, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.680, D_loss_fake = 0.682, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.682, D_loss_fake = 0.681, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.681, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.670, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.667, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.663, D_loss_fake = 0.703, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.663, D_loss_fake = 0.696, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.667, D_loss_fake = 0.710, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.664, D_loss_fake = 0.721, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.670, D_loss_fake = 0.716, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.713, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.692, D_loss_fake = 0.670, G_loss =
0.752
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.707, D_loss_fake = 0.655, G_loss =

```



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0.755
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.716, D_loss_fake = 0.699, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.692, D_loss_fake = 0.710, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.699, D_loss_fake = 0.710, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.711, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.704, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.683, D_loss_fake = 0.666, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.660, G_loss =
0.757
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.660, G_loss =
0.753
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.709, D_loss_fake = 0.664, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.700, D_loss_fake = 0.673, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.719, D_loss_fake = 0.710, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.680, D_loss_fake = 0.711, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.720, G_loss =

```

```

0.680
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.733, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.695, D_loss_fake = 0.735, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.707, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.664, G_loss =
0.760
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.709, D_loss_fake = 0.649, G_loss =
0.769
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.720, D_loss_fake = 0.649, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.720, D_loss_fake = 0.653, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.712, D_loss_fake = 0.654, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.713, D_loss_fake = 0.668, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.723, D_loss_fake = 0.675, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.702, D_loss_fake = 0.690, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.701, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.705, G_loss =

```

```

0.687
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.712, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.708, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.670, D_loss_fake = 0.709, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.675, D_loss_fake = 0.717, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.702, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.704, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.666, D_loss_fake = 0.705, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.666, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.665, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.667, D_loss_fake = 0.698, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.663, D_loss_fake = 0.698, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.666, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.660, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.656, D_loss_fake = 0.711, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.664, D_loss_fake = 0.710, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.675, D_loss_fake = 0.717, G_loss =

```

```

0.680
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.664, D_loss_fake = 0.736, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.671, D_loss_fake = 0.738, G_loss =
0.660
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.661, D_loss_fake = 0.727, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.663, D_loss_fake = 0.739, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.667, D_loss_fake = 0.741, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.678, D_loss_fake = 0.721, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.682, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.680, D_loss_fake = 0.668, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.669, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.683, D_loss_fake = 0.674, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.680, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.673, D_loss_fake = 0.684, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.675, D_loss_fake = 0.731, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.676, D_loss_fake = 0.742, G_loss =

```

```

0.664
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.668, D_loss_fake = 0.748, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.684, D_loss_fake = 0.727, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.724, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.674, D_loss_fake = 0.713, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.680, D_loss_fake = 0.671, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.687, D_loss_fake = 0.652, G_loss =
0.786
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.650, G_loss =
0.774
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.634, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.726, D_loss_fake = 0.652, G_loss =
0.756
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.670, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.729, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.712, D_loss_fake = 0.718, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.700, D_loss_fake = 0.745, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.703, D_loss_fake = 0.722, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.703, D_loss_fake = 0.703, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss =

```

```

0.726
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.671, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.701, D_loss_fake = 0.661, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.697, D_loss_fake = 0.668, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.695, D_loss_fake = 0.667, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.706, D_loss_fake = 0.684, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.713, D_loss_fake = 0.685, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.711, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.719, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.678, D_loss_fake = 0.720, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.665, D_loss_fake = 0.724, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.675, D_loss_fake = 0.732, G_loss =
0.667
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.657, D_loss_fake = 0.734, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.657, D_loss_fake = 0.728, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.664, D_loss_fake = 0.736, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.661, D_loss_fake = 0.720, G_loss =

```

```

0.667
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.660, D_loss_fake = 0.734, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.661, D_loss_fake = 0.727, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.647, D_loss_fake = 0.727, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.663, D_loss_fake = 0.724, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.653, D_loss_fake = 0.723, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.663, D_loss_fake = 0.724, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.673, D_loss_fake = 0.719, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.661, D_loss_fake = 0.715, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.667, D_loss_fake = 0.711, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.670, D_loss_fake = 0.719, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.705, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.694, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss =

```

```

0.713
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.687, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.701, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss =

```



```

0.690
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.677, D_loss_fake = 0.704, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.681, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.696, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.677, D_loss_fake = 0.694, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.683, D_loss_fake = 0.691, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.679, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.690, D_loss_fake = 0.702, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.695, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.664, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.659, D_loss_fake = 0.731, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.705, G_loss =

```

```

0.680
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.678, D_loss_fake = 0.743, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.727, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.717, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss =
0.757
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.721, D_loss_fake = 0.651, G_loss =
0.769
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.730, D_loss_fake = 0.655, G_loss =
0.759
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.738, D_loss_fake = 0.668, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.735, D_loss_fake = 0.669, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.711, D_loss_fake = 0.688, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.710, D_loss_fake = 0.686, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.705, D_loss_fake = 0.696, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.702, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss =

```

```

0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.683, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.676, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.671, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.675, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.705, D_loss_fake = 0.671, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.673, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.694, D_loss_fake = 0.676, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.680, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.670, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.675, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss =

```

```

0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.696, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.676, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.671, D_loss_fake = 0.700, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.682, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.697, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.699, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.678, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.671, D_loss_fake = 0.705, G_loss =

```

```

0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.677, D_loss_fake = 0.696, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.676, D_loss_fake = 0.697, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.674, D_loss_fake = 0.697, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.674, D_loss_fake = 0.713, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.671, D_loss_fake = 0.703, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.678, D_loss_fake = 0.697, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.698, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.698, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.682, D_loss_fake = 0.682, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.681, D_loss_fake = 0.686, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.673, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.681, D_loss_fake = 0.676, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.678, D_loss_fake = 0.674, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.673, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.672, G_loss =

```

```

0.717
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.667, D_loss_fake = 0.702, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.674, D_loss_fake = 0.720, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.673, D_loss_fake = 0.720, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.714, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.654, D_loss_fake = 0.728, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.655, D_loss_fake = 0.743, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.665, D_loss_fake = 0.734, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.678, D_loss_fake = 0.743, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.725, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.665, D_loss_fake = 0.695, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.693, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.674, D_loss_fake = 0.683, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.675, G_loss =

```

```

0.726
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.673, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.670, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.673, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.705, D_loss_fake = 0.693, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.674, D_loss_fake = 0.716, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.740, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.729, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.737, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.684, D_loss_fake = 0.712, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.708, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.683, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.694, D_loss_fake = 0.663, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.636, G_loss =
0.781
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.661, G_loss =
0.792
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.709, D_loss_fake = 0.645, G_loss =

```

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0.783
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.721, D_loss_fake = 0.656, G_loss =
0.763
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.722, D_loss_fake = 0.667, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.715, D_loss_fake = 0.678, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.679, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.704, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.715, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.686, D_loss_fake = 0.707, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.677, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.675, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.669, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.687, D_loss_fake = 0.664, G_loss =

```



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0.733
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.670, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.675, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.670, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.691, D_loss_fake = 0.709, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.701, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.668, D_loss_fake = 0.700, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.669, D_loss_fake = 0.709, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.660, D_loss_fake = 0.720, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.664, D_loss_fake = 0.720, G_loss =

```

```

0.683
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.715, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.662, D_loss_fake = 0.715, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.650, D_loss_fake = 0.712, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.657, D_loss_fake = 0.711, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.719, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.640, D_loss_fake = 0.721, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.660, D_loss_fake = 0.754, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.671, D_loss_fake = 0.735, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.760, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.681, D_loss_fake = 0.743, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.697, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.709, D_loss_fake = 0.674, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.716, D_loss_fake = 0.664, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.724, D_loss_fake = 0.659, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.718, D_loss_fake = 0.663, G_loss =
0.753
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.719, D_loss_fake = 0.662, G_loss =

```

```

0.739
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.719, D_loss_fake = 0.667, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.723, D_loss_fake = 0.673, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.713, D_loss_fake = 0.680, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.680, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.704, D_loss_fake = 0.683, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.701, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss =

```

```

0.693
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.699, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.707, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.675, D_loss_fake = 0.701, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.698, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.665, D_loss_fake = 0.698, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.672, D_loss_fake = 0.727, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.672, D_loss_fake = 0.726, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.676, D_loss_fake = 0.715, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.667, D_loss_fake = 0.719, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.710, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.682, D_loss_fake = 0.710, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss =

```

```

0.701
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.688, D_loss_fake = 0.684, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.678, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.669, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.693, D_loss_fake = 0.710, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.709, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.699, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.683, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.678, D_loss_fake = 0.679, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.684, G_loss =

```

```

0.711
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.707, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.684, D_loss_fake = 0.728, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.683, D_loss_fake = 0.721, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.722, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.722, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.701, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.711, D_loss_fake = 0.670, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.662, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.716, D_loss_fake = 0.661, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.715, D_loss_fake = 0.674, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.668, G_loss =

```

```

0.729
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.716, D_loss_fake = 0.682, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.718, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss =

```

```

0.699
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.702, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.703, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.708, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.713, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.676, D_loss_fake = 0.714, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.676, D_loss_fake = 0.705, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.676, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.707, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.679, D_loss_fake = 0.703, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.681, D_loss_fake = 0.697, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.700, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss =

```



```

0.695
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.679, D_loss_fake = 0.699, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.697, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.677, D_loss_fake = 0.699, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.702, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.677, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.678, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.680, D_loss_fake = 0.702, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.702, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.678, D_loss_fake = 0.709, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.679, D_loss_fake = 0.705, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.709, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.706, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss =

```

```

0.695
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.711, D_loss_fake = 0.671, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.720, D_loss_fake = 0.666, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.718, D_loss_fake = 0.656, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.729, D_loss_fake = 0.662, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.722, D_loss_fake = 0.658, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.730, D_loss_fake = 0.659, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.732, D_loss_fake = 0.668, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.722, D_loss_fake = 0.669, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.721, D_loss_fake = 0.674, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.718, D_loss_fake = 0.671, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.713, D_loss_fake = 0.675, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.711, D_loss_fake = 0.678, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss =

```

```

0.706
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.671, D_loss_fake = 0.696, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.665, D_loss_fake = 0.699, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.663, D_loss_fake = 0.707, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.707, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.641, D_loss_fake = 0.714, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.657, D_loss_fake = 0.711, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.650, D_loss_fake = 0.719, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.710, G_loss =

```

```

0.689
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.667, D_loss_fake = 0.724, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.656, D_loss_fake = 0.703, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.670, D_loss_fake = 0.709, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.646, D_loss_fake = 0.720, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.669, D_loss_fake = 0.735, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.728, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.675, D_loss_fake = 0.726, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.706, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.705, D_loss_fake = 0.678, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.704, D_loss_fake = 0.651, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.715, D_loss_fake = 0.633, G_loss =
0.788
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.736, D_loss_fake = 0.642, G_loss =
0.773
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.738, D_loss_fake = 0.625, G_loss =
0.751
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.743, D_loss_fake = 0.657, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.728, D_loss_fake = 0.664, G_loss =

```

```

0.736
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.723, D_loss_fake = 0.675, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.739, D_loss_fake = 0.677, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.730, D_loss_fake = 0.682, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.725, D_loss_fake = 0.686, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.712, D_loss_fake = 0.691, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.710, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.715, D_loss_fake = 0.692, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.693, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.688, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =

```

```

0.704
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.683, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.684, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.676, D_loss_fake = 0.691, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.669, D_loss_fake = 0.695, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.665, D_loss_fake = 0.706, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.675, D_loss_fake = 0.710, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.665, D_loss_fake = 0.708, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.664, D_loss_fake = 0.715, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.660, D_loss_fake = 0.734, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.666, D_loss_fake = 0.729, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.668, D_loss_fake = 0.737, G_loss =

```

```

0.674
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.671, D_loss_fake = 0.720, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.667, D_loss_fake = 0.726, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.678, D_loss_fake = 0.733, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.678, D_loss_fake = 0.716, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.708, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.712, D_loss_fake = 0.662, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.716, D_loss_fake = 0.657, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.724, D_loss_fake = 0.654, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.727, D_loss_fake = 0.659, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.718, D_loss_fake = 0.656, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.726, D_loss_fake = 0.661, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.723, D_loss_fake = 0.667, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.721, D_loss_fake = 0.672, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.721, D_loss_fake = 0.677, G_loss =

```

```

0.715
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.720, D_loss_fake = 0.677, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.711, D_loss_fake = 0.681, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.714, D_loss_fake = 0.681, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.679, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.680, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss =
0.695
2/2 [=====] - 0s 13ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.662, D_loss_fake = 0.695, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.665, D_loss_fake = 0.699, G_loss =

```



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0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.656, D_loss_fake = 0.703, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.714, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.654, D_loss_fake = 0.701, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.654, D_loss_fake = 0.711, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.646, D_loss_fake = 0.702, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.638, D_loss_fake = 0.729, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.636, D_loss_fake = 0.724, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.636, D_loss_fake = 0.750, G_loss =
0.668
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.626, D_loss_fake = 0.789, G_loss =
0.650
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.648, D_loss_fake = 0.795, G_loss =
0.644
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.654, D_loss_fake = 0.788, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.669, D_loss_fake = 0.745, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.717, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss =
0.761
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.710, D_loss_fake = 0.661, G_loss =

```

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0.787
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.721, D_loss_fake = 0.649, G_loss =
0.787
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.721, D_loss_fake = 0.650, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.731, D_loss_fake = 0.653, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.746, D_loss_fake = 0.679, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.734, D_loss_fake = 0.698, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.722, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.687, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.707, D_loss_fake = 0.699, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.714, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.708, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.703, D_loss_fake = 0.690, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.710, D_loss_fake = 0.681, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.707, D_loss_fake = 0.687, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss =

```

```

0.710
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.678, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.669, D_loss_fake = 0.704, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.674, D_loss_fake = 0.715, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.712, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.657, D_loss_fake = 0.721, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.647, D_loss_fake = 0.731, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.648, D_loss_fake = 0.736, G_loss =
0.660
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.649, D_loss_fake = 0.747, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.627, D_loss_fake = 0.753, G_loss =
0.640
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.637, D_loss_fake = 0.758, G_loss =
0.644
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.647, D_loss_fake = 0.758, G_loss =

```

```

0.656
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.657, D_loss_fake = 0.768, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.657, D_loss_fake = 0.743, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.662, D_loss_fake = 0.719, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.709, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.675, D_loss_fake = 0.696, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.674, D_loss_fake = 0.680, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.671, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.654, G_loss =
0.761
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.651, G_loss =
0.761
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.711, D_loss_fake = 0.660, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.707, D_loss_fake = 0.671, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.706, D_loss_fake = 0.674, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.707, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.706, D_loss_fake = 0.718, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.690, D_loss_fake = 0.716, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.738, G_loss =

```

```

0.660
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.676, D_loss_fake = 0.755, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.741, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.675, D_loss_fake = 0.731, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.725, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.671, D_loss_fake = 0.708, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.672, D_loss_fake = 0.694, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.676, D_loss_fake = 0.689, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.681, D_loss_fake = 0.680, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.682, D_loss_fake = 0.688, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.694, D_loss_fake = 0.709, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.680, D_loss_fake = 0.714, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.700, D_loss_fake = 0.715, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.690, D_loss_fake = 0.723, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss =

```

```

0.680
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.687, D_loss_fake = 0.713, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.679, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.693, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.682, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.686, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.677, D_loss_fake = 0.690, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.673, D_loss_fake = 0.687, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =

```

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0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.675, D_loss_fake = 0.694, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.671, D_loss_fake = 0.695, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.672, D_loss_fake = 0.731, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.669, D_loss_fake = 0.727, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.679, D_loss_fake = 0.726, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.674, D_loss_fake = 0.725, G_loss =
0.661
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.685, D_loss_fake = 0.719, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.679, D_loss_fake = 0.726, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.716, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.712, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.688, D_loss_fake = 0.674, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.678, G_loss =

```

```

0.725
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.665, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.707, D_loss_fake = 0.660, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.717, D_loss_fake = 0.680, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.715, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.702, D_loss_fake = 0.680, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.712, D_loss_fake = 0.685, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.715, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.692, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss =

```



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0.703
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.680, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.679, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.693, D_loss_fake = 0.682, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss =

```

```

0.692
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.704, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.714, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.709, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.708, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.680, D_loss_fake = 0.707, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 5ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.671, D_loss_fake = 0.709, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.683, D_loss_fake = 0.710, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.674, D_loss_fake = 0.703, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.678, D_loss_fake = 0.707, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.677, D_loss_fake = 0.720, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.686, D_loss_fake = 0.711, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.711, G_loss =

```

```

0.683
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.682, D_loss_fake = 0.709, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.711, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.684, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.681, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.676, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.706, D_loss_fake = 0.675, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.673, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.711, D_loss_fake = 0.674, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.718, D_loss_fake = 0.675, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.713, D_loss_fake = 0.673, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.678, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss =

```

```

0.710
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.709, D_loss_fake = 0.683, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.706, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.689, D_loss_fake = 0.684, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.682, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.688, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.676, D_loss_fake = 0.690, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.675, D_loss_fake = 0.699, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.657, D_loss_fake = 0.708, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.665, D_loss_fake = 0.710, G_loss =

```

```

0.690
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.670, D_loss_fake = 0.713, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.667, D_loss_fake = 0.714, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.664, D_loss_fake = 0.723, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.667, D_loss_fake = 0.717, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.718, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.686, D_loss_fake = 0.704, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.707, D_loss_fake = 0.682, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.708, D_loss_fake = 0.673, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.711, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.716, D_loss_fake = 0.675, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.710, D_loss_fake = 0.669, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss =

```

```

0.718
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.694, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.670, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.670, G_loss =

```

```

0.723
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.683, D_loss_fake = 0.669, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.677, D_loss_fake = 0.668, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.678, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.680, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.673, D_loss_fake = 0.695, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.704, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.660, D_loss_fake = 0.727, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.656, D_loss_fake = 0.738, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.660, D_loss_fake = 0.726, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.664, D_loss_fake = 0.722, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.660, D_loss_fake = 0.732, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.662, D_loss_fake = 0.727, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.665, D_loss_fake = 0.728, G_loss =
0.655
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.657, D_loss_fake = 0.760, G_loss =
0.654
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.672, D_loss_fake = 0.787, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.714, G_loss =

```

```

0.694
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.708, D_loss_fake = 0.672, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.709, D_loss_fake = 0.659, G_loss =
0.767
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.710, D_loss_fake = 0.631, G_loss =
0.793
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.725, D_loss_fake = 0.643, G_loss =
0.792
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.727, D_loss_fake = 0.643, G_loss =
0.763
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.724, D_loss_fake = 0.647, G_loss =
0.759
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.743, D_loss_fake = 0.665, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.732, D_loss_fake = 0.668, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.725, D_loss_fake = 0.674, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.718, D_loss_fake = 0.674, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.710, D_loss_fake = 0.680, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.692, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss =

```



```

0.697
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.702, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.722, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.713, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.732, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.667, D_loss_fake = 0.720, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.656, D_loss_fake = 0.738, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.733, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.662, D_loss_fake = 0.788, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.678, D_loss_fake = 0.704, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.723, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 6/46, D_loss_real = 0.688, D_loss_fake = 0.678, G_loss =
0.769
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.671, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss =

```

```

0.719
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.698, D_loss_fake = 0.700, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.713, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.712, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.716, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.711, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.712, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.664, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.659, G_loss =
0.746
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.708, D_loss_fake = 0.663, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 21/46, D_loss_real = 0.711, D_loss_fake = 0.680, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 22/46, D_loss_real = 0.708, D_loss_fake = 0.688, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss =

```

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0.691
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.705, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.693, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.680, D_loss_fake = 0.683, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 39/46, D_loss_real = 0.680, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss =

```

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0.686
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.667, D_loss_fake = 0.709, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.669, D_loss_fake = 0.718, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.673, D_loss_fake = 0.733, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 44/46, D_loss_real = 0.660, D_loss_fake = 0.741, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.667, D_loss_fake = 0.743, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.672, D_loss_fake = 0.734, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.676, D_loss_fake = 0.732, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.715, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.680, D_loss_fake = 0.686, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.664, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 7/46, D_loss_real = 0.698, D_loss_fake = 0.659, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.662, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.670, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss =

```

```

0.715
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.713, D_loss_fake = 0.687, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.694, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.701, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.693, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.693, D_loss_fake = 0.704, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.706, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.708, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.682, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.677, D_loss_fake = 0.706, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.701, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 24/46, D_loss_real = 0.680, D_loss_fake = 0.691, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.675, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.675, D_loss_fake = 0.681, G_loss =

```

```

0.737
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.674, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.665, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.703, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 30/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.674, D_loss_fake = 0.735, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.712, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.689, D_loss_fake = 0.712, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.687, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.689, D_loss_fake = 0.681, G_loss =
0.719
2/2 [=====] - 0s 5ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss =

```

```

0.702
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.705, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.699, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.670, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.676, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.678, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.679, D_loss_fake = 0.693, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.670, D_loss_fake = 0.691, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 9/46, D_loss_real = 0.663, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.697, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 11/46, D_loss_real = 0.656, D_loss_fake = 0.706, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.668, D_loss_fake = 0.715, G_loss =

```

```

0.685
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.662, D_loss_fake = 0.713, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.662, D_loss_fake = 0.721, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.659, D_loss_fake = 0.740, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.728, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.722, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.697, D_loss_fake = 0.666, G_loss =
0.734
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.696, D_loss_fake = 0.658, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.668, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.718, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.722, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.728, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 25/46, D_loss_real = 0.710, D_loss_fake = 0.703, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.719, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 27/46, D_loss_real = 0.698, D_loss_fake = 0.708, G_loss =
0.703
2/2 [=====] - 0s 12ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss =

```



```

0.715
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.690, D_loss_fake = 0.670, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.665, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.648, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.669, G_loss =
0.734
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.699, D_loss_fake = 0.697, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.701, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.705, D_loss_fake = 0.699, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.714, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.710, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.709, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.713, D_loss_fake = 0.683, G_loss =

```

```

0.715
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.714, D_loss_fake = 0.683, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.719, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.718, D_loss_fake = 0.674, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 2/46, D_loss_real = 0.725, D_loss_fake = 0.672, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.670, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.718, D_loss_fake = 0.669, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.713, D_loss_fake = 0.668, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.718, D_loss_fake = 0.663, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.719, D_loss_fake = 0.663, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.718, D_loss_fake = 0.671, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.709, D_loss_fake = 0.670, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.715, D_loss_fake = 0.671, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 11/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 12/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.682, G_loss =

```

```

0.708
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 18/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.694, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.675, D_loss_fake = 0.698, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.676, D_loss_fake = 0.703, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.671, D_loss_fake = 0.716, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.716, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.675, D_loss_fake = 0.720, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.722, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 28/46, D_loss_real = 0.677, D_loss_fake = 0.718, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 29/46, D_loss_real = 0.681, D_loss_fake = 0.710, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss =

```

```

0.710
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.673, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.663, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.659, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.709, D_loss_fake = 0.666, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.711, D_loss_fake = 0.681, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.716, D_loss_fake = 0.688, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.696, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 44/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.692, G_loss =

```

```

0.693
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.704, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.704, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.696, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.677, D_loss_fake = 0.705, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.672, D_loss_fake = 0.705, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.667, D_loss_fake = 0.704, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.715, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.728, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.675, D_loss_fake = 0.699, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.730, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.740, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.714, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.672, G_loss =

```

```

0.739
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.686, D_loss_fake = 0.654, G_loss =
0.783
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.658, G_loss =
0.781
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.689, D_loss_fake = 0.663, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.678, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.677, D_loss_fake = 0.694, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.679, D_loss_fake = 0.713, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.684, D_loss_fake = 0.710, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.716, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.704, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 29/46, D_loss_real = 0.713, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 30/46, D_loss_real = 0.710, D_loss_fake = 0.705, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 31/46, D_loss_real = 0.716, D_loss_fake = 0.677, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.712, D_loss_fake = 0.672, G_loss =

```

```

0.729
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.719, D_loss_fake = 0.663, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.725, D_loss_fake = 0.670, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.716, D_loss_fake = 0.660, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 36/46, D_loss_real = 0.737, D_loss_fake = 0.665, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.723, D_loss_fake = 0.662, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.728, D_loss_fake = 0.675, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.722, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.677, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.720, D_loss_fake = 0.671, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.676, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.711, D_loss_fake = 0.681, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.681, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.676, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 1/46, D_loss_real = 0.710, D_loss_fake = 0.676, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss =

```

```

0.711
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.678, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.703, D_loss_fake = 0.679, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.682, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.692, D_loss_fake = 0.680, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.683, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.688, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.672, D_loss_fake = 0.686, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 15/46, D_loss_real = 0.675, D_loss_fake = 0.687, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 16/46, D_loss_real = 0.661, D_loss_fake = 0.693, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 17/46, D_loss_real = 0.662, D_loss_fake = 0.699, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.660, D_loss_fake = 0.698, G_loss =

```



```

0.692
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.659, D_loss_fake = 0.698, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.640, D_loss_fake = 0.711, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.637, D_loss_fake = 0.725, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 22/46, D_loss_real = 0.627, D_loss_fake = 0.750, G_loss =
0.656
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.638, D_loss_fake = 0.772, G_loss =
0.640
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.645, D_loss_fake = 0.773, G_loss =
0.630
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.644, D_loss_fake = 0.800, G_loss =
0.632
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.658, D_loss_fake = 0.777, G_loss =
0.644
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.661, D_loss_fake = 0.751, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.714, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.656, G_loss =
0.794
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.625, G_loss =
0.816
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 31/46, D_loss_real = 0.711, D_loss_fake = 0.620, G_loss =
0.795
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 32/46, D_loss_real = 0.719, D_loss_fake = 0.646, G_loss =
0.767
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 33/46, D_loss_real = 0.727, D_loss_fake = 0.659, G_loss =
0.751
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.737, D_loss_fake = 0.666, G_loss =

```

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0.736
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.735, D_loss_fake = 0.670, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.723, D_loss_fake = 0.666, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.712, D_loss_fake = 0.674, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 38/46, D_loss_real = 0.714, D_loss_fake = 0.670, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.712, D_loss_fake = 0.680, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.683, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.673, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.679, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.691, D_loss_fake = 0.689, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.671, D_loss_fake = 0.682, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.674, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 1/46, D_loss_real = 0.661, D_loss_fake = 0.687, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 2/46, D_loss_real = 0.650, D_loss_fake = 0.691, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 3/46, D_loss_real = 0.634, D_loss_fake = 0.691, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.612, D_loss_fake = 0.710, G_loss =

```

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0.672
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.622, D_loss_fake = 0.740, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.614, D_loss_fake = 0.732, G_loss =
0.644
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.580, D_loss_fake = 0.770, G_loss =
0.639
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 8/46, D_loss_real = 0.614, D_loss_fake = 0.832, G_loss =
0.613
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.583, D_loss_fake = 0.863, G_loss =
0.601
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.642, D_loss_fake = 0.828, G_loss =
0.603
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.648, D_loss_fake = 0.843, G_loss =
0.637
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.673, D_loss_fake = 0.778, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.702, D_loss_fake = 0.643, G_loss =
0.823
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.733, D_loss_fake = 0.592, G_loss =
0.926
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.740, D_loss_fake = 0.598, G_loss =
0.861
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.790, D_loss_fake = 0.618, G_loss =
0.809
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 17/46, D_loss_real = 0.750, D_loss_fake = 0.636, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 18/46, D_loss_real = 0.762, D_loss_fake = 0.659, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 19/46, D_loss_real = 0.736, D_loss_fake = 0.659, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.751, D_loss_fake = 0.667, G_loss =

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0.721
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.747, D_loss_fake = 0.672, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.731, D_loss_fake = 0.676, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.722, D_loss_fake = 0.680, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 24/46, D_loss_real = 0.724, D_loss_fake = 0.682, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.708, D_loss_fake = 0.680, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.720, D_loss_fake = 0.682, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.708, D_loss_fake = 0.684, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.680, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 33/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss =

```

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0.699
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.689, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.672, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 40/46, D_loss_real = 0.682, D_loss_fake = 0.663, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.678, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.680, D_loss_fake = 0.696, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.681, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.686, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 4/46, D_loss_real = 0.676, D_loss_fake = 0.686, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 5/46, D_loss_real = 0.679, D_loss_fake = 0.684, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.682, G_loss =

```

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0.707
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.673, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.667, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 10/46, D_loss_real = 0.665, D_loss_fake = 0.689, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.677, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.685, D_loss_fake = 0.674, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.673, D_loss_fake = 0.684, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.677, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.681, D_loss_fake = 0.711, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.719, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 20/46, D_loss_real = 0.683, D_loss_fake = 0.722, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.710, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss =

```

```

0.694
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.681, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 26/46, D_loss_real = 0.698, D_loss_fake = 0.671, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.705, D_loss_fake = 0.685, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.718, D_loss_fake = 0.698, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.719, D_loss_fake = 0.689, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.710, D_loss_fake = 0.670, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.714, D_loss_fake = 0.667, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.726, D_loss_fake = 0.668, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.660, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.715, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 36/46, D_loss_real = 0.716, D_loss_fake = 0.685, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.702, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.680, D_loss_fake = 0.712, G_loss =

```

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0.706
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.697, D_loss_fake = 0.698, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.757, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.703, D_loss_fake = 0.672, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.659, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.653, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.709, D_loss_fake = 0.666, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.707, D_loss_fake = 0.669, G_loss =
0.747
2/2 [=====] - 0s 5ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.711, D_loss_fake = 0.690, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.706, D_loss_fake = 0.692, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 5/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 7/46, D_loss_real = 0.699, D_loss_fake = 0.693, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss =

```



```

0.702
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.698, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.684, D_loss_fake = 0.705, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.703, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.694, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.690, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss =

```

```

0.709
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.695, D_loss_fake = 0.673, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.681, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.704, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.701, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 39/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss =

```

```

0.709
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.684, D_loss_fake = 0.691, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.684, D_loss_fake = 0.687, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.683, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.679, D_loss_fake = 0.692, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.678, D_loss_fake = 0.695, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 8/46, D_loss_real = 0.681, D_loss_fake = 0.716, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.720, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.674, D_loss_fake = 0.730, G_loss =

```

```

0.674
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.678, D_loss_fake = 0.715, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.662, D_loss_fake = 0.727, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.714, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.728, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.654, D_loss_fake = 0.808, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.685, D_loss_fake = 0.712, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.684, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.681, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.715, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.719, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.743, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 24/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.670, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.705, D_loss_fake = 0.639, G_loss =

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0.784
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.708, D_loss_fake = 0.620, G_loss =
0.805
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.713, D_loss_fake = 0.641, G_loss =
0.794
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.720, D_loss_fake = 0.666, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 30/46, D_loss_real = 0.721, D_loss_fake = 0.672, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.721, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.708, D_loss_fake = 0.693, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.712, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.673, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.674, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 39/46, D_loss_real = 0.678, D_loss_fake = 0.696, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 40/46, D_loss_real = 0.670, D_loss_fake = 0.707, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 41/46, D_loss_real = 0.664, D_loss_fake = 0.701, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.663, D_loss_fake = 0.694, G_loss =

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0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.658, D_loss_fake = 0.702, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.667, D_loss_fake = 0.690, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.655, D_loss_fake = 0.697, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 46/46, D_loss_real = 0.669, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 76ms/step

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet
to be built. `model.compile_metrics` will be empty until you train or evaluate
the model.

>Saved: plot_008x008-faded.png, g_model_008x008-faded.sav,
d_model_008x008-faded.sav and gan_model_008x008-faded.sav
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 1/46, D_loss_real = 0.660, D_loss_fake = 0.707, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 2/46, D_loss_real = 0.665, D_loss_fake = 0.732, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 3/46, D_loss_real = 0.646, D_loss_fake = 0.764, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 4/46, D_loss_real = 0.655, D_loss_fake = 0.793, G_loss =
0.651
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 5/46, D_loss_real = 0.673, D_loss_fake = 0.791, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.727, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 7/46, D_loss_real = 0.712, D_loss_fake = 0.681, G_loss =
0.772
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 8/46, D_loss_real = 0.709, D_loss_fake = 0.631, G_loss =
0.798
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 9/46, D_loss_real = 0.718, D_loss_fake = 0.628, G_loss =
0.808
2/2 [=====] - 0s 3ms/step

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Epoch 1, Iteration 10/46, D_loss_real = 0.718, D_loss_fake = 0.642, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 11/46, D_loss_real = 0.729, D_loss_fake = 0.657, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 12/46, D_loss_real = 0.722, D_loss_fake = 0.677, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 13/46, D_loss_real = 0.732, D_loss_fake = 0.674, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 14/46, D_loss_real = 0.720, D_loss_fake = 0.671, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 15/46, D_loss_real = 0.724, D_loss_fake = 0.679, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 16/46, D_loss_real = 0.714, D_loss_fake = 0.668, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 17/46, D_loss_real = 0.709, D_loss_fake = 0.665, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 18/46, D_loss_real = 0.711, D_loss_fake = 0.665, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 19/46, D_loss_real = 0.711, D_loss_fake = 0.665, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.672, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.672, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 22/46, D_loss_real = 0.714, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 24/46, D_loss_real = 0.707, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 25/46, D_loss_real = 0.699, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 1, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.697, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 28/46, D_loss_real = 0.697, D_loss_fake = 0.710, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.706, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 30/46, D_loss_real = 0.696, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 31/46, D_loss_real = 0.696, D_loss_fake = 0.705, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.667, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 36/46, D_loss_real = 0.704, D_loss_fake = 0.664, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 37/46, D_loss_real = 0.704, D_loss_fake = 0.661, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 38/46, D_loss_real = 0.712, D_loss_fake = 0.670, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 39/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 40/46, D_loss_real = 0.713, D_loss_fake = 0.678, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 41/46, D_loss_real = 0.705, D_loss_fake = 0.673, G_loss = 0.713
2/2 [=====] - 0s 4ms/step

Epoch 1, Iteration 42/46, D_loss_real = 0.708, D_loss_fake = 0.683, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 44/46, D_loss_real = 0.706, D_loss_fake = 0.675, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 45/46, D_loss_real = 0.706, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 46/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 11/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 12/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 13/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 15/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 17/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.679, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 21/46, D_loss_real = 0.711, D_loss_fake = 0.673, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 22/46, D_loss_real = 0.715, D_loss_fake = 0.682, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 23/46, D_loss_real = 0.713, D_loss_fake = 0.676, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 24/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 25/46, D_loss_real = 0.706, D_loss_fake = 0.675, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.679, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 27/46, D_loss_real = 0.713, D_loss_fake = 0.677, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 28/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 29/46, D_loss_real = 0.712, D_loss_fake = 0.678, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 30/46, D_loss_real = 0.711, D_loss_fake = 0.678, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 31/46, D_loss_real = 0.712, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 32/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 33/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 34/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 36/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.682, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 40/46, D_loss_real = 0.683, D_loss_fake = 0.690, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 41/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 43/46, D_loss_real = 0.680, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 44/46, D_loss_real = 0.680, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.688, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 1/46, D_loss_real = 0.675, D_loss_fake = 0.713, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 2/46, D_loss_real = 0.672, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 3/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 4/46, D_loss_real = 0.680, D_loss_fake = 0.712, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.718, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 6/46, D_loss_real = 0.679, D_loss_fake = 0.708, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.723, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.716, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.705, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 12/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 13/46, D_loss_real = 0.707, D_loss_fake = 0.683, G_loss = 0.716
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 14/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 15/46, D_loss_real = 0.712, D_loss_fake = 0.668, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 16/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 17/46, D_loss_real = 0.712, D_loss_fake = 0.670, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 18/46, D_loss_real = 0.728, D_loss_fake = 0.669, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 19/46, D_loss_real = 0.729, D_loss_fake = 0.672, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 20/46, D_loss_real = 0.722, D_loss_fake = 0.667, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 21/46, D_loss_real = 0.713, D_loss_fake = 0.671, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 22/46, D_loss_real = 0.707, D_loss_fake = 0.671, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 23/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 24/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 25/46, D_loss_real = 0.702, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 26/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 27/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 28/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.694, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 30/46, D_loss_real = 0.706, D_loss_fake = 0.696, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 31/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 32/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 35/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 42/46, D_loss_real = 0.683, D_loss_fake = 0.689, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.681, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 44/46, D_loss_real = 0.683, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 4ms/step

Epoch 3, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.722, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.708, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.714, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.705, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 7/46, D_loss_real = 0.700, D_loss_fake = 0.692, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 8/46, D_loss_real = 0.704, D_loss_fake = 0.663, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 9/46, D_loss_real = 0.713, D_loss_fake = 0.667, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 10/46, D_loss_real = 0.712, D_loss_fake = 0.676, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 11/46, D_loss_real = 0.717, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 12/46, D_loss_real = 0.715, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 14/46, D_loss_real = 0.703, D_loss_fake = 0.691, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 4ms/step

Epoch 4, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 28/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step

Epoch 4, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 34/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 35/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 36/46, D_loss_real = 0.705, D_loss_fake = 0.678, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.673, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 38/46, D_loss_real = 0.707, D_loss_fake = 0.673, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 39/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 40/46, D_loss_real = 0.710, D_loss_fake = 0.678, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 42/46, D_loss_real = 0.715, D_loss_fake = 0.684, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 45/46, D_loss_real = 0.713, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 1/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 2/46, D_loss_real = 0.709, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 3/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 4/46, D_loss_real = 0.706, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.678, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 11/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.676, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 15/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 24/46, D_loss_real = 0.690, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 4ms/step

Epoch 5, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss = 0.706
2/2 [=====] - 0s 5ms/step
Epoch 5, Iteration 42/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 44/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.678, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 46/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 1/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 2/46, D_loss_real = 0.706, D_loss_fake = 0.678, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.686, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 4/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.680, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 6/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 7/46, D_loss_real = 0.717, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 8/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 9/46, D_loss_real = 0.705, D_loss_fake = 0.686, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 10/46, D_loss_real = 0.707, D_loss_fake = 0.683, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.680, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 12/46, D_loss_real = 0.704, D_loss_fake = 0.681, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.680, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 14/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 15/46, D_loss_real = 0.705, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 16/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 17/46, D_loss_real = 0.704, D_loss_fake = 0.686, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 19/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 20/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 23/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 28/46, D_loss_real = 0.685, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.685, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.701, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.705, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.713, D_loss_fake = 0.676, G_loss = 0.725
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 6/46, D_loss_real = 0.713, D_loss_fake = 0.675, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.710, D_loss_fake = 0.672, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.719, D_loss_fake = 0.678, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.708, D_loss_fake = 0.673, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.710, D_loss_fake = 0.672, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.707, D_loss_fake = 0.674, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.708, D_loss_fake = 0.679, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.702, D_loss_fake = 0.676, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.690, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.682, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.681, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 22/46, D_loss_real = 0.681, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.686, D_loss_fake = 0.706, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.729, G_loss = 0.686
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.707, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.717, G_loss = 0.680
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.710, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.702, D_loss_fake = 0.675, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.706, D_loss_fake = 0.670, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.672, G_loss = 0.724
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 38/46, D_loss_real = 0.707, D_loss_fake = 0.668, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.724, D_loss_fake = 0.670, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.711, D_loss_fake = 0.675, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.680, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.706, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 8/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 24/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.703, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.698, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.682, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.703, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.712, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.681, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 26/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.704, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.703, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 42/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.696, D_loss_fake = 0.685, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 44/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.701, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.706, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.689
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.702, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 4ms/step

Epoch 11, Iteration 46/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.689, D_loss_fake = 0.703, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.710
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.701, D_loss_fake = 0.680, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.675, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.706, D_loss_fake = 0.681, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.702, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.710, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 32/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.703, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.688, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.696, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 4/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.680, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 36/46, D_loss_real = 0.675, D_loss_fake = 0.702, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.703, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.677, D_loss_fake = 0.716, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.687, D_loss_fake = 0.706, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.702, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.680, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.691, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.681, D_loss_fake = 0.706, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.687, D_loss_fake = 0.707, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.714, G_loss = 0.683
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.710, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.681, D_loss_fake = 0.705, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.702, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.702, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.705, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.706, D_loss_fake = 0.682, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.703, D_loss_fake = 0.695, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.702, D_loss_fake = 0.697, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.703, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.708, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 8/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 12ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 24/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss = 0.685
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss = 0.680
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.711, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.713, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.715, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.680, D_loss_fake = 0.721, G_loss = 0.681
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.719, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss = 0.709
2/2 [=====] - 0s 6ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.682, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.703, D_loss_fake = 0.687, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.683, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.686, D_loss_fake = 0.703, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.682, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.691, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.707, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 44/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 12ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.685, D_loss_fake = 0.694, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.707, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.681, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.697, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.683, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.697, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.686, D_loss_fake = 0.703, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.707, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.704, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.700, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.706, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.704, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 8ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.703, D_loss_fake = 0.676, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.682, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.700, D_loss_fake = 0.686, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.702, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 18/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss = 0.703
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.684, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.694, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.685, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.712, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.711, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.693, D_loss_fake = 0.710, G_loss = 0.678
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.710, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.706, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.707, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 4/46, D_loss_real = 0.704, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.693, D_loss_fake = 0.705, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.688, D_loss_fake = 0.704, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.682, D_loss_fake = 0.707, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.705, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.706, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.677, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.703, D_loss_fake = 0.680, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.706, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.705, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.706, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.696, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.703, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.703, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 10ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 38/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.680, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.667, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.716, D_loss_fake = 0.670, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.708, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.713, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.697, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.697, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 8/46, D_loss_real = 0.682, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.679, D_loss_fake = 0.700, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.676, D_loss_fake = 0.695, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.683, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.679, D_loss_fake = 0.706, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.704, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.704, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.709, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.710, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.687, D_loss_fake = 0.708, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.689, D_loss_fake = 0.702, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.699, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.683, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.708, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.687, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.704, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.692
2/2 [=====] - 0s 4ms/step

Epoch 25, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.692, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.691, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.698, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.680, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.683, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.702, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.707, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.703, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.710, G_loss = 0.682
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.708, G_loss = 0.682
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.680, D_loss_fake = 0.708, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.714, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.706, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.677, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.705, D_loss_fake = 0.693, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.710, D_loss_fake = 0.673, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.703, D_loss_fake = 0.683, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.676, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.705, D_loss_fake = 0.685, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 12/46, D_loss_real = 0.711, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.703, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.703, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.676, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 28/46, D_loss_real = 0.679, D_loss_fake = 0.688, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.686, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.676, D_loss_fake = 0.703, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.680, D_loss_fake = 0.710, G_loss = 0.686
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.681, D_loss_fake = 0.713, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.717, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.679, D_loss_fake = 0.728, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.686, D_loss_fake = 0.716, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.710, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.708, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.703, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.695, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.685, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.700, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.669, D_loss_fake = 0.710, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.716, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.708, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.701, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.676, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.714, D_loss_fake = 0.691, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.708, D_loss_fake = 0.689, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.698, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.706, D_loss_fake = 0.693, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.682, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.707, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.698, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.695, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.690, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.707, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.684, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.681, D_loss_fake = 0.713, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.688, D_loss_fake = 0.705, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.682, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.693, D_loss_fake = 0.696, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.683, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.698, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.698, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 4ms/step

Epoch 30, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.702, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.692, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.686, D_loss_fake = 0.685, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 12/46, D_loss_real = 0.681, D_loss_fake = 0.691, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.701, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.685, D_loss_fake = 0.707, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.700, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.699, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.698, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.681, D_loss_fake = 0.702, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.706, G_loss = 0.687
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 22/46, D_loss_real = 0.678, D_loss_fake = 0.711, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 23/46, D_loss_real = 0.683, D_loss_fake = 0.713, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.682, D_loss_fake = 0.707, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.686, D_loss_fake = 0.706, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.682, D_loss_fake = 0.706, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.686
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.687, D_loss_fake = 0.703, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.696, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.685, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.675, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.681, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.685, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.697, D_loss_fake = 0.694, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 8/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 9/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.705, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.705, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.695, D_loss_fake = 0.684, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.697, D_loss_fake = 0.708, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.701, D_loss_fake = 0.721, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.749, G_loss = 0.663
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.731, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss = 0.735
2/2 [=====] - 0s 3ms/step

Epoch 32, Iteration 24/46, D_loss_real = 0.703, D_loss_fake = 0.668, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 25/46, D_loss_real = 0.704, D_loss_fake = 0.669, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.707, D_loss_fake = 0.675, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.716, D_loss_fake = 0.682, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.710, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 32, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 41/46, D_loss_real = 0.679, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 46/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.684, D_loss_fake = 0.689, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.682, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.697, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.700, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.703, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.689, D_loss_fake = 0.705, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.682, D_loss_fake = 0.705, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.701, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 4ms/step

Epoch 33, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 27/46, D_loss_real = 0.701, D_loss_fake = 0.675, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.676, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.673, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.710, D_loss_fake = 0.677, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.704, D_loss_fake = 0.666, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.677, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.709, D_loss_fake = 0.672, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.700, D_loss_fake = 0.674, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.708, D_loss_fake = 0.680, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.704, G_loss = 0.688
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.710, G_loss = 0.685
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.707, G_loss = 0.681
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.712, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.686, D_loss_fake = 0.713, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.716, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.709, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.688, D_loss_fake = 0.682, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.688, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 13/46, D_loss_real = 0.680, D_loss_fake = 0.690, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.668, D_loss_fake = 0.708, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.673, D_loss_fake = 0.696, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.674, D_loss_fake = 0.709, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.664, D_loss_fake = 0.728, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 18/46, D_loss_real = 0.682, D_loss_fake = 0.706, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.675, D_loss_fake = 0.702, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.677, D_loss_fake = 0.703, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.677, D_loss_fake = 0.714, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.666, D_loss_fake = 0.697, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.677, D_loss_fake = 0.712, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.715, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.708
2/2 [=====] - 0s 5ms/step
Epoch 34, Iteration 27/46, D_loss_real = 0.695, D_loss_fake = 0.679, G_loss = 0.725
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 28/46, D_loss_real = 0.708, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.709, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 5ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.690, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.702, D_loss_fake = 0.682, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 43/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.683, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.696, G_loss = 0.684
2/2 [=====] - 0s 8ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.682, D_loss_fake = 0.706, G_loss = 0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.683, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.707, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss = 0.709
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 15/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.695, D_loss_fake = 0.680, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 35, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 31/46, D_loss_real = 0.703, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.680, D_loss_fake = 0.699, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.703, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.709, G_loss = 0.684
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.700, G_loss = 0.683
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 1/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.719, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.677, D_loss_fake = 0.708, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.716, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.682, D_loss_fake = 0.709, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.713, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.701, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.715, D_loss_fake = 0.670, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.704, D_loss_fake = 0.669, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.701, D_loss_fake = 0.675, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.712, D_loss_fake = 0.682, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.713, D_loss_fake = 0.687, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 15/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 17/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.688, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.676, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.691, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.690, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 31/46, D_loss_real = 0.662, D_loss_fake = 0.709, G_loss = 0.693
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 32/46, D_loss_real = 0.677, D_loss_fake = 0.700, G_loss = 0.683
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 33/46, D_loss_real = 0.682, D_loss_fake = 0.725, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.673, D_loss_fake = 0.717, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.714, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.728, G_loss = 0.669
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.726, G_loss = 0.669
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.719, G_loss = 0.679
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.717, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.655, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.705, D_loss_fake = 0.656, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.673, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.675, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 4ms/step

Epoch 37, Iteration 2/46, D_loss_real = 0.700, D_loss_fake = 0.699, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.707, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.708, D_loss_fake = 0.693, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.705
2/2 [=====] - 0s 7ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.689, D_loss_fake = 0.683, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 33/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss = 0.698
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 34/46, D_loss_real = 0.693, D_loss_fake = 0.700, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 35/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.709, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 40/46, D_loss_real = 0.706, D_loss_fake = 0.676, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.716, D_loss_fake = 0.682, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.721, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.703, D_loss_fake = 0.690, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 3/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 5/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.699, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.676, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.676, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.677, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.685, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.687, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.689, D_loss_fake = 0.690, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.680, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.672, D_loss_fake = 0.696, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 19/46, D_loss_real = 0.675, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.712, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.680, D_loss_fake = 0.710, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.664, D_loss_fake = 0.707, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.673, D_loss_fake = 0.723, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.717, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 26/46, D_loss_real = 0.675, D_loss_fake = 0.716, G_loss = 0.671
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.720, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.691, D_loss_fake = 0.715, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.686, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.673, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.661, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.711, D_loss_fake = 0.661, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 35/46, D_loss_real = 0.722, D_loss_fake = 0.685, G_loss = 0.708
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 36/46, D_loss_real = 0.725, D_loss_fake = 0.685, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 37/46, D_loss_real = 0.724, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.717, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.689, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.707, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 42/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.685, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.703, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.680, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.686, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.674, D_loss_fake = 0.704, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.681, D_loss_fake = 0.702, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.699, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.712, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.710, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.683, D_loss_fake = 0.714, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.687, D_loss_fake = 0.713, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.701, G_loss = 0.695
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 22/46, D_loss_real = 0.693, D_loss_fake = 0.700, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.703, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.680, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.676, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.675, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.681, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss = 0.700
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.696, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.684, D_loss_fake = 0.688, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.704, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 40, Iteration 8/46, D_loss_real = 0.691, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.697, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.697, G_loss = 0.697
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.687, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.682, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.693, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.693, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.708, G_loss = 0.693
2/2 [=====] - 0s 3ms/step


```

Epoch 40, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.712, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.706, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.697, D_loss_fake = 0.676, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =
0.707
2/2 [=====] - 0s 4ms/step

```

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

```

>Saved: plot_008x008-tuned.png, g_model_008x008-tuned.sav,
d_model_008x008-tuned.sav and gan_model_008x008-tuned.sav
Scaled Data (6000, 16, 16, 3)

```

```

2/2 [=====] - 0s 5ms/step
Epoch 1, Iteration 1/46, D_loss_real = 0.705, D_loss_fake = 0.679, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 4/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 5/46, D_loss_real = 0.688, D_loss_fake = 0.685, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 6/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss =

```

```

0.688
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.700, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 9/46, D_loss_real = 0.679, D_loss_fake = 0.703, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 10/46, D_loss_real = 0.679, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.714, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 12/46, D_loss_real = 0.682, D_loss_fake = 0.718, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 13/46, D_loss_real = 0.683, D_loss_fake = 0.712, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.715, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 15/46, D_loss_real = 0.682, D_loss_fake = 0.710, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 16/46, D_loss_real = 0.688, D_loss_fake = 0.714, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 17/46, D_loss_real = 0.689, D_loss_fake = 0.706, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 20/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss =

```

```

0.700
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 25/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 26/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 32/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 33/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 34/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 35/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.691, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.694, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 38/46, D_loss_real = 0.688, D_loss_fake = 0.691, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.697, G_loss =

```

```

0.692
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 40/46, D_loss_real = 0.678, D_loss_fake = 0.713, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 41/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 42/46, D_loss_real = 0.676, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 43/46, D_loss_real = 0.677, D_loss_fake = 0.704, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 44/46, D_loss_real = 0.677, D_loss_fake = 0.707, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.710, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 46/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 1/46, D_loss_real = 0.676, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.714, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 5/46, D_loss_real = 0.679, D_loss_fake = 0.683, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.677, G_loss =

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```

0.713
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.677, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.680, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 15/46, D_loss_real = 0.679, D_loss_fake = 0.718, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.723, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 17/46, D_loss_real = 0.685, D_loss_fake = 0.725, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.729, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 19/46, D_loss_real = 0.683, D_loss_fake = 0.728, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 20/46, D_loss_real = 0.687, D_loss_fake = 0.726, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.718, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.715, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.697, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 24/46, D_loss_real = 0.684, D_loss_fake = 0.682, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.680, G_loss =

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0.730
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.673, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 27/46, D_loss_real = 0.686, D_loss_fake = 0.672, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 28/46, D_loss_real = 0.683, D_loss_fake = 0.679, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 29/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 30/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.711, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.713, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.697, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.688, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 41/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss =

```

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0.704
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.699, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 44/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 46/46, D_loss_real = 0.680, D_loss_fake = 0.710, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 1/46, D_loss_real = 0.685, D_loss_fake = 0.705, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.717, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 3/46, D_loss_real = 0.679, D_loss_fake = 0.720, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 4/46, D_loss_real = 0.674, D_loss_fake = 0.715, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 5/46, D_loss_real = 0.672, D_loss_fake = 0.709, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 6/46, D_loss_real = 0.675, D_loss_fake = 0.703, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.704, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.688, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 10/46, D_loss_real = 0.679, D_loss_fake = 0.692, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.689, G_loss =

```

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0.713
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 13/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.690, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 16/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.691, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 23/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 25/46, D_loss_real = 0.690, D_loss_fake = 0.679, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 26/46, D_loss_real = 0.697, D_loss_fake = 0.678, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss =

```



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0.712
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.694, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 30/46, D_loss_real = 0.703, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.692, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.679, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.661, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.661, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.670, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.668, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 39/46, D_loss_real = 0.678, D_loss_fake = 0.696, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 41/46, D_loss_real = 0.672, D_loss_fake = 0.712, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 42/46, D_loss_real = 0.660, D_loss_fake = 0.725, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.716, G_loss =

```

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0.693
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.707, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 45/46, D_loss_real = 0.672, D_loss_fake = 0.698, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 46/46, D_loss_real = 0.662, D_loss_fake = 0.709, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 1/46, D_loss_real = 0.673, D_loss_fake = 0.730, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.707, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.707, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 5/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 6/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.677, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.681, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.675, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 10/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 11/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.673, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.676, G_loss =

```

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0.721
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 14/46, D_loss_real = 0.677, D_loss_fake = 0.685, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 15/46, D_loss_real = 0.673, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 16/46, D_loss_real = 0.671, D_loss_fake = 0.703, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 17/46, D_loss_real = 0.675, D_loss_fake = 0.763, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 18/46, D_loss_real = 0.683, D_loss_fake = 0.754, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 19/46, D_loss_real = 0.681, D_loss_fake = 0.757, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.675, G_loss =
0.773
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.642, G_loss =
0.817
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 22/46, D_loss_real = 0.703, D_loss_fake = 0.658, G_loss =
0.782
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.667, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 24/46, D_loss_real = 0.724, D_loss_fake = 0.687, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.706, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.712, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss =

```

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0.706
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 32/46, D_loss_real = 0.691, D_loss_fake = 0.670, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 33/46, D_loss_real = 0.681, D_loss_fake = 0.664, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.670, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.672, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.675, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.697, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 38/46, D_loss_real = 0.673, D_loss_fake = 0.704, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 39/46, D_loss_real = 0.681, D_loss_fake = 0.700, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 40/46, D_loss_real = 0.678, D_loss_fake = 0.734, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.738, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.720, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 43/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 44/46, D_loss_real = 0.709, D_loss_fake = 0.681, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 45/46, D_loss_real = 0.707, D_loss_fake = 0.702, G_loss =

```

```

0.707
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.695, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 1/46, D_loss_real = 0.712, D_loss_fake = 0.694, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 2/46, D_loss_real = 0.705, D_loss_fake = 0.692, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 3/46, D_loss_real = 0.691, D_loss_fake = 0.696, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 4/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.686, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 9/46, D_loss_real = 0.683, D_loss_fake = 0.674, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 10/46, D_loss_real = 0.685, D_loss_fake = 0.666, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.674, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 12/46, D_loss_real = 0.678, D_loss_fake = 0.677, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 13/46, D_loss_real = 0.694, D_loss_fake = 0.678, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 15/46, D_loss_real = 0.701, D_loss_fake = 0.690, G_loss =

```

```

0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 16/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.697, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 20/46, D_loss_real = 0.697, D_loss_fake = 0.686, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.685, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.681, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.669, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.677, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 26/46, D_loss_real = 0.691, D_loss_fake = 0.668, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 27/46, D_loss_real = 0.706, D_loss_fake = 0.694, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 28/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 29/46, D_loss_real = 0.709, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 30/46, D_loss_real = 0.714, D_loss_fake = 0.713, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.712, G_loss =

```

```

0.707
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 32/46, D_loss_real = 0.703, D_loss_fake = 0.690, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 33/46, D_loss_real = 0.702, D_loss_fake = 0.663, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 34/46, D_loss_real = 0.717, D_loss_fake = 0.674, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 35/46, D_loss_real = 0.709, D_loss_fake = 0.671, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 36/46, D_loss_real = 0.701, D_loss_fake = 0.673, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.700, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 42/46, D_loss_real = 0.675, D_loss_fake = 0.693, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 43/46, D_loss_real = 0.670, D_loss_fake = 0.697, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 44/46, D_loss_real = 0.636, D_loss_fake = 0.703, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.703, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 46/46, D_loss_real = 0.645, D_loss_fake = 0.703, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 1/46, D_loss_real = 0.646, D_loss_fake = 0.707, G_loss =

```

```

0.689
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 2/46, D_loss_real = 0.646, D_loss_fake = 0.713, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 3/46, D_loss_real = 0.639, D_loss_fake = 0.709, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 4/46, D_loss_real = 0.666, D_loss_fake = 0.704, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 5/46, D_loss_real = 0.664, D_loss_fake = 0.706, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 6/46, D_loss_real = 0.663, D_loss_fake = 0.709, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.711, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 8/46, D_loss_real = 0.672, D_loss_fake = 0.690, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 9/46, D_loss_real = 0.677, D_loss_fake = 0.695, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 10/46, D_loss_real = 0.669, D_loss_fake = 0.674, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.715, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.699, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.707, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 15/46, D_loss_real = 0.713, D_loss_fake = 0.695, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 16/46, D_loss_real = 0.703, D_loss_fake = 0.713, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.716, G_loss =

```



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0.698
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 18/46, D_loss_real = 0.669, D_loss_fake = 0.729, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.710, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.707, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 22/46, D_loss_real = 0.699, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 23/46, D_loss_real = 0.705, D_loss_fake = 0.702, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 26/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.651, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 28/46, D_loss_real = 0.706, D_loss_fake = 0.671, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 29/46, D_loss_real = 0.710, D_loss_fake = 0.668, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.690, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss =

```

```

0.700
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.700, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 36/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.679, D_loss_fake = 0.689, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.707, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.677, D_loss_fake = 0.693, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.702, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.678, D_loss_fake = 0.706, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.705, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.710, G_loss =

```

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0.684
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.677, D_loss_fake = 0.716, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.691, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.694, D_loss_fake = 0.671, G_loss =
0.751
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.669, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.703, D_loss_fake = 0.677, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.719, D_loss_fake = 0.686, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.712, D_loss_fake = 0.680, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.710, D_loss_fake = 0.686, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.711, D_loss_fake = 0.682, G_loss =

```

```

0.720
2/2 [=====] - 0s 9ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.672, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.672, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.713, D_loss_fake = 0.670, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.708, D_loss_fake = 0.676, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.690, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.689, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.690, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.682, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.687, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.700, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.694, G_loss =

```

```

0.695
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.711, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.663, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.710, D_loss_fake = 0.652, G_loss =
0.765
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.710, D_loss_fake = 0.654, G_loss =
0.772
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.724, D_loss_fake = 0.669, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.710, D_loss_fake = 0.662, G_loss =
0.738
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.716, D_loss_fake = 0.692, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.710, D_loss_fake = 0.692, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.711, D_loss_fake = 0.697, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.701, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.705, D_loss_fake = 0.690, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.697, D_loss_fake = 0.674, G_loss =

```

```

0.723
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.706, D_loss_fake = 0.676, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.716, D_loss_fake = 0.685, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.691, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.707, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.703, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.685, D_loss_fake = 0.705, G_loss =

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```

0.685
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.711, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.688, D_loss_fake = 0.713, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.708, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.678, D_loss_fake = 0.709, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.686, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.680, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.681, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss =

```

```

0.712
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.704, D_loss_fake = 0.685, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.705, D_loss_fake = 0.692, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.692, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.683, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.706, D_loss_fake = 0.674, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.706, D_loss_fake = 0.656, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.714, D_loss_fake = 0.659, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.715, D_loss_fake = 0.648, G_loss =

```



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0.746
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.714, D_loss_fake = 0.659, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.709, D_loss_fake = 0.671, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.679, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.686, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.690, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.675, D_loss_fake = 0.697, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.681, D_loss_fake = 0.698, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.699, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.683, D_loss_fake = 0.704, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =

```

```

0.699
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.708, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.674, D_loss_fake = 0.705, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.678, D_loss_fake = 0.720, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.681, D_loss_fake = 0.741, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.703, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.713, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.694, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.689, D_loss_fake = 0.683, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.695, D_loss_fake = 0.679, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.708, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss =

```

```

0.697
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.682, D_loss_fake = 0.696, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.680, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.677, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.687, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.708, D_loss_fake = 0.683, G_loss =

```

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0.715
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.703, D_loss_fake = 0.674, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.705, D_loss_fake = 0.668, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.703, D_loss_fake = 0.660, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.708, D_loss_fake = 0.662, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.710, D_loss_fake = 0.665, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.712, D_loss_fake = 0.669, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.721, D_loss_fake = 0.668, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.704, D_loss_fake = 0.665, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.673, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.704, D_loss_fake = 0.672, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.712, D_loss_fake = 0.676, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.682, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss =

```

```

0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.698, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.678, D_loss_fake = 0.694, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.687, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.678, D_loss_fake = 0.682, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.681, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.682, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.688, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.709, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.688, D_loss_fake = 0.706, G_loss =

```

```

0.700
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.699, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.717, D_loss_fake = 0.665, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.708, D_loss_fake = 0.663, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.711, D_loss_fake = 0.656, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.721, D_loss_fake = 0.668, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.713, D_loss_fake = 0.678, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.707, D_loss_fake = 0.674, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.715, D_loss_fake = 0.679, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.704, D_loss_fake = 0.678, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.712, D_loss_fake = 0.671, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.703, D_loss_fake = 0.671, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.699, D_loss_fake = 0.682, G_loss =

```

```

0.717
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.685, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.678, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.689, D_loss_fake = 0.681, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.683, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.671, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.665, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.675, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.677, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.682, D_loss_fake = 0.673, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.681, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.682, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.712, D_loss_fake = 0.685, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.706, D_loss_fake = 0.684, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =

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0.716
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.686, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.708, D_loss_fake = 0.670, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.716, D_loss_fake = 0.659, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.707, D_loss_fake = 0.664, G_loss =
0.753
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.717, D_loss_fake = 0.664, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.703, D_loss_fake = 0.673, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.713, D_loss_fake = 0.664, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.705, D_loss_fake = 0.668, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.675, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.673, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.682, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.683, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.672, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.660, D_loss_fake = 0.696, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.664, D_loss_fake = 0.701, G_loss =

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0.705
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.656, D_loss_fake = 0.708, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.733, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.681, D_loss_fake = 0.697, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.671, D_loss_fake = 0.706, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.699, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.681, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.682, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.708, D_loss_fake = 0.678, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.715, D_loss_fake = 0.681, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.712, D_loss_fake = 0.688, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.723, D_loss_fake = 0.672, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.723, D_loss_fake = 0.671, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.672, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.675, G_loss =

```

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0.729
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.668, D_loss_fake = 0.672, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.673, D_loss_fake = 0.703, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.711, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.644, D_loss_fake = 0.744, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.662, D_loss_fake = 0.755, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.642, D_loss_fake = 0.749, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.648, D_loss_fake = 0.741, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.650, D_loss_fake = 0.746, G_loss =
0.662
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.659, D_loss_fake = 0.737, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.670, D_loss_fake = 0.710, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.673, D_loss_fake = 0.695, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.682, D_loss_fake = 0.667, G_loss =
0.762
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.671, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss =
0.734
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss =

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0.714
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.679, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.672, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.673, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.682, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.674, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.712, D_loss_fake = 0.669, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.706, D_loss_fake = 0.666, G_loss =

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0.741
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.661, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.707, D_loss_fake = 0.661, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.664, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.664, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.693, D_loss_fake = 0.682, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.675, D_loss_fake = 0.694, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.685, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.686, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.664, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.660, G_loss =
0.753
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.695, D_loss_fake = 0.658, G_loss =
0.764
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.658, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.699, G_loss =
0.748
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.717, D_loss_fake = 0.704, G_loss =

```

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0.716
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.688, G_loss =
0.715
2/2 [=====] - 0s 5ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.709, D_loss_fake = 0.686, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.704, D_loss_fake = 0.671, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.701, D_loss_fake = 0.653, G_loss =
0.784
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.703, D_loss_fake = 0.630, G_loss =
0.793
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.707, D_loss_fake = 0.653, G_loss =
0.771
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.716, D_loss_fake = 0.665, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.681, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.678, D_loss_fake = 0.677, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.678, D_loss_fake = 0.709, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.673, D_loss_fake = 0.711, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.674, D_loss_fake = 0.708, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.703, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.676, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.681, D_loss_fake = 0.672, G_loss =
0.757
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.672, D_loss_fake = 0.644, G_loss =

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0.788
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.641, G_loss =
0.818
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.629, G_loss =
0.817
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.736, D_loss_fake = 0.635, G_loss =
0.815
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.727, D_loss_fake = 0.670, G_loss =
0.775
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.710, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.755, D_loss_fake = 0.725, G_loss =
0.771
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.736, D_loss_fake = 0.633, G_loss =
0.856
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.725, D_loss_fake = 0.600, G_loss =
0.884
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.721, D_loss_fake = 0.617, G_loss =
0.829
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.742, D_loss_fake = 0.657, G_loss =
0.781
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.670, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.694, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.670, D_loss_fake = 0.702, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.657, D_loss_fake = 0.716, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.661, D_loss_fake = 0.700, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.643, D_loss_fake = 0.684, G_loss =

```

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0.729
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.666, D_loss_fake = 0.716, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.714, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.678, D_loss_fake = 0.688, G_loss =
0.760
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.682, D_loss_fake = 0.680, G_loss =
0.787
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.708, D_loss_fake = 0.642, G_loss =
0.790
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.705, D_loss_fake = 0.661, G_loss =
0.795
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.725, D_loss_fake = 0.666, G_loss =
0.774
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.684, G_loss =
0.774
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.677, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.684, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.705, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.671, D_loss_fake = 0.702, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.687, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.653, D_loss_fake = 0.685, G_loss =

```

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0.738
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.645, D_loss_fake = 0.677, G_loss =
0.760
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.674, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.641, D_loss_fake = 0.675, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.638, D_loss_fake = 0.696, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.649, D_loss_fake = 0.689, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.658, D_loss_fake = 0.736, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.639, D_loss_fake = 0.767, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.670, D_loss_fake = 0.684, G_loss =
0.765
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss =
0.793
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.705, D_loss_fake = 0.661, G_loss =
0.814
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.712, D_loss_fake = 0.631, G_loss =
0.836
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.633, G_loss =
0.817
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.659, G_loss =
0.785
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.735, D_loss_fake = 0.675, G_loss =
0.785
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.743, D_loss_fake = 0.660, G_loss =
0.780
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.739, D_loss_fake = 0.649, G_loss =

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0.797
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.707, D_loss_fake = 0.666, G_loss =
0.796
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.738, D_loss_fake = 0.670, G_loss =
0.773
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.693, D_loss_fake = 0.704, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.716, D_loss_fake = 0.722, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.671, G_loss =
0.800
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.709, D_loss_fake = 0.642, G_loss =
0.807
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.621, G_loss =
0.827
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.708, D_loss_fake = 0.606, G_loss =
0.875
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.705, D_loss_fake = 0.625, G_loss =
0.853
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.626, G_loss =
0.802
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.656, D_loss_fake = 0.648, G_loss =
0.772
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.623, D_loss_fake = 0.679, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.583, D_loss_fake = 0.742, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.586, D_loss_fake = 0.792, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.613, D_loss_fake = 0.799, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.606, D_loss_fake = 0.819, G_loss =

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0.723
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.677, D_loss_fake = 0.770, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.736, D_loss_fake = 0.675, G_loss =
0.798
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.762, D_loss_fake = 0.639, G_loss =
0.848
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.751, D_loss_fake = 0.625, G_loss =
0.842
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.784, D_loss_fake = 0.589, G_loss =
0.879
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.777, D_loss_fake = 0.592, G_loss =
0.865
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.784, D_loss_fake = 0.595, G_loss =
0.827
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.793, D_loss_fake = 0.602, G_loss =
0.851
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.762, D_loss_fake = 0.586, G_loss =
0.885
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.766, D_loss_fake = 0.579, G_loss =
0.870
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.754, D_loss_fake = 0.622, G_loss =
0.787
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.727, D_loss_fake = 0.640, G_loss =
0.760
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.675, D_loss_fake = 0.654, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.626, D_loss_fake = 0.679, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.606, D_loss_fake = 0.692, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.607, D_loss_fake = 0.697, G_loss =

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0.751
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.578, D_loss_fake = 0.673, G_loss =
0.761
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.583, D_loss_fake = 0.666, G_loss =
0.808
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.543, D_loss_fake = 0.672, G_loss =
0.799
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.595, D_loss_fake = 0.749, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.607, D_loss_fake = 0.906, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.657, D_loss_fake = 0.973, G_loss =
0.752
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.723, G_loss =
1.398
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.791, D_loss_fake = 0.472, G_loss =
1.107
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.789, D_loss_fake = 0.554, G_loss =
0.901
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.796, D_loss_fake = 0.609, G_loss =
0.790
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.730, D_loss_fake = 0.663, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.671, D_loss_fake = 0.707, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.657, D_loss_fake = 0.733, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.633, D_loss_fake = 0.750, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.668, D_loss_fake = 0.705, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.649, D_loss_fake = 0.716, G_loss =

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0.724
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.688, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.665, D_loss_fake = 0.677, G_loss =
0.763
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.676, D_loss_fake = 0.673, G_loss =
0.781
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.679, D_loss_fake = 0.656, G_loss =
0.789
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.678, D_loss_fake = 0.640, G_loss =
0.784
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.642, G_loss =
0.787
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.673, D_loss_fake = 0.653, G_loss =
0.773
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.645, G_loss =
0.774
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.659, D_loss_fake = 0.662, G_loss =
0.760
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.629, D_loss_fake = 0.715, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.634, D_loss_fake = 0.723, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.636, D_loss_fake = 0.699, G_loss =
0.767
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.654, D_loss_fake = 0.693, G_loss =
0.792
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.691, D_loss_fake = 0.657, G_loss =
0.855
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.682, D_loss_fake = 0.658, G_loss =
0.858
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.726, D_loss_fake = 0.622, G_loss =

```

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0.876
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.731, D_loss_fake = 0.641, G_loss =
0.896
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.712, D_loss_fake = 0.589, G_loss =
0.892
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.735, D_loss_fake = 0.601, G_loss =
0.875
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.713, D_loss_fake = 0.622, G_loss =
0.871
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.699, D_loss_fake = 0.631, G_loss =
0.857
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.723, D_loss_fake = 0.623, G_loss =
0.800
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.704, D_loss_fake = 0.704, G_loss =
0.767
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.748, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.704, D_loss_fake = 0.722, G_loss =
0.759
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss =
0.896
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.722, D_loss_fake = 0.594, G_loss =
0.945
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.736, D_loss_fake = 0.646, G_loss =
0.846
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.686, D_loss_fake = 0.672, G_loss =
0.770
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.665, D_loss_fake = 0.682, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.666, G_loss =
0.776
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.636, D_loss_fake = 0.658, G_loss =

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0.818
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.663, D_loss_fake = 0.660, G_loss =
0.805
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.638, D_loss_fake = 0.689, G_loss =
0.782
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.680, D_loss_fake = 0.684, G_loss =
0.790
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.657, D_loss_fake = 0.668, G_loss =
0.822
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.645, G_loss =
0.873
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.736, D_loss_fake = 0.618, G_loss =
0.866
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.721, D_loss_fake = 0.601, G_loss =
0.864
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.711, D_loss_fake = 0.604, G_loss =
0.852
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.708, D_loss_fake = 0.618, G_loss =
0.833
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.645, G_loss =
0.806
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.617, D_loss_fake = 0.680, G_loss =
0.769
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.640, D_loss_fake = 0.699, G_loss =
0.765
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.645, D_loss_fake = 0.730, G_loss =
0.810
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.647, G_loss =
0.847
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.632, G_loss =
0.893
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.603, G_loss =

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0.912
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.677, D_loss_fake = 0.589, G_loss =
0.901
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.653, G_loss =
0.889
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.732, D_loss_fake = 0.666, G_loss =
0.867
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.661, G_loss =
0.939
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.730, D_loss_fake = 0.646, G_loss =
0.915
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.794, D_loss_fake = 0.665, G_loss =
0.942
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.760, D_loss_fake = 0.545, G_loss =
0.994
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.722, D_loss_fake = 0.546, G_loss =
0.939
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.712, D_loss_fake = 0.600, G_loss =
0.850
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.658, G_loss =
0.779
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.708, G_loss =
0.767
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.624, D_loss_fake = 0.678, G_loss =
0.808
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.641, D_loss_fake = 0.642, G_loss =
0.849
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.633, D_loss_fake = 0.655, G_loss =
0.838
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.610, D_loss_fake = 0.654, G_loss =
0.814
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.604, D_loss_fake = 0.694, G_loss =

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0.772
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.599, D_loss_fake = 0.732, G_loss =
0.795
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.589, D_loss_fake = 0.776, G_loss =
0.840
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.651, D_loss_fake = 0.652, G_loss =
0.857
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.675, D_loss_fake = 0.673, G_loss =
0.843
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.667, D_loss_fake = 0.681, G_loss =
0.887
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.684, D_loss_fake = 0.613, G_loss =
0.903
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.617, G_loss =
0.899
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.660, D_loss_fake = 0.641, G_loss =
0.900
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.640, D_loss_fake = 0.671, G_loss =
0.911
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.690, D_loss_fake = 0.631, G_loss =
0.978
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.739, D_loss_fake = 0.613, G_loss =
0.970
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.728, D_loss_fake = 0.653, G_loss =
0.940
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.727, G_loss =
0.833
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.642, D_loss_fake = 0.709, G_loss =
0.810
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.669, D_loss_fake = 0.699, G_loss =
0.790
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.648, D_loss_fake = 0.676, G_loss =

```



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0.944
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.654, D_loss_fake = 0.666, G_loss =
0.936
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.736, D_loss_fake = 0.623, G_loss =
0.978
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.743, D_loss_fake = 0.570, G_loss =
0.983
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.744, D_loss_fake = 0.573, G_loss =
0.886
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.685, D_loss_fake = 0.629, G_loss =
0.826
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.636, D_loss_fake = 0.661, G_loss =
0.821
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.638, D_loss_fake = 0.663, G_loss =
0.825
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.598, D_loss_fake = 0.673, G_loss =
0.811
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.616, D_loss_fake = 0.669, G_loss =
0.780
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.607, D_loss_fake = 0.787, G_loss =
0.835
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.652, D_loss_fake = 0.670, G_loss =
0.886
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.703, D_loss_fake = 0.654, G_loss =
0.919
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.715, D_loss_fake = 0.638, G_loss =
0.894
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.726, D_loss_fake = 0.653, G_loss =
0.828
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.711, G_loss =
0.761
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.657, D_loss_fake = 0.793, G_loss =

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0.843
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.687, D_loss_fake = 0.724, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.704, D_loss_fake = 0.862, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.707, D_loss_fake = 0.819, G_loss =
0.779
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.705, D_loss_fake = 0.685, G_loss =
0.753
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.746, D_loss_fake = 0.726, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.725, D_loss_fake = 0.715, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.733, D_loss_fake = 0.708, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.702, D_loss_fake = 0.728, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.742, G_loss =
0.660
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.671, D_loss_fake = 0.777, G_loss =
0.657
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.671, D_loss_fake = 0.752, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.740, G_loss =
0.668
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.659, D_loss_fake = 0.731, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.744, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.736, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.687, D_loss_fake = 0.724, G_loss =

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0.671
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.679, D_loss_fake = 0.725, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.671, D_loss_fake = 0.730, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.665, D_loss_fake = 0.736, G_loss =
0.660
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.660, D_loss_fake = 0.745, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.743, G_loss =
0.659
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.672, D_loss_fake = 0.740, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.649, D_loss_fake = 0.730, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.670, D_loss_fake = 0.731, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.665, D_loss_fake = 0.721, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.655, D_loss_fake = 0.728, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.659, D_loss_fake = 0.729, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.655, D_loss_fake = 0.736, G_loss =
0.666
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.660, D_loss_fake = 0.731, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.652, D_loss_fake = 0.724, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.667, D_loss_fake = 0.723, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.670, D_loss_fake = 0.712, G_loss =

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0.684
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.663, D_loss_fake = 0.712, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.667, D_loss_fake = 0.713, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.668, D_loss_fake = 0.718, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.670, D_loss_fake = 0.709, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.671, D_loss_fake = 0.718, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.673, D_loss_fake = 0.712, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.671, D_loss_fake = 0.722, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.664, D_loss_fake = 0.719, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.663, D_loss_fake = 0.720, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.655, D_loss_fake = 0.724, G_loss =
0.673
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.649, D_loss_fake = 0.721, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.649, D_loss_fake = 0.730, G_loss =
0.660
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.655, D_loss_fake = 0.736, G_loss =
0.666
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.644, D_loss_fake = 0.738, G_loss =

```

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0.663
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.644, D_loss_fake = 0.733, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.651, D_loss_fake = 0.737, G_loss =
0.659
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.647, D_loss_fake = 0.739, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.650, D_loss_fake = 0.733, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.645, D_loss_fake = 0.728, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.645, D_loss_fake = 0.736, G_loss =
0.662
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.640, D_loss_fake = 0.742, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.653, D_loss_fake = 0.736, G_loss =
0.667
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.645, D_loss_fake = 0.734, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.654, D_loss_fake = 0.720, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.646, D_loss_fake = 0.735, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.642, D_loss_fake = 0.726, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.648, D_loss_fake = 0.729, G_loss =
0.664
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.649, D_loss_fake = 0.728, G_loss =
0.669
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.644, D_loss_fake = 0.735, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.646, D_loss_fake = 0.726, G_loss =

```

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0.671
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.642, D_loss_fake = 0.734, G_loss =
0.655
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.639, D_loss_fake = 0.724, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.652, D_loss_fake = 0.727, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.641, D_loss_fake = 0.733, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.648, D_loss_fake = 0.729, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.650, D_loss_fake = 0.735, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.648, D_loss_fake = 0.733, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.645, D_loss_fake = 0.735, G_loss =
0.659
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.643, D_loss_fake = 0.738, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.644, D_loss_fake = 0.731, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.638, D_loss_fake = 0.736, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.739, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.636, D_loss_fake = 0.739, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.639, D_loss_fake = 0.735, G_loss =
0.658
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.633, D_loss_fake = 0.743, G_loss =
0.657
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.628, D_loss_fake = 0.750, G_loss =

```

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0.667
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.646, D_loss_fake = 0.728, G_loss =
0.654
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.640, D_loss_fake = 0.761, G_loss =
0.656
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.623, D_loss_fake = 0.750, G_loss =
0.654
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.650, D_loss_fake = 0.762, G_loss =
0.659
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.662, D_loss_fake = 0.762, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.683, D_loss_fake = 0.685, G_loss =
0.802
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.694, D_loss_fake = 0.651, G_loss =
0.781
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.718, D_loss_fake = 0.690, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.712, D_loss_fake = 0.695, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.709, D_loss_fake = 0.687, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.667, D_loss_fake = 0.704, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.668, D_loss_fake = 0.703, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.663, D_loss_fake = 0.703, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.663, D_loss_fake = 0.698, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.715, G_loss =

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```

0.685
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.659, D_loss_fake = 0.718, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.654, D_loss_fake = 0.737, G_loss =
0.673
2/2 [=====] - 0s 12ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.663, D_loss_fake = 0.742, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.655, D_loss_fake = 0.738, G_loss =
0.656
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.651, D_loss_fake = 0.756, G_loss =
0.664
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.650, D_loss_fake = 0.745, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.664, D_loss_fake = 0.725, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.675, D_loss_fake = 0.704, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.675, G_loss =
0.757
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.667, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.704, D_loss_fake = 0.670, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.716, D_loss_fake = 0.685, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.706, D_loss_fake = 0.703, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.696, D_loss_fake = 0.704, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.691, D_loss_fake = 0.715, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.708, G_loss =

```



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0.688
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.677, D_loss_fake = 0.711, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.667, D_loss_fake = 0.716, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.670, D_loss_fake = 0.716, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.647, D_loss_fake = 0.732, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.663, D_loss_fake = 0.736, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.663, D_loss_fake = 0.736, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.665, D_loss_fake = 0.731, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.650, D_loss_fake = 0.741, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.742, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.691, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.695, D_loss_fake = 0.675, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.678, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss =

```

```

0.719
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.686, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.683, D_loss_fake = 0.685, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.677, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.668, D_loss_fake = 0.707, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.676, D_loss_fake = 0.692, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.667, D_loss_fake = 0.699, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.667, D_loss_fake = 0.713, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.654, D_loss_fake = 0.728, G_loss =
0.666
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.753, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.673, D_loss_fake = 0.747, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.670, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.653, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.716, D_loss_fake = 0.675, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.717, D_loss_fake = 0.688, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.712, D_loss_fake = 0.697, G_loss =

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0.703
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.702, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.702, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.705, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.679, D_loss_fake = 0.705, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.672, D_loss_fake = 0.701, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.716, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.652, D_loss_fake = 0.713, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.711, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.644, D_loss_fake = 0.727, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.636, D_loss_fake = 0.732, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.643, D_loss_fake = 0.731, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.636, D_loss_fake = 0.754, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.658, D_loss_fake = 0.733, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.756, G_loss =

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0.678
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.753, G_loss =
0.666
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.674, D_loss_fake = 0.741, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.688, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.686, D_loss_fake = 0.677, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.681, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.682, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.678, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.682, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.698, D_loss_fake = 0.694, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.695, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.692, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.708, D_loss_fake = 0.682, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.691, G_loss =

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0.711
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.677, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.677, D_loss_fake = 0.685, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.690, D_loss_fake = 0.685, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.672, D_loss_fake = 0.698, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.683, D_loss_fake = 0.700, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.674, D_loss_fake = 0.704, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.681, D_loss_fake = 0.706, G_loss =

```

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0.686
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.675, D_loss_fake = 0.710, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.719, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.713, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.661, D_loss_fake = 0.718, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.673, D_loss_fake = 0.719, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.691, D_loss_fake = 0.674, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.706, D_loss_fake = 0.661, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.710, D_loss_fake = 0.655, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.715, D_loss_fake = 0.657, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.715, D_loss_fake = 0.667, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.712, D_loss_fake = 0.669, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.670, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.707, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.677, G_loss =

```

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0.720
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.701, D_loss_fake = 0.673, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.691, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.676, D_loss_fake = 0.698, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.674, D_loss_fake = 0.696, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.665, D_loss_fake = 0.707, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.667, D_loss_fake = 0.697, G_loss =

```

```

0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.651, D_loss_fake = 0.706, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.653, D_loss_fake = 0.704, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.660, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.670, D_loss_fake = 0.709, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.650, D_loss_fake = 0.722, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.651, D_loss_fake = 0.706, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.663, D_loss_fake = 0.715, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.661, D_loss_fake = 0.706, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.662, D_loss_fake = 0.709, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.668, D_loss_fake = 0.704, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.704, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.659, D_loss_fake = 0.709, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.666, D_loss_fake = 0.722, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.672, D_loss_fake = 0.722, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.672, D_loss_fake = 0.720, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.744, G_loss =

```



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0.690
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.672, D_loss_fake = 0.723, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.696, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.705, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.715, D_loss_fake = 0.704, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.704, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.711, D_loss_fake = 0.683, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.719, D_loss_fake = 0.677, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.714, D_loss_fake = 0.682, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.720, D_loss_fake = 0.685, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.709, D_loss_fake = 0.683, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.678, D_loss_fake = 0.690, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.681, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.688, G_loss =

```

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0.704
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.690, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.670, D_loss_fake = 0.698, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.673, D_loss_fake = 0.694, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.663, D_loss_fake = 0.704, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.665, D_loss_fake = 0.708, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.680, D_loss_fake = 0.713, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.728, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.688, D_loss_fake = 0.699, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.684, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.701, D_loss_fake = 0.671, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.706, D_loss_fake = 0.686, G_loss =

```

```

0.709
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.684, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.713, D_loss_fake = 0.690, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.694, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.682, D_loss_fake = 0.694, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.670, D_loss_fake = 0.693, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.676, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.674, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.663, D_loss_fake = 0.709, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.667, D_loss_fake = 0.709, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.667, D_loss_fake = 0.718, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.667, D_loss_fake = 0.717, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.661, D_loss_fake = 0.721, G_loss =
0.674
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.670, D_loss_fake = 0.712, G_loss =

```

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0.690
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.669, D_loss_fake = 0.713, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.676, D_loss_fake = 0.697, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.673, D_loss_fake = 0.697, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.683, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.707, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.683, D_loss_fake = 0.706, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.678, D_loss_fake = 0.722, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.666, D_loss_fake = 0.727, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.742, G_loss =
0.670
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.730, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.697, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.691, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.707, D_loss_fake = 0.692, G_loss =

```

```

0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.719, D_loss_fake = 0.678, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.703, D_loss_fake = 0.674, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.719, D_loss_fake = 0.688, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.714, D_loss_fake = 0.676, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.675, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.713, D_loss_fake = 0.688, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.712, D_loss_fake = 0.700, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.693, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.681, D_loss_fake = 0.697, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.709, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.669, D_loss_fake = 0.710, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.672, D_loss_fake = 0.746, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.671, D_loss_fake = 0.736, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.674, D_loss_fake = 0.713, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.674, D_loss_fake = 0.706, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss =

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0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.679, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.689, D_loss_fake = 0.686, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.689, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.688, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.688, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss =

```

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0.709
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.700, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.702, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.678, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.674, D_loss_fake = 0.690, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.678, D_loss_fake = 0.695, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss =

```

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0.684
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.722, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.659, D_loss_fake = 0.720, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.659, D_loss_fake = 0.726, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.661, D_loss_fake = 0.722, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.660, D_loss_fake = 0.720, G_loss =
0.675
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.654, D_loss_fake = 0.715, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.650, D_loss_fake = 0.748, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.652, D_loss_fake = 0.740, G_loss =
0.665
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.664, D_loss_fake = 0.753, G_loss =
0.674
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.671, D_loss_fake = 0.730, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.671, D_loss_fake = 0.729, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.701, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.681, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.687, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.698, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.679, D_loss_fake = 0.698, G_loss =

```



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0.697
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.667, D_loss_fake = 0.716, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.671, D_loss_fake = 0.726, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.722, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.661, D_loss_fake = 0.736, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.679, D_loss_fake = 0.716, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.706, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.687, D_loss_fake = 0.702, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.688, D_loss_fake = 0.679, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.676, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.681, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.670, D_loss_fake = 0.687, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.675, D_loss_fake = 0.689, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.667, D_loss_fake = 0.699, G_loss =

```

```

0.693
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.665, D_loss_fake = 0.713, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.725, G_loss =
0.676
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.652, D_loss_fake = 0.747, G_loss =
0.672
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.753, G_loss =
0.669
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.678, D_loss_fake = 0.725, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.679, D_loss_fake = 0.714, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.707, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.707, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.708, D_loss_fake = 0.685, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss =

```

```

0.716
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.710, D_loss_fake = 0.677, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.697, D_loss_fake = 0.675, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.705, D_loss_fake = 0.664, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.712, D_loss_fake = 0.664, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.715, D_loss_fake = 0.661, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.712, D_loss_fake = 0.664, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.710, D_loss_fake = 0.665, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.682, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.693, D_loss_fake = 0.707, G_loss =

```

```

0.698
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.699, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.700, D_loss_fake = 0.697, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.700, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.697, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.683, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.690, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.693, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.692, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.693, G_loss =

```

```

0.699
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.677, D_loss_fake = 0.701, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.705, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.705, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.682, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.692, D_loss_fake = 0.671, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.670, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.693, D_loss_fake = 0.677, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.681, G_loss =

```

```

0.715
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.707, D_loss_fake = 0.684, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.687, D_loss_fake = 0.698, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.696, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.666, D_loss_fake = 0.703, G_loss =
0.684
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.667, D_loss_fake = 0.713, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.668, D_loss_fake = 0.728, G_loss =
0.672
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.659, D_loss_fake = 0.733, G_loss =
0.664
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.661, D_loss_fake = 0.745, G_loss =
0.660
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.654, D_loss_fake = 0.752, G_loss =

```

```

0.671
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.672, D_loss_fake = 0.732, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.695, G_loss =
0.739
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.711, D_loss_fake = 0.688, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.709, D_loss_fake = 0.683, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.683, D_loss_fake = 0.691, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.670, D_loss_fake = 0.704, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.658, D_loss_fake = 0.715, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.659, D_loss_fake = 0.719, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.652, D_loss_fake = 0.738, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.652, D_loss_fake = 0.758, G_loss =
0.667
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.658, D_loss_fake = 0.766, G_loss =
0.671
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.677, D_loss_fake = 0.743, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss =

```

```

0.709
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.695, D_loss_fake = 0.675, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.701, D_loss_fake = 0.662, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.707, D_loss_fake = 0.664, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.667, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.671, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.672, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.706, D_loss_fake = 0.680, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.691, D_loss_fake = 0.698, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss =

```


0.706
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.711, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.676, D_loss_fake = 0.716, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.677, D_loss_fake = 0.722, G_loss = 0.678
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.670, D_loss_fake = 0.723, G_loss = 0.669
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.675, D_loss_fake = 0.714, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.676, D_loss_fake = 0.720, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.680, D_loss_fake = 0.717, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.676, D_loss_fake = 0.719, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.713, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.701, D_loss_fake = 0.686, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss =

```

0.704
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.697, D_loss_fake = 0.698, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.700, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.661, D_loss_fake = 0.719, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.664, D_loss_fake = 0.735, G_loss =
0.675
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.748, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.674, D_loss_fake = 0.736, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.733, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.719, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.709, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.691, D_loss_fake = 0.702, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss =

```

```

0.706
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.692, D_loss_fake = 0.682, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.693, D_loss_fake = 0.683, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.681, D_loss_fake = 0.685, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.679, D_loss_fake = 0.685, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.676, D_loss_fake = 0.696, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.706, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.703, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.703, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss =

```

```

0.691
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.700, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.692, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.695, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.709, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.669, D_loss_fake = 0.699, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.666, D_loss_fake = 0.705, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.677, D_loss_fake = 0.709, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.677, D_loss_fake = 0.709, G_loss =

```

```

0.688
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.664, D_loss_fake = 0.710, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.669, D_loss_fake = 0.707, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.660, D_loss_fake = 0.701, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.677, D_loss_fake = 0.716, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.707, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.668, D_loss_fake = 0.709, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.700, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.692, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.688, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.683, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.689, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.685, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.693, D_loss_fake = 0.685, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.698, G_loss =

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```

0.695
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.701, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.683, D_loss_fake = 0.700, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.674, D_loss_fake = 0.699, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.660, D_loss_fake = 0.705, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.640, D_loss_fake = 0.700, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.644, D_loss_fake = 0.719, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.612, D_loss_fake = 0.714, G_loss =
0.675
2/2 [=====] - 0s 5ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.622, D_loss_fake = 0.730, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.615, D_loss_fake = 0.742, G_loss =
0.676
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.607, D_loss_fake = 0.790, G_loss =
0.673
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.631, D_loss_fake = 0.765, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.659, D_loss_fake = 0.722, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.667, D_loss_fake = 0.724, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.677, D_loss_fake = 0.699, G_loss =
0.774
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.668, G_loss =

```

```

0.772
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.725, D_loss_fake = 0.682, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.742, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.707, D_loss_fake = 0.686, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.703, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.705, D_loss_fake = 0.695, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.680, D_loss_fake = 0.687, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.694, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.702, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.685, D_loss_fake = 0.709, G_loss =

```

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0.692
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 7/46, D_loss_real = 0.677, D_loss_fake = 0.696, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.677, D_loss_fake = 0.693, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.667, D_loss_fake = 0.735, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.694, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.728, G_loss =

```



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0.703
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 22/46, D_loss_real = 0.700, D_loss_fake = 0.666, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 23/46, D_loss_real = 0.707, D_loss_fake = 0.674, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.703, D_loss_fake = 0.674, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.704, D_loss_fake = 0.676, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.693, D_loss_fake = 0.672, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.679, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.689, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.680, D_loss_fake = 0.697, G_loss =
0.700
2/2 [=====] - 0s 5ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.671, D_loss_fake = 0.702, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.669, D_loss_fake = 0.705, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.676, D_loss_fake = 0.709, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.655, D_loss_fake = 0.706, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.660, D_loss_fake = 0.732, G_loss =

```

```

0.687
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.680, D_loss_fake = 0.713, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.710, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 38/46, D_loss_real = 0.677, D_loss_fake = 0.716, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 39/46, D_loss_real = 0.666, D_loss_fake = 0.711, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.688, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.666, D_loss_fake = 0.686, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.669, D_loss_fake = 0.686, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.664, D_loss_fake = 0.690, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.761, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.660, D_loss_fake = 0.783, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.685, D_loss_fake = 0.717, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.705, D_loss_fake = 0.666, G_loss =
0.785
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.733, D_loss_fake = 0.665, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.747, D_loss_fake = 0.686, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss =

```

```

0.709
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.696, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 7/46, D_loss_real = 0.673, D_loss_fake = 0.683, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 8/46, D_loss_real = 0.669, D_loss_fake = 0.698, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 9/46, D_loss_real = 0.661, D_loss_fake = 0.707, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.661, D_loss_fake = 0.717, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.664, D_loss_fake = 0.711, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.669, D_loss_fake = 0.697, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.673, D_loss_fake = 0.704, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.702, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.679, D_loss_fake = 0.660, G_loss =
0.774
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.668, D_loss_fake = 0.656, G_loss =
0.773
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.670, D_loss_fake = 0.663, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.655, D_loss_fake = 0.676, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.655, D_loss_fake = 0.713, G_loss =

```

```

0.705
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.662, D_loss_fake = 0.718, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 23/46, D_loss_real = 0.645, D_loss_fake = 0.736, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 24/46, D_loss_real = 0.649, D_loss_fake = 0.721, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 25/46, D_loss_real = 0.677, D_loss_fake = 0.711, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss =
0.754
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.650, G_loss =
0.808
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.695, D_loss_fake = 0.635, G_loss =
0.851
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.658, G_loss =
0.779
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 30/46, D_loss_real = 0.709, D_loss_fake = 0.687, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.719, D_loss_fake = 0.721, G_loss =
0.691
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.681, D_loss_fake = 0.726, G_loss =
0.675
2/2 [=====] - 0s 5ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.662, D_loss_fake = 0.725, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.673, D_loss_fake = 0.726, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.684, G_loss =
0.774
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.701, D_loss_fake = 0.676, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.710, D_loss_fake = 0.691, G_loss =

```

```

0.717
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.703, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.707, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 41/46, D_loss_real = 0.676, D_loss_fake = 0.703, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.668, D_loss_fake = 0.709, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.692, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.658, D_loss_fake = 0.701, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.659, D_loss_fake = 0.702, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 46/46, D_loss_real = 0.660, D_loss_fake = 0.686, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.654, D_loss_fake = 0.678, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.666, D_loss_fake = 0.709, G_loss =
0.754
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.682, D_loss_fake = 0.722, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.691, D_loss_fake = 0.708, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.739, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.738, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.719, D_loss_fake = 0.692, G_loss =

```

```

0.746
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.731, D_loss_fake = 0.697, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 9/46, D_loss_real = 0.727, D_loss_fake = 0.682, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 10/46, D_loss_real = 0.730, D_loss_fake = 0.681, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 11/46, D_loss_real = 0.722, D_loss_fake = 0.691, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.680, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.684, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 16/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.686, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.675, D_loss_fake = 0.687, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.688, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.696, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.698, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.683, D_loss_fake = 0.674, G_loss =

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0.725
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.690, D_loss_fake = 0.674, G_loss =
0.736
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 25/46, D_loss_real = 0.701, D_loss_fake = 0.658, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 26/46, D_loss_real = 0.705, D_loss_fake = 0.667, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 27/46, D_loss_real = 0.701, D_loss_fake = 0.668, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.676, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.672, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.720, D_loss_fake = 0.676, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.715, D_loss_fake = 0.669, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.715, D_loss_fake = 0.678, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.676, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.666, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.667, G_loss =

```

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0.752
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.712, D_loss_fake = 0.661, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.658, G_loss =
0.766
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 42/46, D_loss_real = 0.703, D_loss_fake = 0.657, G_loss =
0.783
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.651, G_loss =
0.772
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.731, D_loss_fake = 0.698, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.706, D_loss_fake = 0.760, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.711, D_loss_fake = 0.707, G_loss =
0.748
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.715, D_loss_fake = 0.694, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 2/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss =
0.759
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.751, D_loss_fake = 0.685, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.716, D_loss_fake = 0.683, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.719, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.709, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.696, D_loss_fake = 0.698, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.675, D_loss_fake = 0.699, G_loss =

```



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0.710
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.682, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 12/46, D_loss_real = 0.681, D_loss_fake = 0.679, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 13/46, D_loss_real = 0.682, D_loss_fake = 0.678, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.683, D_loss_fake = 0.672, G_loss =
0.740
2/2 [=====] - 0s 5ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.675, D_loss_fake = 0.675, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.692, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.673, D_loss_fake = 0.688, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 18/46, D_loss_real = 0.669, D_loss_fake = 0.700, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.667, D_loss_fake = 0.709, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.653, D_loss_fake = 0.735, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.750, G_loss =
0.668
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.658, D_loss_fake = 0.741, G_loss =
0.671
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.664, D_loss_fake = 0.745, G_loss =
0.663
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.668, D_loss_fake = 0.739, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.684, D_loss_fake = 0.740, G_loss =

```

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0.694
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.678, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 28/46, D_loss_real = 0.706, D_loss_fake = 0.673, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 29/46, D_loss_real = 0.714, D_loss_fake = 0.672, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.723, D_loss_fake = 0.669, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.734, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.718, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.709, D_loss_fake = 0.681, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 34/46, D_loss_real = 0.706, D_loss_fake = 0.680, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.668, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.663, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.708, D_loss_fake = 0.684, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.703, D_loss_fake = 0.686, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.666, G_loss =

```

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0.715
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.690, D_loss_fake = 0.672, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 43/46, D_loss_real = 0.690, D_loss_fake = 0.682, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 44/46, D_loss_real = 0.687, D_loss_fake = 0.682, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 45/46, D_loss_real = 0.686, D_loss_fake = 0.668, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.668, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.678, D_loss_fake = 0.685, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.679, D_loss_fake = 0.674, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.667, D_loss_fake = 0.681, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 4/46, D_loss_real = 0.674, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.664, D_loss_fake = 0.712, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.716, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.708, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.688, D_loss_fake = 0.707, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.705, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.689, D_loss_fake = 0.708, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.700, D_loss_fake = 0.700, G_loss =

```

```

0.716
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.706, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.686, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 14/46, D_loss_real = 0.710, D_loss_fake = 0.692, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 15/46, D_loss_real = 0.706, D_loss_fake = 0.682, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.679, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.678, D_loss_fake = 0.685, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.684, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.674, D_loss_fake = 0.688, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.683, D_loss_fake = 0.688, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.674, D_loss_fake = 0.702, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.704, G_loss =
0.681
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.662, D_loss_fake = 0.731, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.749, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.673, D_loss_fake = 0.715, G_loss =

```

```

0.699
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.706, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 30/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.679, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.672, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.678, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.684, D_loss_fake = 0.672, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.677, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.670, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.689, D_loss_fake = 0.669, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.708, D_loss_fake = 0.694, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.697, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.697, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.708, G_loss =
0.682
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.721, G_loss =

```

```

0.686
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.697, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.679, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.684, D_loss_fake = 0.674, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.677, D_loss_fake = 0.674, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 6/46, D_loss_real = 0.668, D_loss_fake = 0.678, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.679, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.685, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.651, D_loss_fake = 0.706, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.635, D_loss_fake = 0.706, G_loss =
0.677
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.628, D_loss_fake = 0.747, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.616, D_loss_fake = 0.756, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.621, D_loss_fake = 0.785, G_loss =

```

```

0.650
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.613, D_loss_fake = 0.755, G_loss =
0.662
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 15/46, D_loss_real = 0.633, D_loss_fake = 0.739, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 16/46, D_loss_real = 0.640, D_loss_fake = 0.694, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 17/46, D_loss_real = 0.641, D_loss_fake = 0.696, G_loss =
0.782
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.667, D_loss_fake = 0.711, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.728, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.711, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.681, G_loss =
0.755
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss =
0.763
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.715, D_loss_fake = 0.656, G_loss =
0.774
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.716, D_loss_fake = 0.669, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.722, D_loss_fake = 0.682, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.715, D_loss_fake = 0.670, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.704, D_loss_fake = 0.665, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss =
0.729
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.710, D_loss_fake = 0.696, G_loss =

```

```

0.719
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.714, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.718, G_loss =
0.679
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.713, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.689, D_loss_fake = 0.736, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.684, D_loss_fake = 0.721, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.713, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.695, D_loss_fake = 0.687, G_loss =
0.746
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.689, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.705, D_loss_fake = 0.686, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.705, D_loss_fake = 0.691, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.714, D_loss_fake = 0.680, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.709, D_loss_fake = 0.659, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.708, D_loss_fake = 0.673, G_loss =

```



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0.730
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.718, D_loss_fake = 0.675, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 1/46, D_loss_real = 0.717, D_loss_fake = 0.674, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.712, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.701, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.708, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.708, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.705, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.684, D_loss_fake = 0.707, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.705, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.689, D_loss_fake = 0.712, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.687, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.674, D_loss_fake = 0.692, G_loss =

```

```

0.701
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.677, D_loss_fake = 0.707, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 17/46, D_loss_real = 0.673, D_loss_fake = 0.698, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 18/46, D_loss_real = 0.659, D_loss_fake = 0.708, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 19/46, D_loss_real = 0.666, D_loss_fake = 0.708, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.651, D_loss_fake = 0.722, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.719, G_loss =
0.681
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.655, D_loss_fake = 0.742, G_loss =
0.680
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.661, D_loss_fake = 0.754, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 24/46, D_loss_real = 0.672, D_loss_fake = 0.766, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.678, D_loss_fake = 0.726, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.714, D_loss_fake = 0.680, G_loss =
0.759
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.725, D_loss_fake = 0.692, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.720, D_loss_fake = 0.692, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.714, D_loss_fake = 0.684, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.693, D_loss_fake = 0.689, G_loss =

```

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0.709
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.674, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.675, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 40/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss =
0.734
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.701, D_loss_fake = 0.668, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.710, D_loss_fake = 0.655, G_loss =
0.750
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.720, D_loss_fake = 0.663, G_loss =
0.749
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.713, D_loss_fake = 0.666, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.718, D_loss_fake = 0.660, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.727, D_loss_fake = 0.660, G_loss =

```

```

0.746
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.725, D_loss_fake = 0.652, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 3/46, D_loss_real = 0.732, D_loss_fake = 0.657, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 4/46, D_loss_real = 0.715, D_loss_fake = 0.647, G_loss =
0.762
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 5/46, D_loss_real = 0.721, D_loss_fake = 0.638, G_loss =
0.765
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.728, D_loss_fake = 0.648, G_loss =
0.760
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.713, D_loss_fake = 0.647, G_loss =
0.780
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.714, D_loss_fake = 0.657, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.716, D_loss_fake = 0.657, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 10/46, D_loss_real = 0.726, D_loss_fake = 0.666, G_loss =
0.742
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.712, D_loss_fake = 0.659, G_loss =
0.740
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.705, D_loss_fake = 0.661, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.698, D_loss_fake = 0.673, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.671, G_loss =
0.734
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.701, D_loss_fake = 0.689, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.708, G_loss =

```

```

0.706
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.715, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 19/46, D_loss_real = 0.677, D_loss_fake = 0.712, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 20/46, D_loss_real = 0.677, D_loss_fake = 0.722, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.709, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.673, D_loss_fake = 0.697, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.680, D_loss_fake = 0.706, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.677, D_loss_fake = 0.706, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 26/46, D_loss_real = 0.676, D_loss_fake = 0.702, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.661, D_loss_fake = 0.698, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.675, D_loss_fake = 0.666, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.661, G_loss =
0.765
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.676, D_loss_fake = 0.671, G_loss =
0.760
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.670, D_loss_fake = 0.653, G_loss =
0.762
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.672, D_loss_fake = 0.671, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.659, D_loss_fake = 0.693, G_loss =

```

```

0.721
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.661, D_loss_fake = 0.715, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 35/46, D_loss_real = 0.641, D_loss_fake = 0.729, G_loss =
0.678
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 36/46, D_loss_real = 0.634, D_loss_fake = 0.763, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 37/46, D_loss_real = 0.632, D_loss_fake = 0.735, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.634, D_loss_fake = 0.722, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.660, D_loss_fake = 0.739, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.650, D_loss_fake = 0.725, G_loss =
0.680
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.656, D_loss_fake = 0.749, G_loss =
0.663
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 42/46, D_loss_real = 0.653, D_loss_fake = 0.777, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.652, D_loss_fake = 0.802, G_loss =
0.683
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.716, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.676, D_loss_fake = 0.689, G_loss =
0.751
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.699, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.710, G_loss =

```

```

0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.682, D_loss_fake = 0.714, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 5/46, D_loss_real = 0.676, D_loss_fake = 0.753, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 6/46, D_loss_real = 0.683, D_loss_fake = 0.735, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.738, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.645, G_loss =
0.822
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.711, D_loss_fake = 0.633, G_loss =
0.817
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.734, D_loss_fake = 0.659, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 12/46, D_loss_real = 0.758, D_loss_fake = 0.684, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.689, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.691, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.701, D_loss_fake = 0.693, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss =

```

```

0.714
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.687, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 22/46, D_loss_real = 0.703, D_loss_fake = 0.672, G_loss =
0.746
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.670, G_loss =
0.756
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.660, G_loss =
0.765
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.711, D_loss_fake = 0.650, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.718, D_loss_fake = 0.649, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.730, D_loss_fake = 0.675, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 28/46, D_loss_real = 0.720, D_loss_fake = 0.692, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.710, D_loss_fake = 0.703, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.691, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.692, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.703, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.681, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss =

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0.721
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.671, D_loss_fake = 0.681, G_loss =
0.732
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.688, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 38/46, D_loss_real = 0.678, D_loss_fake = 0.694, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 39/46, D_loss_real = 0.676, D_loss_fake = 0.698, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.672, D_loss_fake = 0.719, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.664, D_loss_fake = 0.724, G_loss =
0.679
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.653, D_loss_fake = 0.729, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.679, D_loss_fake = 0.740, G_loss =
0.681
2/2 [=====] - 0s 11ms/step
Epoch 39, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.722, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.690, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.667, D_loss_fake = 0.703, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.662, D_loss_fake = 0.684, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.669, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.672, D_loss_fake = 0.684, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.680, D_loss_fake = 0.715, G_loss =

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0.714
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.695, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.698, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 8/46, D_loss_real = 0.715, D_loss_fake = 0.709, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.667, G_loss =
0.765
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.710, D_loss_fake = 0.642, G_loss =
0.793
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.729, D_loss_fake = 0.659, G_loss =
0.763
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.710, D_loss_fake = 0.659, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.726, D_loss_fake = 0.668, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.672, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.679, D_loss_fake = 0.687, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.669, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.633, D_loss_fake = 0.728, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.629, D_loss_fake = 0.761, G_loss =
0.665
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.609, D_loss_fake = 0.769, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.625, D_loss_fake = 0.763, G_loss =
0.654
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.637, D_loss_fake = 0.799, G_loss =

```

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0.659
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.636, D_loss_fake = 0.782, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 23/46, D_loss_real = 0.642, D_loss_fake = 0.766, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 24/46, D_loss_real = 0.665, D_loss_fake = 0.726, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 25/46, D_loss_real = 0.673, D_loss_fake = 0.707, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.727, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.672, D_loss_fake = 0.700, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.690, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.718, D_loss_fake = 0.676, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 30/46, D_loss_real = 0.708, D_loss_fake = 0.663, G_loss =
0.765
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.721, D_loss_fake = 0.659, G_loss =
0.752
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.728, D_loss_fake = 0.667, G_loss =
0.760
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.724, D_loss_fake = 0.662, G_loss =
0.753
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.728, D_loss_fake = 0.652, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.741, D_loss_fake = 0.638, G_loss =
0.778
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.729, D_loss_fake = 0.634, G_loss =
0.774
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.737, D_loss_fake = 0.630, G_loss =

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0.779
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.756, D_loss_fake = 0.632, G_loss =
0.772
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 39/46, D_loss_real = 0.735, D_loss_fake = 0.641, G_loss =
0.761
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 40/46, D_loss_real = 0.741, D_loss_fake = 0.642, G_loss =
0.758
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 41/46, D_loss_real = 0.732, D_loss_fake = 0.645, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.723, D_loss_fake = 0.655, G_loss =
0.752
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.720, D_loss_fake = 0.654, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.715, D_loss_fake = 0.664, G_loss =
0.738
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.665, G_loss =
0.728
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 46/46, D_loss_real = 0.692, D_loss_fake = 0.681, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 2/46, D_loss_real = 0.669, D_loss_fake = 0.699, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 3/46, D_loss_real = 0.667, D_loss_fake = 0.691, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 4/46, D_loss_real = 0.644, D_loss_fake = 0.720, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 5/46, D_loss_real = 0.671, D_loss_fake = 0.732, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.737, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 7/46, D_loss_real = 0.674, D_loss_fake = 0.732, G_loss =

```

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0.683
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 8/46, D_loss_real = 0.674, D_loss_fake = 0.735, G_loss =
0.694
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 9/46, D_loss_real = 0.674, D_loss_fake = 0.732, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.709, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 11/46, D_loss_real = 0.683, D_loss_fake = 0.703, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.706, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.706, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.715, G_loss =
0.693
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.713, G_loss =
0.689
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 16/46, D_loss_real = 0.684, D_loss_fake = 0.701, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.685, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.698, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.683, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.679, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.682, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 23/46, D_loss_real = 0.682, D_loss_fake = 0.686, G_loss =

```

```

0.720
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 24/46, D_loss_real = 0.684, D_loss_fake = 0.679, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 25/46, D_loss_real = 0.683, D_loss_fake = 0.682, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 27/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 28/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.682, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.676, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.668, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.663, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.651, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 35/46, D_loss_real = 0.706, D_loss_fake = 0.652, G_loss =
0.755
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 36/46, D_loss_real = 0.702, D_loss_fake = 0.665, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 37/46, D_loss_real = 0.689, D_loss_fake = 0.661, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.665, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 39/46, D_loss_real = 0.677, D_loss_fake = 0.688, G_loss =

```

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0.708
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.710, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 41/46, D_loss_real = 0.683, D_loss_fake = 0.708, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.708, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 43/46, D_loss_real = 0.706, D_loss_fake = 0.707, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.697, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 1/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 2/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.684, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 4/46, D_loss_real = 0.668, D_loss_fake = 0.721, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.710, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.678, G_loss =
0.731
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 7/46, D_loss_real = 0.707, D_loss_fake = 0.687, G_loss =
0.780
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 8/46, D_loss_real = 0.671, D_loss_fake = 0.705, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 9/46, D_loss_real = 0.670, D_loss_fake = 0.719, G_loss =

```

```

0.688
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 10/46, D_loss_real = 0.702, D_loss_fake = 0.737, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 11/46, D_loss_real = 0.681, D_loss_fake = 0.742, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.713, G_loss =
0.697
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 13/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 14/46, D_loss_real = 0.701, D_loss_fake = 0.703, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 15/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.671, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.672, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 20/46, D_loss_real = 0.664, D_loss_fake = 0.672, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 21/46, D_loss_real = 0.655, D_loss_fake = 0.668, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 22/46, D_loss_real = 0.632, D_loss_fake = 0.666, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 23/46, D_loss_real = 0.606, D_loss_fake = 0.700, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 24/46, D_loss_real = 0.610, D_loss_fake = 0.713, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 25/46, D_loss_real = 0.575, D_loss_fake = 0.757, G_loss =

```



```

0.682
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 26/46, D_loss_real = 0.566, D_loss_fake = 0.800, G_loss =
0.656
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 27/46, D_loss_real = 0.576, D_loss_fake = 0.841, G_loss =
0.661
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 28/46, D_loss_real = 0.596, D_loss_fake = 0.784, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 29/46, D_loss_real = 0.610, D_loss_fake = 0.806, G_loss =
0.686
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 30/46, D_loss_real = 0.642, D_loss_fake = 0.756, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.708, G_loss =
0.763
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 32/46, D_loss_real = 0.677, D_loss_fake = 0.673, G_loss =
0.846
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss =
0.790
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 34/46, D_loss_real = 0.725, D_loss_fake = 0.700, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 35/46, D_loss_real = 0.732, D_loss_fake = 0.687, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 36/46, D_loss_real = 0.670, D_loss_fake = 0.671, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 37/46, D_loss_real = 0.671, D_loss_fake = 0.673, G_loss =
0.736
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 38/46, D_loss_real = 0.622, D_loss_fake = 0.696, G_loss =
0.730
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 39/46, D_loss_real = 0.620, D_loss_fake = 0.683, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 40/46, D_loss_real = 0.619, D_loss_fake = 0.741, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 41/46, D_loss_real = 0.642, D_loss_fake = 0.831, G_loss =

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0.633
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 42/46, D_loss_real = 0.596, D_loss_fake = 0.867, G_loss =
0.641
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 43/46, D_loss_real = 0.643, D_loss_fake = 0.793, G_loss =
0.643
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 44/46, D_loss_real = 0.662, D_loss_fake = 0.771, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.739, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.695, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 1/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 2/46, D_loss_real = 0.715, D_loss_fake = 0.696, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 3/46, D_loss_real = 0.719, D_loss_fake = 0.691, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 4/46, D_loss_real = 0.710, D_loss_fake = 0.678, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 5/46, D_loss_real = 0.710, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 6/46, D_loss_real = 0.711, D_loss_fake = 0.680, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 7/46, D_loss_real = 0.711, D_loss_fake = 0.678, G_loss =
0.719
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 8/46, D_loss_real = 0.721, D_loss_fake = 0.687, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 9/46, D_loss_real = 0.709, D_loss_fake = 0.686, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.678, G_loss =

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0.716
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 12/46, D_loss_real = 0.701, D_loss_fake = 0.683, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 13/46, D_loss_real = 0.703, D_loss_fake = 0.681, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.675, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.678, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 16/46, D_loss_real = 0.689, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 18/46, D_loss_real = 0.689, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.681, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 20/46, D_loss_real = 0.683, D_loss_fake = 0.686, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 21/46, D_loss_real = 0.675, D_loss_fake = 0.684, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.683, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 23/46, D_loss_real = 0.676, D_loss_fake = 0.686, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 24/46, D_loss_real = 0.674, D_loss_fake = 0.686, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.694, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 26/46, D_loss_real = 0.698, D_loss_fake = 0.702, G_loss =
0.720
2/2 [=====] - 0s 5ms/step
Epoch 43, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss =

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0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.700, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.693, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 31/46, D_loss_real = 0.703, D_loss_fake = 0.692, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 32/46, D_loss_real = 0.705, D_loss_fake = 0.673, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.670, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 34/46, D_loss_real = 0.701, D_loss_fake = 0.685, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.685, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 36/46, D_loss_real = 0.703, D_loss_fake = 0.694, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.690, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.698, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.704, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.693, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.675, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.669, G_loss =

```

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0.747
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.694, G_loss =
0.732
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.690, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 46/46, D_loss_real = 0.662, D_loss_fake = 0.730, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 1/46, D_loss_real = 0.683, D_loss_fake = 0.736, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.720, G_loss =
0.712
2/2 [=====] - 0s 6ms/step
Epoch 44, Iteration 3/46, D_loss_real = 0.684, D_loss_fake = 0.712, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 4/46, D_loss_real = 0.706, D_loss_fake = 0.711, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.689, G_loss =
0.696
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 7/46, D_loss_real = 0.678, D_loss_fake = 0.703, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 8/46, D_loss_real = 0.681, D_loss_fake = 0.722, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 9/46, D_loss_real = 0.667, D_loss_fake = 0.714, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.736, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 11/46, D_loss_real = 0.661, D_loss_fake = 0.714, G_loss =
0.691
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.718, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 13/46, D_loss_real = 0.666, D_loss_fake = 0.693, G_loss =

```

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0.710
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 14/46, D_loss_real = 0.675, D_loss_fake = 0.691, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 15/46, D_loss_real = 0.669, D_loss_fake = 0.687, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.669, G_loss =
0.750
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.664, G_loss =
0.752
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 18/46, D_loss_real = 0.681, D_loss_fake = 0.668, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 19/46, D_loss_real = 0.696, D_loss_fake = 0.684, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.702, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 22/46, D_loss_real = 0.675, D_loss_fake = 0.707, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 23/46, D_loss_real = 0.669, D_loss_fake = 0.711, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 24/46, D_loss_real = 0.679, D_loss_fake = 0.691, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 25/46, D_loss_real = 0.682, D_loss_fake = 0.688, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 26/46, D_loss_real = 0.678, D_loss_fake = 0.671, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.666, G_loss =
0.745
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.665, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.656, G_loss =

```

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0.759
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.660, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.666, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.707, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 34/46, D_loss_real = 0.702, D_loss_fake = 0.697, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.715, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 37/46, D_loss_real = 0.692, D_loss_fake = 0.719, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.717, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.691, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 40/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.676, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 42/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.679, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 44/46, D_loss_real = 0.702, D_loss_fake = 0.686, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 45/46, D_loss_real = 0.695, D_loss_fake = 0.696, G_loss =

```

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0.702
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.702, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 1/46, D_loss_real = 0.698, D_loss_fake = 0.698, G_loss =
0.702
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 2/46, D_loss_real = 0.684, D_loss_fake = 0.703, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 3/46, D_loss_real = 0.664, D_loss_fake = 0.705, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 4/46, D_loss_real = 0.685, D_loss_fake = 0.704, G_loss =
0.687
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.712, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.720, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 7/46, D_loss_real = 0.655, D_loss_fake = 0.721, G_loss =
0.689
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 8/46, D_loss_real = 0.652, D_loss_fake = 0.721, G_loss =
0.688
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 9/46, D_loss_real = 0.655, D_loss_fake = 0.720, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 10/46, D_loss_real = 0.666, D_loss_fake = 0.717, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 11/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 12/46, D_loss_real = 0.671, D_loss_fake = 0.707, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 13/46, D_loss_real = 0.673, D_loss_fake = 0.716, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 14/46, D_loss_real = 0.663, D_loss_fake = 0.712, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.691, G_loss =

```



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0.722
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.683, G_loss =
0.729
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.690, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 18/46, D_loss_real = 0.709, D_loss_fake = 0.679, G_loss =
0.769
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 19/46, D_loss_real = 0.715, D_loss_fake = 0.649, G_loss =
0.778
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 20/46, D_loss_real = 0.708, D_loss_fake = 0.642, G_loss =
0.798
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 21/46, D_loss_real = 0.716, D_loss_fake = 0.632, G_loss =
0.777
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.649, G_loss =
0.753
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 23/46, D_loss_real = 0.709, D_loss_fake = 0.687, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.693, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 25/46, D_loss_real = 0.696, D_loss_fake = 0.704, G_loss =
0.701
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.709, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 27/46, D_loss_real = 0.678, D_loss_fake = 0.690, G_loss =
0.735
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 28/46, D_loss_real = 0.679, D_loss_fake = 0.683, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.681, G_loss =
0.712
2/2 [=====] - 0s 5ms/step
Epoch 45, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.700, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss =

```

```

0.695
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 32/46, D_loss_real = 0.665, D_loss_fake = 0.713, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 33/46, D_loss_real = 0.661, D_loss_fake = 0.721, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 34/46, D_loss_real = 0.650, D_loss_fake = 0.724, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 35/46, D_loss_real = 0.671, D_loss_fake = 0.729, G_loss =
0.692
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 36/46, D_loss_real = 0.667, D_loss_fake = 0.738, G_loss =
0.677
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 37/46, D_loss_real = 0.681, D_loss_fake = 0.728, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.712, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.677, G_loss =
0.789
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 40/46, D_loss_real = 0.722, D_loss_fake = 0.668, G_loss =
0.765
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 41/46, D_loss_real = 0.722, D_loss_fake = 0.674, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 42/46, D_loss_real = 0.731, D_loss_fake = 0.682, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.678, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.683, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.693, G_loss =

```

```

0.706
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 2/46, D_loss_real = 0.681, D_loss_fake = 0.695, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.699, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.706, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.683, G_loss =
0.725
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.667, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 9/46, D_loss_real = 0.700, D_loss_fake = 0.673, G_loss =
0.739
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 10/46, D_loss_real = 0.709, D_loss_fake = 0.663, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 11/46, D_loss_real = 0.706, D_loss_fake = 0.653, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 12/46, D_loss_real = 0.718, D_loss_fake = 0.671, G_loss =
0.764
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 13/46, D_loss_real = 0.722, D_loss_fake = 0.676, G_loss =
0.763
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 14/46, D_loss_real = 0.723, D_loss_fake = 0.646, G_loss =
0.757
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 15/46, D_loss_real = 0.736, D_loss_fake = 0.657, G_loss =
0.755
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 16/46, D_loss_real = 0.727, D_loss_fake = 0.653, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 17/46, D_loss_real = 0.721, D_loss_fake = 0.660, G_loss =

```

```

0.743
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 18/46, D_loss_real = 0.727, D_loss_fake = 0.656, G_loss =
0.738
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 19/46, D_loss_real = 0.729, D_loss_fake = 0.664, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 20/46, D_loss_real = 0.712, D_loss_fake = 0.672, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 21/46, D_loss_real = 0.722, D_loss_fake = 0.676, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 22/46, D_loss_real = 0.713, D_loss_fake = 0.671, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 23/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.673, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 25/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss =
0.706
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.682, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 29/46, D_loss_real = 0.708, D_loss_fake = 0.686, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 30/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 31/46, D_loss_real = 0.714, D_loss_fake = 0.667, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 32/46, D_loss_real = 0.721, D_loss_fake = 0.672, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 33/46, D_loss_real = 0.716, D_loss_fake = 0.677, G_loss =

```

```

0.726
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 34/46, D_loss_real = 0.723, D_loss_fake = 0.683, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 35/46, D_loss_real = 0.723, D_loss_fake = 0.689, G_loss =
0.708
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 36/46, D_loss_real = 0.712, D_loss_fake = 0.690, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 37/46, D_loss_real = 0.715, D_loss_fake = 0.693, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 38/46, D_loss_real = 0.717, D_loss_fake = 0.689, G_loss =
0.702
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.689, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 40/46, D_loss_real = 0.715, D_loss_fake = 0.691, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 41/46, D_loss_real = 0.710, D_loss_fake = 0.695, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.678, G_loss =
0.717
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.673, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 44/46, D_loss_real = 0.696, D_loss_fake = 0.676, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.667, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 46/46, D_loss_real = 0.708, D_loss_fake = 0.658, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 1/46, D_loss_real = 0.709, D_loss_fake = 0.664, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 2/46, D_loss_real = 0.705, D_loss_fake = 0.662, G_loss =
0.740
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 3/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss =

```

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0.734
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 4/46, D_loss_real = 0.718, D_loss_fake = 0.672, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 5/46, D_loss_real = 0.706, D_loss_fake = 0.680, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.681, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 8/46, D_loss_real = 0.711, D_loss_fake = 0.687, G_loss =
0.748
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 9/46, D_loss_real = 0.709, D_loss_fake = 0.660, G_loss =
0.757
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 10/46, D_loss_real = 0.722, D_loss_fake = 0.633, G_loss =
0.777
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 11/46, D_loss_real = 0.718, D_loss_fake = 0.650, G_loss =
0.768
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 12/46, D_loss_real = 0.724, D_loss_fake = 0.648, G_loss =
0.757
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 13/46, D_loss_real = 0.715, D_loss_fake = 0.687, G_loss =
0.745
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 14/46, D_loss_real = 0.712, D_loss_fake = 0.688, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 15/46, D_loss_real = 0.724, D_loss_fake = 0.712, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 16/46, D_loss_real = 0.714, D_loss_fake = 0.747, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 17/46, D_loss_real = 0.716, D_loss_fake = 0.674, G_loss =
0.778
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 18/46, D_loss_real = 0.715, D_loss_fake = 0.683, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.688, G_loss =

```

```

0.706
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 20/46, D_loss_real = 0.695, D_loss_fake = 0.688, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.695, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.698, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 23/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.702, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.700, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.696, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 27/46, D_loss_real = 0.675, D_loss_fake = 0.698, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 28/46, D_loss_real = 0.676, D_loss_fake = 0.692, G_loss =
0.699
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 30/46, D_loss_real = 0.666, D_loss_fake = 0.695, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 31/46, D_loss_real = 0.664, D_loss_fake = 0.701, G_loss =
0.707
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 32/46, D_loss_real = 0.676, D_loss_fake = 0.681, G_loss =
0.719
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 33/46, D_loss_real = 0.663, D_loss_fake = 0.689, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 34/46, D_loss_real = 0.676, D_loss_fake = 0.693, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 35/46, D_loss_real = 0.667, D_loss_fake = 0.697, G_loss =

```

```

0.708
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.702, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 37/46, D_loss_real = 0.689, D_loss_fake = 0.732, G_loss =
0.686
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 38/46, D_loss_real = 0.679, D_loss_fake = 0.735, G_loss =
0.685
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 39/46, D_loss_real = 0.677, D_loss_fake = 0.730, G_loss =
0.715
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.695, G_loss =
0.787
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 41/46, D_loss_real = 0.711, D_loss_fake = 0.661, G_loss =
0.785
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 42/46, D_loss_real = 0.720, D_loss_fake = 0.666, G_loss =
0.753
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 43/46, D_loss_real = 0.724, D_loss_fake = 0.679, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 44/46, D_loss_real = 0.718, D_loss_fake = 0.682, G_loss =
0.723
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 45/46, D_loss_real = 0.717, D_loss_fake = 0.685, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 46/46, D_loss_real = 0.721, D_loss_fake = 0.670, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 1/46, D_loss_real = 0.715, D_loss_fake = 0.665, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 2/46, D_loss_real = 0.714, D_loss_fake = 0.653, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 3/46, D_loss_real = 0.706, D_loss_fake = 0.646, G_loss =
0.749
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 4/46, D_loss_real = 0.707, D_loss_fake = 0.644, G_loss =
0.751
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.650, G_loss =

```



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0.746
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 6/46, D_loss_real = 0.707, D_loss_fake = 0.648, G_loss =
0.747
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 7/46, D_loss_real = 0.683, D_loss_fake = 0.660, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 8/46, D_loss_real = 0.682, D_loss_fake = 0.668, G_loss =
0.742
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.674, G_loss =
0.724
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 10/46, D_loss_real = 0.680, D_loss_fake = 0.677, G_loss =
0.723
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.695, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 12/46, D_loss_real = 0.671, D_loss_fake = 0.706, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 13/46, D_loss_real = 0.672, D_loss_fake = 0.716, G_loss =
0.709
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 14/46, D_loss_real = 0.683, D_loss_fake = 0.697, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 15/46, D_loss_real = 0.690, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 16/46, D_loss_real = 0.679, D_loss_fake = 0.689, G_loss =
0.714
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss =
0.709
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 18/46, D_loss_real = 0.701, D_loss_fake = 0.701, G_loss =
0.722
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.685, G_loss =
0.722
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 20/46, D_loss_real = 0.682, D_loss_fake = 0.701, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 21/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss =

```

```

0.704
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 23/46, D_loss_real = 0.679, D_loss_fake = 0.715, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 24/46, D_loss_real = 0.666, D_loss_fake = 0.731, G_loss =
0.670
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 25/46, D_loss_real = 0.657, D_loss_fake = 0.743, G_loss =
0.673
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.747, G_loss =
0.684
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.702, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.676, G_loss =
0.744
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.665, G_loss =
0.754
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.677, G_loss =
0.730
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 31/46, D_loss_real = 0.708, D_loss_fake = 0.686, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.704, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 33/46, D_loss_real = 0.682, D_loss_fake = 0.733, G_loss =
0.687
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.733, G_loss =
0.696
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 35/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss =
0.692
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.702, G_loss =
0.695
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 37/46, D_loss_real = 0.670, D_loss_fake = 0.708, G_loss =

```

```

0.712
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 38/46, D_loss_real = 0.672, D_loss_fake = 0.693, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 39/46, D_loss_real = 0.665, D_loss_fake = 0.693, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.704, G_loss =
0.716
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 41/46, D_loss_real = 0.668, D_loss_fake = 0.699, G_loss =
0.710
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 42/46, D_loss_real = 0.671, D_loss_fake = 0.688, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 43/46, D_loss_real = 0.672, D_loss_fake = 0.709, G_loss =
0.706
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 44/46, D_loss_real = 0.695, D_loss_fake = 0.708, G_loss =
0.698
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 45/46, D_loss_real = 0.689, D_loss_fake = 0.718, G_loss =
0.704
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.682, G_loss =
0.716
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 2/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss =
0.724
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.680, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 4/46, D_loss_real = 0.702, D_loss_fake = 0.684, G_loss =
0.737
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 5/46, D_loss_real = 0.701, D_loss_fake = 0.672, G_loss =
0.741
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 6/46, D_loss_real = 0.709, D_loss_fake = 0.671, G_loss =
0.744
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 7/46, D_loss_real = 0.721, D_loss_fake = 0.672, G_loss =

```

```

0.746
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 8/46, D_loss_real = 0.724, D_loss_fake = 0.664, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 9/46, D_loss_real = 0.719, D_loss_fake = 0.671, G_loss =
0.733
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 10/46, D_loss_real = 0.717, D_loss_fake = 0.678, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 11/46, D_loss_real = 0.716, D_loss_fake = 0.669, G_loss =
0.721
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 12/46, D_loss_real = 0.707, D_loss_fake = 0.676, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 13/46, D_loss_real = 0.711, D_loss_fake = 0.677, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 14/46, D_loss_real = 0.708, D_loss_fake = 0.689, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.692, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 16/46, D_loss_real = 0.693, D_loss_fake = 0.693, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 17/46, D_loss_real = 0.681, D_loss_fake = 0.693, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.699, G_loss =
0.712
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.691, G_loss =
0.711
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 20/46, D_loss_real = 0.663, D_loss_fake = 0.692, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 21/46, D_loss_real = 0.649, D_loss_fake = 0.703, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 22/46, D_loss_real = 0.645, D_loss_fake = 0.694, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.709, G_loss =

```

```

0.702
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 24/46, D_loss_real = 0.626, D_loss_fake = 0.751, G_loss =
0.683
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 25/46, D_loss_real = 0.633, D_loss_fake = 0.755, G_loss =
0.688
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 26/46, D_loss_real = 0.637, D_loss_fake = 0.762, G_loss =
0.705
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 27/46, D_loss_real = 0.663, D_loss_fake = 0.705, G_loss =
0.726
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 28/46, D_loss_real = 0.688, D_loss_fake = 0.690, G_loss =
0.731
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 29/46, D_loss_real = 0.712, D_loss_fake = 0.708, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.713, G_loss =
0.693
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.719, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.709, G_loss =
0.698
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.713, G_loss =
0.690
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 34/46, D_loss_real = 0.675, D_loss_fake = 0.720, G_loss =
0.700
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.716, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.700, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.677, G_loss =
0.735
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.664, G_loss =
0.743
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.658, G_loss =

```

```

0.756
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 40/46, D_loss_real = 0.666, D_loss_fake = 0.646, G_loss =
0.774
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.642, G_loss =
0.780
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.650, G_loss =
0.781
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.649, G_loss =
0.781
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.658, G_loss =
0.727
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 45/46, D_loss_real = 0.670, D_loss_fake = 0.714, G_loss =
0.708
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.796, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 1/46, D_loss_real = 0.707, D_loss_fake = 0.733, G_loss =
0.682
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 2/46, D_loss_real = 0.714, D_loss_fake = 0.736, G_loss =
0.738
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.675, G_loss =
0.809
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 4/46, D_loss_real = 0.708, D_loss_fake = 0.621, G_loss =
0.852
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 5/46, D_loss_real = 0.741, D_loss_fake = 0.667, G_loss =
0.752
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 6/46, D_loss_real = 0.733, D_loss_fake = 0.684, G_loss =
0.713
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 7/46, D_loss_real = 0.732, D_loss_fake = 0.681, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 8/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss =
0.717
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 9/46, D_loss_real = 0.678, D_loss_fake = 0.691, G_loss =

```

```

0.718
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.706, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.696, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 12/46, D_loss_real = 0.682, D_loss_fake = 0.714, G_loss =
0.714
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.672, G_loss =
0.725
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.685, G_loss =
0.726
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 16/46, D_loss_real = 0.663, D_loss_fake = 0.715, G_loss =
0.720
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 17/46, D_loss_real = 0.660, D_loss_fake = 0.736, G_loss =
0.703
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 18/46, D_loss_real = 0.674, D_loss_fake = 0.736, G_loss =
0.743
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 19/46, D_loss_real = 0.679, D_loss_fake = 0.705, G_loss =
0.782
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.687, G_loss =
0.746
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss =
0.707
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss =
0.704
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.712, G_loss =
0.699
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 24/46, D_loss_real = 0.681, D_loss_fake = 0.695, G_loss =
0.701
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 25/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss =

```

```

0.703
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.692, G_loss =
0.705
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 27/46, D_loss_real = 0.671, D_loss_fake = 0.694, G_loss =
0.711
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 28/46, D_loss_real = 0.674, D_loss_fake = 0.691, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 29/46, D_loss_real = 0.681, D_loss_fake = 0.689, G_loss =
0.712
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 30/46, D_loss_real = 0.674, D_loss_fake = 0.705, G_loss =
0.713
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 31/46, D_loss_real = 0.667, D_loss_fake = 0.702, G_loss =
0.721
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.696, G_loss =
0.715
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 33/46, D_loss_real = 0.680, D_loss_fake = 0.697, G_loss =
0.710
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 34/46, D_loss_real = 0.674, D_loss_fake = 0.698, G_loss =
0.703
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 35/46, D_loss_real = 0.670, D_loss_fake = 0.731, G_loss =
0.694
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 36/46, D_loss_real = 0.676, D_loss_fake = 0.711, G_loss =
0.697
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 37/46, D_loss_real = 0.673, D_loss_fake = 0.709, G_loss =
0.685
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 38/46, D_loss_real = 0.672, D_loss_fake = 0.733, G_loss =
0.678
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 39/46, D_loss_real = 0.673, D_loss_fake = 0.732, G_loss =
0.690
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.720, G_loss =
0.700
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 41/46, D_loss_real = 0.682, D_loss_fake = 0.711, G_loss =

```



```

0.714
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.680, G_loss =
0.718
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss =
0.734
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 44/46, D_loss_real = 0.704, D_loss_fake = 0.664, G_loss =
0.737
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.686, G_loss =
0.733
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss =
0.735
2/2 [=====] - 0s 90ms/step

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet
to be built. `model.compile_metrics` will be empty until you train or evaluate
the model.

>Saved: plot_016x016-faded.png, g_model_016x016-faded.sav,
d_model_016x016-faded.sav and gan_model_016x016-faded.sav
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 1/46, D_loss_real = 0.709, D_loss_fake = 0.646, G_loss =
0.781
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 2/46, D_loss_real = 0.712, D_loss_fake = 0.645, G_loss =
0.772
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 3/46, D_loss_real = 0.706, D_loss_fake = 0.644, G_loss =
0.758
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 4/46, D_loss_real = 0.705, D_loss_fake = 0.671, G_loss =
0.741
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.661, G_loss =
0.728
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss =
0.727
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 7/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss =
0.720
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 8/46, D_loss_real = 0.700, D_loss_fake = 0.676, G_loss =
0.730
2/2 [=====] - 0s 3ms/step

```

Epoch 1, Iteration 9/46, D_loss_real = 0.671, D_loss_fake = 0.677, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.698, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 11/46, D_loss_real = 0.657, D_loss_fake = 0.692, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.709, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.739, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.752, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.779, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 16/46, D_loss_real = 0.701, D_loss_fake = 0.700, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.661, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 18/46, D_loss_real = 0.700, D_loss_fake = 0.654, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.668, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 21/46, D_loss_real = 0.708, D_loss_fake = 0.700, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 23/46, D_loss_real = 0.703, D_loss_fake = 0.700, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 24/46, D_loss_real = 0.684, D_loss_fake = 0.698, G_loss = 0.712
2/2 [=====] - 0s 3ms/step

Epoch 1, Iteration 25/46, D_loss_real = 0.687, D_loss_fake = 0.689, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.678, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 27/46, D_loss_real = 0.698, D_loss_fake = 0.680, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.677, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.673, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 30/46, D_loss_real = 0.703, D_loss_fake = 0.665, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.692, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 32/46, D_loss_real = 0.674, D_loss_fake = 0.683, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 33/46, D_loss_real = 0.681, D_loss_fake = 0.706, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.712, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.697, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 37/46, D_loss_real = 0.675, D_loss_fake = 0.701, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 38/46, D_loss_real = 0.660, D_loss_fake = 0.710, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 1, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.721, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 40/46, D_loss_real = 0.686, D_loss_fake = 0.714, G_loss = 0.687
2/2 [=====] - 0s 3ms/step

Epoch 1, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.757, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 42/46, D_loss_real = 0.707, D_loss_fake = 0.732, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 43/46, D_loss_real = 0.716, D_loss_fake = 0.690, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 44/46, D_loss_real = 0.723, D_loss_fake = 0.634, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 45/46, D_loss_real = 0.731, D_loss_fake = 0.656, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 1, Iteration 46/46, D_loss_real = 0.748, D_loss_fake = 0.674, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 1/46, D_loss_real = 0.742, D_loss_fake = 0.690, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 2/46, D_loss_real = 0.718, D_loss_fake = 0.681, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 3/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 4/46, D_loss_real = 0.710, D_loss_fake = 0.682, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.675, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.672, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.673, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.673, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.666, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 10/46, D_loss_real = 0.692, D_loss_fake = 0.668, G_loss = 0.739
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.680, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 12/46, D_loss_real = 0.700, D_loss_fake = 0.674, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 13/46, D_loss_real = 0.726, D_loss_fake = 0.697, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 14/46, D_loss_real = 0.700, D_loss_fake = 0.693, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.699, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.710, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 17/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.670, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 20/46, D_loss_real = 0.696, D_loss_fake = 0.660, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 22/46, D_loss_real = 0.698, D_loss_fake = 0.676, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 23/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 24/46, D_loss_real = 0.712, D_loss_fake = 0.688, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.698, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.712, G_loss = 0.699
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.689, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 30/46, D_loss_real = 0.700, D_loss_fake = 0.670, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 31/46, D_loss_real = 0.703, D_loss_fake = 0.670, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.665, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 33/46, D_loss_real = 0.705, D_loss_fake = 0.664, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.674, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 36/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.681, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 39/46, D_loss_real = 0.691, D_loss_fake = 0.690, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.706, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.707, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.716, G_loss = 0.749
2/2 [=====] - 0s 3ms/step

Epoch 2, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.666, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 44/46, D_loss_real = 0.714, D_loss_fake = 0.674, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 2, Iteration 45/46, D_loss_real = 0.725, D_loss_fake = 0.676, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 2, Iteration 46/46, D_loss_real = 0.725, D_loss_fake = 0.680, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 1/46, D_loss_real = 0.724, D_loss_fake = 0.678, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 2/46, D_loss_real = 0.714, D_loss_fake = 0.673, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 3/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.679, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 5/46, D_loss_real = 0.711, D_loss_fake = 0.680, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 7/46, D_loss_real = 0.690, D_loss_fake = 0.673, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 8/46, D_loss_real = 0.694, D_loss_fake = 0.697, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 9/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.687, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.698, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.715
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 13/46, D_loss_real = 0.678, D_loss_fake = 0.683, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.671, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.677, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 16/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 17/46, D_loss_real = 0.676, D_loss_fake = 0.717, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.709, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 21/46, D_loss_real = 0.700, D_loss_fake = 0.700, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.705, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 23/46, D_loss_real = 0.707, D_loss_fake = 0.688, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.686, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 25/46, D_loss_real = 0.715, D_loss_fake = 0.686, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 26/46, D_loss_real = 0.730, D_loss_fake = 0.684, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 27/46, D_loss_real = 0.711, D_loss_fake = 0.680, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 28/46, D_loss_real = 0.716, D_loss_fake = 0.693, G_loss = 0.709
2/2 [=====] - 0s 4ms/step

Epoch 3, Iteration 29/46, D_loss_real = 0.710, D_loss_fake = 0.686, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.697, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 31/46, D_loss_real = 0.710, D_loss_fake = 0.690, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 32/46, D_loss_real = 0.715, D_loss_fake = 0.683, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 33/46, D_loss_real = 0.706, D_loss_fake = 0.672, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.673, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 35/46, D_loss_real = 0.708, D_loss_fake = 0.675, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 36/46, D_loss_real = 0.697, D_loss_fake = 0.671, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 37/46, D_loss_real = 0.697, D_loss_fake = 0.672, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.671, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 39/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 41/46, D_loss_real = 0.681, D_loss_fake = 0.685, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 3, Iteration 42/46, D_loss_real = 0.674, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 43/46, D_loss_real = 0.676, D_loss_fake = 0.686, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 44/46, D_loss_real = 0.680, D_loss_fake = 0.697, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 3, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.687, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 3, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.700, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 1/46, D_loss_real = 0.673, D_loss_fake = 0.692, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 2/46, D_loss_real = 0.687, D_loss_fake = 0.694, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 3/46, D_loss_real = 0.676, D_loss_fake = 0.695, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 4/46, D_loss_real = 0.671, D_loss_fake = 0.721, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 5/46, D_loss_real = 0.682, D_loss_fake = 0.729, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 6/46, D_loss_real = 0.676, D_loss_fake = 0.722, G_loss = 0.671
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 7/46, D_loss_real = 0.676, D_loss_fake = 0.732, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 8/46, D_loss_real = 0.661, D_loss_fake = 0.724, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 9/46, D_loss_real = 0.656, D_loss_fake = 0.697, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 10/46, D_loss_real = 0.663, D_loss_fake = 0.689, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 11/46, D_loss_real = 0.664, D_loss_fake = 0.684, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 12/46, D_loss_real = 0.673, D_loss_fake = 0.707, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.721, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.733, G_loss = 0.667
2/2 [=====] - 0s 4ms/step

Epoch 4, Iteration 15/46, D_loss_real = 0.683, D_loss_fake = 0.747, G_loss = 0.682
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.714, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.707, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 18/46, D_loss_real = 0.699, D_loss_fake = 0.696, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 19/46, D_loss_real = 0.719, D_loss_fake = 0.689, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 20/46, D_loss_real = 0.716, D_loss_fake = 0.690, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 21/46, D_loss_real = 0.694, D_loss_fake = 0.683, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.680, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 24/46, D_loss_real = 0.709, D_loss_fake = 0.672, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 25/46, D_loss_real = 0.697, D_loss_fake = 0.678, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 26/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 27/46, D_loss_real = 0.692, D_loss_fake = 0.678, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.672, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.715
2/2 [=====] - 0s 4ms/step

Epoch 4, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 32/46, D_loss_real = 0.696, D_loss_fake = 0.707, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.702, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.700, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 35/46, D_loss_real = 0.677, D_loss_fake = 0.687, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.688, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.693, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 38/46, D_loss_real = 0.683, D_loss_fake = 0.684, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.675, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.673, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 4, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.669, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 4, Iteration 46/46, D_loss_real = 0.695, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 4ms/step

Epoch 5, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 3/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 4/46, D_loss_real = 0.681, D_loss_fake = 0.687, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.684, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.687, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.685, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 9/46, D_loss_real = 0.678, D_loss_fake = 0.687, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 10/46, D_loss_real = 0.670, D_loss_fake = 0.691, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 11/46, D_loss_real = 0.675, D_loss_fake = 0.699, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 13/46, D_loss_real = 0.659, D_loss_fake = 0.694, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 14/46, D_loss_real = 0.683, D_loss_fake = 0.694, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.685, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 16/46, D_loss_real = 0.664, D_loss_fake = 0.690, G_loss = 0.717
2/2 [=====] - 0s 4ms/step

Epoch 5, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.685, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 18/46, D_loss_real = 0.708, D_loss_fake = 0.680, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 19/46, D_loss_real = 0.710, D_loss_fake = 0.682, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 21/46, D_loss_real = 0.708, D_loss_fake = 0.674, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 22/46, D_loss_real = 0.705, D_loss_fake = 0.655, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.655, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 24/46, D_loss_real = 0.708, D_loss_fake = 0.655, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.665, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 26/46, D_loss_real = 0.694, D_loss_fake = 0.670, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.680, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 28/46, D_loss_real = 0.700, D_loss_fake = 0.684, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 30/46, D_loss_real = 0.683, D_loss_fake = 0.684, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.683, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 32/46, D_loss_real = 0.706, D_loss_fake = 0.666, G_loss = 0.778
2/2 [=====] - 0s 3ms/step

Epoch 5, Iteration 33/46, D_loss_real = 0.697, D_loss_fake = 0.643, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 34/46, D_loss_real = 0.707, D_loss_fake = 0.633, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 35/46, D_loss_real = 0.722, D_loss_fake = 0.658, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 36/46, D_loss_real = 0.713, D_loss_fake = 0.644, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 5, Iteration 37/46, D_loss_real = 0.706, D_loss_fake = 0.642, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.640, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.666, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 40/46, D_loss_real = 0.683, D_loss_fake = 0.690, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 41/46, D_loss_real = 0.687, D_loss_fake = 0.710, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 42/46, D_loss_real = 0.666, D_loss_fake = 0.720, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.701, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.668, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.686, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 5, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.727, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 2/46, D_loss_real = 0.697, D_loss_fake = 0.737, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 3/46, D_loss_real = 0.672, D_loss_fake = 0.740, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 4/46, D_loss_real = 0.696, D_loss_fake = 0.731, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.731, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.715, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 7/46, D_loss_real = 0.690, D_loss_fake = 0.696, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 8/46, D_loss_real = 0.694, D_loss_fake = 0.648, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.657, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 10/46, D_loss_real = 0.703, D_loss_fake = 0.663, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 11/46, D_loss_real = 0.708, D_loss_fake = 0.668, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.687, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 13/46, D_loss_real = 0.663, D_loss_fake = 0.694, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 14/46, D_loss_real = 0.720, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.695, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.727, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.702, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.701, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 21/46, D_loss_real = 0.680, D_loss_fake = 0.684, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 22/46, D_loss_real = 0.659, D_loss_fake = 0.687, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 23/46, D_loss_real = 0.659, D_loss_fake = 0.662, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 24/46, D_loss_real = 0.665, D_loss_fake = 0.651, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 25/46, D_loss_real = 0.648, D_loss_fake = 0.640, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 26/46, D_loss_real = 0.664, D_loss_fake = 0.651, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 27/46, D_loss_real = 0.645, D_loss_fake = 0.773, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 28/46, D_loss_real = 0.670, D_loss_fake = 0.728, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 29/46, D_loss_real = 0.687, D_loss_fake = 0.729, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.762, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 31/46, D_loss_real = 0.720, D_loss_fake = 0.701, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.722, D_loss_fake = 0.637, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.737, D_loss_fake = 0.668, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.720, D_loss_fake = 0.692, G_loss = 0.710
2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 35/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 36/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.686, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.669, D_loss_fake = 0.693, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.664, D_loss_fake = 0.683, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.660, D_loss_fake = 0.695, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.655, D_loss_fake = 0.690, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.655, D_loss_fake = 0.715, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.672, D_loss_fake = 0.708, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.637, D_loss_fake = 0.732, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.676, D_loss_fake = 0.741, G_loss = 0.667
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.663, D_loss_fake = 0.762, G_loss = 0.672
2/2 [=====] - 0s 8ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.666, D_loss_fake = 0.757, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.649, D_loss_fake = 0.728, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.657, D_loss_fake = 0.729, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.661, D_loss_fake = 0.721, G_loss = 0.692
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 5/46, D_loss_real = 0.670, D_loss_fake = 0.719, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.691, D_loss_fake = 0.707, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.704, D_loss_fake = 0.690, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.704, D_loss_fake = 0.684, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.700, D_loss_fake = 0.670, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.688, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.700, D_loss_fake = 0.692, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.699, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.709, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.709, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.694, D_loss_fake = 0.699, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.699, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.707, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss = 0.709
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.690, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.686, D_loss_fake = 0.683, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.701, D_loss_fake = 0.677, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.713, D_loss_fake = 0.681, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.711, D_loss_fake = 0.674, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.715, D_loss_fake = 0.657, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.712, D_loss_fake = 0.653, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.720, D_loss_fake = 0.646, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.727, D_loss_fake = 0.637, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.723, D_loss_fake = 0.631, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.721, D_loss_fake = 0.663, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.720, D_loss_fake = 0.652, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.716, D_loss_fake = 0.678, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.715, D_loss_fake = 0.678, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.716, D_loss_fake = 0.710, G_loss = 0.709
2/2 [=====] - 0s 4ms/step

Epoch 7, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.711, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.705, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.662, D_loss_fake = 0.713, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.687, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.664, D_loss_fake = 0.683, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.633, D_loss_fake = 0.687, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.701, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.653, D_loss_fake = 0.723, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.610, D_loss_fake = 0.738, G_loss = 0.674
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.618, D_loss_fake = 0.758, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.638, D_loss_fake = 0.750, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.629, D_loss_fake = 0.734, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.658, D_loss_fake = 0.724, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.647, D_loss_fake = 0.712, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.663, D_loss_fake = 0.681, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.657, D_loss_fake = 0.674, G_loss = 0.761
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.659, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.689, D_loss_fake = 0.658, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.705, D_loss_fake = 0.662, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.704, D_loss_fake = 0.674, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.727, D_loss_fake = 0.692, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.726, D_loss_fake = 0.709, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.697, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.674, D_loss_fake = 0.708, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.711, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.665, D_loss_fake = 0.705, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.705, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.657, D_loss_fake = 0.710, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.655, D_loss_fake = 0.702, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.644, D_loss_fake = 0.709, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.642, D_loss_fake = 0.729, G_loss = 0.713
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 23/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.664, D_loss_fake = 0.714, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.665, D_loss_fake = 0.732, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.721, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.668, D_loss_fake = 0.700, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.706, D_loss_fake = 0.679, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.737, D_loss_fake = 0.687, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.707, D_loss_fake = 0.692, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.706, D_loss_fake = 0.692, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.711, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.665, D_loss_fake = 0.709, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.694, D_loss_fake = 0.705, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.700, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.676, D_loss_fake = 0.716, G_loss = 0.697
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 39/46, D_loss_real = 0.662, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.702, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.665, D_loss_fake = 0.713, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.667, D_loss_fake = 0.702, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.670, D_loss_fake = 0.727, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.663, D_loss_fake = 0.701, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.674, D_loss_fake = 0.706, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.693, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.667, D_loss_fake = 0.686, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.703, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.714, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.694, D_loss_fake = 0.678, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.690, G_loss = 0.715
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.688, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.703, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.684, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.750, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.671, D_loss_fake = 0.697, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.685, D_loss_fake = 0.684, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.669, D_loss_fake = 0.714, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.678, D_loss_fake = 0.716, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.672, D_loss_fake = 0.706, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.709, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.676, D_loss_fake = 0.725, G_loss = 0.682
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 25/46, D_loss_real = 0.669, D_loss_fake = 0.717, G_loss = 0.680
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.678, D_loss_fake = 0.731, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.678, D_loss_fake = 0.733, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.713, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.697, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.697, D_loss_fake = 0.705, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.699, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.699, D_loss_fake = 0.697, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.697, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.689, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.673, G_loss = 0.729
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 41/46, D_loss_real = 0.680, D_loss_fake = 0.677, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.688, D_loss_fake = 0.675, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.667, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.661, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.671, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.675, D_loss_fake = 0.676, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.694, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.692, D_loss_fake = 0.696, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.706, D_loss_fake = 0.687, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.701, D_loss_fake = 0.697, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.718, D_loss_fake = 0.686, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.707, D_loss_fake = 0.686, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.715, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.710, D_loss_fake = 0.667, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.722, D_loss_fake = 0.658, G_loss = 0.758
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 11/46, D_loss_real = 0.713, D_loss_fake = 0.645, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.639, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.731, D_loss_fake = 0.649, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.720, D_loss_fake = 0.642, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.725, D_loss_fake = 0.652, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.726, D_loss_fake = 0.653, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.742, D_loss_fake = 0.668, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.720, D_loss_fake = 0.665, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.724, D_loss_fake = 0.670, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.721, D_loss_fake = 0.671, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.666, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.669, D_loss_fake = 0.678, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.671, D_loss_fake = 0.720, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.671, D_loss_fake = 0.714, G_loss = 0.724
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 27/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.664, D_loss_fake = 0.696, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.692, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.719, D_loss_fake = 0.685, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.715, D_loss_fake = 0.685, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.709, D_loss_fake = 0.678, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.705, D_loss_fake = 0.660, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.709, D_loss_fake = 0.653, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.660, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.693, D_loss_fake = 0.660, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.666, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.682, D_loss_fake = 0.667, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.678, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.673, D_loss_fake = 0.681, G_loss = 0.722
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 43/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.669, D_loss_fake = 0.706, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.672, D_loss_fake = 0.698, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.656, D_loss_fake = 0.704, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.671, D_loss_fake = 0.718, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.719, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.650, D_loss_fake = 0.716, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.668, D_loss_fake = 0.719, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.665, D_loss_fake = 0.702, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.661, D_loss_fake = 0.694, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.668, D_loss_fake = 0.693, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.658, D_loss_fake = 0.705, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.672, D_loss_fake = 0.681, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.709, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.692, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.703, G_loss = 0.725
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 13/46, D_loss_real = 0.701, D_loss_fake = 0.671, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.705, D_loss_fake = 0.670, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.705, D_loss_fake = 0.681, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.708, D_loss_fake = 0.683, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.692, D_loss_fake = 0.690, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.691, D_loss_fake = 0.691, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.684, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.680, D_loss_fake = 0.677, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.676, D_loss_fake = 0.674, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.667, D_loss_fake = 0.662, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.666, D_loss_fake = 0.653, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.657, D_loss_fake = 0.654, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.632, D_loss_fake = 0.672, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.630, D_loss_fake = 0.690, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.645, D_loss_fake = 0.774, G_loss = 0.723
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 29/46, D_loss_real = 0.618, D_loss_fake = 0.727, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.636, D_loss_fake = 0.740, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.636, D_loss_fake = 0.737, G_loss = 0.661
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.627, D_loss_fake = 0.781, G_loss = 0.653
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.641, D_loss_fake = 0.775, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.623, D_loss_fake = 0.807, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.660, D_loss_fake = 0.723, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.660, D_loss_fake = 0.679, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.656, D_loss_fake = 0.653, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.663, D_loss_fake = 0.682, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.628, D_loss_fake = 0.738, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.657, D_loss_fake = 0.718, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.631, D_loss_fake = 0.792, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.645, D_loss_fake = 0.732, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.662, D_loss_fake = 0.724, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.665, D_loss_fake = 0.704, G_loss = 0.718
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 45/46, D_loss_real = 0.661, D_loss_fake = 0.705, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.647, D_loss_fake = 0.677, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.659, D_loss_fake = 0.688, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.660, D_loss_fake = 0.704, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.661, D_loss_fake = 0.738, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.662, D_loss_fake = 0.737, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.636, D_loss_fake = 0.717, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.663, D_loss_fake = 0.708, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.664, D_loss_fake = 0.716, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.655, D_loss_fake = 0.730, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.639, D_loss_fake = 0.733, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.641, D_loss_fake = 0.724, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.624, D_loss_fake = 0.722, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.656, D_loss_fake = 0.665, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.667, D_loss_fake = 0.624, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.672, D_loss_fake = 0.688, G_loss = 0.769
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 15/46, D_loss_real = 0.659, D_loss_fake = 0.715, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.656, D_loss_fake = 0.761, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.678, D_loss_fake = 0.697, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.677, D_loss_fake = 0.689, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.725, D_loss_fake = 0.730, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.682, D_loss_fake = 0.724, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.760, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.744, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.678, D_loss_fake = 0.729, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.705, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.714, D_loss_fake = 0.681, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.706, D_loss_fake = 0.672, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.728, D_loss_fake = 0.684, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.711, D_loss_fake = 0.687, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.687, D_loss_fake = 0.680, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss = 0.734
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.672, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.674, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.679, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.672, D_loss_fake = 0.690, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.687, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.662, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.652, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.644, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.644, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.715, D_loss_fake = 0.649, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.657, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.660, D_loss_fake = 0.660, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.662, D_loss_fake = 0.670, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.693, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.692, D_loss_fake = 0.709, G_loss = 0.712
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.710, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.702, D_loss_fake = 0.695, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.686, D_loss_fake = 0.657, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.676, D_loss_fake = 0.627, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.678, D_loss_fake = 0.661, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.700, D_loss_fake = 0.629, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.725, D_loss_fake = 0.684, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.717, D_loss_fake = 0.700, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.707, D_loss_fake = 0.713, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.706, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.708, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.700, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.671, D_loss_fake = 0.669, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.665, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.643, D_loss_fake = 0.677, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.629, D_loss_fake = 0.687, G_loss = 0.770
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 17/46, D_loss_real = 0.608, D_loss_fake = 0.660, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.593, D_loss_fake = 0.715, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.613, D_loss_fake = 0.793, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.643, D_loss_fake = 0.801, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.726, D_loss_fake = 0.710, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.752, D_loss_fake = 0.698, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.751, D_loss_fake = 0.675, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.768, D_loss_fake = 0.673, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.703, D_loss_fake = 0.659, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.726, D_loss_fake = 0.653, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.719, D_loss_fake = 0.649, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.684, D_loss_fake = 0.652, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.668, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.671, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.663, D_loss_fake = 0.662, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.671, D_loss_fake = 0.648, G_loss = 0.759
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 33/46, D_loss_real = 0.660, D_loss_fake = 0.662, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.668, D_loss_fake = 0.676, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.655, D_loss_fake = 0.709, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.668, D_loss_fake = 0.789, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.657, D_loss_fake = 0.717, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.667, D_loss_fake = 0.673, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.665, D_loss_fake = 0.711, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.668, D_loss_fake = 0.725, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.658, D_loss_fake = 0.768, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.678, D_loss_fake = 0.681, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.651, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.707, D_loss_fake = 0.699, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.688, D_loss_fake = 0.703, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.700, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.650, D_loss_fake = 0.690, G_loss = 0.741
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.672, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.704, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.674, D_loss_fake = 0.719, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.699, D_loss_fake = 0.652, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.659, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.722, D_loss_fake = 0.639, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.731, D_loss_fake = 0.692, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.729, D_loss_fake = 0.663, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.718, D_loss_fake = 0.684, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.709, D_loss_fake = 0.703, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.733, D_loss_fake = 0.701, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.732, D_loss_fake = 0.689, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.708, D_loss_fake = 0.708, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.709, D_loss_fake = 0.700, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.671, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.662, G_loss = 0.751
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 19/46, D_loss_real = 0.707, D_loss_fake = 0.663, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.677, D_loss_fake = 0.667, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.697, D_loss_fake = 0.664, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.674, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.697, D_loss_fake = 0.691, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.719, D_loss_fake = 0.706, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.701, D_loss_fake = 0.725, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.717, D_loss_fake = 0.741, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.717, D_loss_fake = 0.719, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.734, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.722, D_loss_fake = 0.695, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.718, D_loss_fake = 0.689, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.729, D_loss_fake = 0.665, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.669, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.729, D_loss_fake = 0.661, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.705, D_loss_fake = 0.656, G_loss = 0.751
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.651, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.689, D_loss_fake = 0.648, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.712, D_loss_fake = 0.650, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.658, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.644, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.687, D_loss_fake = 0.658, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.679, D_loss_fake = 0.648, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.674, D_loss_fake = 0.661, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.667, D_loss_fake = 0.660, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.650, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.676, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.666, D_loss_fake = 0.699, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.653, D_loss_fake = 0.701, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.643, D_loss_fake = 0.739, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.658, D_loss_fake = 0.735, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.642, D_loss_fake = 0.709, G_loss = 0.712
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 5/46, D_loss_real = 0.672, D_loss_fake = 0.693, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.663, D_loss_fake = 0.719, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.678, D_loss_fake = 0.705, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.687, D_loss_fake = 0.696, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.662, D_loss_fake = 0.694, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.660, D_loss_fake = 0.716, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.648, D_loss_fake = 0.710, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.654, D_loss_fake = 0.704, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.651, D_loss_fake = 0.708, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.657, D_loss_fake = 0.714, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.724, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.672, D_loss_fake = 0.712, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.710, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.685, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.685, D_loss_fake = 0.676, G_loss = 0.741
2/2 [=====] - 0s 3ms/step

Epoch 15, Iteration 21/46, D_loss_real = 0.705, D_loss_fake = 0.690, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.720, D_loss_fake = 0.684, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.716, D_loss_fake = 0.698, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.716, D_loss_fake = 0.701, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.708, D_loss_fake = 0.707, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.707, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.711, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.733, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.710, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.675, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.709, D_loss_fake = 0.676, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.702, D_loss_fake = 0.642, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.683, D_loss_fake = 0.635, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.637, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.689, D_loss_fake = 0.673, G_loss = 0.804
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.666, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.670, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.688, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.692, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.713, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.666, D_loss_fake = 0.708, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.681, D_loss_fake = 0.698, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.697, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.711, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.671, D_loss_fake = 0.742, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.715, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.718, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.712, D_loss_fake = 0.719, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.711, D_loss_fake = 0.688, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.726, D_loss_fake = 0.666, G_loss = 0.768
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 7/46, D_loss_real = 0.719, D_loss_fake = 0.649, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.648, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.699, D_loss_fake = 0.629, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.702, D_loss_fake = 0.646, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.647, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.701, D_loss_fake = 0.646, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.697, D_loss_fake = 0.663, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.678, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.664, D_loss_fake = 0.693, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.669, D_loss_fake = 0.746, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.677, D_loss_fake = 0.768, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.707, D_loss_fake = 0.742, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.717, D_loss_fake = 0.663, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.712, D_loss_fake = 0.648, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.708, D_loss_fake = 0.661, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.713, D_loss_fake = 0.660, G_loss = 0.733
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 23/46, D_loss_real = 0.706, D_loss_fake = 0.664, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.659, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.676, D_loss_fake = 0.665, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.671, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.691, D_loss_fake = 0.657, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.676, D_loss_fake = 0.679, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.677, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.672, D_loss_fake = 0.659, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.704, D_loss_fake = 0.682, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.669, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.679, D_loss_fake = 0.669, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.666, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.687, D_loss_fake = 0.656, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.652, D_loss_fake = 0.694, G_loss = 0.750
2/2 [=====] - 0s 3ms/step

Epoch 16, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.717, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.676, D_loss_fake = 0.680, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.677, D_loss_fake = 0.667, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.667, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.688, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.650, D_loss_fake = 0.673, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.639, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.629, D_loss_fake = 0.722, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.618, D_loss_fake = 0.742, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.648, D_loss_fake = 0.796, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.642, D_loss_fake = 0.765, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.655, D_loss_fake = 0.778, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.673, D_loss_fake = 0.701, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.712, D_loss_fake = 0.679, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.726, D_loss_fake = 0.695, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.732, D_loss_fake = 0.668, G_loss = 0.744
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 9/46, D_loss_real = 0.719, D_loss_fake = 0.699, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.732, D_loss_fake = 0.671, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.735, D_loss_fake = 0.689, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.718, D_loss_fake = 0.665, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.720, D_loss_fake = 0.654, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.731, D_loss_fake = 0.647, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.735, D_loss_fake = 0.638, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.720, D_loss_fake = 0.661, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.667, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.682, D_loss_fake = 0.700, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.676, D_loss_fake = 0.719, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.695, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.643, D_loss_fake = 0.702, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.649, D_loss_fake = 0.707, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.613, D_loss_fake = 0.703, G_loss = 0.698
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 25/46, D_loss_real = 0.632, D_loss_fake = 0.733, G_loss = 0.679
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.633, D_loss_fake = 0.824, G_loss = 0.668
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.579, D_loss_fake = 0.857, G_loss = 0.672
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.625, D_loss_fake = 0.783, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.736, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.646, D_loss_fake = 0.717, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.714, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.714, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.710, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.675, D_loss_fake = 0.701, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.664, D_loss_fake = 0.706, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.665, D_loss_fake = 0.702, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.665, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.670, D_loss_fake = 0.700, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.683, D_loss_fake = 0.698, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.661, D_loss_fake = 0.681, G_loss = 0.738
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.675, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.702, D_loss_fake = 0.688, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.705, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.699, D_loss_fake = 0.709, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.723, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.701, D_loss_fake = 0.722, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.728, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.712, G_loss = 0.686
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.715, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.677, D_loss_fake = 0.694, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.672, D_loss_fake = 0.696, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.685, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.673, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.672, D_loss_fake = 0.664, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.670, D_loss_fake = 0.670, G_loss = 0.735
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 11/46, D_loss_real = 0.663, D_loss_fake = 0.678, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.656, D_loss_fake = 0.696, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.673, D_loss_fake = 0.695, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.660, D_loss_fake = 0.712, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.666, D_loss_fake = 0.703, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.670, D_loss_fake = 0.689, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.674, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.702, D_loss_fake = 0.704, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.671, D_loss_fake = 0.692, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.672, D_loss_fake = 0.731, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.671, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.661, D_loss_fake = 0.681, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.700, D_loss_fake = 0.683, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.707, D_loss_fake = 0.697, G_loss = 0.721
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 27/46, D_loss_real = 0.714, D_loss_fake = 0.687, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.715, D_loss_fake = 0.661, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.709, D_loss_fake = 0.658, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.728, D_loss_fake = 0.667, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.660, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.723, D_loss_fake = 0.660, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.725, D_loss_fake = 0.651, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.728, D_loss_fake = 0.670, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.712, D_loss_fake = 0.664, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.691, D_loss_fake = 0.660, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.716, D_loss_fake = 0.658, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.727, D_loss_fake = 0.660, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.736, D_loss_fake = 0.625, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.715, D_loss_fake = 0.645, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.733, D_loss_fake = 0.628, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.729, D_loss_fake = 0.666, G_loss = 0.804
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 43/46, D_loss_real = 0.729, D_loss_fake = 0.633, G_loss = 0.810
2/2 [=====] - 0s 7ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.740, D_loss_fake = 0.619, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.720, D_loss_fake = 0.625, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.716, D_loss_fake = 0.641, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.713, D_loss_fake = 0.654, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.701, D_loss_fake = 0.659, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.705, D_loss_fake = 0.663, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.707, D_loss_fake = 0.668, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.662, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.662, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.690, D_loss_fake = 0.661, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.667, D_loss_fake = 0.671, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.635, D_loss_fake = 0.675, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.669, D_loss_fake = 0.706, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.671, D_loss_fake = 0.732, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.646, D_loss_fake = 0.702, G_loss = 0.725
2/2 [=====] - 0s 4ms/step

Epoch 19, Iteration 13/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.663, D_loss_fake = 0.725, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.670, D_loss_fake = 0.712, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.683, D_loss_fake = 0.689, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.657, D_loss_fake = 0.679, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.669, D_loss_fake = 0.671, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.663, D_loss_fake = 0.645, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.669, D_loss_fake = 0.641, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.668, D_loss_fake = 0.662, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.677, D_loss_fake = 0.691, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.668, D_loss_fake = 0.655, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.675, D_loss_fake = 0.683, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.631, D_loss_fake = 0.705, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.670, D_loss_fake = 0.713, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.651, D_loss_fake = 0.795, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.724, G_loss = 0.739
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 29/46, D_loss_real = 0.658, D_loss_fake = 0.683, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.727, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.733, D_loss_fake = 0.630, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.631, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.646, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.644, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.708, D_loss_fake = 0.671, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.689, D_loss_fake = 0.722, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.704, D_loss_fake = 0.702, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.673, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.706, D_loss_fake = 0.704, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.716, D_loss_fake = 0.702, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.700, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.708, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.711, D_loss_fake = 0.732, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.716, D_loss_fake = 0.686, G_loss = 0.731
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.669, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.721, D_loss_fake = 0.671, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.731, D_loss_fake = 0.678, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.702, D_loss_fake = 0.678, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.676, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.704, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.696, D_loss_fake = 0.685, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.709, D_loss_fake = 0.697, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.672, D_loss_fake = 0.695, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.667, D_loss_fake = 0.683, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.679, D_loss_fake = 0.702, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.666, D_loss_fake = 0.692, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.686, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.663, D_loss_fake = 0.695, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss = 0.678
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.647, D_loss_fake = 0.760, G_loss = 0.672
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 15/46, D_loss_real = 0.646, D_loss_fake = 0.830, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.695, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.708, D_loss_fake = 0.626, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.720, D_loss_fake = 0.619, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.718, D_loss_fake = 0.643, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.735, D_loss_fake = 0.648, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.732, D_loss_fake = 0.648, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.741, D_loss_fake = 0.655, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.739, D_loss_fake = 0.670, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.719, D_loss_fake = 0.673, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.705, D_loss_fake = 0.694, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.712, D_loss_fake = 0.701, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.707, D_loss_fake = 0.664, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.655, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.693, D_loss_fake = 0.663, G_loss = 0.757
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 31/46, D_loss_real = 0.699, D_loss_fake = 0.663, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.710, D_loss_fake = 0.661, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.685, D_loss_fake = 0.671, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.675, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.668, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.653, D_loss_fake = 0.693, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.658, D_loss_fake = 0.695, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.650, D_loss_fake = 0.702, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.645, D_loss_fake = 0.709, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.628, D_loss_fake = 0.705, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.636, D_loss_fake = 0.711, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.654, D_loss_fake = 0.711, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.642, D_loss_fake = 0.712, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.641, D_loss_fake = 0.726, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.644, D_loss_fake = 0.702, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.631, D_loss_fake = 0.690, G_loss = 0.713
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 1/46, D_loss_real = 0.612, D_loss_fake = 0.722, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.630, D_loss_fake = 0.823, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.589, D_loss_fake = 0.716, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.608, D_loss_fake = 0.755, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.627, D_loss_fake = 0.773, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.650, D_loss_fake = 0.712, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.666, D_loss_fake = 0.722, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.643, D_loss_fake = 0.727, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.652, D_loss_fake = 0.715, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.650, D_loss_fake = 0.695, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.644, D_loss_fake = 0.709, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.652, D_loss_fake = 0.728, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.669, D_loss_fake = 0.693, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.681, D_loss_fake = 0.708, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.656, D_loss_fake = 0.721, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.675, D_loss_fake = 0.692, G_loss = 0.735
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 17/46, D_loss_real = 0.702, D_loss_fake = 0.701, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.656, D_loss_fake = 0.672, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.708, D_loss_fake = 0.711, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.671, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.710, D_loss_fake = 0.654, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.721, D_loss_fake = 0.648, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.721, D_loss_fake = 0.651, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.718, D_loss_fake = 0.649, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.700, D_loss_fake = 0.652, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.720, D_loss_fake = 0.663, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.722, D_loss_fake = 0.666, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.662, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.665, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.682, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.679, D_loss_fake = 0.668, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.675, D_loss_fake = 0.700, G_loss = 0.737
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.724, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.699, D_loss_fake = 0.650, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.714, D_loss_fake = 0.589, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.727, D_loss_fake = 0.606, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.729, D_loss_fake = 0.623, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.734, D_loss_fake = 0.632, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.717, D_loss_fake = 0.657, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.714, D_loss_fake = 0.649, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.655, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.676, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.681, D_loss_fake = 0.660, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.667, D_loss_fake = 0.656, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.637, D_loss_fake = 0.670, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.657, D_loss_fake = 0.655, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.645, D_loss_fake = 0.694, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.644, D_loss_fake = 0.687, G_loss = 0.751
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.681, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.650, D_loss_fake = 0.695, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.668, D_loss_fake = 0.679, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.725, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.720, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.707, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.702, D_loss_fake = 0.702, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.659, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.697, D_loss_fake = 0.648, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.702, D_loss_fake = 0.643, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.640, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.676, D_loss_fake = 0.625, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.709, D_loss_fake = 0.652, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.659, D_loss_fake = 0.686, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.696, D_loss_fake = 0.678, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.739, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.688, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.713, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.709, D_loss_fake = 0.698, G_loss = 0.741
2/2 [=====] - 0s 12ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.703, D_loss_fake = 0.684, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.711, D_loss_fake = 0.641, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.709, D_loss_fake = 0.673, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.721, D_loss_fake = 0.650, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.712, D_loss_fake = 0.658, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.664, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.689, D_loss_fake = 0.655, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.666, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.662, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.656, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.680, D_loss_fake = 0.663, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.634, D_loss_fake = 0.689, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.645, D_loss_fake = 0.697, G_loss = 0.702
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 35/46, D_loss_real = 0.663, D_loss_fake = 0.736, G_loss = 0.670
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.667, D_loss_fake = 0.766, G_loss = 0.673
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.658, D_loss_fake = 0.781, G_loss = 0.657
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.663, D_loss_fake = 0.766, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.647, D_loss_fake = 0.756, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.688, D_loss_fake = 0.703, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.666, D_loss_fake = 0.686, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.655, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.648, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.644, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.677, D_loss_fake = 0.635, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.670, D_loss_fake = 0.629, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.664, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.665, G_loss = 0.780
2/2 [=====] - 0s 6ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.664, D_loss_fake = 0.665, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.716, D_loss_fake = 0.658, G_loss = 0.804
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 5/46, D_loss_real = 0.641, D_loss_fake = 0.696, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.675, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.695, D_loss_fake = 0.713, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.732, D_loss_fake = 0.696, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.713, D_loss_fake = 0.694, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.766, D_loss_fake = 0.699, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.732, D_loss_fake = 0.695, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.739, D_loss_fake = 0.720, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.697, D_loss_fake = 0.743, G_loss = 0.679
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.728, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.720, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.676, D_loss_fake = 0.705, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.662, D_loss_fake = 0.680, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.624, D_loss_fake = 0.668, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.618, D_loss_fake = 0.702, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.603, D_loss_fake = 0.684, G_loss = 0.742
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 21/46, D_loss_real = 0.555, D_loss_fake = 0.700, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.602, D_loss_fake = 0.783, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.598, D_loss_fake = 0.764, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.569, D_loss_fake = 0.832, G_loss = 0.663
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.613, D_loss_fake = 0.783, G_loss = 0.647
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.654, D_loss_fake = 0.796, G_loss = 0.676
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.681, D_loss_fake = 0.779, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.664, D_loss_fake = 0.682, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.649, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.651, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.727, D_loss_fake = 0.643, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.748, D_loss_fake = 0.642, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.740, D_loss_fake = 0.624, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.738, D_loss_fake = 0.637, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.717, D_loss_fake = 0.630, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.724, D_loss_fake = 0.660, G_loss = 0.763
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 37/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.708, D_loss_fake = 0.665, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.710, D_loss_fake = 0.664, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.729, D_loss_fake = 0.680, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.711, D_loss_fake = 0.686, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.700, D_loss_fake = 0.680, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.674, D_loss_fake = 0.674, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.671, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.683, D_loss_fake = 0.674, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.681, D_loss_fake = 0.659, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.675, D_loss_fake = 0.669, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.681, D_loss_fake = 0.655, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.674, D_loss_fake = 0.664, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.631, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.715, D_loss_fake = 0.702, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.669, G_loss = 0.777
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.665, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.709, D_loss_fake = 0.667, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.729, D_loss_fake = 0.673, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.700, D_loss_fake = 0.664, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.668, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.675, D_loss_fake = 0.672, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.689, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.694, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.710, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.680, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.666, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.720, D_loss_fake = 0.656, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.643, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.653, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.710, D_loss_fake = 0.661, G_loss = 0.776
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 23/46, D_loss_real = 0.726, D_loss_fake = 0.668, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.667, D_loss_fake = 0.662, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.654, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.710, D_loss_fake = 0.658, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.720, D_loss_fake = 0.649, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.664, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.748, D_loss_fake = 0.659, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.721, D_loss_fake = 0.694, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.707, D_loss_fake = 0.707, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.718, D_loss_fake = 0.721, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.695, D_loss_fake = 0.700, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.715, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.737, D_loss_fake = 0.675, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.726, D_loss_fake = 0.672, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.716, D_loss_fake = 0.667, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.731, D_loss_fake = 0.683, G_loss = 0.747
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 39/46, D_loss_real = 0.735, D_loss_fake = 0.657, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.717, D_loss_fake = 0.655, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.722, D_loss_fake = 0.641, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.727, D_loss_fake = 0.636, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.721, D_loss_fake = 0.641, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.717, D_loss_fake = 0.637, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.708, D_loss_fake = 0.656, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.655, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.670, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.668, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.678, D_loss_fake = 0.674, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.678, D_loss_fake = 0.672, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.674, D_loss_fake = 0.674, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.681, D_loss_fake = 0.671, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.669, D_loss_fake = 0.670, G_loss = 0.779
2/2 [=====] - 0s 4ms/step

Epoch 25, Iteration 9/46, D_loss_real = 0.680, D_loss_fake = 0.634, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.650, D_loss_fake = 0.646, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.652, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.677, D_loss_fake = 0.644, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.682, D_loss_fake = 0.646, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.634, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.701, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.749, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.707, D_loss_fake = 0.714, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.732, D_loss_fake = 0.707, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.715, D_loss_fake = 0.645, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.733, D_loss_fake = 0.640, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.703, D_loss_fake = 0.654, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.694, D_loss_fake = 0.667, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.688, G_loss = 0.735
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 25/46, D_loss_real = 0.669, D_loss_fake = 0.693, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.647, D_loss_fake = 0.723, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.654, D_loss_fake = 0.724, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.599, D_loss_fake = 0.728, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.628, D_loss_fake = 0.729, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.606, D_loss_fake = 0.731, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.609, D_loss_fake = 0.732, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.623, D_loss_fake = 0.732, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.606, D_loss_fake = 0.700, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.634, D_loss_fake = 0.681, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.630, D_loss_fake = 0.701, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.632, D_loss_fake = 0.760, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.650, D_loss_fake = 0.724, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.663, D_loss_fake = 0.741, G_loss = 0.683
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.650, D_loss_fake = 0.733, G_loss = 0.667
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.644, D_loss_fake = 0.727, G_loss = 0.679
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 41/46, D_loss_real = 0.673, D_loss_fake = 0.745, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.648, D_loss_fake = 0.737, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.721, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.739, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.673, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.702, D_loss_fake = 0.669, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.673, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.713, D_loss_fake = 0.729, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.719, D_loss_fake = 0.687, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.706, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.675, D_loss_fake = 0.695, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.648, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.715, D_loss_fake = 0.639, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.679, D_loss_fake = 0.652, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.710, D_loss_fake = 0.645, G_loss = 0.782
2/2 [=====] - 0s 4ms/step

Epoch 26, Iteration 11/46, D_loss_real = 0.725, D_loss_fake = 0.665, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.722, D_loss_fake = 0.683, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.712, D_loss_fake = 0.695, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.705, D_loss_fake = 0.737, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.714, D_loss_fake = 0.708, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.708, D_loss_fake = 0.684, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.722, D_loss_fake = 0.663, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.722, D_loss_fake = 0.657, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.725, D_loss_fake = 0.645, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.738, D_loss_fake = 0.632, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.735, D_loss_fake = 0.623, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.715, D_loss_fake = 0.634, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.662, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.715, D_loss_fake = 0.670, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.685, D_loss_fake = 0.673, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.714
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.678, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.694, D_loss_fake = 0.717, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.675, D_loss_fake = 0.719, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.657, D_loss_fake = 0.721, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.656, D_loss_fake = 0.725, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.744, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.661, D_loss_fake = 0.756, G_loss = 0.667
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.663, D_loss_fake = 0.749, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.656, D_loss_fake = 0.722, G_loss = 0.670
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.674, D_loss_fake = 0.737, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.664, D_loss_fake = 0.708, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.669, D_loss_fake = 0.684, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.668, D_loss_fake = 0.683, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.656, D_loss_fake = 0.676, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.663, D_loss_fake = 0.679, G_loss = 0.724
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 43/46, D_loss_real = 0.666, D_loss_fake = 0.691, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.696, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.673, D_loss_fake = 0.709, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.651, D_loss_fake = 0.706, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.667, D_loss_fake = 0.699, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.692, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.687, D_loss_fake = 0.676, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.661, D_loss_fake = 0.659, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.694, D_loss_fake = 0.665, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.657, D_loss_fake = 0.704, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.704, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.709, D_loss_fake = 0.704, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.725, D_loss_fake = 0.693, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.727, D_loss_fake = 0.670, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.743, D_loss_fake = 0.686, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.673, G_loss = 0.747
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 13/46, D_loss_real = 0.730, D_loss_fake = 0.659, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.735, D_loss_fake = 0.655, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.718, D_loss_fake = 0.666, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.707, D_loss_fake = 0.681, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.724, D_loss_fake = 0.673, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.729, D_loss_fake = 0.649, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.723, D_loss_fake = 0.663, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.660, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.717, D_loss_fake = 0.664, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.664, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.671, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.682, D_loss_fake = 0.697, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.670, D_loss_fake = 0.689, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.673, D_loss_fake = 0.699, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.690, G_loss = 0.714
2/2 [=====] - 0s 4ms/step

Epoch 27, Iteration 29/46, D_loss_real = 0.655, D_loss_fake = 0.686, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.662, D_loss_fake = 0.686, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.679, D_loss_fake = 0.689, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.665, D_loss_fake = 0.686, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.677, D_loss_fake = 0.681, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.658, D_loss_fake = 0.711, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.649, D_loss_fake = 0.695, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.710, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.662, D_loss_fake = 0.667, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.645, D_loss_fake = 0.679, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.634, D_loss_fake = 0.685, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.639, D_loss_fake = 0.703, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.627, D_loss_fake = 0.737, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.622, D_loss_fake = 0.710, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.638, D_loss_fake = 0.780, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.651, D_loss_fake = 0.680, G_loss = 0.766
2/2 [=====] - 0s 4ms/step

Epoch 27, Iteration 45/46, D_loss_real = 0.672, D_loss_fake = 0.697, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.678, D_loss_fake = 0.642, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.692, D_loss_fake = 0.638, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.682, D_loss_fake = 0.636, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.702, D_loss_fake = 0.629, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.693, D_loss_fake = 0.657, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.645, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.654, D_loss_fake = 0.646, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.646, D_loss_fake = 0.666, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.595, D_loss_fake = 0.683, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.608, D_loss_fake = 0.727, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.616, D_loss_fake = 0.737, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.605, D_loss_fake = 0.775, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.814, G_loss = 0.681
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.613, D_loss_fake = 0.818, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.649, D_loss_fake = 0.818, G_loss = 0.698
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 15/46, D_loss_real = 0.671, D_loss_fake = 0.739, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.667, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.729, D_loss_fake = 0.648, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.699, D_loss_fake = 0.621, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.722, D_loss_fake = 0.643, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.715, D_loss_fake = 0.671, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.741, D_loss_fake = 0.680, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.733, D_loss_fake = 0.655, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.716, D_loss_fake = 0.685, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.698, D_loss_fake = 0.678, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.683, D_loss_fake = 0.685, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.658, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.670, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.699, D_loss_fake = 0.679, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.683, D_loss_fake = 0.662, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.675, D_loss_fake = 0.682, G_loss = 0.737
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.701, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.674, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.655, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.696, D_loss_fake = 0.649, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.689, D_loss_fake = 0.644, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.675, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.706, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.709, D_loss_fake = 0.655, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.707, D_loss_fake = 0.645, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.650, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.640, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.690, D_loss_fake = 0.644, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.672, D_loss_fake = 0.659, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.657, D_loss_fake = 0.686, G_loss = 0.762
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 1/46, D_loss_real = 0.653, D_loss_fake = 0.672, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.652, D_loss_fake = 0.714, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.711, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.705, D_loss_fake = 0.716, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.668, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.722, D_loss_fake = 0.660, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.719, D_loss_fake = 0.646, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.710, D_loss_fake = 0.621, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.610, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.676, D_loss_fake = 0.605, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.682, D_loss_fake = 0.624, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.663, D_loss_fake = 0.630, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.671, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.711, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.638, D_loss_fake = 0.757, G_loss = 0.701
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 17/46, D_loss_real = 0.659, D_loss_fake = 0.735, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.641, D_loss_fake = 0.754, G_loss = 0.685
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.672, D_loss_fake = 0.746, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.664, D_loss_fake = 0.686, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.681, D_loss_fake = 0.660, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.649, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.728, D_loss_fake = 0.689, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.697, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.711, D_loss_fake = 0.674, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.655, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.664, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.692, D_loss_fake = 0.654, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.671, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.696, D_loss_fake = 0.662, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.715, D_loss_fake = 0.697, G_loss = 0.764
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 33/46, D_loss_real = 0.670, D_loss_fake = 0.678, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.680, D_loss_fake = 0.675, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.679, D_loss_fake = 0.670, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.663, D_loss_fake = 0.666, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.663, D_loss_fake = 0.684, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.655, D_loss_fake = 0.673, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.647, D_loss_fake = 0.682, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.678, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.626, D_loss_fake = 0.690, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.649, D_loss_fake = 0.720, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.675, D_loss_fake = 0.691, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.621, D_loss_fake = 0.738, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.655, D_loss_fake = 0.686, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.668, D_loss_fake = 0.671, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.649, D_loss_fake = 0.661, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.637, D_loss_fake = 0.670, G_loss = 0.789
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 3/46, D_loss_real = 0.671, D_loss_fake = 0.732, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.647, D_loss_fake = 0.647, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.676, D_loss_fake = 0.663, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.775, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.703, D_loss_fake = 0.663, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.589, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.619, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.695, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.720, D_loss_fake = 0.709, G_loss = 0.757
2/2 [=====] - 0s 12ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.707, D_loss_fake = 0.694, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.714, D_loss_fake = 0.629, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.733, D_loss_fake = 0.658, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.737, D_loss_fake = 0.685, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.747, D_loss_fake = 0.673, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.709, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.695, D_loss_fake = 0.679, G_loss = 0.738
2/2 [=====] - 0s 4ms/step

Epoch 30, Iteration 19/46, D_loss_real = 0.679, D_loss_fake = 0.672, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.662, D_loss_fake = 0.679, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.643, D_loss_fake = 0.713, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.661, D_loss_fake = 0.706, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.670, D_loss_fake = 0.748, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.654, D_loss_fake = 0.716, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.668, D_loss_fake = 0.724, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.745, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.685, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.670, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.697, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.727, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.682, D_loss_fake = 0.711, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.674, D_loss_fake = 0.688, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.690, D_loss_fake = 0.725, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.719, G_loss = 0.707
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 35/46, D_loss_real = 0.697, D_loss_fake = 0.685, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.670, D_loss_fake = 0.680, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.680, D_loss_fake = 0.671, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.687, D_loss_fake = 0.681, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.702, D_loss_fake = 0.663, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.644, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.718, D_loss_fake = 0.658, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.674, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.710, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.711, D_loss_fake = 0.716, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.699, D_loss_fake = 0.722, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.704, D_loss_fake = 0.719, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.721, D_loss_fake = 0.675, G_loss = 0.731
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.674, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 6/46, D_loss_real = 0.706, D_loss_fake = 0.695, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 7/46, D_loss_real = 0.698, D_loss_fake = 0.688, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.697, D_loss_fake = 0.676, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.672, D_loss_fake = 0.687, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.682, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 12/46, D_loss_real = 0.653, D_loss_fake = 0.672, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.663, D_loss_fake = 0.730, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.701, D_loss_fake = 0.652, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.702, D_loss_fake = 0.639, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.635, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.623, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.687, D_loss_fake = 0.626, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.705, D_loss_fake = 0.645, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.668, G_loss = 0.746
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.667, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 22/46, D_loss_real = 0.683, D_loss_fake = 0.674, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 23/46, D_loss_real = 0.717, D_loss_fake = 0.701, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.715, D_loss_fake = 0.663, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.717, D_loss_fake = 0.657, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.709, D_loss_fake = 0.643, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.705, D_loss_fake = 0.642, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 28/46, D_loss_real = 0.697, D_loss_fake = 0.638, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.700, D_loss_fake = 0.642, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.676, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.671, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.677, D_loss_fake = 0.733, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.641, D_loss_fake = 0.714, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.653, D_loss_fake = 0.757, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.653, D_loss_fake = 0.709, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.680, D_loss_fake = 0.673, G_loss = 0.724
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 38/46, D_loss_real = 0.672, D_loss_fake = 0.672, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 39/46, D_loss_real = 0.668, D_loss_fake = 0.683, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.661, D_loss_fake = 0.712, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.658, D_loss_fake = 0.695, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.662, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.666, D_loss_fake = 0.673, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 44/46, D_loss_real = 0.680, D_loss_fake = 0.708, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.651, D_loss_fake = 0.716, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.672, D_loss_fake = 0.711, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.643, D_loss_fake = 0.677, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.658, D_loss_fake = 0.692, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.653, D_loss_fake = 0.701, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.603, D_loss_fake = 0.726, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.636, D_loss_fake = 0.753, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.623, D_loss_fake = 0.694, G_loss = 0.745
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 7/46, D_loss_real = 0.654, D_loss_fake = 0.673, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 8/46, D_loss_real = 0.660, D_loss_fake = 0.671, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.618, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.630, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.696, D_loss_fake = 0.636, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.704, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.693, D_loss_fake = 0.668, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.681, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.734, D_loss_fake = 0.703, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.710, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.692, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.640, D_loss_fake = 0.672, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.655, D_loss_fake = 0.704, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.657, D_loss_fake = 0.724, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.651, D_loss_fake = 0.748, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.674, G_loss = 0.743
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 23/46, D_loss_real = 0.648, D_loss_fake = 0.687, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 24/46, D_loss_real = 0.641, D_loss_fake = 0.659, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 25/46, D_loss_real = 0.638, D_loss_fake = 0.668, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.639, D_loss_fake = 0.684, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.639, D_loss_fake = 0.682, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.689, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.705, D_loss_fake = 0.664, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 30/46, D_loss_real = 0.700, D_loss_fake = 0.637, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.740, D_loss_fake = 0.622, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.737, D_loss_fake = 0.618, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.722, D_loss_fake = 0.635, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.720, D_loss_fake = 0.627, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.715, D_loss_fake = 0.615, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.730, D_loss_fake = 0.615, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.663, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.698, D_loss_fake = 0.665, G_loss = 0.747
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 40/46, D_loss_real = 0.694, D_loss_fake = 0.711, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 41/46, D_loss_real = 0.700, D_loss_fake = 0.738, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.765, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.659, D_loss_fake = 0.725, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.673, D_loss_fake = 0.672, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.654, D_loss_fake = 0.727, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 46/46, D_loss_real = 0.650, D_loss_fake = 0.698, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.693, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.659, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.672, D_loss_fake = 0.715, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.708, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.597, D_loss_fake = 0.704, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.653, D_loss_fake = 0.780, G_loss = 0.690
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.776, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.677, D_loss_fake = 0.728, G_loss = 0.698
2/2 [=====] - 0s 4ms/step

Epoch 33, Iteration 9/46, D_loss_real = 0.656, D_loss_fake = 0.723, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 10/46, D_loss_real = 0.662, D_loss_fake = 0.719, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.732, G_loss = 0.681
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.749, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.678, D_loss_fake = 0.714, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.662, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.709, D_loss_fake = 0.668, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.674, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.703, D_loss_fake = 0.723, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.709, D_loss_fake = 0.703, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.709, D_loss_fake = 0.696, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.692, D_loss_fake = 0.702, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.711, G_loss = 0.723
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 25/46, D_loss_real = 0.669, D_loss_fake = 0.716, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 26/46, D_loss_real = 0.692, D_loss_fake = 0.699, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 27/46, D_loss_real = 0.690, D_loss_fake = 0.684, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.671, D_loss_fake = 0.683, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.649, D_loss_fake = 0.699, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.642, D_loss_fake = 0.703, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 32/46, D_loss_real = 0.659, D_loss_fake = 0.672, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.658, D_loss_fake = 0.699, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.648, D_loss_fake = 0.688, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.669, D_loss_fake = 0.670, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.711, D_loss_fake = 0.656, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.688, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.689, D_loss_fake = 0.678, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.725, D_loss_fake = 0.679, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.711, D_loss_fake = 0.702, G_loss = 0.705
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 41/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.701, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.750, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.731, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.673, D_loss_fake = 0.716, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.664, D_loss_fake = 0.720, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.668, D_loss_fake = 0.752, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 2/46, D_loss_real = 0.664, D_loss_fake = 0.703, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.668, D_loss_fake = 0.699, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.699, D_loss_fake = 0.736, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.690, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.727, D_loss_fake = 0.684, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.729, D_loss_fake = 0.689, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.717, D_loss_fake = 0.693, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.714, D_loss_fake = 0.691, G_loss = 0.719
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.689, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 12/46, D_loss_real = 0.672, D_loss_fake = 0.699, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 13/46, D_loss_real = 0.636, D_loss_fake = 0.695, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.640, D_loss_fake = 0.712, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.622, D_loss_fake = 0.758, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.629, D_loss_fake = 0.735, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.646, D_loss_fake = 0.720, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.695, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.689, D_loss_fake = 0.702, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.705, D_loss_fake = 0.666, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.666, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.714, D_loss_fake = 0.652, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.711, D_loss_fake = 0.647, G_loss = 0.787
2/2 [=====] - 0s 7ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.736, D_loss_fake = 0.663, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.720, D_loss_fake = 0.671, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.721, D_loss_fake = 0.647, G_loss = 0.757
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 27/46, D_loss_real = 0.727, D_loss_fake = 0.657, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.712, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 29/46, D_loss_real = 0.709, D_loss_fake = 0.665, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.705, D_loss_fake = 0.682, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.718, D_loss_fake = 0.676, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.706, D_loss_fake = 0.662, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.721, D_loss_fake = 0.675, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.680, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.719, D_loss_fake = 0.684, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.693, D_loss_fake = 0.684, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.680, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.680, D_loss_fake = 0.688, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.685, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.681, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.707, D_loss_fake = 0.690, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.672, G_loss = 0.732
2/2 [=====] - 0s 4ms/step

Epoch 34, Iteration 43/46, D_loss_real = 0.696, D_loss_fake = 0.670, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 44/46, D_loss_real = 0.686, D_loss_fake = 0.680, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.672, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.691, D_loss_fake = 0.665, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.682, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.691, D_loss_fake = 0.674, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.691, D_loss_fake = 0.665, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 4/46, D_loss_real = 0.660, D_loss_fake = 0.691, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.655, D_loss_fake = 0.678, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.638, D_loss_fake = 0.733, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.632, D_loss_fake = 0.770, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.656, D_loss_fake = 0.707, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.690, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.670, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.670, D_loss_fake = 0.706, G_loss = 0.745
2/2 [=====] - 0s 3ms/step

Epoch 35, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.695, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 14/46, D_loss_real = 0.718, D_loss_fake = 0.668, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.707, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.662, D_loss_fake = 0.689, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.671, D_loss_fake = 0.702, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.727, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.661, D_loss_fake = 0.718, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 20/46, D_loss_real = 0.664, D_loss_fake = 0.706, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.655, D_loss_fake = 0.715, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.645, D_loss_fake = 0.720, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.638, D_loss_fake = 0.711, G_loss = 0.680
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.666, D_loss_fake = 0.763, G_loss = 0.677
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.778, G_loss = 0.681
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.751, G_loss = 0.674
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.730, G_loss = 0.700
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.675, D_loss_fake = 0.717, G_loss = 0.716
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 29/46, D_loss_real = 0.693, D_loss_fake = 0.687, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 30/46, D_loss_real = 0.686, D_loss_fake = 0.742, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.709, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.705, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.713, D_loss_fake = 0.690, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.726, D_loss_fake = 0.674, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.729, D_loss_fake = 0.673, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 36/46, D_loss_real = 0.718, D_loss_fake = 0.637, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.743, D_loss_fake = 0.612, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.742, D_loss_fake = 0.617, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.737, D_loss_fake = 0.635, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.749, D_loss_fake = 0.638, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.751, D_loss_fake = 0.640, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.757, D_loss_fake = 0.643, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.752, D_loss_fake = 0.658, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.650, G_loss = 0.750
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 45/46, D_loss_real = 0.720, D_loss_fake = 0.650, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.655, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 1/46, D_loss_real = 0.683, D_loss_fake = 0.660, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.668, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.673, D_loss_fake = 0.687, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.660, D_loss_fake = 0.682, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.670, D_loss_fake = 0.767, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 6/46, D_loss_real = 0.673, D_loss_fake = 0.765, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.667, D_loss_fake = 0.687, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.711, D_loss_fake = 0.659, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.704, D_loss_fake = 0.679, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.708, D_loss_fake = 0.685, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.709, D_loss_fake = 0.681, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.698, D_loss_fake = 0.686, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.699, D_loss_fake = 0.671, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.684, G_loss = 0.727
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 15/46, D_loss_real = 0.707, D_loss_fake = 0.716, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.669, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.671, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.712, D_loss_fake = 0.659, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.660, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.648, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 22/46, D_loss_real = 0.707, D_loss_fake = 0.645, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.654, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.653, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.700, D_loss_fake = 0.656, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.710, D_loss_fake = 0.675, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.693, D_loss_fake = 0.676, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.705, D_loss_fake = 0.666, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.717, D_loss_fake = 0.672, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.690, G_loss = 0.726
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.679, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 32/46, D_loss_real = 0.704, D_loss_fake = 0.674, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 33/46, D_loss_real = 0.686, D_loss_fake = 0.663, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.707, D_loss_fake = 0.663, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.676, D_loss_fake = 0.682, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.695, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.681, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 38/46, D_loss_real = 0.674, D_loss_fake = 0.676, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.677, D_loss_fake = 0.674, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.656, D_loss_fake = 0.684, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.657, D_loss_fake = 0.689, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.659, D_loss_fake = 0.692, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.652, D_loss_fake = 0.675, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.653, D_loss_fake = 0.678, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.655, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss = 0.739
2/2 [=====] - 0s 4ms/step

Epoch 37, Iteration 1/46, D_loss_real = 0.712, D_loss_fake = 0.691, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 2/46, D_loss_real = 0.691, D_loss_fake = 0.790, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 3/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.702, D_loss_fake = 0.680, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.716, D_loss_fake = 0.677, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.726, D_loss_fake = 0.666, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 8/46, D_loss_real = 0.714, D_loss_fake = 0.670, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.697, D_loss_fake = 0.671, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.669, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.691, D_loss_fake = 0.675, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.665, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.685, D_loss_fake = 0.669, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.689, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.682, D_loss_fake = 0.717, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.673, D_loss_fake = 0.722, G_loss = 0.728
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.694, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.687, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.670, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.682, D_loss_fake = 0.678, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.696, D_loss_fake = 0.695, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 24/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.673, D_loss_fake = 0.689, G_loss = 0.706
2/2 [=====] - 0s 5ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.682, D_loss_fake = 0.705, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.674, D_loss_fake = 0.744, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.748, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.672, D_loss_fake = 0.715, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.672, D_loss_fake = 0.727, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.694, G_loss = 0.716
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 33/46, D_loss_real = 0.661, D_loss_fake = 0.703, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 34/46, D_loss_real = 0.677, D_loss_fake = 0.707, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 35/46, D_loss_real = 0.661, D_loss_fake = 0.693, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.669, D_loss_fake = 0.704, G_loss = 0.700
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.709, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.670, D_loss_fake = 0.726, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.649, D_loss_fake = 0.690, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.659, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.664, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.698, D_loss_fake = 0.642, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.694, D_loss_fake = 0.671, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.719, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.668, D_loss_fake = 0.729, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.701, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.675, D_loss_fake = 0.749, G_loss = 0.676
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.675, D_loss_fake = 0.732, G_loss = 0.678
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.734, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 4/46, D_loss_real = 0.677, D_loss_fake = 0.696, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.686, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.693, D_loss_fake = 0.682, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.664, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.680, D_loss_fake = 0.681, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 10/46, D_loss_real = 0.687, D_loss_fake = 0.658, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.669, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.669, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.645, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.704, D_loss_fake = 0.670, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.665, D_loss_fake = 0.664, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.669, D_loss_fake = 0.664, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.646, D_loss_fake = 0.682, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.674, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 19/46, D_loss_real = 0.673, D_loss_fake = 0.729, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 20/46, D_loss_real = 0.667, D_loss_fake = 0.719, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.717, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.716, D_loss_fake = 0.640, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.714, D_loss_fake = 0.656, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.731, D_loss_fake = 0.667, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 26/46, D_loss_real = 0.713, D_loss_fake = 0.679, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.731, D_loss_fake = 0.671, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.709, D_loss_fake = 0.652, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.649, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.665, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.710, D_loss_fake = 0.668, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.708, D_loss_fake = 0.699, G_loss = 0.735
2/2 [=====] - 0s 4ms/step

Epoch 38, Iteration 35/46, D_loss_real = 0.707, D_loss_fake = 0.674, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 36/46, D_loss_real = 0.723, D_loss_fake = 0.673, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 37/46, D_loss_real = 0.707, D_loss_fake = 0.682, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.673, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.710, D_loss_fake = 0.670, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.666, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.696, D_loss_fake = 0.681, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.684, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.719, D_loss_fake = 0.677, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.730, D_loss_fake = 0.694, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.718, D_loss_fake = 0.666, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.729, D_loss_fake = 0.660, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.716, D_loss_fake = 0.677, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.717, D_loss_fake = 0.679, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.718, D_loss_fake = 0.659, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.707, D_loss_fake = 0.667, G_loss = 0.735
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 5/46, D_loss_real = 0.720, D_loss_fake = 0.661, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 6/46, D_loss_real = 0.715, D_loss_fake = 0.660, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.656, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.706, D_loss_fake = 0.655, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.710, D_loss_fake = 0.667, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.691, D_loss_fake = 0.662, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.693, D_loss_fake = 0.664, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 12/46, D_loss_real = 0.689, D_loss_fake = 0.683, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.687, D_loss_fake = 0.681, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.671, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.682, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.650, D_loss_fake = 0.704, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.649, D_loss_fake = 0.688, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.645, D_loss_fake = 0.678, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.652, D_loss_fake = 0.699, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.661, D_loss_fake = 0.664, G_loss = 0.740
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.674, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 22/46, D_loss_real = 0.648, D_loss_fake = 0.705, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 23/46, D_loss_real = 0.651, D_loss_fake = 0.714, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.680, D_loss_fake = 0.785, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.662, D_loss_fake = 0.706, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.681, D_loss_fake = 0.666, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.676, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.693, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.663, D_loss_fake = 0.714, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.692, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.711, D_loss_fake = 0.717, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.725, D_loss_fake = 0.648, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.724, D_loss_fake = 0.638, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.727, D_loss_fake = 0.619, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.728, D_loss_fake = 0.653, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.727, D_loss_fake = 0.660, G_loss = 0.745
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 37/46, D_loss_real = 0.724, D_loss_fake = 0.663, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 38/46, D_loss_real = 0.696, D_loss_fake = 0.677, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.681, D_loss_fake = 0.686, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.702, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.663, D_loss_fake = 0.730, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.718, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.727, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.719, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.681, D_loss_fake = 0.719, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.693, D_loss_fake = 0.700, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.713, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.674, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.708, D_loss_fake = 0.670, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.714, D_loss_fake = 0.679, G_loss = 0.734
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.672, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 8/46, D_loss_real = 0.707, D_loss_fake = 0.665, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 9/46, D_loss_real = 0.709, D_loss_fake = 0.662, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.705, D_loss_fake = 0.661, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.705, D_loss_fake = 0.635, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.680, D_loss_fake = 0.667, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.718, D_loss_fake = 0.696, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 14/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.707, D_loss_fake = 0.709, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.703, D_loss_fake = 0.696, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.675, D_loss_fake = 0.723, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.708, D_loss_fake = 0.680, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.682, D_loss_fake = 0.677, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.668, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.672, D_loss_fake = 0.689, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.658, D_loss_fake = 0.713, G_loss = 0.728
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 23/46, D_loss_real = 0.643, D_loss_fake = 0.702, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 24/46, D_loss_real = 0.664, D_loss_fake = 0.711, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 25/46, D_loss_real = 0.636, D_loss_fake = 0.725, G_loss = 0.686
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.641, D_loss_fake = 0.744, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.648, D_loss_fake = 0.738, G_loss = 0.684
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.631, D_loss_fake = 0.751, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.622, D_loss_fake = 0.722, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.733, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.673, D_loss_fake = 0.826, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.676, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.728, D_loss_fake = 0.661, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.751, D_loss_fake = 0.647, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.732, D_loss_fake = 0.654, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.731, D_loss_fake = 0.671, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.708, D_loss_fake = 0.697, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.720, D_loss_fake = 0.687, G_loss = 0.739
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 39/46, D_loss_real = 0.733, D_loss_fake = 0.675, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 40/46, D_loss_real = 0.725, D_loss_fake = 0.693, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 41/46, D_loss_real = 0.717, D_loss_fake = 0.675, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.717, D_loss_fake = 0.687, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.673, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.711, D_loss_fake = 0.676, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.724, D_loss_fake = 0.664, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.655, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 1/46, D_loss_real = 0.733, D_loss_fake = 0.642, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 2/46, D_loss_real = 0.691, D_loss_fake = 0.642, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.657, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.656, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 5/46, D_loss_real = 0.691, D_loss_fake = 0.685, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.691, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 7/46, D_loss_real = 0.709, D_loss_fake = 0.641, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 8/46, D_loss_real = 0.726, D_loss_fake = 0.634, G_loss = 0.789
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 9/46, D_loss_real = 0.737, D_loss_fake = 0.644, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 10/46, D_loss_real = 0.728, D_loss_fake = 0.652, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 11/46, D_loss_real = 0.754, D_loss_fake = 0.659, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 12/46, D_loss_real = 0.727, D_loss_fake = 0.659, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 13/46, D_loss_real = 0.712, D_loss_fake = 0.675, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 14/46, D_loss_real = 0.708, D_loss_fake = 0.684, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 15/46, D_loss_real = 0.708, D_loss_fake = 0.687, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 17/46, D_loss_real = 0.694, D_loss_fake = 0.700, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 18/46, D_loss_real = 0.713, D_loss_fake = 0.707, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 19/46, D_loss_real = 0.717, D_loss_fake = 0.692, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.692, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.693, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.713, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 23/46, D_loss_real = 0.670, D_loss_fake = 0.689, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.686, G_loss = 0.709
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 25/46, D_loss_real = 0.676, D_loss_fake = 0.704, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 26/46, D_loss_real = 0.690, D_loss_fake = 0.672, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 27/46, D_loss_real = 0.703, D_loss_fake = 0.662, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.687, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 29/46, D_loss_real = 0.698, D_loss_fake = 0.700, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.681, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.677, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.697, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 33/46, D_loss_real = 0.684, D_loss_fake = 0.692, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.680, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 35/46, D_loss_real = 0.677, D_loss_fake = 0.683, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 36/46, D_loss_real = 0.671, D_loss_fake = 0.674, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 37/46, D_loss_real = 0.663, D_loss_fake = 0.693, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 38/46, D_loss_real = 0.658, D_loss_fake = 0.698, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 39/46, D_loss_real = 0.628, D_loss_fake = 0.736, G_loss = 0.687
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 40/46, D_loss_real = 0.624, D_loss_fake = 0.800, G_loss = 0.694
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.777, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.723, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.712, G_loss = 0.695
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 44/46, D_loss_real = 0.693, D_loss_fake = 0.725, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 45/46, D_loss_real = 0.663, D_loss_fake = 0.712, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 46/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.674, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 2/46, D_loss_real = 0.703, D_loss_fake = 0.662, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.663, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 4/46, D_loss_real = 0.697, D_loss_fake = 0.665, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 5/46, D_loss_real = 0.692, D_loss_fake = 0.664, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 6/46, D_loss_real = 0.691, D_loss_fake = 0.663, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.656, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 8/46, D_loss_real = 0.698, D_loss_fake = 0.654, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.656, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 10/46, D_loss_real = 0.703, D_loss_fake = 0.689, G_loss = 0.726
2/2 [=====] - 0s 4ms/step

Epoch 42, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.664, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 12/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 13/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.678, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 15/46, D_loss_real = 0.713, D_loss_fake = 0.692, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.678, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 17/46, D_loss_real = 0.692, D_loss_fake = 0.687, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 18/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.675, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 21/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 22/46, D_loss_real = 0.711, D_loss_fake = 0.687, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 23/46, D_loss_real = 0.693, D_loss_fake = 0.681, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 25/46, D_loss_real = 0.695, D_loss_fake = 0.699, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 26/46, D_loss_real = 0.680, D_loss_fake = 0.703, G_loss = 0.701
2/2 [=====] - 0s 3ms/step

Epoch 42, Iteration 27/46, D_loss_real = 0.688, D_loss_fake = 0.701, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 28/46, D_loss_real = 0.669, D_loss_fake = 0.698, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.714, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 30/46, D_loss_real = 0.685, D_loss_fake = 0.706, G_loss = 0.683
2/2 [=====] - 0s 5ms/step
Epoch 42, Iteration 31/46, D_loss_real = 0.667, D_loss_fake = 0.714, G_loss = 0.697
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 33/46, D_loss_real = 0.678, D_loss_fake = 0.703, G_loss = 0.690
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.717, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 35/46, D_loss_real = 0.674, D_loss_fake = 0.694, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.701, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.688, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.671, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 40/46, D_loss_real = 0.684, D_loss_fake = 0.664, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.676, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 42/46, D_loss_real = 0.684, D_loss_fake = 0.666, G_loss = 0.739
2/2 [=====] - 0s 3ms/step

Epoch 42, Iteration 43/46, D_loss_real = 0.707, D_loss_fake = 0.670, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.671, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 45/46, D_loss_real = 0.674, D_loss_fake = 0.692, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 46/46, D_loss_real = 0.659, D_loss_fake = 0.677, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.698, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 2/46, D_loss_real = 0.673, D_loss_fake = 0.701, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 3/46, D_loss_real = 0.644, D_loss_fake = 0.705, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 4/46, D_loss_real = 0.653, D_loss_fake = 0.702, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 5/46, D_loss_real = 0.662, D_loss_fake = 0.724, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 6/46, D_loss_real = 0.679, D_loss_fake = 0.722, G_loss = 0.691
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 7/46, D_loss_real = 0.655, D_loss_fake = 0.722, G_loss = 0.677
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 8/46, D_loss_real = 0.648, D_loss_fake = 0.755, G_loss = 0.683
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 9/46, D_loss_real = 0.662, D_loss_fake = 0.739, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 10/46, D_loss_real = 0.664, D_loss_fake = 0.735, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 11/46, D_loss_real = 0.687, D_loss_fake = 0.711, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.678, G_loss = 0.738
2/2 [=====] - 0s 4ms/step

Epoch 43, Iteration 13/46, D_loss_real = 0.696, D_loss_fake = 0.653, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 14/46, D_loss_real = 0.690, D_loss_fake = 0.655, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 15/46, D_loss_real = 0.658, D_loss_fake = 0.652, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 16/46, D_loss_real = 0.692, D_loss_fake = 0.678, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 17/46, D_loss_real = 0.675, D_loss_fake = 0.667, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 18/46, D_loss_real = 0.688, D_loss_fake = 0.695, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 19/46, D_loss_real = 0.700, D_loss_fake = 0.667, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.688, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 21/46, D_loss_real = 0.714, D_loss_fake = 0.704, G_loss = 0.712
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 22/46, D_loss_real = 0.715, D_loss_fake = 0.698, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 23/46, D_loss_real = 0.698, D_loss_fake = 0.710, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 24/46, D_loss_real = 0.704, D_loss_fake = 0.700, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 25/46, D_loss_real = 0.702, D_loss_fake = 0.706, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 26/46, D_loss_real = 0.700, D_loss_fake = 0.694, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 27/46, D_loss_real = 0.706, D_loss_fake = 0.693, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 28/46, D_loss_real = 0.708, D_loss_fake = 0.687, G_loss = 0.717
2/2 [=====] - 0s 3ms/step

Epoch 43, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.684, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 30/46, D_loss_real = 0.687, D_loss_fake = 0.691, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 31/46, D_loss_real = 0.686, D_loss_fake = 0.676, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 32/46, D_loss_real = 0.689, D_loss_fake = 0.691, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 33/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.692, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 35/46, D_loss_real = 0.671, D_loss_fake = 0.688, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.710, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 38/46, D_loss_real = 0.668, D_loss_fake = 0.680, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.694, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 40/46, D_loss_real = 0.681, D_loss_fake = 0.693, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.704, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 42/46, D_loss_real = 0.678, D_loss_fake = 0.701, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 43/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 44/46, D_loss_real = 0.675, D_loss_fake = 0.709, G_loss = 0.706
2/2 [=====] - 0s 3ms/step

Epoch 43, Iteration 45/46, D_loss_real = 0.684, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 46/46, D_loss_real = 0.675, D_loss_fake = 0.701, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 1/46, D_loss_real = 0.664, D_loss_fake = 0.722, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 2/46, D_loss_real = 0.675, D_loss_fake = 0.704, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.698, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.690, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 5/46, D_loss_real = 0.651, D_loss_fake = 0.709, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.697, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.675, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 8/46, D_loss_real = 0.672, D_loss_fake = 0.696, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 9/46, D_loss_real = 0.651, D_loss_fake = 0.694, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.700, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 11/46, D_loss_real = 0.636, D_loss_fake = 0.710, G_loss = 0.684
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 12/46, D_loss_real = 0.654, D_loss_fake = 0.725, G_loss = 0.687
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 13/46, D_loss_real = 0.647, D_loss_fake = 0.718, G_loss = 0.671
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 14/46, D_loss_real = 0.659, D_loss_fake = 0.761, G_loss = 0.675
2/2 [=====] - 0s 3ms/step

Epoch 44, Iteration 15/46, D_loss_real = 0.659, D_loss_fake = 0.748, G_loss = 0.694
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 16/46, D_loss_real = 0.660, D_loss_fake = 0.721, G_loss = 0.689
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 17/46, D_loss_real = 0.647, D_loss_fake = 0.715, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 18/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss = 0.692
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 19/46, D_loss_real = 0.633, D_loss_fake = 0.708, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 20/46, D_loss_real = 0.662, D_loss_fake = 0.730, G_loss = 0.675
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 21/46, D_loss_real = 0.677, D_loss_fake = 0.738, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 22/46, D_loss_real = 0.711, D_loss_fake = 0.693, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 23/46, D_loss_real = 0.710, D_loss_fake = 0.687, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 24/46, D_loss_real = 0.724, D_loss_fake = 0.662, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 25/46, D_loss_real = 0.672, D_loss_fake = 0.674, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 26/46, D_loss_real = 0.701, D_loss_fake = 0.684, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 27/46, D_loss_real = 0.723, D_loss_fake = 0.703, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 28/46, D_loss_real = 0.715, D_loss_fake = 0.687, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 29/46, D_loss_real = 0.719, D_loss_fake = 0.669, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 30/46, D_loss_real = 0.725, D_loss_fake = 0.617, G_loss = 0.781
2/2 [=====] - 0s 3ms/step

Epoch 44, Iteration 31/46, D_loss_real = 0.724, D_loss_fake = 0.634, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 32/46, D_loss_real = 0.713, D_loss_fake = 0.598, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 33/46, D_loss_real = 0.713, D_loss_fake = 0.593, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 34/46, D_loss_real = 0.710, D_loss_fake = 0.605, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 35/46, D_loss_real = 0.732, D_loss_fake = 0.612, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.619, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.641, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 38/46, D_loss_real = 0.712, D_loss_fake = 0.674, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 39/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.714, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 41/46, D_loss_real = 0.685, D_loss_fake = 0.696, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 42/46, D_loss_real = 0.679, D_loss_fake = 0.719, G_loss = 0.696
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 43/46, D_loss_real = 0.693, D_loss_fake = 0.717, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 44/46, D_loss_real = 0.674, D_loss_fake = 0.733, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 45/46, D_loss_real = 0.683, D_loss_fake = 0.695, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.679, G_loss = 0.734
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.666, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 2/46, D_loss_real = 0.665, D_loss_fake = 0.684, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 3/46, D_loss_real = 0.669, D_loss_fake = 0.705, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 4/46, D_loss_real = 0.624, D_loss_fake = 0.695, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 5/46, D_loss_real = 0.633, D_loss_fake = 0.732, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 6/46, D_loss_real = 0.635, D_loss_fake = 0.778, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 7/46, D_loss_real = 0.653, D_loss_fake = 0.731, G_loss = 0.689
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 8/46, D_loss_real = 0.670, D_loss_fake = 0.721, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.702, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 10/46, D_loss_real = 0.675, D_loss_fake = 0.696, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 11/46, D_loss_real = 0.682, D_loss_fake = 0.637, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 12/46, D_loss_real = 0.664, D_loss_fake = 0.661, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 13/46, D_loss_real = 0.674, D_loss_fake = 0.670, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 14/46, D_loss_real = 0.672, D_loss_fake = 0.681, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.688, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.676, G_loss = 0.729
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.673, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 18/46, D_loss_real = 0.677, D_loss_fake = 0.692, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.683, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.713, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 21/46, D_loss_real = 0.663, D_loss_fake = 0.695, G_loss = 0.711
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 22/46, D_loss_real = 0.670, D_loss_fake = 0.707, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.682, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.679, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 26/46, D_loss_real = 0.705, D_loss_fake = 0.675, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 27/46, D_loss_real = 0.705, D_loss_fake = 0.658, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.666, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 29/46, D_loss_real = 0.702, D_loss_fake = 0.659, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.654, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.651, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 32/46, D_loss_real = 0.673, D_loss_fake = 0.649, G_loss = 0.772
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.644, G_loss = 0.767
 2/2 [=====] - 0s 4ms/step
 Epoch 45, Iteration 34/46, D_loss_real = 0.684, D_loss_fake = 0.637, G_loss = 0.764
 2/2 [=====] - 0s 4ms/step
 Epoch 45, Iteration 35/46, D_loss_real = 0.676, D_loss_fake = 0.633, G_loss = 0.771
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 36/46, D_loss_real = 0.673, D_loss_fake = 0.638, G_loss = 0.773
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.635, G_loss = 0.764
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 38/46, D_loss_real = 0.653, D_loss_fake = 0.673, G_loss = 0.754
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 39/46, D_loss_real = 0.676, D_loss_fake = 0.676, G_loss = 0.726
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 40/46, D_loss_real = 0.674, D_loss_fake = 0.699, G_loss = 0.720
 2/2 [=====] - 0s 4ms/step
 Epoch 45, Iteration 41/46, D_loss_real = 0.652, D_loss_fake = 0.737, G_loss = 0.714
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 42/46, D_loss_real = 0.671, D_loss_fake = 0.726, G_loss = 0.725
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 43/46, D_loss_real = 0.691, D_loss_fake = 0.704, G_loss = 0.696
 2/2 [=====] - 0s 4ms/step
 Epoch 45, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.791, G_loss = 0.720
 2/2 [=====] - 0s 3ms/step
 Epoch 45, Iteration 45/46, D_loss_real = 0.718, D_loss_fake = 0.676, G_loss = 0.773
 2/2 [=====] - 0s 4ms/step
 Epoch 45, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss = 0.766
 2/2 [=====] - 0s 3ms/step
 Epoch 46, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.660, G_loss = 0.749
 2/2 [=====] - 0s 3ms/step
 Epoch 46, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.685, G_loss = 0.732
 2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 3/46, D_loss_real = 0.654, D_loss_fake = 0.672, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 4/46, D_loss_real = 0.651, D_loss_fake = 0.683, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 5/46, D_loss_real = 0.637, D_loss_fake = 0.668, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 6/46, D_loss_real = 0.651, D_loss_fake = 0.672, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 7/46, D_loss_real = 0.670, D_loss_fake = 0.660, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 8/46, D_loss_real = 0.643, D_loss_fake = 0.655, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 9/46, D_loss_real = 0.642, D_loss_fake = 0.664, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 10/46, D_loss_real = 0.618, D_loss_fake = 0.696, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 11/46, D_loss_real = 0.622, D_loss_fake = 0.764, G_loss = 0.705
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 12/46, D_loss_real = 0.636, D_loss_fake = 0.769, G_loss = 0.695
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 13/46, D_loss_real = 0.649, D_loss_fake = 0.963, G_loss = 0.696
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 14/46, D_loss_real = 0.667, D_loss_fake = 0.716, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.740, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.701, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 17/46, D_loss_real = 0.705, D_loss_fake = 0.677, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.654, G_loss = 0.754
2/2 [=====] - 0s 4ms/step

Epoch 46, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.670, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 21/46, D_loss_real = 0.680, D_loss_fake = 0.681, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 22/46, D_loss_real = 0.681, D_loss_fake = 0.671, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.704, G_loss = 0.693
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 24/46, D_loss_real = 0.636, D_loss_fake = 0.739, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 25/46, D_loss_real = 0.630, D_loss_fake = 0.766, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 26/46, D_loss_real = 0.630, D_loss_fake = 0.721, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 27/46, D_loss_real = 0.662, D_loss_fake = 0.722, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 28/46, D_loss_real = 0.623, D_loss_fake = 0.726, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 29/46, D_loss_real = 0.664, D_loss_fake = 0.669, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.666, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 31/46, D_loss_real = 0.708, D_loss_fake = 0.701, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.683, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 33/46, D_loss_real = 0.677, D_loss_fake = 0.676, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 34/46, D_loss_real = 0.723, D_loss_fake = 0.750, G_loss = 0.773
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 35/46, D_loss_real = 0.737, D_loss_fake = 0.660, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 36/46, D_loss_real = 0.737, D_loss_fake = 0.720, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 37/46, D_loss_real = 0.719, D_loss_fake = 0.631, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 38/46, D_loss_real = 0.727, D_loss_fake = 0.601, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 39/46, D_loss_real = 0.717, D_loss_fake = 0.600, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 40/46, D_loss_real = 0.716, D_loss_fake = 0.619, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 41/46, D_loss_real = 0.733, D_loss_fake = 0.635, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.644, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 43/46, D_loss_real = 0.691, D_loss_fake = 0.638, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.652, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 45/46, D_loss_real = 0.658, D_loss_fake = 0.665, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.670, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 1/46, D_loss_real = 0.661, D_loss_fake = 0.666, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 2/46, D_loss_real = 0.644, D_loss_fake = 0.674, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 3/46, D_loss_real = 0.669, D_loss_fake = 0.666, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 4/46, D_loss_real = 0.626, D_loss_fake = 0.687, G_loss = 0.746
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 5/46, D_loss_real = 0.660, D_loss_fake = 0.698, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 6/46, D_loss_real = 0.647, D_loss_fake = 0.677, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 7/46, D_loss_real = 0.642, D_loss_fake = 0.672, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 8/46, D_loss_real = 0.663, D_loss_fake = 0.680, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 9/46, D_loss_real = 0.677, D_loss_fake = 0.679, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 10/46, D_loss_real = 0.670, D_loss_fake = 0.691, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 11/46, D_loss_real = 0.654, D_loss_fake = 0.713, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 12/46, D_loss_real = 0.664, D_loss_fake = 0.703, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 13/46, D_loss_real = 0.645, D_loss_fake = 0.694, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 14/46, D_loss_real = 0.655, D_loss_fake = 0.710, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 15/46, D_loss_real = 0.665, D_loss_fake = 0.697, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 16/46, D_loss_real = 0.679, D_loss_fake = 0.704, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 18/46, D_loss_real = 0.703, D_loss_fake = 0.671, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 19/46, D_loss_real = 0.723, D_loss_fake = 0.693, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 20/46, D_loss_real = 0.702, D_loss_fake = 0.671, G_loss = 0.758
2/2 [=====] - 0s 3ms/step

Epoch 47, Iteration 21/46, D_loss_real = 0.722, D_loss_fake = 0.674, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 22/46, D_loss_real = 0.725, D_loss_fake = 0.633, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 23/46, D_loss_real = 0.708, D_loss_fake = 0.632, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 24/46, D_loss_real = 0.730, D_loss_fake = 0.643, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 25/46, D_loss_real = 0.717, D_loss_fake = 0.650, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 26/46, D_loss_real = 0.711, D_loss_fake = 0.636, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.666, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 28/46, D_loss_real = 0.675, D_loss_fake = 0.686, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.662, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.690, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 31/46, D_loss_real = 0.688, D_loss_fake = 0.694, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.733, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 33/46, D_loss_real = 0.650, D_loss_fake = 0.703, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 34/46, D_loss_real = 0.674, D_loss_fake = 0.690, G_loss = 0.709
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 35/46, D_loss_real = 0.640, D_loss_fake = 0.697, G_loss = 0.705
2/2 [=====] - 0s 7ms/step
Epoch 47, Iteration 36/46, D_loss_real = 0.658, D_loss_fake = 0.705, G_loss = 0.695
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 37/46, D_loss_real = 0.625, D_loss_fake = 0.719, G_loss = 0.699
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 38/46, D_loss_real = 0.646, D_loss_fake = 0.731, G_loss = 0.688
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 39/46, D_loss_real = 0.649, D_loss_fake = 0.727, G_loss = 0.688
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 40/46, D_loss_real = 0.670, D_loss_fake = 0.725, G_loss = 0.692
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.707, G_loss = 0.706
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 42/46, D_loss_real = 0.669, D_loss_fake = 0.720, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 43/46, D_loss_real = 0.664, D_loss_fake = 0.709, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 44/46, D_loss_real = 0.650, D_loss_fake = 0.716, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.699, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 46/46, D_loss_real = 0.643, D_loss_fake = 0.679, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 1/46, D_loss_real = 0.668, D_loss_fake = 0.694, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 2/46, D_loss_real = 0.674, D_loss_fake = 0.688, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 3/46, D_loss_real = 0.624, D_loss_fake = 0.677, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 4/46, D_loss_real = 0.653, D_loss_fake = 0.688, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 5/46, D_loss_real = 0.659, D_loss_fake = 0.735, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 6/46, D_loss_real = 0.687, D_loss_fake = 0.660, G_loss = 0.771
2/2 [=====] - 0s 4ms/step

Epoch 48, Iteration 7/46, D_loss_real = 0.654, D_loss_fake = 0.683, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.645, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 9/46, D_loss_real = 0.683, D_loss_fake = 0.764, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 10/46, D_loss_real = 0.736, D_loss_fake = 0.641, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 11/46, D_loss_real = 0.746, D_loss_fake = 0.636, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 12/46, D_loss_real = 0.769, D_loss_fake = 0.626, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 13/46, D_loss_real = 0.734, D_loss_fake = 0.631, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 14/46, D_loss_real = 0.736, D_loss_fake = 0.654, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 15/46, D_loss_real = 0.757, D_loss_fake = 0.642, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 16/46, D_loss_real = 0.718, D_loss_fake = 0.669, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 17/46, D_loss_real = 0.724, D_loss_fake = 0.667, G_loss = 0.756
2/2 [=====] - 0s 13ms/step
Epoch 48, Iteration 18/46, D_loss_real = 0.707, D_loss_fake = 0.677, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 19/46, D_loss_real = 0.706, D_loss_fake = 0.665, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.676, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 21/46, D_loss_real = 0.695, D_loss_fake = 0.678, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss = 0.724
2/2 [=====] - 0s 4ms/step

Epoch 48, Iteration 23/46, D_loss_real = 0.699, D_loss_fake = 0.677, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 24/46, D_loss_real = 0.699, D_loss_fake = 0.698, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 25/46, D_loss_real = 0.672, D_loss_fake = 0.670, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 26/46, D_loss_real = 0.696, D_loss_fake = 0.656, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 27/46, D_loss_real = 0.714, D_loss_fake = 0.677, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 28/46, D_loss_real = 0.723, D_loss_fake = 0.686, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 30/46, D_loss_real = 0.722, D_loss_fake = 0.674, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 31/46, D_loss_real = 0.713, D_loss_fake = 0.667, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 32/46, D_loss_real = 0.700, D_loss_fake = 0.689, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 33/46, D_loss_real = 0.691, D_loss_fake = 0.679, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 34/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 35/46, D_loss_real = 0.665, D_loss_fake = 0.683, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.682, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 37/46, D_loss_real = 0.671, D_loss_fake = 0.667, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 38/46, D_loss_real = 0.666, D_loss_fake = 0.704, G_loss = 0.728
2/2 [=====] - 0s 3ms/step

Epoch 48, Iteration 39/46, D_loss_real = 0.656, D_loss_fake = 0.699, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 40/46, D_loss_real = 0.665, D_loss_fake = 0.665, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 41/46, D_loss_real = 0.676, D_loss_fake = 0.684, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 42/46, D_loss_real = 0.653, D_loss_fake = 0.684, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 43/46, D_loss_real = 0.658, D_loss_fake = 0.690, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 44/46, D_loss_real = 0.654, D_loss_fake = 0.689, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 45/46, D_loss_real = 0.654, D_loss_fake = 0.675, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 46/46, D_loss_real = 0.659, D_loss_fake = 0.692, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 1/46, D_loss_real = 0.644, D_loss_fake = 0.714, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 2/46, D_loss_real = 0.666, D_loss_fake = 0.687, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 3/46, D_loss_real = 0.660, D_loss_fake = 0.681, G_loss = 0.720
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 4/46, D_loss_real = 0.661, D_loss_fake = 0.711, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 5/46, D_loss_real = 0.692, D_loss_fake = 0.738, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 6/46, D_loss_real = 0.691, D_loss_fake = 0.708, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 7/46, D_loss_real = 0.718, D_loss_fake = 0.701, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 8/46, D_loss_real = 0.708, D_loss_fake = 0.695, G_loss = 0.761
2/2 [=====] - 0s 3ms/step

Epoch 49, Iteration 9/46, D_loss_real = 0.699, D_loss_fake = 0.683, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.660, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 11/46, D_loss_real = 0.695, D_loss_fake = 0.694, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 12/46, D_loss_real = 0.689, D_loss_fake = 0.687, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 13/46, D_loss_real = 0.670, D_loss_fake = 0.693, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.672, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 16/46, D_loss_real = 0.696, D_loss_fake = 0.673, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 17/46, D_loss_real = 0.671, D_loss_fake = 0.715, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.719, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 19/46, D_loss_real = 0.669, D_loss_fake = 0.708, G_loss = 0.710
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 20/46, D_loss_real = 0.673, D_loss_fake = 0.722, G_loss = 0.717
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 21/46, D_loss_real = 0.701, D_loss_fake = 0.677, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 22/46, D_loss_real = 0.665, D_loss_fake = 0.652, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 23/46, D_loss_real = 0.674, D_loss_fake = 0.661, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 24/46, D_loss_real = 0.672, D_loss_fake = 0.677, G_loss = 0.739
2/2 [=====] - 0s 3ms/step

Epoch 49, Iteration 25/46, D_loss_real = 0.652, D_loss_fake = 0.677, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 26/46, D_loss_real = 0.688, D_loss_fake = 0.682, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 27/46, D_loss_real = 0.666, D_loss_fake = 0.681, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 28/46, D_loss_real = 0.687, D_loss_fake = 0.747, G_loss = 0.729
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 29/46, D_loss_real = 0.697, D_loss_fake = 0.696, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 30/46, D_loss_real = 0.709, D_loss_fake = 0.665, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 31/46, D_loss_real = 0.713, D_loss_fake = 0.646, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.638, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 33/46, D_loss_real = 0.722, D_loss_fake = 0.635, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 34/46, D_loss_real = 0.701, D_loss_fake = 0.647, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 35/46, D_loss_real = 0.713, D_loss_fake = 0.651, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 36/46, D_loss_real = 0.703, D_loss_fake = 0.647, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 37/46, D_loss_real = 0.703, D_loss_fake = 0.664, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 38/46, D_loss_real = 0.690, D_loss_fake = 0.660, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 39/46, D_loss_real = 0.690, D_loss_fake = 0.671, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 40/46, D_loss_real = 0.686, D_loss_fake = 0.677, G_loss = 0.745
2/2 [=====] - 0s 4ms/step

Epoch 49, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.671, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 42/46, D_loss_real = 0.675, D_loss_fake = 0.675, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 43/46, D_loss_real = 0.685, D_loss_fake = 0.645, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 44/46, D_loss_real = 0.681, D_loss_fake = 0.624, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 45/46, D_loss_real = 0.689, D_loss_fake = 0.640, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 46/46, D_loss_real = 0.699, D_loss_fake = 0.630, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 1/46, D_loss_real = 0.657, D_loss_fake = 0.658, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 2/46, D_loss_real = 0.642, D_loss_fake = 0.676, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 3/46, D_loss_real = 0.657, D_loss_fake = 0.707, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 4/46, D_loss_real = 0.664, D_loss_fake = 0.711, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 5/46, D_loss_real = 0.680, D_loss_fake = 0.665, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 6/46, D_loss_real = 0.717, D_loss_fake = 0.619, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 7/46, D_loss_real = 0.730, D_loss_fake = 0.651, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 8/46, D_loss_real = 0.721, D_loss_fake = 0.670, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 9/46, D_loss_real = 0.712, D_loss_fake = 0.661, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.659, G_loss = 0.751
2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 11/46, D_loss_real = 0.687, D_loss_fake = 0.692, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 12/46, D_loss_real = 0.665, D_loss_fake = 0.674, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 13/46, D_loss_real = 0.623, D_loss_fake = 0.674, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 14/46, D_loss_real = 0.614, D_loss_fake = 0.676, G_loss = 0.732
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 15/46, D_loss_real = 0.597, D_loss_fake = 0.701, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 16/46, D_loss_real = 0.547, D_loss_fake = 0.795, G_loss = 0.698
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 17/46, D_loss_real = 0.570, D_loss_fake = 0.784, G_loss = 0.682
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 18/46, D_loss_real = 0.600, D_loss_fake = 0.840, G_loss = 0.699
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 19/46, D_loss_real = 0.657, D_loss_fake = 0.746, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 20/46, D_loss_real = 0.658, D_loss_fake = 0.683, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.676, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 22/46, D_loss_real = 0.701, D_loss_fake = 0.691, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 23/46, D_loss_real = 0.690, D_loss_fake = 0.672, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 24/46, D_loss_real = 0.679, D_loss_fake = 0.678, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 25/46, D_loss_real = 0.680, D_loss_fake = 0.677, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 26/46, D_loss_real = 0.660, D_loss_fake = 0.667, G_loss = 0.788
2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 27/46, D_loss_real = 0.680, D_loss_fake = 0.666, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.670, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 29/46, D_loss_real = 0.687, D_loss_fake = 0.639, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 30/46, D_loss_real = 0.678, D_loss_fake = 0.635, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.667, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.650, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 33/46, D_loss_real = 0.667, D_loss_fake = 0.659, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 34/46, D_loss_real = 0.711, D_loss_fake = 0.682, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 35/46, D_loss_real = 0.658, D_loss_fake = 0.688, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 36/46, D_loss_real = 0.645, D_loss_fake = 0.695, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 37/46, D_loss_real = 0.642, D_loss_fake = 0.696, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 38/46, D_loss_real = 0.654, D_loss_fake = 0.675, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 39/46, D_loss_real = 0.665, D_loss_fake = 0.679, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.656, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.675, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 42/46, D_loss_real = 0.664, D_loss_fake = 0.660, G_loss = 0.838
2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 43/46, D_loss_real = 0.715, D_loss_fake = 0.680, G_loss = 0.834

2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.707, G_loss = 0.776

2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 45/46, D_loss_real = 0.703, D_loss_fake = 0.711, G_loss = 0.745

2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 46/46, D_loss_real = 0.689, D_loss_fake = 0.719, G_loss = 0.717

2/2 [=====] - 0s 4ms/step

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

Kết quả truyền trực tuyến bị cắt bất đến 5000 dòng cuối.

2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 32/46, D_loss_real = 0.656, D_loss_fake = 0.725, G_loss = 0.711

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 33/46, D_loss_real = 0.618, D_loss_fake = 0.723, G_loss = 0.723

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 34/46, D_loss_real = 0.614, D_loss_fake = 0.771, G_loss = 0.752

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 35/46, D_loss_real = 0.636, D_loss_fake = 0.714, G_loss = 0.757

2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.686, G_loss = 0.762

2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 37/46, D_loss_real = 0.669, D_loss_fake = 0.678, G_loss = 0.798

2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 38/46, D_loss_real = 0.671, D_loss_fake = 0.645, G_loss = 0.790

2/2 [=====] - 0s 4ms/step

Epoch 6, Iteration 39/46, D_loss_real = 0.686, D_loss_fake = 0.665, G_loss = 0.796

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.631, G_loss = 0.788

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 41/46, D_loss_real = 0.715, D_loss_fake = 0.671, G_loss = 0.766

2/2 [=====] - 0s 3ms/step

Epoch 6, Iteration 42/46, D_loss_real = 0.724, D_loss_fake = 0.673, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.719, D_loss_fake = 0.679, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.719, D_loss_fake = 0.677, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.728, D_loss_fake = 0.677, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.714, D_loss_fake = 0.697, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.701, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.716, D_loss_fake = 0.643, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.637, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.648, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.713, D_loss_fake = 0.654, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.673, D_loss_fake = 0.654, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.666, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.663, D_loss_fake = 0.664, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.672, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.646, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.653, D_loss_fake = 0.701, G_loss = 0.746
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 12/46, D_loss_real = 0.676, D_loss_fake = 0.658, G_loss = 0.740
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.659, D_loss_fake = 0.688, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.672, D_loss_fake = 0.695, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.670, D_loss_fake = 0.661, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.658, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.711, D_loss_fake = 0.658, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.714, D_loss_fake = 0.654, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.718, D_loss_fake = 0.673, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.645, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.645, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 23/46, D_loss_real = 0.718, D_loss_fake = 0.656, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.653, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.670, D_loss_fake = 0.682, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.663, D_loss_fake = 0.647, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.657, D_loss_fake = 0.659, G_loss = 0.756
2/2 [=====] - 0s 3ms/step

Epoch 7, Iteration 28/46, D_loss_real = 0.646, D_loss_fake = 0.656, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.628, D_loss_fake = 0.664, G_loss = 0.722
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.675, D_loss_fake = 0.715, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.652, D_loss_fake = 0.734, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.628, D_loss_fake = 0.699, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.681, D_loss_fake = 0.672, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.691, D_loss_fake = 0.665, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.726, D_loss_fake = 0.666, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.689, D_loss_fake = 0.671, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.662, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.681, D_loss_fake = 0.690, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 39/46, D_loss_real = 0.644, D_loss_fake = 0.684, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.630, D_loss_fake = 0.713, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.594, D_loss_fake = 0.712, G_loss = 0.701
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.604, D_loss_fake = 0.773, G_loss = 0.706
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.636, D_loss_fake = 0.717, G_loss = 0.714
2/2 [=====] - 0s 4ms/step

Epoch 7, Iteration 44/46, D_loss_real = 0.595, D_loss_fake = 0.736, G_loss = 0.708
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.626, D_loss_fake = 0.739, G_loss = 0.702
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.614, D_loss_fake = 0.708, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.644, D_loss_fake = 0.784, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.645, D_loss_fake = 0.742, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.651, D_loss_fake = 0.696, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.671, D_loss_fake = 0.676, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.719, D_loss_fake = 0.632, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.706, D_loss_fake = 0.654, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.721, D_loss_fake = 0.656, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.726, D_loss_fake = 0.619, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 9/46, D_loss_real = 0.734, D_loss_fake = 0.639, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.723, D_loss_fake = 0.647, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.713, D_loss_fake = 0.633, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.681, D_loss_fake = 0.675, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.646, D_loss_fake = 0.642, G_loss = 0.791
2/2 [=====] - 0s 4ms/step

Epoch 8, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.646, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.629, D_loss_fake = 0.646, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.621, D_loss_fake = 0.627, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.587, D_loss_fake = 0.713, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.611, D_loss_fake = 0.762, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.644, D_loss_fake = 0.794, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.653, D_loss_fake = 0.740, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.660, D_loss_fake = 0.724, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.656, D_loss_fake = 0.688, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.716, D_loss_fake = 0.616, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.748, D_loss_fake = 0.656, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 25/46, D_loss_real = 0.763, D_loss_fake = 0.661, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.717, D_loss_fake = 0.655, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.729, D_loss_fake = 0.650, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.651, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.663, G_loss = 0.746
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 30/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.687, D_loss_fake = 0.668, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.647, G_loss = 0.751
2/2 [=====] - 0s 5ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.686, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.708, D_loss_fake = 0.669, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.681, D_loss_fake = 0.669, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.697, G_loss = 0.737
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.670, D_loss_fake = 0.693, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.659, D_loss_fake = 0.677, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.692, D_loss_fake = 0.675, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 41/46, D_loss_real = 0.703, D_loss_fake = 0.685, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.686, D_loss_fake = 0.695, G_loss = 0.722
2/2 [=====] - 0s 5ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.716, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.656, G_loss = 0.756
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 46/46, D_loss_real = 0.701, D_loss_fake = 0.663, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.671, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.647, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.685, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.676, D_loss_fake = 0.679, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.690, D_loss_fake = 0.653, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.651, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.633, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.658, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.640, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.681, D_loss_fake = 0.679, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 11/46, D_loss_real = 0.657, D_loss_fake = 0.675, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.657, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.673, D_loss_fake = 0.638, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.665, D_loss_fake = 0.702, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.678, D_loss_fake = 0.690, G_loss = 0.769
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 16/46, D_loss_real = 0.677, D_loss_fake = 0.648, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.705, D_loss_fake = 0.644, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.664, D_loss_fake = 0.682, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.713, D_loss_fake = 0.648, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.639, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.682, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.685, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.667, D_loss_fake = 0.683, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.641, D_loss_fake = 0.657, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.633, D_loss_fake = 0.693, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.611, D_loss_fake = 0.744, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 27/46, D_loss_real = 0.638, D_loss_fake = 0.691, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.694, D_loss_fake = 0.783, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.683, D_loss_fake = 0.681, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.663, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.707, D_loss_fake = 0.668, G_loss = 0.759
2/2 [=====] - 0s 3ms/step

Epoch 9, Iteration 32/46, D_loss_real = 0.674, D_loss_fake = 0.681, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.651, D_loss_fake = 0.656, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.622, D_loss_fake = 0.658, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.660, D_loss_fake = 0.678, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.615, D_loss_fake = 0.721, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.637, D_loss_fake = 0.733, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.583, D_loss_fake = 0.752, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.606, D_loss_fake = 0.717, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.629, D_loss_fake = 0.712, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.655, D_loss_fake = 0.683, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.633, D_loss_fake = 0.731, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 43/46, D_loss_real = 0.650, D_loss_fake = 0.674, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.641, D_loss_fake = 0.659, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.666, D_loss_fake = 0.687, G_loss = 0.752
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.661, D_loss_fake = 0.753, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.662, D_loss_fake = 0.685, G_loss = 0.744
2/2 [=====] - 0s 3ms/step

Epoch 10, Iteration 2/46, D_loss_real = 0.702, D_loss_fake = 0.693, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.685, D_loss_fake = 0.704, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.707, D_loss_fake = 0.639, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.730, D_loss_fake = 0.643, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.689, D_loss_fake = 0.677, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.688, D_loss_fake = 0.698, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.707, D_loss_fake = 0.723, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.679, D_loss_fake = 0.705, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.678, D_loss_fake = 0.684, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.646, D_loss_fake = 0.649, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.658, D_loss_fake = 0.673, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 13/46, D_loss_real = 0.639, D_loss_fake = 0.627, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.623, D_loss_fake = 0.730, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.606, D_loss_fake = 0.742, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.635, D_loss_fake = 0.720, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.636, D_loss_fake = 0.730, G_loss = 0.773
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 18/46, D_loss_real = 0.685, D_loss_fake = 0.707, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.653, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.647, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.709, D_loss_fake = 0.650, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.694, D_loss_fake = 0.633, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.709, D_loss_fake = 0.628, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.721, D_loss_fake = 0.639, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.725, D_loss_fake = 0.661, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.685, D_loss_fake = 0.616, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.656, D_loss_fake = 0.663, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 29/46, D_loss_real = 0.637, D_loss_fake = 0.670, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.643, D_loss_fake = 0.674, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.630, D_loss_fake = 0.766, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.630, D_loss_fake = 0.763, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.633, D_loss_fake = 0.769, G_loss = 0.799
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 34/46, D_loss_real = 0.660, D_loss_fake = 0.663, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.624, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.710, D_loss_fake = 0.623, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.701, D_loss_fake = 0.666, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.626, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.679, D_loss_fake = 0.611, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.655, D_loss_fake = 0.636, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.669, D_loss_fake = 0.631, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.672, D_loss_fake = 0.711, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.678, D_loss_fake = 0.691, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.665, D_loss_fake = 0.693, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 45/46, D_loss_real = 0.657, D_loss_fake = 0.718, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.697, D_loss_fake = 0.741, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.680, D_loss_fake = 0.684, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.717, D_loss_fake = 0.695, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.720, D_loss_fake = 0.652, G_loss = 0.798
2/2 [=====] - 0s 4ms/step

Epoch 11, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.642, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.701, D_loss_fake = 0.588, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.666, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.706, D_loss_fake = 0.656, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.617, D_loss_fake = 0.607, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.611, D_loss_fake = 0.637, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.591, D_loss_fake = 0.703, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.627, D_loss_fake = 0.791, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.671, D_loss_fake = 0.780, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.709, D_loss_fake = 0.672, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.651, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 15/46, D_loss_real = 0.691, D_loss_fake = 0.632, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.668, D_loss_fake = 0.685, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.685, D_loss_fake = 0.670, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.672, D_loss_fake = 0.643, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.715, D_loss_fake = 0.615, G_loss = 0.806
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.651, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.666, D_loss_fake = 0.672, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.689, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.660, D_loss_fake = 0.640, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.707, D_loss_fake = 0.623, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.699, D_loss_fake = 0.636, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.579, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.585, G_loss = 0.894
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.621, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.721, D_loss_fake = 0.601, G_loss = 0.869
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.622, D_loss_fake = 0.637, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 31/46, D_loss_real = 0.606, D_loss_fake = 0.702, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.689, D_loss_fake = 0.626, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.673, D_loss_fake = 0.661, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.650, D_loss_fake = 0.698, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.667, D_loss_fake = 0.802, G_loss = 0.733
2/2 [=====] - 0s 4ms/step

Epoch 11, Iteration 36/46, D_loss_real = 0.690, D_loss_fake = 0.797, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.715, D_loss_fake = 0.669, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.694, D_loss_fake = 0.638, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.703, D_loss_fake = 0.625, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.683, D_loss_fake = 0.622, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.646, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.606, D_loss_fake = 0.643, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.615, D_loss_fake = 0.647, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.572, D_loss_fake = 0.701, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.549, D_loss_fake = 0.705, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.569, D_loss_fake = 0.814, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 1/46, D_loss_real = 0.566, D_loss_fake = 0.823, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.582, D_loss_fake = 0.688, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.630, D_loss_fake = 0.627, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.706, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.676, G_loss = 0.770
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 6/46, D_loss_real = 0.664, D_loss_fake = 0.750, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.703, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.693, D_loss_fake = 0.728, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.691, D_loss_fake = 0.643, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.716, D_loss_fake = 0.592, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.715, D_loss_fake = 0.564, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.645, D_loss_fake = 0.609, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.608, D_loss_fake = 0.597, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.566, D_loss_fake = 0.635, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.504, D_loss_fake = 0.764, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.575, D_loss_fake = 0.829, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 17/46, D_loss_real = 0.571, D_loss_fake = 0.801, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.658, D_loss_fake = 0.827, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.649, D_loss_fake = 0.768, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.661, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.675, D_loss_fake = 0.728, G_loss = 0.828
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 22/46, D_loss_real = 0.704, D_loss_fake = 0.638, G_loss = 0.905
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.741, D_loss_fake = 0.601, G_loss = 0.944
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.751, D_loss_fake = 0.595, G_loss = 0.943
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.756, D_loss_fake = 0.594, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.776, D_loss_fake = 0.636, G_loss = 0.790
2/2 [=====] - 0s 7ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.759, D_loss_fake = 0.628, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.746, D_loss_fake = 0.640, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.719, D_loss_fake = 0.655, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.674, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.644, D_loss_fake = 0.668, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.658, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 33/46, D_loss_real = 0.639, D_loss_fake = 0.638, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.664, D_loss_fake = 0.651, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.653, D_loss_fake = 0.688, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.661, D_loss_fake = 0.619, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.591, G_loss = 0.891
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 38/46, D_loss_real = 0.731, D_loss_fake = 0.637, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.659, D_loss_fake = 0.605, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.710, D_loss_fake = 0.639, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.628, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.652, D_loss_fake = 0.616, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.677, D_loss_fake = 0.653, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.707, D_loss_fake = 0.640, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.712, D_loss_fake = 0.600, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.733, D_loss_fake = 0.603, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.679, D_loss_fake = 0.597, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.662, D_loss_fake = 0.709, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 3/46, D_loss_real = 0.606, D_loss_fake = 0.688, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.687, D_loss_fake = 0.707, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.648, D_loss_fake = 0.793, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.711, D_loss_fake = 0.498, G_loss = 1.085
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.770, D_loss_fake = 0.510, G_loss = 0.961
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 8/46, D_loss_real = 0.665, D_loss_fake = 0.591, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.642, D_loss_fake = 0.731, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.567, D_loss_fake = 0.682, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.611, D_loss_fake = 0.778, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.568, D_loss_fake = 0.717, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.638, D_loss_fake = 0.681, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.620, D_loss_fake = 0.597, G_loss = 0.931
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.612, D_loss_fake = 0.689, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.554, D_loss_fake = 0.786, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.611, D_loss_fake = 0.772, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.567, D_loss_fake = 0.826, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 19/46, D_loss_real = 0.616, D_loss_fake = 0.794, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.649, D_loss_fake = 0.604, G_loss = 0.918
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.699, D_loss_fake = 0.597, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.665, D_loss_fake = 0.605, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.663, D_loss_fake = 0.661, G_loss = 0.813
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 24/46, D_loss_real = 0.617, D_loss_fake = 0.677, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.634, D_loss_fake = 0.670, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.587, D_loss_fake = 0.669, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.607, D_loss_fake = 0.695, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.622, D_loss_fake = 0.601, G_loss = 0.888
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.635, D_loss_fake = 0.603, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.610, D_loss_fake = 0.562, G_loss = 0.985
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.561, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.707, D_loss_fake = 0.606, G_loss = 0.891
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.659, D_loss_fake = 0.691, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.663, D_loss_fake = 0.609, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 35/46, D_loss_real = 0.694, D_loss_fake = 0.692, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.644, D_loss_fake = 0.749, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.681, D_loss_fake = 0.769, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.635, D_loss_fake = 0.665, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.643, G_loss = 0.836
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 40/46, D_loss_real = 0.628, D_loss_fake = 0.671, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.660, D_loss_fake = 0.651, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.595, D_loss_fake = 0.822, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.738, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.664, D_loss_fake = 0.665, G_loss = 0.925
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.743, D_loss_fake = 0.591, G_loss = 0.959
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.734, D_loss_fake = 0.575, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.730, D_loss_fake = 0.612, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.661, D_loss_fake = 0.649, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.662, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.650, D_loss_fake = 0.673, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 5/46, D_loss_real = 0.638, D_loss_fake = 0.673, G_loss = 0.779
2/2 [=====] - 0s 5ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.629, D_loss_fake = 0.687, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.618, D_loss_fake = 0.721, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.592, D_loss_fake = 0.821, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.580, D_loss_fake = 0.763, G_loss = 0.728
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 10/46, D_loss_real = 0.652, D_loss_fake = 0.746, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.660, D_loss_fake = 0.727, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.655, D_loss_fake = 0.551, G_loss = 1.058
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.732, D_loss_fake = 0.506, G_loss = 1.068
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.553, G_loss = 0.971
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.633, D_loss_fake = 0.640, G_loss = 0.924
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.671, D_loss_fake = 0.704, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.606, D_loss_fake = 0.858, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.618, D_loss_fake = 0.870, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.702, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.582, G_loss = 0.942
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.597, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.630, D_loss_fake = 0.606, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.576, D_loss_fake = 0.710, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.593, D_loss_fake = 0.702, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.669, D_loss_fake = 0.686, G_loss = 0.825
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.631, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.649, D_loss_fake = 0.598, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.627, D_loss_fake = 0.616, G_loss = 0.933
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.603, D_loss_fake = 0.548, G_loss = 0.947
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.607, D_loss_fake = 0.545, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.630, D_loss_fake = 0.625, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.682, D_loss_fake = 0.831, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.596, D_loss_fake = 0.814, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.607, D_loss_fake = 0.746, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.766, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.736, D_loss_fake = 0.796, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 37/46, D_loss_real = 0.647, D_loss_fake = 0.798, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.710, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.693, D_loss_fake = 0.625, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.701, D_loss_fake = 0.601, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.720, D_loss_fake = 0.622, G_loss = 0.964
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 42/46, D_loss_real = 0.711, D_loss_fake = 0.609, G_loss = 0.905
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.740, D_loss_fake = 0.632, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.761, D_loss_fake = 0.638, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.722, D_loss_fake = 0.584, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.711, D_loss_fake = 0.642, G_loss = 0.927
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.708, D_loss_fake = 0.576, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.686, D_loss_fake = 0.609, G_loss = 0.906
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.714, D_loss_fake = 0.644, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.585, D_loss_fake = 0.649, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.601, D_loss_fake = 0.735, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.629, D_loss_fake = 0.791, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 7/46, D_loss_real = 0.616, D_loss_fake = 0.806, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.645, D_loss_fake = 0.688, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.691, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.671, D_loss_fake = 0.614, G_loss = 0.920
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.704, D_loss_fake = 0.608, G_loss = 0.940
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 12/46, D_loss_real = 0.745, D_loss_fake = 0.584, G_loss = 0.927
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.724, D_loss_fake = 0.634, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.707, D_loss_fake = 0.650, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.645, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.661, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.728, D_loss_fake = 0.666, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.662, D_loss_fake = 0.648, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.674, D_loss_fake = 0.717, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.691, D_loss_fake = 0.634, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.625, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.705, D_loss_fake = 0.660, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 23/46, D_loss_real = 0.666, D_loss_fake = 0.650, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.675, D_loss_fake = 0.670, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.722, D_loss_fake = 0.730, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.703, D_loss_fake = 0.682, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.645, G_loss = 0.840
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 28/46, D_loss_real = 0.639, D_loss_fake = 0.641, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.671, D_loss_fake = 0.633, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.646, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.740, D_loss_fake = 0.694, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.682, D_loss_fake = 0.704, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.734, D_loss_fake = 0.654, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.706, D_loss_fake = 0.662, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.752, D_loss_fake = 0.639, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.706, D_loss_fake = 0.633, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.662, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.691, D_loss_fake = 0.666, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 39/46, D_loss_real = 0.696, D_loss_fake = 0.652, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.675, D_loss_fake = 0.620, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.681, D_loss_fake = 0.648, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.722, D_loss_fake = 0.641, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.677, D_loss_fake = 0.638, G_loss = 0.827
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 44/46, D_loss_real = 0.698, D_loss_fake = 0.628, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.689, D_loss_fake = 0.689, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.718, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.716, D_loss_fake = 0.701, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.670, D_loss_fake = 0.685, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.773, D_loss_fake = 0.641, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.781, D_loss_fake = 0.692, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.708, D_loss_fake = 0.687, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.703, D_loss_fake = 0.717, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.706, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.687, D_loss_fake = 0.686, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 9/46, D_loss_real = 0.712, D_loss_fake = 0.676, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.683, D_loss_fake = 0.680, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.707, D_loss_fake = 0.680, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.678, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.675, D_loss_fake = 0.706, G_loss = 0.738
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 14/46, D_loss_real = 0.681, D_loss_fake = 0.682, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.661, D_loss_fake = 0.674, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.676, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.683, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.663, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.715, D_loss_fake = 0.704, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.710, D_loss_fake = 0.713, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.728, D_loss_fake = 0.731, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.707, D_loss_fake = 0.736, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.693, D_loss_fake = 0.721, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.665, D_loss_fake = 0.670, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 25/46, D_loss_real = 0.693, D_loss_fake = 0.661, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.647, D_loss_fake = 0.655, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.658, D_loss_fake = 0.638, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.647, D_loss_fake = 0.681, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.625, D_loss_fake = 0.724, G_loss = 0.747
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 30/46, D_loss_real = 0.611, D_loss_fake = 0.720, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.661, D_loss_fake = 0.704, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.691, D_loss_fake = 0.646, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.681, D_loss_fake = 0.626, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.656, D_loss_fake = 0.641, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.667, D_loss_fake = 0.663, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.680, D_loss_fake = 0.688, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.641, D_loss_fake = 0.711, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.674, D_loss_fake = 0.716, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.677, D_loss_fake = 0.726, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.659, D_loss_fake = 0.707, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 41/46, D_loss_real = 0.663, D_loss_fake = 0.684, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.689, D_loss_fake = 0.637, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.631, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.693, D_loss_fake = 0.606, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.677, D_loss_fake = 0.633, G_loss = 0.782
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 46/46, D_loss_real = 0.683, D_loss_fake = 0.651, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.651, D_loss_fake = 0.682, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.679, D_loss_fake = 0.673, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.694, D_loss_fake = 0.669, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.661, D_loss_fake = 0.640, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.661, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.714, D_loss_fake = 0.644, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.688, D_loss_fake = 0.641, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.644, D_loss_fake = 0.635, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.641, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.646, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 11/46, D_loss_real = 0.736, D_loss_fake = 0.590, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.727, D_loss_fake = 0.612, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.705, D_loss_fake = 0.655, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.705, D_loss_fake = 0.659, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.691, D_loss_fake = 0.684, G_loss = 0.785
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 16/46, D_loss_real = 0.649, D_loss_fake = 0.673, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.677, D_loss_fake = 0.678, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.685, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.684, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.670, D_loss_fake = 0.657, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.623, D_loss_fake = 0.629, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.637, D_loss_fake = 0.671, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.636, D_loss_fake = 0.670, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.577, D_loss_fake = 0.688, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.576, D_loss_fake = 0.699, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.561, D_loss_fake = 0.769, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 27/46, D_loss_real = 0.599, D_loss_fake = 0.702, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.642, D_loss_fake = 0.692, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.629, D_loss_fake = 0.642, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.688, D_loss_fake = 0.689, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.655, D_loss_fake = 0.676, G_loss = 0.873
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 32/46, D_loss_real = 0.662, D_loss_fake = 0.684, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.660, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.663, D_loss_fake = 0.634, G_loss = 0.815
2/2 [=====] - 0s 5ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.631, D_loss_fake = 0.684, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.679, D_loss_fake = 0.635, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.698, D_loss_fake = 0.649, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.651, D_loss_fake = 0.645, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.640, D_loss_fake = 0.723, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.650, D_loss_fake = 0.698, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.705, D_loss_fake = 0.680, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.674, D_loss_fake = 0.680, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 43/46, D_loss_real = 0.719, D_loss_fake = 0.621, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.726, D_loss_fake = 0.607, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.712, D_loss_fake = 0.620, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.685, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.686, G_loss = 0.771
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 2/46, D_loss_real = 0.673, D_loss_fake = 0.701, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.669, D_loss_fake = 0.749, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.700, D_loss_fake = 0.647, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.714, D_loss_fake = 0.621, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.600, G_loss = 0.903
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.722, D_loss_fake = 0.586, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.613, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.701, D_loss_fake = 0.638, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.680, D_loss_fake = 0.629, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.698, D_loss_fake = 0.694, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.648, D_loss_fake = 0.700, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 13/46, D_loss_real = 0.683, D_loss_fake = 0.649, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.686, D_loss_fake = 0.628, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.613, D_loss_fake = 0.637, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.589, D_loss_fake = 0.644, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.604, D_loss_fake = 0.615, G_loss = 0.848
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 18/46, D_loss_real = 0.576, D_loss_fake = 0.715, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.625, D_loss_fake = 0.700, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.705, D_loss_fake = 0.770, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.712, D_loss_fake = 0.839, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.700, D_loss_fake = 0.747, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.734, D_loss_fake = 0.638, G_loss = 0.971
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.751, D_loss_fake = 0.555, G_loss = 1.112
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.716, D_loss_fake = 0.579, G_loss = 0.938
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.728, D_loss_fake = 0.619, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.637, D_loss_fake = 0.640, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.593, D_loss_fake = 0.652, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 29/46, D_loss_real = 0.638, D_loss_fake = 0.703, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.541, D_loss_fake = 0.716, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.525, D_loss_fake = 0.762, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.504, D_loss_fake = 0.776, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.540, D_loss_fake = 0.782, G_loss = 0.728
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 34/46, D_loss_real = 0.602, D_loss_fake = 0.793, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.644, D_loss_fake = 0.678, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.674, D_loss_fake = 0.650, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.559, G_loss = 1.019
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.741, D_loss_fake = 0.581, G_loss = 0.916
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.731, D_loss_fake = 0.643, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.670, D_loss_fake = 0.610, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.639, D_loss_fake = 0.615, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.604, D_loss_fake = 0.631, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.520, D_loss_fake = 0.751, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.551, D_loss_fake = 0.713, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 45/46, D_loss_real = 0.500, D_loss_fake = 0.760, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.610, D_loss_fake = 0.775, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.617, D_loss_fake = 0.688, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.680, D_loss_fake = 0.636, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.678, D_loss_fake = 0.647, G_loss = 0.896
2/2 [=====] - 0s 4ms/step

Epoch 19, Iteration 4/46, D_loss_real = 0.726, D_loss_fake = 0.611, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.670, D_loss_fake = 0.604, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.626, D_loss_fake = 0.627, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.674, D_loss_fake = 0.620, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.692, D_loss_fake = 0.653, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.633, D_loss_fake = 0.640, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.625, D_loss_fake = 0.622, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.606, D_loss_fake = 0.642, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.541, D_loss_fake = 0.675, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.469, D_loss_fake = 0.842, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.482, D_loss_fake = 0.898, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 15/46, D_loss_real = 0.567, D_loss_fake = 0.727, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.669, D_loss_fake = 0.654, G_loss = 0.950
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.744, D_loss_fake = 0.619, G_loss = 0.963
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.742, D_loss_fake = 0.605, G_loss = 0.973
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.801, D_loss_fake = 0.592, G_loss = 1.012
2/2 [=====] - 0s 4ms/step

Epoch 19, Iteration 20/46, D_loss_real = 0.767, D_loss_fake = 0.548, G_loss = 1.009
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.768, D_loss_fake = 0.551, G_loss = 0.921
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.686, D_loss_fake = 0.572, G_loss = 0.904
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.617, D_loss_fake = 0.630, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.666, D_loss_fake = 0.640, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.603, D_loss_fake = 0.761, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.639, D_loss_fake = 0.656, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.668, D_loss_fake = 0.690, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.722, D_loss_fake = 0.684, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.740, D_loss_fake = 0.635, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.733, D_loss_fake = 0.597, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 31/46, D_loss_real = 0.766, D_loss_fake = 0.612, G_loss = 0.887
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.698, D_loss_fake = 0.648, G_loss = 0.889
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.753, D_loss_fake = 0.604, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.750, D_loss_fake = 0.621, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.722, D_loss_fake = 0.570, G_loss = 0.851
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 36/46, D_loss_real = 0.683, D_loss_fake = 0.626, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.702, D_loss_fake = 0.636, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.676, D_loss_fake = 0.637, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.573, D_loss_fake = 0.623, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.577, D_loss_fake = 0.678, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.526, D_loss_fake = 0.680, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.589, D_loss_fake = 0.706, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.577, D_loss_fake = 0.665, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.593, D_loss_fake = 0.651, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.593, D_loss_fake = 0.681, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.586, D_loss_fake = 0.771, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 1/46, D_loss_real = 0.571, D_loss_fake = 0.800, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.589, D_loss_fake = 0.832, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.647, D_loss_fake = 0.643, G_loss = 0.962
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.737, D_loss_fake = 0.607, G_loss = 0.930
2/2 [=====] - 0s 3ms/step

Epoch 20, Iteration 6/46, D_loss_real = 0.671, D_loss_fake = 0.645, G_loss = 0.872
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.655, D_loss_fake = 0.628, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.634, D_loss_fake = 0.645, G_loss = 0.845
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.663, D_loss_fake = 0.704, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.628, D_loss_fake = 0.676, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.666, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.650, D_loss_fake = 0.654, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.599, D_loss_fake = 0.755, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.567, D_loss_fake = 0.706, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.568, D_loss_fake = 0.703, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.578, D_loss_fake = 0.713, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 17/46, D_loss_real = 0.686, D_loss_fake = 0.955, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.658, D_loss_fake = 0.891, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.754, D_loss_fake = 0.670, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.727, D_loss_fake = 0.602, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.707, D_loss_fake = 0.628, G_loss = 0.820
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 22/46, D_loss_real = 0.687, D_loss_fake = 0.611, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.674, D_loss_fake = 0.640, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.609, D_loss_fake = 0.648, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.654, D_loss_fake = 0.665, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.643, D_loss_fake = 0.716, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.606, D_loss_fake = 0.649, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.670, D_loss_fake = 0.724, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.636, D_loss_fake = 0.640, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.665, D_loss_fake = 0.710, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.665, D_loss_fake = 0.720, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.746, D_loss_fake = 0.704, G_loss = 0.950
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 33/46, D_loss_real = 0.773, D_loss_fake = 0.653, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.820, D_loss_fake = 0.647, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.786, D_loss_fake = 0.585, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.787, D_loss_fake = 0.586, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.745, D_loss_fake = 0.589, G_loss = 0.873
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 38/46, D_loss_real = 0.734, D_loss_fake = 0.604, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.622, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.622, D_loss_fake = 0.652, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.666, D_loss_fake = 0.618, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.639, D_loss_fake = 0.789, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.639, D_loss_fake = 0.722, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.668, D_loss_fake = 0.692, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.615, D_loss_fake = 0.746, G_loss = 0.807
2/2 [=====] - 0s 5ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.619, D_loss_fake = 0.685, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.600, D_loss_fake = 0.712, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.604, D_loss_fake = 0.696, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 3/46, D_loss_real = 0.694, D_loss_fake = 0.723, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.704, D_loss_fake = 0.683, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.668, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.745, D_loss_fake = 0.633, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.713, D_loss_fake = 0.630, G_loss = 0.820
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 8/46, D_loss_real = 0.708, D_loss_fake = 0.633, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.733, D_loss_fake = 0.637, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.695, D_loss_fake = 0.625, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.742, D_loss_fake = 0.643, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.615, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.667, D_loss_fake = 0.645, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.719, D_loss_fake = 0.693, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.748, D_loss_fake = 0.653, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.720, D_loss_fake = 0.682, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.727, D_loss_fake = 0.628, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.712, D_loss_fake = 0.617, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 19/46, D_loss_real = 0.720, D_loss_fake = 0.591, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.713, D_loss_fake = 0.605, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.676, D_loss_fake = 0.613, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.691, D_loss_fake = 0.610, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.671, D_loss_fake = 0.677, G_loss = 0.780
2/2 [=====] - 0s 4ms/step

Epoch 21, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.697, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.656, D_loss_fake = 0.677, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.627, D_loss_fake = 0.693, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.675, D_loss_fake = 0.693, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.663, D_loss_fake = 0.686, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.624, D_loss_fake = 0.684, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.653, D_loss_fake = 0.722, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.670, D_loss_fake = 0.734, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.579, D_loss_fake = 0.729, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.590, D_loss_fake = 0.729, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.615, D_loss_fake = 0.717, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 35/46, D_loss_real = 0.679, D_loss_fake = 0.733, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.635, D_loss_fake = 0.731, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.681, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.659, D_loss_fake = 0.667, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.650, D_loss_fake = 0.671, G_loss = 0.812
2/2 [=====] - 0s 4ms/step

Epoch 21, Iteration 40/46, D_loss_real = 0.706, D_loss_fake = 0.655, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.753, D_loss_fake = 0.678, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.737, D_loss_fake = 0.657, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.729, D_loss_fake = 0.669, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.745, D_loss_fake = 0.701, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.711, D_loss_fake = 0.658, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.640, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.714, D_loss_fake = 0.653, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.708, D_loss_fake = 0.673, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.712, D_loss_fake = 0.667, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.713, D_loss_fake = 0.699, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 5/46, D_loss_real = 0.668, D_loss_fake = 0.694, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.684, D_loss_fake = 0.722, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.666, D_loss_fake = 0.680, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.655, D_loss_fake = 0.667, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.715, D_loss_fake = 0.664, G_loss = 0.773
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 10/46, D_loss_real = 0.683, D_loss_fake = 0.659, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.701, D_loss_fake = 0.637, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.706, D_loss_fake = 0.632, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.721, D_loss_fake = 0.646, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.706, D_loss_fake = 0.627, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.712, D_loss_fake = 0.662, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.723, D_loss_fake = 0.656, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.728, D_loss_fake = 0.629, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.742, D_loss_fake = 0.597, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.736, D_loss_fake = 0.618, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.690, D_loss_fake = 0.670, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 21/46, D_loss_real = 0.727, D_loss_fake = 0.613, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.721, D_loss_fake = 0.633, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.712, D_loss_fake = 0.660, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.713, D_loss_fake = 0.629, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.724, D_loss_fake = 0.648, G_loss = 0.767
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 26/46, D_loss_real = 0.717, D_loss_fake = 0.659, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.672, D_loss_fake = 0.689, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.692, D_loss_fake = 0.700, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.655, D_loss_fake = 0.690, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.647, D_loss_fake = 0.676, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.669, D_loss_fake = 0.660, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.616, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.675, D_loss_fake = 0.615, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.677, D_loss_fake = 0.671, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.690, D_loss_fake = 0.618, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.677, D_loss_fake = 0.669, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 37/46, D_loss_real = 0.657, D_loss_fake = 0.638, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.721, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.642, D_loss_fake = 0.729, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.687, D_loss_fake = 0.717, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.715, D_loss_fake = 0.650, G_loss = 0.818
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 42/46, D_loss_real = 0.713, D_loss_fake = 0.672, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.697, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.710, D_loss_fake = 0.663, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.729, D_loss_fake = 0.653, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.703, D_loss_fake = 0.644, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.695, D_loss_fake = 0.630, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.721, D_loss_fake = 0.650, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.693, D_loss_fake = 0.601, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.656, G_loss = 0.765
2/2 [=====] - 0s 5ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.642, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.678, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.680, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.664, D_loss_fake = 0.656, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.668, D_loss_fake = 0.665, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.683, D_loss_fake = 0.641, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.663, D_loss_fake = 0.652, G_loss = 0.806
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 12/46, D_loss_real = 0.728, D_loss_fake = 0.657, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.691, D_loss_fake = 0.666, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.661, D_loss_fake = 0.654, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.624, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.706, D_loss_fake = 0.647, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.667, D_loss_fake = 0.645, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.673, D_loss_fake = 0.650, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.667, D_loss_fake = 0.654, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.676, D_loss_fake = 0.671, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.719, D_loss_fake = 0.676, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.679, D_loss_fake = 0.673, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 23/46, D_loss_real = 0.708, D_loss_fake = 0.625, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.705, D_loss_fake = 0.636, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.630, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.640, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.680, D_loss_fake = 0.639, G_loss = 0.800
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 28/46, D_loss_real = 0.676, D_loss_fake = 0.653, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.691, D_loss_fake = 0.665, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.684, D_loss_fake = 0.714, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.648, D_loss_fake = 0.705, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.631, D_loss_fake = 0.692, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.638, D_loss_fake = 0.737, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.612, D_loss_fake = 0.680, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.619, D_loss_fake = 0.687, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.671, D_loss_fake = 0.727, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.644, D_loss_fake = 0.663, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.677, D_loss_fake = 0.658, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 39/46, D_loss_real = 0.673, D_loss_fake = 0.756, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.686, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.722, D_loss_fake = 0.663, G_loss = 0.779
2/2 [=====] - 0s 5ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.692, D_loss_fake = 0.617, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.636, G_loss = 0.811
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 44/46, D_loss_real = 0.687, D_loss_fake = 0.636, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.672, D_loss_fake = 0.644, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.649, D_loss_fake = 0.708, G_loss = 0.746
2/2 [=====] - 0s 5ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.792, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.672, D_loss_fake = 0.676, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.721, D_loss_fake = 0.632, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.674, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.648, D_loss_fake = 0.689, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.641, D_loss_fake = 0.671, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.616, D_loss_fake = 0.636, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.635, D_loss_fake = 0.717, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 9/46, D_loss_real = 0.631, D_loss_fake = 0.710, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.581, D_loss_fake = 0.693, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.617, D_loss_fake = 0.692, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.635, D_loss_fake = 0.640, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.672, D_loss_fake = 0.607, G_loss = 0.866
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 14/46, D_loss_real = 0.667, D_loss_fake = 0.683, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.671, D_loss_fake = 0.655, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.743, D_loss_fake = 0.730, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.728, D_loss_fake = 0.662, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.721, D_loss_fake = 0.592, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.689, D_loss_fake = 0.594, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.692, D_loss_fake = 0.666, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.648, D_loss_fake = 0.704, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.718, D_loss_fake = 0.768, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.633, D_loss_fake = 0.734, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.677, D_loss_fake = 0.718, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 25/46, D_loss_real = 0.638, D_loss_fake = 0.729, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.679, D_loss_fake = 0.675, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.633, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.644, D_loss_fake = 0.641, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.640, D_loss_fake = 0.678, G_loss = 0.800
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 30/46, D_loss_real = 0.588, D_loss_fake = 0.633, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.594, D_loss_fake = 0.642, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.594, D_loss_fake = 0.733, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.641, D_loss_fake = 0.722, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.612, D_loss_fake = 0.703, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.642, D_loss_fake = 0.687, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.663, D_loss_fake = 0.724, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.681, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.664, D_loss_fake = 0.680, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.700, D_loss_fake = 0.712, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.700, D_loss_fake = 0.726, G_loss = 0.685
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 41/46, D_loss_real = 0.716, D_loss_fake = 0.701, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.704, D_loss_fake = 0.739, G_loss = 0.702
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.731, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.747, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.683, G_loss = 0.763
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 46/46, D_loss_real = 0.659, D_loss_fake = 0.676, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.658, D_loss_fake = 0.653, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.598, D_loss_fake = 0.653, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.586, D_loss_fake = 0.679, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.570, D_loss_fake = 0.686, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.581, D_loss_fake = 0.726, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.569, D_loss_fake = 0.676, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.642, D_loss_fake = 0.763, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.589, D_loss_fake = 0.691, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.655, D_loss_fake = 0.717, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.661, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 11/46, D_loss_real = 0.748, D_loss_fake = 0.636, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.710, D_loss_fake = 0.617, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.646, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.747, D_loss_fake = 0.621, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.749, D_loss_fake = 0.596, G_loss = 0.804
2/2 [=====] - 0s 4ms/step

Epoch 25, Iteration 16/46, D_loss_real = 0.749, D_loss_fake = 0.642, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.764, D_loss_fake = 0.629, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.739, D_loss_fake = 0.620, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.739, D_loss_fake = 0.631, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.739, D_loss_fake = 0.624, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.715, D_loss_fake = 0.616, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.704, D_loss_fake = 0.614, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.715, D_loss_fake = 0.635, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.624, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.724, D_loss_fake = 0.605, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.711, D_loss_fake = 0.629, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.635, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.667, D_loss_fake = 0.645, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.660, D_loss_fake = 0.648, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.584, D_loss_fake = 0.661, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.607, D_loss_fake = 0.695, G_loss = 0.764
2/2 [=====] - 0s 4ms/step

Epoch 25, Iteration 32/46, D_loss_real = 0.570, D_loss_fake = 0.712, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.561, D_loss_fake = 0.749, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.624, D_loss_fake = 0.705, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.644, D_loss_fake = 0.635, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.663, D_loss_fake = 0.585, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.664, D_loss_fake = 0.654, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.683, D_loss_fake = 0.696, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.663, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.744, D_loss_fake = 0.684, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.709, D_loss_fake = 0.615, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.710, D_loss_fake = 0.644, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 43/46, D_loss_real = 0.714, D_loss_fake = 0.683, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.775, D_loss_fake = 0.746, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.701, D_loss_fake = 0.657, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.656, D_loss_fake = 0.654, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.694, D_loss_fake = 0.636, G_loss = 0.838
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 2/46, D_loss_real = 0.659, D_loss_fake = 0.681, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.724, D_loss_fake = 0.606, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.712, D_loss_fake = 0.589, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.692, D_loss_fake = 0.599, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.680, D_loss_fake = 0.686, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.701, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.663, D_loss_fake = 0.695, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.737, D_loss_fake = 0.701, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.669, D_loss_fake = 0.696, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.729, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.717, D_loss_fake = 0.688, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 13/46, D_loss_real = 0.731, D_loss_fake = 0.699, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.726, D_loss_fake = 0.674, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.722, D_loss_fake = 0.643, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.712, D_loss_fake = 0.637, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.719, D_loss_fake = 0.613, G_loss = 0.797
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 18/46, D_loss_real = 0.684, D_loss_fake = 0.633, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.666, D_loss_fake = 0.646, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.583, D_loss_fake = 0.686, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.632, D_loss_fake = 0.687, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.585, D_loss_fake = 0.742, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.603, D_loss_fake = 0.746, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.602, D_loss_fake = 0.692, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.609, D_loss_fake = 0.730, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.685, D_loss_fake = 0.689, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.739, D_loss_fake = 0.605, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.745, D_loss_fake = 0.578, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 29/46, D_loss_real = 0.703, D_loss_fake = 0.575, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.719, D_loss_fake = 0.635, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.692, D_loss_fake = 0.659, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.654, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.629, D_loss_fake = 0.662, G_loss = 0.743
2/2 [=====] - 0s 4ms/step

Epoch 26, Iteration 34/46, D_loss_real = 0.606, D_loss_fake = 0.679, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.614, D_loss_fake = 0.691, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.567, D_loss_fake = 0.690, G_loss = 0.716
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.592, D_loss_fake = 0.719, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.574, D_loss_fake = 0.795, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.559, D_loss_fake = 0.734, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.606, D_loss_fake = 0.826, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.620, D_loss_fake = 0.669, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.710, D_loss_fake = 0.591, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.699, D_loss_fake = 0.668, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.721, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 45/46, D_loss_real = 0.707, D_loss_fake = 0.779, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.680, D_loss_fake = 0.725, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.703, D_loss_fake = 0.632, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.733, D_loss_fake = 0.594, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.731, D_loss_fake = 0.631, G_loss = 0.866
2/2 [=====] - 0s 4ms/step

Epoch 27, Iteration 4/46, D_loss_real = 0.748, D_loss_fake = 0.620, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.716, D_loss_fake = 0.632, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.674, D_loss_fake = 0.657, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.708, D_loss_fake = 0.652, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.686, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.714, D_loss_fake = 0.725, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.636, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.674, D_loss_fake = 0.671, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.697, D_loss_fake = 0.727, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.726, D_loss_fake = 0.624, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.764, D_loss_fake = 0.585, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 15/46, D_loss_real = 0.718, D_loss_fake = 0.639, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.721, D_loss_fake = 0.626, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.631, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.651, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.695, D_loss_fake = 0.645, G_loss = 0.792
2/2 [=====] - 0s 4ms/step

Epoch 27, Iteration 20/46, D_loss_real = 0.689, D_loss_fake = 0.618, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.646, D_loss_fake = 0.632, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.689, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.657, D_loss_fake = 0.683, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.668, D_loss_fake = 0.654, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.678, D_loss_fake = 0.638, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.663, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.667, D_loss_fake = 0.692, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.666, D_loss_fake = 0.663, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.768, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.673, D_loss_fake = 0.622, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 31/46, D_loss_real = 0.698, D_loss_fake = 0.628, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.724, D_loss_fake = 0.613, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.679, D_loss_fake = 0.611, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.695, D_loss_fake = 0.674, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.655, D_loss_fake = 0.719, G_loss = 0.761
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 36/46, D_loss_real = 0.644, D_loss_fake = 0.725, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.609, D_loss_fake = 0.748, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.618, D_loss_fake = 0.700, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.624, D_loss_fake = 0.676, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.671, D_loss_fake = 0.723, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.634, D_loss_fake = 0.711, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.625, D_loss_fake = 0.683, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.626, D_loss_fake = 0.657, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.631, D_loss_fake = 0.772, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.678, D_loss_fake = 0.731, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.703, D_loss_fake = 0.717, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 1/46, D_loss_real = 0.754, D_loss_fake = 0.676, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.767, D_loss_fake = 0.608, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.746, D_loss_fake = 0.608, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.593, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.608, G_loss = 0.813
2/2 [=====] - 0s 4ms/step

Epoch 28, Iteration 6/46, D_loss_real = 0.670, D_loss_fake = 0.637, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.628, D_loss_fake = 0.681, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.559, D_loss_fake = 0.707, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.569, D_loss_fake = 0.744, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.568, D_loss_fake = 0.747, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.526, D_loss_fake = 0.726, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.513, D_loss_fake = 0.775, G_loss = 0.721
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.555, D_loss_fake = 0.733, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.528, D_loss_fake = 0.659, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.513, D_loss_fake = 0.688, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.533, D_loss_fake = 0.710, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 17/46, D_loss_real = 0.529, D_loss_fake = 0.819, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.658, D_loss_fake = 0.725, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.637, D_loss_fake = 0.602, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.631, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.687, D_loss_fake = 0.629, G_loss = 0.767
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 22/46, D_loss_real = 0.769, D_loss_fake = 0.630, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.713, D_loss_fake = 0.781, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.720, D_loss_fake = 0.769, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.720, D_loss_fake = 0.632, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.741, D_loss_fake = 0.627, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.723, D_loss_fake = 0.655, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.693, D_loss_fake = 0.656, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.704, D_loss_fake = 0.698, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.659, D_loss_fake = 0.696, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.642, D_loss_fake = 0.669, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.621, D_loss_fake = 0.687, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 33/46, D_loss_real = 0.666, D_loss_fake = 0.716, G_loss = 0.701
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.637, D_loss_fake = 0.784, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.641, D_loss_fake = 0.731, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.703, D_loss_fake = 0.643, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.688, D_loss_fake = 0.647, G_loss = 0.831
2/2 [=====] - 0s 4ms/step

Epoch 28, Iteration 38/46, D_loss_real = 0.724, D_loss_fake = 0.663, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.728, D_loss_fake = 0.644, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.720, D_loss_fake = 0.649, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.669, D_loss_fake = 0.673, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.647, D_loss_fake = 0.662, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.679, D_loss_fake = 0.664, G_loss = 0.771
2/2 [=====] - 0s 5ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.630, D_loss_fake = 0.656, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.641, D_loss_fake = 0.685, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.619, D_loss_fake = 0.735, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.586, D_loss_fake = 0.718, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.605, D_loss_fake = 0.721, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 3/46, D_loss_real = 0.632, D_loss_fake = 0.719, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.642, D_loss_fake = 0.702, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.658, D_loss_fake = 0.726, G_loss = 0.723
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.722, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.652, D_loss_fake = 0.689, G_loss = 0.796
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 8/46, D_loss_real = 0.701, D_loss_fake = 0.621, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.750, D_loss_fake = 0.625, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.750, D_loss_fake = 0.628, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.709, D_loss_fake = 0.614, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.747, D_loss_fake = 0.634, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.757, D_loss_fake = 0.611, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.727, D_loss_fake = 0.614, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.738, D_loss_fake = 0.606, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.770, D_loss_fake = 0.627, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.703, D_loss_fake = 0.648, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.668, D_loss_fake = 0.671, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 19/46, D_loss_real = 0.702, D_loss_fake = 0.657, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.623, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.662, D_loss_fake = 0.689, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.681, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.670, D_loss_fake = 0.657, G_loss = 0.765
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 24/46, D_loss_real = 0.651, D_loss_fake = 0.651, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.602, D_loss_fake = 0.704, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.652, D_loss_fake = 0.695, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.649, D_loss_fake = 0.703, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.666, D_loss_fake = 0.633, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.643, D_loss_fake = 0.634, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.672, D_loss_fake = 0.660, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.621, D_loss_fake = 0.637, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.667, D_loss_fake = 0.686, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.717, D_loss_fake = 0.651, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.709, D_loss_fake = 0.653, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 35/46, D_loss_real = 0.721, D_loss_fake = 0.668, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.662, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.715, D_loss_fake = 0.663, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.714, D_loss_fake = 0.657, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.654, G_loss = 0.799
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 40/46, D_loss_real = 0.734, D_loss_fake = 0.658, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.708, D_loss_fake = 0.652, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.724, D_loss_fake = 0.653, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.745, D_loss_fake = 0.685, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.716, D_loss_fake = 0.647, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.742, D_loss_fake = 0.631, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.735, D_loss_fake = 0.640, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.711, D_loss_fake = 0.638, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.732, D_loss_fake = 0.652, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.743, D_loss_fake = 0.662, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.718, D_loss_fake = 0.684, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 5/46, D_loss_real = 0.720, D_loss_fake = 0.727, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.696, D_loss_fake = 0.659, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.662, D_loss_fake = 0.663, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.661, D_loss_fake = 0.654, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.688, D_loss_fake = 0.664, G_loss = 0.823
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 10/46, D_loss_real = 0.692, D_loss_fake = 0.648, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.666, D_loss_fake = 0.635, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.668, D_loss_fake = 0.607, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.697, D_loss_fake = 0.606, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.706, D_loss_fake = 0.631, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.695, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.683, D_loss_fake = 0.672, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.667, D_loss_fake = 0.669, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.724, D_loss_fake = 0.679, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.726, D_loss_fake = 0.691, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 21/46, D_loss_real = 0.693, D_loss_fake = 0.661, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.704, D_loss_fake = 0.666, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.681, D_loss_fake = 0.707, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.643, D_loss_fake = 0.711, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.675, D_loss_fake = 0.712, G_loss = 0.765
2/2 [=====] - 0s 4ms/step

Epoch 30, Iteration 26/46, D_loss_real = 0.633, D_loss_fake = 0.718, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.668, D_loss_fake = 0.706, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.667, D_loss_fake = 0.697, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.656, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.699, D_loss_fake = 0.626, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.625, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.678, D_loss_fake = 0.630, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.626, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.672, D_loss_fake = 0.629, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.650, D_loss_fake = 0.675, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.651, D_loss_fake = 0.637, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 37/46, D_loss_real = 0.656, D_loss_fake = 0.679, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.622, D_loss_fake = 0.657, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.636, D_loss_fake = 0.710, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.722, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.657, D_loss_fake = 0.735, G_loss = 0.745
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 42/46, D_loss_real = 0.673, D_loss_fake = 0.725, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.707, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.716, D_loss_fake = 0.688, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.664, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.669, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.676, D_loss_fake = 0.673, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.710, D_loss_fake = 0.687, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.707, D_loss_fake = 0.660, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.674, D_loss_fake = 0.648, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 5/46, D_loss_real = 0.699, D_loss_fake = 0.681, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.665, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 7/46, D_loss_real = 0.663, D_loss_fake = 0.641, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.662, D_loss_fake = 0.650, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.678, D_loss_fake = 0.684, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.633, D_loss_fake = 0.681, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.580, D_loss_fake = 0.700, G_loss = 0.746
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 12/46, D_loss_real = 0.610, D_loss_fake = 0.720, G_loss = 0.705
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.579, D_loss_fake = 0.827, G_loss = 0.710
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.603, D_loss_fake = 0.725, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.627, D_loss_fake = 0.659, G_loss = 0.771
2/2 [=====] - 0s 5ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.664, D_loss_fake = 0.686, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.697, D_loss_fake = 0.710, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.668, D_loss_fake = 0.672, G_loss = 0.781
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.723, D_loss_fake = 0.653, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.674, D_loss_fake = 0.684, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.673, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.656, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 23/46, D_loss_real = 0.676, D_loss_fake = 0.667, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.647, D_loss_fake = 0.703, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.633, D_loss_fake = 0.771, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.713, D_loss_fake = 0.633, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.699, D_loss_fake = 0.572, G_loss = 0.931
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 28/46, D_loss_real = 0.713, D_loss_fake = 0.543, G_loss = 0.924
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.734, D_loss_fake = 0.590, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.702, D_loss_fake = 0.673, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.662, D_loss_fake = 0.653, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.760, D_loss_fake = 0.638, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.720, D_loss_fake = 0.632, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.739, D_loss_fake = 0.686, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.712, D_loss_fake = 0.611, G_loss = 0.878
2/2 [=====] - 0s 5ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.744, D_loss_fake = 0.591, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 37/46, D_loss_real = 0.740, D_loss_fake = 0.613, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 38/46, D_loss_real = 0.756, D_loss_fake = 0.614, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 39/46, D_loss_real = 0.766, D_loss_fake = 0.647, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.755, D_loss_fake = 0.639, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.714, D_loss_fake = 0.653, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.661, G_loss = 0.747
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 44/46, D_loss_real = 0.641, D_loss_fake = 0.662, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.668, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.633, D_loss_fake = 0.689, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.570, D_loss_fake = 0.707, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.577, D_loss_fake = 0.684, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.575, D_loss_fake = 0.714, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.550, D_loss_fake = 0.694, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.532, D_loss_fake = 0.797, G_loss = 0.734
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.561, D_loss_fake = 0.837, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 7/46, D_loss_real = 0.575, D_loss_fake = 0.740, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 8/46, D_loss_real = 0.636, D_loss_fake = 0.670, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 9/46, D_loss_real = 0.654, D_loss_fake = 0.689, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.645, D_loss_fake = 0.573, G_loss = 0.904
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.648, G_loss = 0.897
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.619, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.663, D_loss_fake = 0.625, G_loss = 0.824
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 14/46, D_loss_real = 0.658, D_loss_fake = 0.710, G_loss = 0.779
2/2 [=====] - 0s 5ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.689, D_loss_fake = 0.680, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.650, D_loss_fake = 0.697, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.674, D_loss_fake = 0.764, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.648, D_loss_fake = 0.632, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.706, D_loss_fake = 0.592, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.658, D_loss_fake = 0.646, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.668, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.669, D_loss_fake = 0.675, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 23/46, D_loss_real = 0.677, D_loss_fake = 0.701, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 24/46, D_loss_real = 0.606, D_loss_fake = 0.692, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 25/46, D_loss_real = 0.607, D_loss_fake = 0.688, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.640, D_loss_fake = 0.714, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.656, D_loss_fake = 0.679, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.633, D_loss_fake = 0.709, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.679, D_loss_fake = 0.673, G_loss = 0.758
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.700, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.636, D_loss_fake = 0.751, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.748, D_loss_fake = 0.638, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.756, D_loss_fake = 0.648, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.747, D_loss_fake = 0.607, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.757, D_loss_fake = 0.639, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.749, D_loss_fake = 0.637, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.718, D_loss_fake = 0.621, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.615, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 39/46, D_loss_real = 0.685, D_loss_fake = 0.646, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.659, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 41/46, D_loss_real = 0.650, D_loss_fake = 0.641, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.667, D_loss_fake = 0.666, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.654, D_loss_fake = 0.647, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.672, D_loss_fake = 0.643, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.659, D_loss_fake = 0.661, G_loss = 0.794
2/2 [=====] - 0s 4ms/step

Epoch 32, Iteration 46/46, D_loss_real = 0.668, D_loss_fake = 0.668, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.642, D_loss_fake = 0.663, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.598, D_loss_fake = 0.792, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.622, D_loss_fake = 0.728, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.636, D_loss_fake = 0.715, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.685, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.643, D_loss_fake = 0.630, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.649, D_loss_fake = 0.653, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.676, D_loss_fake = 0.665, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 9/46, D_loss_real = 0.677, D_loss_fake = 0.708, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 10/46, D_loss_real = 0.619, D_loss_fake = 0.692, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 11/46, D_loss_real = 0.652, D_loss_fake = 0.711, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.739, D_loss_fake = 0.637, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.698, D_loss_fake = 0.681, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.713, D_loss_fake = 0.602, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.748, D_loss_fake = 0.609, G_loss = 0.834
2/2 [=====] - 0s 5ms/step

Epoch 33, Iteration 16/46, D_loss_real = 0.725, D_loss_fake = 0.604, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.746, D_loss_fake = 0.619, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.606, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.664, D_loss_fake = 0.629, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.631, D_loss_fake = 0.650, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.642, D_loss_fake = 0.660, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.588, D_loss_fake = 0.695, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.640, D_loss_fake = 0.749, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.610, D_loss_fake = 0.734, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 25/46, D_loss_real = 0.619, D_loss_fake = 0.730, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 26/46, D_loss_real = 0.635, D_loss_fake = 0.742, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.730, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.654, D_loss_fake = 0.730, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.660, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.672, D_loss_fake = 0.636, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.685, D_loss_fake = 0.703, G_loss = 0.783
2/2 [=====] - 0s 4ms/step

Epoch 33, Iteration 32/46, D_loss_real = 0.682, D_loss_fake = 0.672, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.672, D_loss_fake = 0.660, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.660, D_loss_fake = 0.633, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.623, D_loss_fake = 0.641, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.639, D_loss_fake = 0.648, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.606, D_loss_fake = 0.710, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.577, D_loss_fake = 0.657, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.656, D_loss_fake = 0.667, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.685, D_loss_fake = 0.685, G_loss = 0.822
2/2 [=====] - 0s 5ms/step
Epoch 33, Iteration 41/46, D_loss_real = 0.650, D_loss_fake = 0.655, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 42/46, D_loss_real = 0.674, D_loss_fake = 0.733, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 43/46, D_loss_real = 0.700, D_loss_fake = 0.696, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.771, D_loss_fake = 0.551, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.722, D_loss_fake = 0.575, G_loss = 0.925
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.716, D_loss_fake = 0.616, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.696, D_loss_fake = 0.629, G_loss = 0.803
2/2 [=====] - 0s 4ms/step

Epoch 34, Iteration 2/46, D_loss_real = 0.698, D_loss_fake = 0.655, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.679, D_loss_fake = 0.696, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.690, D_loss_fake = 0.690, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.661, D_loss_fake = 0.666, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.702, D_loss_fake = 0.663, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.656, D_loss_fake = 0.673, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.673, D_loss_fake = 0.679, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.724, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.659, D_loss_fake = 0.654, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 11/46, D_loss_real = 0.668, D_loss_fake = 0.692, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 12/46, D_loss_real = 0.669, D_loss_fake = 0.683, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 13/46, D_loss_real = 0.688, D_loss_fake = 0.668, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.697, D_loss_fake = 0.711, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.636, D_loss_fake = 0.656, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.687, D_loss_fake = 0.705, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.671, D_loss_fake = 0.674, G_loss = 0.796
2/2 [=====] - 0s 4ms/step

Epoch 34, Iteration 18/46, D_loss_real = 0.724, D_loss_fake = 0.650, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.697, D_loss_fake = 0.627, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.674, D_loss_fake = 0.590, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.665, D_loss_fake = 0.592, G_loss = 0.897
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.695, D_loss_fake = 0.589, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.615, D_loss_fake = 0.697, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.637, D_loss_fake = 0.689, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.645, D_loss_fake = 0.711, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.675, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 27/46, D_loss_real = 0.602, D_loss_fake = 0.707, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 28/46, D_loss_real = 0.616, D_loss_fake = 0.723, G_loss = 0.744
2/2 [=====] - 0s 5ms/step
Epoch 34, Iteration 29/46, D_loss_real = 0.635, D_loss_fake = 0.690, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.651, D_loss_fake = 0.693, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.679, D_loss_fake = 0.727, G_loss = 0.733
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.707, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.652, D_loss_fake = 0.708, G_loss = 0.733
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 34/46, D_loss_real = 0.603, D_loss_fake = 0.755, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.627, D_loss_fake = 0.709, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.590, D_loss_fake = 0.681, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.671, D_loss_fake = 0.657, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.633, D_loss_fake = 0.620, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.585, D_loss_fake = 0.666, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.554, D_loss_fake = 0.645, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.599, D_loss_fake = 0.730, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.582, D_loss_fake = 0.736, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 43/46, D_loss_real = 0.653, D_loss_fake = 0.678, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 44/46, D_loss_real = 0.723, D_loss_fake = 0.636, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 45/46, D_loss_real = 0.711, D_loss_fake = 0.657, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.688, D_loss_fake = 0.665, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.662, D_loss_fake = 0.710, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.609, D_loss_fake = 0.666, G_loss = 0.817
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 4/46, D_loss_real = 0.688, D_loss_fake = 0.687, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.652, D_loss_fake = 0.671, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.655, G_loss = 0.874
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.685, D_loss_fake = 0.583, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.630, D_loss_fake = 0.610, G_loss = 0.872
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.569, D_loss_fake = 0.711, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.603, D_loss_fake = 0.778, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.626, D_loss_fake = 0.723, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.758, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 13/46, D_loss_real = 0.782, D_loss_fake = 0.640, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 14/46, D_loss_real = 0.745, D_loss_fake = 0.580, G_loss = 0.888
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 15/46, D_loss_real = 0.746, D_loss_fake = 0.588, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.722, D_loss_fake = 0.598, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.695, D_loss_fake = 0.618, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.626, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.635, G_loss = 0.821
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.625, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.641, D_loss_fake = 0.615, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.646, D_loss_fake = 0.744, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.688, D_loss_fake = 0.673, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.696, D_loss_fake = 0.676, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.639, D_loss_fake = 0.733, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.682, D_loss_fake = 0.637, G_loss = 0.869
2/2 [=====] - 0s 8ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.719, D_loss_fake = 0.627, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.636, D_loss_fake = 0.634, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 29/46, D_loss_real = 0.610, D_loss_fake = 0.670, G_loss = 0.774
2/2 [=====] - 0s 5ms/step
Epoch 35, Iteration 30/46, D_loss_real = 0.627, D_loss_fake = 0.711, G_loss = 0.750
2/2 [=====] - 0s 5ms/step
Epoch 35, Iteration 31/46, D_loss_real = 0.593, D_loss_fake = 0.744, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.663, D_loss_fake = 0.691, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.689, D_loss_fake = 0.720, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.722, D_loss_fake = 0.654, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.746, D_loss_fake = 0.615, G_loss = 0.858
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 36/46, D_loss_real = 0.781, D_loss_fake = 0.599, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.755, D_loss_fake = 0.598, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.760, D_loss_fake = 0.604, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.737, D_loss_fake = 0.626, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.723, D_loss_fake = 0.617, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.731, D_loss_fake = 0.629, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.721, D_loss_fake = 0.611, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.635, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.704, D_loss_fake = 0.662, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 45/46, D_loss_real = 0.680, D_loss_fake = 0.651, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 46/46, D_loss_real = 0.704, D_loss_fake = 0.676, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 1/46, D_loss_real = 0.671, D_loss_fake = 0.663, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.711, D_loss_fake = 0.679, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.702, D_loss_fake = 0.677, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.693, D_loss_fake = 0.644, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.735, D_loss_fake = 0.644, G_loss = 0.787
2/2 [=====] - 0s 4ms/step

Epoch 36, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.677, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.702, D_loss_fake = 0.672, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.678, D_loss_fake = 0.652, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.661, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.698, D_loss_fake = 0.643, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.660, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.660, D_loss_fake = 0.639, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.654, D_loss_fake = 0.675, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.687, D_loss_fake = 0.655, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 15/46, D_loss_real = 0.664, D_loss_fake = 0.644, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 16/46, D_loss_real = 0.678, D_loss_fake = 0.641, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 17/46, D_loss_real = 0.715, D_loss_fake = 0.698, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.704, D_loss_fake = 0.632, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.707, D_loss_fake = 0.641, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.713, D_loss_fake = 0.606, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.599, G_loss = 0.857
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.620, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.710, D_loss_fake = 0.642, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.683, D_loss_fake = 0.620, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.684, D_loss_fake = 0.694, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.683, D_loss_fake = 0.710, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.709, D_loss_fake = 0.688, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.722, D_loss_fake = 0.667, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.712, D_loss_fake = 0.684, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.657, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 31/46, D_loss_real = 0.689, D_loss_fake = 0.636, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 32/46, D_loss_real = 0.681, D_loss_fake = 0.647, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 33/46, D_loss_real = 0.620, D_loss_fake = 0.654, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.642, D_loss_fake = 0.650, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.633, D_loss_fake = 0.630, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.571, D_loss_fake = 0.719, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.575, D_loss_fake = 0.747, G_loss = 0.729
2/2 [=====] - 0s 4ms/step

Epoch 36, Iteration 38/46, D_loss_real = 0.598, D_loss_fake = 0.724, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.652, D_loss_fake = 0.654, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.653, D_loss_fake = 0.654, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.674, D_loss_fake = 0.683, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.637, D_loss_fake = 0.655, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.639, D_loss_fake = 0.683, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.684, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.698, D_loss_fake = 0.615, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.622, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.627, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 2/46, D_loss_real = 0.694, D_loss_fake = 0.622, G_loss = 0.845
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 3/46, D_loss_real = 0.728, D_loss_fake = 0.622, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.653, D_loss_fake = 0.654, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.655, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.698, D_loss_fake = 0.662, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.731, D_loss_fake = 0.707, G_loss = 0.815
2/2 [=====] - 0s 4ms/step

Epoch 37, Iteration 8/46, D_loss_real = 0.744, D_loss_fake = 0.654, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.654, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.669, D_loss_fake = 0.706, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.636, D_loss_fake = 0.691, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.611, D_loss_fake = 0.755, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.582, D_loss_fake = 0.745, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.622, D_loss_fake = 0.681, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.565, D_loss_fake = 0.720, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.554, D_loss_fake = 0.683, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 17/46, D_loss_real = 0.589, D_loss_fake = 0.751, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 18/46, D_loss_real = 0.562, D_loss_fake = 0.707, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 19/46, D_loss_real = 0.592, D_loss_fake = 0.689, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.606, D_loss_fake = 0.723, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.620, D_loss_fake = 0.747, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.626, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.744, D_loss_fake = 0.617, G_loss = 0.903
2/2 [=====] - 0s 4ms/step

Epoch 37, Iteration 24/46, D_loss_real = 0.714, D_loss_fake = 0.692, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.709, D_loss_fake = 0.633, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.733, D_loss_fake = 0.591, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.677, D_loss_fake = 0.613, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.664, D_loss_fake = 0.654, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.645, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.668, D_loss_fake = 0.652, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.585, D_loss_fake = 0.674, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.655, D_loss_fake = 0.661, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 33/46, D_loss_real = 0.632, D_loss_fake = 0.686, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 34/46, D_loss_real = 0.661, D_loss_fake = 0.669, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.719, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.686, D_loss_fake = 0.664, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.668, D_loss_fake = 0.679, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.692, D_loss_fake = 0.669, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.650, D_loss_fake = 0.717, G_loss = 0.774
2/2 [=====] - 0s 4ms/step

Epoch 37, Iteration 40/46, D_loss_real = 0.615, D_loss_fake = 0.704, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.625, D_loss_fake = 0.653, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.708, D_loss_fake = 0.697, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.704, D_loss_fake = 0.642, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.699, D_loss_fake = 0.619, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.671, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.692, D_loss_fake = 0.624, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.664, D_loss_fake = 0.639, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.634, D_loss_fake = 0.647, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 3/46, D_loss_real = 0.683, D_loss_fake = 0.624, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 4/46, D_loss_real = 0.663, D_loss_fake = 0.666, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 5/46, D_loss_real = 0.748, D_loss_fake = 0.594, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.746, D_loss_fake = 0.615, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.669, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.673, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.699, D_loss_fake = 0.677, G_loss = 0.816
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.666, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.654, D_loss_fake = 0.637, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.663, D_loss_fake = 0.665, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.661, D_loss_fake = 0.703, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.684, D_loss_fake = 0.708, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.640, D_loss_fake = 0.696, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.630, D_loss_fake = 0.645, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.606, D_loss_fake = 0.729, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.664, D_loss_fake = 0.665, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 19/46, D_loss_real = 0.617, D_loss_fake = 0.657, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 20/46, D_loss_real = 0.613, D_loss_fake = 0.689, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 21/46, D_loss_real = 0.576, D_loss_fake = 0.700, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.599, D_loss_fake = 0.667, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.617, D_loss_fake = 0.723, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.665, D_loss_fake = 0.676, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.692, D_loss_fake = 0.613, G_loss = 0.911
2/2 [=====] - 0s 4ms/step

Epoch 38, Iteration 26/46, D_loss_real = 0.715, D_loss_fake = 0.602, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.668, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.698, D_loss_fake = 0.642, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.684, D_loss_fake = 0.677, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.641, D_loss_fake = 0.633, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.652, D_loss_fake = 0.740, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.673, D_loss_fake = 0.694, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.655, D_loss_fake = 0.659, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.719, D_loss_fake = 0.646, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 35/46, D_loss_real = 0.658, D_loss_fake = 0.678, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 36/46, D_loss_real = 0.635, D_loss_fake = 0.679, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 37/46, D_loss_real = 0.591, D_loss_fake = 0.778, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.622, D_loss_fake = 0.659, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.661, D_loss_fake = 0.696, G_loss = 0.799
2/2 [=====] - 0s 5ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.646, D_loss_fake = 0.694, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.667, D_loss_fake = 0.705, G_loss = 0.758
2/2 [=====] - 0s 4ms/step

Epoch 38, Iteration 42/46, D_loss_real = 0.676, D_loss_fake = 0.691, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.658, D_loss_fake = 0.659, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.666, D_loss_fake = 0.726, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.725, D_loss_fake = 0.651, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.702, D_loss_fake = 0.634, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.704, D_loss_fake = 0.647, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.660, D_loss_fake = 0.650, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.652, D_loss_fake = 0.680, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.613, D_loss_fake = 0.701, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 5/46, D_loss_real = 0.642, D_loss_fake = 0.739, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.712, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 7/46, D_loss_real = 0.669, D_loss_fake = 0.689, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.733, D_loss_fake = 0.653, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.703, D_loss_fake = 0.658, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.665, D_loss_fake = 0.589, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.616, D_loss_fake = 0.612, G_loss = 0.840
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 12/46, D_loss_real = 0.626, D_loss_fake = 0.625, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.614, D_loss_fake = 0.687, G_loss = 0.755
2/2 [=====] - 0s 5ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.553, D_loss_fake = 0.772, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.655, D_loss_fake = 0.743, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.662, D_loss_fake = 0.717, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.633, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.585, G_loss = 0.922
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.677, D_loss_fake = 0.616, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.676, D_loss_fake = 0.656, G_loss = 0.805
2/2 [=====] - 0s 5ms/step
Epoch 39, Iteration 21/46, D_loss_real = 0.618, D_loss_fake = 0.712, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 22/46, D_loss_real = 0.565, D_loss_fake = 0.757, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 23/46, D_loss_real = 0.532, D_loss_fake = 0.734, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.612, D_loss_fake = 0.755, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.611, D_loss_fake = 0.679, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.720, D_loss_fake = 0.668, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.666, D_loss_fake = 0.592, G_loss = 0.883
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 28/46, D_loss_real = 0.667, D_loss_fake = 0.651, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.682, D_loss_fake = 0.596, G_loss = 0.860
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.667, D_loss_fake = 0.621, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.642, D_loss_fake = 0.610, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.672, D_loss_fake = 0.627, G_loss = 0.926
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.721, D_loss_fake = 0.619, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.639, D_loss_fake = 0.640, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.672, D_loss_fake = 0.681, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.620, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 37/46, D_loss_real = 0.750, D_loss_fake = 0.668, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 38/46, D_loss_real = 0.760, D_loss_fake = 0.717, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 39/46, D_loss_real = 0.718, D_loss_fake = 0.661, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.676, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.689, D_loss_fake = 0.707, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.668, D_loss_fake = 0.687, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.676, D_loss_fake = 0.648, G_loss = 0.798
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 44/46, D_loss_real = 0.651, D_loss_fake = 0.672, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.661, D_loss_fake = 0.636, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.610, D_loss_fake = 0.671, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.532, D_loss_fake = 0.648, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.575, D_loss_fake = 0.690, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.588, D_loss_fake = 0.694, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.644, D_loss_fake = 0.644, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.658, D_loss_fake = 0.617, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.629, D_loss_fake = 0.686, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 7/46, D_loss_real = 0.682, D_loss_fake = 0.675, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.734, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 9/46, D_loss_real = 0.654, D_loss_fake = 0.826, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.729, D_loss_fake = 0.580, G_loss = 1.026
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.752, D_loss_fake = 0.609, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.747, D_loss_fake = 0.591, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.637, D_loss_fake = 0.619, G_loss = 0.818
2/2 [=====] - 0s 3ms/step

Epoch 40, Iteration 14/46, D_loss_real = 0.618, D_loss_fake = 0.613, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.513, D_loss_fake = 0.684, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.466, D_loss_fake = 0.770, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.500, D_loss_fake = 0.810, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.570, D_loss_fake = 0.747, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.704, D_loss_fake = 0.731, G_loss = 0.948
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.784, D_loss_fake = 0.672, G_loss = 0.912
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.771, D_loss_fake = 0.551, G_loss = 0.983
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.709, D_loss_fake = 0.563, G_loss = 0.983
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 23/46, D_loss_real = 0.766, D_loss_fake = 0.544, G_loss = 0.967
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 24/46, D_loss_real = 0.735, D_loss_fake = 0.573, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 25/46, D_loss_real = 0.743, D_loss_fake = 0.626, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.691, D_loss_fake = 0.756, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.750, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.695, D_loss_fake = 0.713, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.728, D_loss_fake = 0.728, G_loss = 0.737
2/2 [=====] - 0s 3ms/step

Epoch 40, Iteration 30/46, D_loss_real = 0.682, D_loss_fake = 0.726, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.631, D_loss_fake = 0.737, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.658, D_loss_fake = 0.668, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.646, D_loss_fake = 0.682, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.592, D_loss_fake = 0.713, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.495, D_loss_fake = 0.754, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.497, D_loss_fake = 0.805, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.591, D_loss_fake = 0.815, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.636, D_loss_fake = 0.640, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 39/46, D_loss_real = 0.670, D_loss_fake = 0.643, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 40/46, D_loss_real = 0.584, D_loss_fake = 0.618, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 41/46, D_loss_real = 0.582, D_loss_fake = 0.667, G_loss = 0.926
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.653, D_loss_fake = 0.641, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.578, D_loss_fake = 0.894, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.633, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.785, D_loss_fake = 0.597, G_loss = 0.928
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 46/46, D_loss_real = 0.798, D_loss_fake = 0.602, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 1/46, D_loss_real = 0.741, D_loss_fake = 0.629, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 2/46, D_loss_real = 0.733, D_loss_fake = 0.608, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 3/46, D_loss_real = 0.664, D_loss_fake = 0.617, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 4/46, D_loss_real = 0.626, D_loss_fake = 0.639, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 5/46, D_loss_real = 0.544, D_loss_fake = 0.665, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 6/46, D_loss_real = 0.438, D_loss_fake = 0.805, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 7/46, D_loss_real = 0.438, D_loss_fake = 0.794, G_loss = 0.724
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 8/46, D_loss_real = 0.454, D_loss_fake = 0.775, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 9/46, D_loss_real = 0.547, D_loss_fake = 0.756, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 10/46, D_loss_real = 0.565, D_loss_fake = 0.805, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 11/46, D_loss_real = 0.632, D_loss_fake = 0.727, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 13/46, D_loss_real = 0.720, D_loss_fake = 0.578, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.629, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 15/46, D_loss_real = 0.683, D_loss_fake = 0.583, G_loss = 0.875
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 16/46, D_loss_real = 0.697, D_loss_fake = 0.714, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.637, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 18/46, D_loss_real = 0.740, D_loss_fake = 0.629, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 19/46, D_loss_real = 0.690, D_loss_fake = 0.637, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 20/46, D_loss_real = 0.703, D_loss_fake = 0.617, G_loss = 0.916
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.562, G_loss = 0.903
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 22/46, D_loss_real = 0.679, D_loss_fake = 0.534, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.579, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 24/46, D_loss_real = 0.607, D_loss_fake = 0.622, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 25/46, D_loss_real = 0.605, D_loss_fake = 0.682, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 26/46, D_loss_real = 0.600, D_loss_fake = 0.773, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 27/46, D_loss_real = 0.634, D_loss_fake = 0.807, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 28/46, D_loss_real = 0.717, D_loss_fake = 0.643, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 29/46, D_loss_real = 0.782, D_loss_fake = 0.652, G_loss = 0.929
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 30/46, D_loss_real = 0.777, D_loss_fake = 0.570, G_loss = 0.975
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 31/46, D_loss_real = 0.770, D_loss_fake = 0.561, G_loss = 1.013
2/2 [=====] - 0s 4ms/step

Epoch 41, Iteration 32/46, D_loss_real = 0.768, D_loss_fake = 0.517, G_loss = 0.976
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 33/46, D_loss_real = 0.840, D_loss_fake = 0.537, G_loss = 0.970
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 34/46, D_loss_real = 0.767, D_loss_fake = 0.545, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 35/46, D_loss_real = 0.749, D_loss_fake = 0.541, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 36/46, D_loss_real = 0.656, D_loss_fake = 0.546, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 37/46, D_loss_real = 0.638, D_loss_fake = 0.547, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 38/46, D_loss_real = 0.630, D_loss_fake = 0.572, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 39/46, D_loss_real = 0.637, D_loss_fake = 0.703, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 40/46, D_loss_real = 0.596, D_loss_fake = 0.662, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 41/46, D_loss_real = 0.584, D_loss_fake = 0.650, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 42/46, D_loss_real = 0.610, D_loss_fake = 0.822, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 43/46, D_loss_real = 0.593, D_loss_fake = 0.697, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 44/46, D_loss_real = 0.611, D_loss_fake = 0.817, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 45/46, D_loss_real = 0.609, D_loss_fake = 0.858, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.669, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 1/46, D_loss_real = 0.691, D_loss_fake = 0.611, G_loss = 0.887
2/2 [=====] - 0s 4ms/step

Epoch 42, Iteration 2/46, D_loss_real = 0.663, D_loss_fake = 0.629, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 3/46, D_loss_real = 0.698, D_loss_fake = 0.599, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 4/46, D_loss_real = 0.721, D_loss_fake = 0.609, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.659, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 6/46, D_loss_real = 0.694, D_loss_fake = 0.619, G_loss = 0.869
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 7/46, D_loss_real = 0.713, D_loss_fake = 0.587, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.623, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 9/46, D_loss_real = 0.711, D_loss_fake = 0.595, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.621, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 11/46, D_loss_real = 0.646, D_loss_fake = 0.677, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 12/46, D_loss_real = 0.699, D_loss_fake = 0.687, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 13/46, D_loss_real = 0.614, D_loss_fake = 0.663, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 14/46, D_loss_real = 0.601, D_loss_fake = 0.672, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 15/46, D_loss_real = 0.606, D_loss_fake = 0.697, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.693, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 17/46, D_loss_real = 0.628, D_loss_fake = 0.654, G_loss = 0.773
2/2 [=====] - 0s 3ms/step

Epoch 42, Iteration 18/46, D_loss_real = 0.646, D_loss_fake = 0.708, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 19/46, D_loss_real = 0.645, D_loss_fake = 0.601, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 20/46, D_loss_real = 0.693, D_loss_fake = 0.671, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 21/46, D_loss_real = 0.679, D_loss_fake = 0.633, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 22/46, D_loss_real = 0.648, D_loss_fake = 0.664, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 23/46, D_loss_real = 0.712, D_loss_fake = 0.630, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 24/46, D_loss_real = 0.747, D_loss_fake = 0.648, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 25/46, D_loss_real = 0.675, D_loss_fake = 0.587, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 26/46, D_loss_real = 0.718, D_loss_fake = 0.590, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 27/46, D_loss_real = 0.702, D_loss_fake = 0.619, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 28/46, D_loss_real = 0.681, D_loss_fake = 0.601, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.628, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 30/46, D_loss_real = 0.664, D_loss_fake = 0.674, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 31/46, D_loss_real = 0.606, D_loss_fake = 0.704, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.699, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 33/46, D_loss_real = 0.570, D_loss_fake = 0.661, G_loss = 0.792
2/2 [=====] - 0s 4ms/step

Epoch 42, Iteration 34/46, D_loss_real = 0.568, D_loss_fake = 0.694, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 35/46, D_loss_real = 0.492, D_loss_fake = 0.695, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 36/46, D_loss_real = 0.523, D_loss_fake = 0.731, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 37/46, D_loss_real = 0.564, D_loss_fake = 0.700, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 38/46, D_loss_real = 0.555, D_loss_fake = 0.754, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 39/46, D_loss_real = 0.613, D_loss_fake = 0.686, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 40/46, D_loss_real = 0.735, D_loss_fake = 0.714, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 41/46, D_loss_real = 0.747, D_loss_fake = 0.692, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 42/46, D_loss_real = 0.677, D_loss_fake = 0.620, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 43/46, D_loss_real = 0.692, D_loss_fake = 0.598, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 44/46, D_loss_real = 0.684, D_loss_fake = 0.659, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 45/46, D_loss_real = 0.714, D_loss_fake = 0.657, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 46/46, D_loss_real = 0.659, D_loss_fake = 0.612, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 1/46, D_loss_real = 0.640, D_loss_fake = 0.745, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.713, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 3/46, D_loss_real = 0.751, D_loss_fake = 0.693, G_loss = 0.810
2/2 [=====] - 0s 3ms/step

Epoch 43, Iteration 4/46, D_loss_real = 0.662, D_loss_fake = 0.710, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 5/46, D_loss_real = 0.676, D_loss_fake = 0.689, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 6/46, D_loss_real = 0.676, D_loss_fake = 0.612, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 7/46, D_loss_real = 0.660, D_loss_fake = 0.610, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 8/46, D_loss_real = 0.576, D_loss_fake = 0.680, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 9/46, D_loss_real = 0.484, D_loss_fake = 0.868, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 10/46, D_loss_real = 0.592, D_loss_fake = 0.862, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 11/46, D_loss_real = 0.662, D_loss_fake = 0.868, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 12/46, D_loss_real = 0.707, D_loss_fake = 0.649, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 13/46, D_loss_real = 0.765, D_loss_fake = 0.605, G_loss = 0.935
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 14/46, D_loss_real = 0.710, D_loss_fake = 0.583, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 15/46, D_loss_real = 0.740, D_loss_fake = 0.562, G_loss = 0.934
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 16/46, D_loss_real = 0.695, D_loss_fake = 0.579, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 17/46, D_loss_real = 0.731, D_loss_fake = 0.599, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 18/46, D_loss_real = 0.695, D_loss_fake = 0.595, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 19/46, D_loss_real = 0.687, D_loss_fake = 0.685, G_loss = 0.818
2/2 [=====] - 0s 4ms/step

Epoch 43, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.718, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 21/46, D_loss_real = 0.683, D_loss_fake = 0.721, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 22/46, D_loss_real = 0.688, D_loss_fake = 0.646, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 23/46, D_loss_real = 0.715, D_loss_fake = 0.605, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 24/46, D_loss_real = 0.772, D_loss_fake = 0.593, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 25/46, D_loss_real = 0.723, D_loss_fake = 0.582, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 26/46, D_loss_real = 0.764, D_loss_fake = 0.623, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 27/46, D_loss_real = 0.737, D_loss_fake = 0.638, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 28/46, D_loss_real = 0.742, D_loss_fake = 0.655, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 29/46, D_loss_real = 0.686, D_loss_fake = 0.642, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.673, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 31/46, D_loss_real = 0.667, D_loss_fake = 0.652, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 32/46, D_loss_real = 0.609, D_loss_fake = 0.730, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 33/46, D_loss_real = 0.619, D_loss_fake = 0.731, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 34/46, D_loss_real = 0.590, D_loss_fake = 0.759, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 35/46, D_loss_real = 0.673, D_loss_fake = 0.701, G_loss = 0.778
2/2 [=====] - 0s 4ms/step

Epoch 43, Iteration 36/46, D_loss_real = 0.661, D_loss_fake = 0.735, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 37/46, D_loss_real = 0.689, D_loss_fake = 0.593, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 38/46, D_loss_real = 0.732, D_loss_fake = 0.579, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 39/46, D_loss_real = 0.713, D_loss_fake = 0.602, G_loss = 0.869
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 40/46, D_loss_real = 0.738, D_loss_fake = 0.623, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.649, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.627, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 43/46, D_loss_real = 0.683, D_loss_fake = 0.639, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 44/46, D_loss_real = 0.713, D_loss_fake = 0.617, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 45/46, D_loss_real = 0.707, D_loss_fake = 0.641, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 46/46, D_loss_real = 0.670, D_loss_fake = 0.662, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 1/46, D_loss_real = 0.617, D_loss_fake = 0.658, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 2/46, D_loss_real = 0.662, D_loss_fake = 0.652, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 3/46, D_loss_real = 0.625, D_loss_fake = 0.705, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 4/46, D_loss_real = 0.641, D_loss_fake = 0.703, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 5/46, D_loss_real = 0.679, D_loss_fake = 0.678, G_loss = 0.808
2/2 [=====] - 0s 3ms/step

Epoch 44, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.677, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 7/46, D_loss_real = 0.701, D_loss_fake = 0.652, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 8/46, D_loss_real = 0.724, D_loss_fake = 0.622, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 9/46, D_loss_real = 0.719, D_loss_fake = 0.637, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 10/46, D_loss_real = 0.673, D_loss_fake = 0.638, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 11/46, D_loss_real = 0.664, D_loss_fake = 0.636, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 12/46, D_loss_real = 0.649, D_loss_fake = 0.645, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 13/46, D_loss_real = 0.640, D_loss_fake = 0.678, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 14/46, D_loss_real = 0.625, D_loss_fake = 0.811, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 15/46, D_loss_real = 0.642, D_loss_fake = 0.721, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 16/46, D_loss_real = 0.641, D_loss_fake = 0.727, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 17/46, D_loss_real = 0.705, D_loss_fake = 0.659, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 18/46, D_loss_real = 0.710, D_loss_fake = 0.648, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 19/46, D_loss_real = 0.707, D_loss_fake = 0.632, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 20/46, D_loss_real = 0.698, D_loss_fake = 0.636, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 21/46, D_loss_real = 0.764, D_loss_fake = 0.645, G_loss = 0.811
2/2 [=====] - 0s 4ms/step

Epoch 44, Iteration 22/46, D_loss_real = 0.739, D_loss_fake = 0.644, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 23/46, D_loss_real = 0.673, D_loss_fake = 0.701, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.628, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 25/46, D_loss_real = 0.695, D_loss_fake = 0.668, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 26/46, D_loss_real = 0.730, D_loss_fake = 0.674, G_loss = 0.727
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 27/46, D_loss_real = 0.664, D_loss_fake = 0.700, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 28/46, D_loss_real = 0.636, D_loss_fake = 0.695, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 29/46, D_loss_real = 0.650, D_loss_fake = 0.705, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 30/46, D_loss_real = 0.615, D_loss_fake = 0.728, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 31/46, D_loss_real = 0.622, D_loss_fake = 0.701, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 32/46, D_loss_real = 0.659, D_loss_fake = 0.714, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 33/46, D_loss_real = 0.636, D_loss_fake = 0.678, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 34/46, D_loss_real = 0.667, D_loss_fake = 0.635, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 35/46, D_loss_real = 0.688, D_loss_fake = 0.612, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 36/46, D_loss_real = 0.706, D_loss_fake = 0.642, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 37/46, D_loss_real = 0.687, D_loss_fake = 0.641, G_loss = 0.786
2/2 [=====] - 0s 4ms/step

Epoch 44, Iteration 38/46, D_loss_real = 0.649, D_loss_fake = 0.723, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 39/46, D_loss_real = 0.711, D_loss_fake = 0.667, G_loss = 0.823
2/2 [=====] - 0s 5ms/step
Epoch 44, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.645, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 41/46, D_loss_real = 0.709, D_loss_fake = 0.630, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 42/46, D_loss_real = 0.704, D_loss_fake = 0.652, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 43/46, D_loss_real = 0.718, D_loss_fake = 0.668, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 44/46, D_loss_real = 0.710, D_loss_fake = 0.677, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 45/46, D_loss_real = 0.702, D_loss_fake = 0.669, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 46/46, D_loss_real = 0.674, D_loss_fake = 0.708, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 1/46, D_loss_real = 0.673, D_loss_fake = 0.740, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 2/46, D_loss_real = 0.653, D_loss_fake = 0.700, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 3/46, D_loss_real = 0.635, D_loss_fake = 0.709, G_loss = 0.743
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 4/46, D_loss_real = 0.615, D_loss_fake = 0.748, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 5/46, D_loss_real = 0.613, D_loss_fake = 0.730, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 6/46, D_loss_real = 0.667, D_loss_fake = 0.675, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 7/46, D_loss_real = 0.654, D_loss_fake = 0.649, G_loss = 0.776
2/2 [=====] - 0s 4ms/step

Epoch 45, Iteration 8/46, D_loss_real = 0.642, D_loss_fake = 0.680, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.646, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 10/46, D_loss_real = 0.711, D_loss_fake = 0.645, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 11/46, D_loss_real = 0.733, D_loss_fake = 0.661, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.686, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 13/46, D_loss_real = 0.649, D_loss_fake = 0.678, G_loss = 0.718
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 14/46, D_loss_real = 0.675, D_loss_fake = 0.741, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 15/46, D_loss_real = 0.590, D_loss_fake = 0.709, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 16/46, D_loss_real = 0.650, D_loss_fake = 0.732, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 17/46, D_loss_real = 0.661, D_loss_fake = 0.701, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.666, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 19/46, D_loss_real = 0.700, D_loss_fake = 0.670, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.650, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 21/46, D_loss_real = 0.707, D_loss_fake = 0.631, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 22/46, D_loss_real = 0.697, D_loss_fake = 0.659, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 23/46, D_loss_real = 0.702, D_loss_fake = 0.650, G_loss = 0.798
2/2 [=====] - 0s 5ms/step

Epoch 45, Iteration 24/46, D_loss_real = 0.693, D_loss_fake = 0.666, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 25/46, D_loss_real = 0.716, D_loss_fake = 0.658, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 26/46, D_loss_real = 0.683, D_loss_fake = 0.687, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 27/46, D_loss_real = 0.730, D_loss_fake = 0.694, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 28/46, D_loss_real = 0.715, D_loss_fake = 0.643, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 29/46, D_loss_real = 0.732, D_loss_fake = 0.663, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 30/46, D_loss_real = 0.690, D_loss_fake = 0.663, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 31/46, D_loss_real = 0.683, D_loss_fake = 0.656, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 32/46, D_loss_real = 0.702, D_loss_fake = 0.656, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.715, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 34/46, D_loss_real = 0.678, D_loss_fake = 0.646, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 35/46, D_loss_real = 0.688, D_loss_fake = 0.693, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 36/46, D_loss_real = 0.660, D_loss_fake = 0.677, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 37/46, D_loss_real = 0.694, D_loss_fake = 0.680, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 38/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 39/46, D_loss_real = 0.680, D_loss_fake = 0.667, G_loss = 0.757
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.662, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 41/46, D_loss_real = 0.670, D_loss_fake = 0.734, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 42/46, D_loss_real = 0.671, D_loss_fake = 0.713, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 43/46, D_loss_real = 0.728, D_loss_fake = 0.659, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 44/46, D_loss_real = 0.735, D_loss_fake = 0.674, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 45/46, D_loss_real = 0.718, D_loss_fake = 0.650, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 46/46, D_loss_real = 0.705, D_loss_fake = 0.673, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 1/46, D_loss_real = 0.741, D_loss_fake = 0.641, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 2/46, D_loss_real = 0.702, D_loss_fake = 0.651, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 3/46, D_loss_real = 0.716, D_loss_fake = 0.661, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 4/46, D_loss_real = 0.714, D_loss_fake = 0.705, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 5/46, D_loss_real = 0.735, D_loss_fake = 0.629, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 6/46, D_loss_real = 0.680, D_loss_fake = 0.618, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 7/46, D_loss_real = 0.698, D_loss_fake = 0.649, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 8/46, D_loss_real = 0.668, D_loss_fake = 0.626, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 9/46, D_loss_real = 0.676, D_loss_fake = 0.653, G_loss = 0.766
2/2 [=====] - 0s 4ms/step

Epoch 46, Iteration 10/46, D_loss_real = 0.645, D_loss_fake = 0.680, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 11/46, D_loss_real = 0.689, D_loss_fake = 0.662, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 12/46, D_loss_real = 0.653, D_loss_fake = 0.675, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.766, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 14/46, D_loss_real = 0.652, D_loss_fake = 0.723, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 15/46, D_loss_real = 0.705, D_loss_fake = 0.662, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.674, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.678, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.680, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 19/46, D_loss_real = 0.674, D_loss_fake = 0.650, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 20/46, D_loss_real = 0.717, D_loss_fake = 0.631, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 21/46, D_loss_real = 0.699, D_loss_fake = 0.633, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 22/46, D_loss_real = 0.685, D_loss_fake = 0.640, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 23/46, D_loss_real = 0.644, D_loss_fake = 0.624, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 24/46, D_loss_real = 0.665, D_loss_fake = 0.687, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 25/46, D_loss_real = 0.707, D_loss_fake = 0.686, G_loss = 0.790
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 26/46, D_loss_real = 0.662, D_loss_fake = 0.685, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 27/46, D_loss_real = 0.692, D_loss_fake = 0.772, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 28/46, D_loss_real = 0.784, D_loss_fake = 0.666, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.664, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 30/46, D_loss_real = 0.717, D_loss_fake = 0.666, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.666, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 32/46, D_loss_real = 0.678, D_loss_fake = 0.692, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 33/46, D_loss_real = 0.667, D_loss_fake = 0.690, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 34/46, D_loss_real = 0.622, D_loss_fake = 0.679, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.670, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 37/46, D_loss_real = 0.652, D_loss_fake = 0.675, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 38/46, D_loss_real = 0.678, D_loss_fake = 0.684, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 39/46, D_loss_real = 0.688, D_loss_fake = 0.745, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 40/46, D_loss_real = 0.695, D_loss_fake = 0.682, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 41/46, D_loss_real = 0.698, D_loss_fake = 0.663, G_loss = 0.764
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 42/46, D_loss_real = 0.674, D_loss_fake = 0.663, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 43/46, D_loss_real = 0.689, D_loss_fake = 0.685, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 44/46, D_loss_real = 0.715, D_loss_fake = 0.688, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 45/46, D_loss_real = 0.662, D_loss_fake = 0.663, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 46/46, D_loss_real = 0.719, D_loss_fake = 0.690, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 1/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 2/46, D_loss_real = 0.714, D_loss_fake = 0.664, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 3/46, D_loss_real = 0.718, D_loss_fake = 0.646, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 4/46, D_loss_real = 0.686, D_loss_fake = 0.639, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 5/46, D_loss_real = 0.724, D_loss_fake = 0.666, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 6/46, D_loss_real = 0.680, D_loss_fake = 0.672, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 7/46, D_loss_real = 0.608, D_loss_fake = 0.712, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 8/46, D_loss_real = 0.651, D_loss_fake = 0.670, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 9/46, D_loss_real = 0.659, D_loss_fake = 0.734, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 10/46, D_loss_real = 0.615, D_loss_fake = 0.761, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 11/46, D_loss_real = 0.660, D_loss_fake = 0.737, G_loss = 0.766
2/2 [=====] - 0s 3ms/step

Epoch 47, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.718, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.646, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 14/46, D_loss_real = 0.709, D_loss_fake = 0.659, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 15/46, D_loss_real = 0.691, D_loss_fake = 0.668, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.669, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 17/46, D_loss_real = 0.741, D_loss_fake = 0.677, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 18/46, D_loss_real = 0.758, D_loss_fake = 0.663, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 19/46, D_loss_real = 0.708, D_loss_fake = 0.648, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 20/46, D_loss_real = 0.707, D_loss_fake = 0.631, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 21/46, D_loss_real = 0.716, D_loss_fake = 0.624, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 22/46, D_loss_real = 0.671, D_loss_fake = 0.646, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 23/46, D_loss_real = 0.703, D_loss_fake = 0.649, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 24/46, D_loss_real = 0.709, D_loss_fake = 0.684, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 25/46, D_loss_real = 0.696, D_loss_fake = 0.632, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 26/46, D_loss_real = 0.704, D_loss_fake = 0.638, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 27/46, D_loss_real = 0.684, D_loss_fake = 0.674, G_loss = 0.761
2/2 [=====] - 0s 3ms/step

Epoch 47, Iteration 28/46, D_loss_real = 0.695, D_loss_fake = 0.671, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.694, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 30/46, D_loss_real = 0.708, D_loss_fake = 0.694, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 31/46, D_loss_real = 0.696, D_loss_fake = 0.680, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 32/46, D_loss_real = 0.693, D_loss_fake = 0.662, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 33/46, D_loss_real = 0.660, D_loss_fake = 0.671, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 34/46, D_loss_real = 0.694, D_loss_fake = 0.668, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 35/46, D_loss_real = 0.671, D_loss_fake = 0.691, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 36/46, D_loss_real = 0.653, D_loss_fake = 0.716, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 37/46, D_loss_real = 0.676, D_loss_fake = 0.695, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 38/46, D_loss_real = 0.716, D_loss_fake = 0.663, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.690, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.689, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 41/46, D_loss_real = 0.703, D_loss_fake = 0.652, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 42/46, D_loss_real = 0.672, D_loss_fake = 0.643, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 43/46, D_loss_real = 0.665, D_loss_fake = 0.656, G_loss = 0.789
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 44/46, D_loss_real = 0.688, D_loss_fake = 0.649, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.608, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 46/46, D_loss_real = 0.631, D_loss_fake = 0.647, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 1/46, D_loss_real = 0.678, D_loss_fake = 0.764, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 2/46, D_loss_real = 0.652, D_loss_fake = 0.717, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 3/46, D_loss_real = 0.720, D_loss_fake = 0.679, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 4/46, D_loss_real = 0.760, D_loss_fake = 0.668, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 5/46, D_loss_real = 0.714, D_loss_fake = 0.652, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 6/46, D_loss_real = 0.704, D_loss_fake = 0.670, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 7/46, D_loss_real = 0.670, D_loss_fake = 0.671, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.675, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 9/46, D_loss_real = 0.666, D_loss_fake = 0.665, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.704, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 11/46, D_loss_real = 0.685, D_loss_fake = 0.670, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 12/46, D_loss_real = 0.713, D_loss_fake = 0.647, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 13/46, D_loss_real = 0.690, D_loss_fake = 0.667, G_loss = 0.780
2/2 [=====] - 0s 4ms/step

Epoch 48, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.667, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 15/46, D_loss_real = 0.697, D_loss_fake = 0.665, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 16/46, D_loss_real = 0.706, D_loss_fake = 0.659, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 17/46, D_loss_real = 0.693, D_loss_fake = 0.682, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.674, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 19/46, D_loss_real = 0.681, D_loss_fake = 0.720, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.678, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 21/46, D_loss_real = 0.678, D_loss_fake = 0.670, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 22/46, D_loss_real = 0.653, D_loss_fake = 0.667, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.672, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 24/46, D_loss_real = 0.668, D_loss_fake = 0.676, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 25/46, D_loss_real = 0.637, D_loss_fake = 0.684, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 26/46, D_loss_real = 0.625, D_loss_fake = 0.679, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 27/46, D_loss_real = 0.666, D_loss_fake = 0.681, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 28/46, D_loss_real = 0.643, D_loss_fake = 0.676, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 29/46, D_loss_real = 0.623, D_loss_fake = 0.692, G_loss = 0.729
2/2 [=====] - 0s 4ms/step

Epoch 48, Iteration 30/46, D_loss_real = 0.679, D_loss_fake = 0.715, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 31/46, D_loss_real = 0.609, D_loss_fake = 0.724, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 32/46, D_loss_real = 0.614, D_loss_fake = 0.718, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 33/46, D_loss_real = 0.651, D_loss_fake = 0.708, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 34/46, D_loss_real = 0.643, D_loss_fake = 0.692, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 35/46, D_loss_real = 0.662, D_loss_fake = 0.710, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 36/46, D_loss_real = 0.621, D_loss_fake = 0.692, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 37/46, D_loss_real = 0.654, D_loss_fake = 0.734, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 38/46, D_loss_real = 0.632, D_loss_fake = 0.712, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 39/46, D_loss_real = 0.673, D_loss_fake = 0.752, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 40/46, D_loss_real = 0.661, D_loss_fake = 0.698, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 41/46, D_loss_real = 0.651, D_loss_fake = 0.686, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 42/46, D_loss_real = 0.616, D_loss_fake = 0.711, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 43/46, D_loss_real = 0.645, D_loss_fake = 0.673, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 44/46, D_loss_real = 0.627, D_loss_fake = 0.706, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 45/46, D_loss_real = 0.652, D_loss_fake = 0.719, G_loss = 0.885
2/2 [=====] - 0s 3ms/step

Epoch 48, Iteration 46/46, D_loss_real = 0.645, D_loss_fake = 0.665, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 1/46, D_loss_real = 0.716, D_loss_fake = 0.612, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 2/46, D_loss_real = 0.732, D_loss_fake = 0.679, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 3/46, D_loss_real = 0.732, D_loss_fake = 0.680, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 4/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 5/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 6/46, D_loss_real = 0.686, D_loss_fake = 0.634, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 7/46, D_loss_real = 0.728, D_loss_fake = 0.640, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 8/46, D_loss_real = 0.692, D_loss_fake = 0.617, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 9/46, D_loss_real = 0.694, D_loss_fake = 0.658, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.662, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 11/46, D_loss_real = 0.683, D_loss_fake = 0.705, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 12/46, D_loss_real = 0.663, D_loss_fake = 0.732, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 13/46, D_loss_real = 0.632, D_loss_fake = 0.758, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 14/46, D_loss_real = 0.652, D_loss_fake = 0.729, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 15/46, D_loss_real = 0.687, D_loss_fake = 0.670, G_loss = 0.793
2/2 [=====] - 0s 3ms/step

Epoch 49, Iteration 16/46, D_loss_real = 0.611, D_loss_fake = 0.656, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 17/46, D_loss_real = 0.598, D_loss_fake = 0.610, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 18/46, D_loss_real = 0.514, D_loss_fake = 0.736, G_loss = 0.872
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 19/46, D_loss_real = 0.607, D_loss_fake = 0.710, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 20/46, D_loss_real = 0.603, D_loss_fake = 0.860, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 21/46, D_loss_real = 0.729, D_loss_fake = 0.746, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 22/46, D_loss_real = 0.719, D_loss_fake = 0.612, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 23/46, D_loss_real = 0.738, D_loss_fake = 0.601, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 24/46, D_loss_real = 0.711, D_loss_fake = 0.591, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 25/46, D_loss_real = 0.714, D_loss_fake = 0.697, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.638, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 27/46, D_loss_real = 0.681, D_loss_fake = 0.680, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 28/46, D_loss_real = 0.680, D_loss_fake = 0.654, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 29/46, D_loss_real = 0.651, D_loss_fake = 0.660, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 30/46, D_loss_real = 0.610, D_loss_fake = 0.689, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 31/46, D_loss_real = 0.697, D_loss_fake = 0.682, G_loss = 0.760
2/2 [=====] - 0s 4ms/step

Epoch 49, Iteration 32/46, D_loss_real = 0.610, D_loss_fake = 0.700, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 33/46, D_loss_real = 0.633, D_loss_fake = 0.723, G_loss = 0.733
2/2 [=====] - 0s 5ms/step
Epoch 49, Iteration 34/46, D_loss_real = 0.605, D_loss_fake = 0.707, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 35/46, D_loss_real = 0.617, D_loss_fake = 0.750, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 36/46, D_loss_real = 0.638, D_loss_fake = 0.752, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 37/46, D_loss_real = 0.639, D_loss_fake = 0.723, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 38/46, D_loss_real = 0.622, D_loss_fake = 0.788, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 39/46, D_loss_real = 0.647, D_loss_fake = 0.715, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.698, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 41/46, D_loss_real = 0.676, D_loss_fake = 0.707, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 42/46, D_loss_real = 0.687, D_loss_fake = 0.662, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 43/46, D_loss_real = 0.695, D_loss_fake = 0.678, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 44/46, D_loss_real = 0.719, D_loss_fake = 0.644, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 45/46, D_loss_real = 0.705, D_loss_fake = 0.643, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 46/46, D_loss_real = 0.690, D_loss_fake = 0.642, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 1/46, D_loss_real = 0.727, D_loss_fake = 0.669, G_loss = 0.749
2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 2/46, D_loss_real = 0.706, D_loss_fake = 0.658, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 3/46, D_loss_real = 0.717, D_loss_fake = 0.632, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 4/46, D_loss_real = 0.664, D_loss_fake = 0.635, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.655, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.681, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 7/46, D_loss_real = 0.628, D_loss_fake = 0.651, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 8/46, D_loss_real = 0.625, D_loss_fake = 0.650, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 9/46, D_loss_real = 0.634, D_loss_fake = 0.718, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 10/46, D_loss_real = 0.616, D_loss_fake = 0.711, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 11/46, D_loss_real = 0.673, D_loss_fake = 0.750, G_loss = 0.735
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.690, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.668, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 14/46, D_loss_real = 0.689, D_loss_fake = 0.706, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 15/46, D_loss_real = 0.698, D_loss_fake = 0.665, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 16/46, D_loss_real = 0.729, D_loss_fake = 0.649, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 17/46, D_loss_real = 0.682, D_loss_fake = 0.625, G_loss = 0.803
2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 18/46, D_loss_real = 0.715, D_loss_fake = 0.678, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 19/46, D_loss_real = 0.718, D_loss_fake = 0.656, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 20/46, D_loss_real = 0.705, D_loss_fake = 0.675, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 21/46, D_loss_real = 0.709, D_loss_fake = 0.658, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 22/46, D_loss_real = 0.674, D_loss_fake = 0.672, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 23/46, D_loss_real = 0.720, D_loss_fake = 0.647, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 24/46, D_loss_real = 0.643, D_loss_fake = 0.650, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 25/46, D_loss_real = 0.660, D_loss_fake = 0.672, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 26/46, D_loss_real = 0.654, D_loss_fake = 0.658, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 27/46, D_loss_real = 0.611, D_loss_fake = 0.676, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 28/46, D_loss_real = 0.598, D_loss_fake = 0.648, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 29/46, D_loss_real = 0.593, D_loss_fake = 0.659, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 30/46, D_loss_real = 0.589, D_loss_fake = 0.753, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 31/46, D_loss_real = 0.612, D_loss_fake = 0.807, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 32/46, D_loss_real = 0.646, D_loss_fake = 0.822, G_loss = 0.726
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 33/46, D_loss_real = 0.688, D_loss_fake = 0.829, G_loss = 0.762
2/2 [=====] - 0s 4ms/step

Epoch 50, Iteration 34/46, D_loss_real = 0.791, D_loss_fake = 0.625, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 35/46, D_loss_real = 0.776, D_loss_fake = 0.607, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 36/46, D_loss_real = 0.750, D_loss_fake = 0.637, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 37/46, D_loss_real = 0.766, D_loss_fake = 0.629, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 38/46, D_loss_real = 0.720, D_loss_fake = 0.629, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 39/46, D_loss_real = 0.698, D_loss_fake = 0.645, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 40/46, D_loss_real = 0.682, D_loss_fake = 0.680, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.719, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 42/46, D_loss_real = 0.609, D_loss_fake = 0.737, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 43/46, D_loss_real = 0.604, D_loss_fake = 0.696, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 44/46, D_loss_real = 0.650, D_loss_fake = 0.763, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 45/46, D_loss_real = 0.695, D_loss_fake = 0.715, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 46/46, D_loss_real = 0.686, D_loss_fake = 0.670, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 1/46, D_loss_real = 0.728, D_loss_fake = 0.652, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 2/46, D_loss_real = 0.699, D_loss_fake = 0.636, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 3/46, D_loss_real = 0.709, D_loss_fake = 0.617, G_loss = 0.824
2/2 [=====] - 0s 3ms/step

Epoch 51, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.616, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 5/46, D_loss_real = 0.664, D_loss_fake = 0.653, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.650, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.708, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 8/46, D_loss_real = 0.634, D_loss_fake = 0.714, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 9/46, D_loss_real = 0.641, D_loss_fake = 0.737, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 10/46, D_loss_real = 0.609, D_loss_fake = 0.686, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 11/46, D_loss_real = 0.616, D_loss_fake = 0.745, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 12/46, D_loss_real = 0.611, D_loss_fake = 0.671, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 13/46, D_loss_real = 0.661, D_loss_fake = 0.739, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 14/46, D_loss_real = 0.618, D_loss_fake = 0.707, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 15/46, D_loss_real = 0.674, D_loss_fake = 0.743, G_loss = 0.729
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 16/46, D_loss_real = 0.645, D_loss_fake = 0.785, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 17/46, D_loss_real = 0.735, D_loss_fake = 0.698, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 18/46, D_loss_real = 0.708, D_loss_fake = 0.649, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 19/46, D_loss_real = 0.723, D_loss_fake = 0.682, G_loss = 0.757
2/2 [=====] - 0s 4ms/step

Epoch 51, Iteration 20/46, D_loss_real = 0.734, D_loss_fake = 0.652, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.659, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 22/46, D_loss_real = 0.720, D_loss_fake = 0.647, G_loss = 0.750
2/2 [=====] - 0s 5ms/step
Epoch 51, Iteration 23/46, D_loss_real = 0.718, D_loss_fake = 0.679, G_loss = 0.753
2/2 [=====] - 0s 5ms/step
Epoch 51, Iteration 24/46, D_loss_real = 0.684, D_loss_fake = 0.654, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 25/46, D_loss_real = 0.718, D_loss_fake = 0.661, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 26/46, D_loss_real = 0.683, D_loss_fake = 0.659, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 27/46, D_loss_real = 0.711, D_loss_fake = 0.618, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 28/46, D_loss_real = 0.704, D_loss_fake = 0.636, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 29/46, D_loss_real = 0.675, D_loss_fake = 0.638, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.642, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 31/46, D_loss_real = 0.694, D_loss_fake = 0.670, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 32/46, D_loss_real = 0.697, D_loss_fake = 0.655, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 33/46, D_loss_real = 0.658, D_loss_fake = 0.675, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 34/46, D_loss_real = 0.648, D_loss_fake = 0.722, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 35/46, D_loss_real = 0.664, D_loss_fake = 0.706, G_loss = 0.779
2/2 [=====] - 0s 3ms/step

Epoch 51, Iteration 36/46, D_loss_real = 0.655, D_loss_fake = 0.679, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 37/46, D_loss_real = 0.659, D_loss_fake = 0.667, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 38/46, D_loss_real = 0.657, D_loss_fake = 0.734, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.642, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.701, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 41/46, D_loss_real = 0.722, D_loss_fake = 0.646, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.636, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 43/46, D_loss_real = 0.667, D_loss_fake = 0.648, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.622, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.657, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 1/46, D_loss_real = 0.683, D_loss_fake = 0.664, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 2/46, D_loss_real = 0.671, D_loss_fake = 0.668, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 3/46, D_loss_real = 0.669, D_loss_fake = 0.663, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 4/46, D_loss_real = 0.680, D_loss_fake = 0.669, G_loss = 0.755
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 5/46, D_loss_real = 0.674, D_loss_fake = 0.647, G_loss = 0.794
2/2 [=====] - 0s 3ms/step

Epoch 52, Iteration 6/46, D_loss_real = 0.673, D_loss_fake = 0.671, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 7/46, D_loss_real = 0.693, D_loss_fake = 0.667, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 8/46, D_loss_real = 0.679, D_loss_fake = 0.649, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 9/46, D_loss_real = 0.674, D_loss_fake = 0.629, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 10/46, D_loss_real = 0.664, D_loss_fake = 0.642, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 11/46, D_loss_real = 0.694, D_loss_fake = 0.719, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 12/46, D_loss_real = 0.696, D_loss_fake = 0.687, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 13/46, D_loss_real = 0.680, D_loss_fake = 0.695, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 14/46, D_loss_real = 0.705, D_loss_fake = 0.638, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.664, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 16/46, D_loss_real = 0.712, D_loss_fake = 0.631, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 17/46, D_loss_real = 0.706, D_loss_fake = 0.659, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 18/46, D_loss_real = 0.663, D_loss_fake = 0.664, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 19/46, D_loss_real = 0.666, D_loss_fake = 0.712, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 20/46, D_loss_real = 0.696, D_loss_fake = 0.676, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.732, G_loss = 0.743
2/2 [=====] - 0s 4ms/step

Epoch 52, Iteration 22/46, D_loss_real = 0.691, D_loss_fake = 0.708, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 23/46, D_loss_real = 0.711, D_loss_fake = 0.724, G_loss = 0.713
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 24/46, D_loss_real = 0.713, D_loss_fake = 0.695, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 25/46, D_loss_real = 0.694, D_loss_fake = 0.696, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 26/46, D_loss_real = 0.707, D_loss_fake = 0.678, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 27/46, D_loss_real = 0.744, D_loss_fake = 0.604, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 28/46, D_loss_real = 0.749, D_loss_fake = 0.618, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 29/46, D_loss_real = 0.722, D_loss_fake = 0.618, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 30/46, D_loss_real = 0.705, D_loss_fake = 0.624, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 31/46, D_loss_real = 0.703, D_loss_fake = 0.635, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 32/46, D_loss_real = 0.666, D_loss_fake = 0.634, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 33/46, D_loss_real = 0.635, D_loss_fake = 0.672, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 34/46, D_loss_real = 0.679, D_loss_fake = 0.699, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 35/46, D_loss_real = 0.623, D_loss_fake = 0.702, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 36/46, D_loss_real = 0.649, D_loss_fake = 0.767, G_loss = 0.708
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 37/46, D_loss_real = 0.631, D_loss_fake = 0.742, G_loss = 0.803
2/2 [=====] - 0s 3ms/step

Epoch 52, Iteration 38/46, D_loss_real = 0.682, D_loss_fake = 0.668, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 39/46, D_loss_real = 0.668, D_loss_fake = 0.678, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 40/46, D_loss_real = 0.698, D_loss_fake = 0.616, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 41/46, D_loss_real = 0.694, D_loss_fake = 0.636, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 42/46, D_loss_real = 0.665, D_loss_fake = 0.652, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 43/46, D_loss_real = 0.677, D_loss_fake = 0.684, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 44/46, D_loss_real = 0.661, D_loss_fake = 0.687, G_loss = 0.780
2/2 [=====] - 0s 5ms/step
Epoch 52, Iteration 45/46, D_loss_real = 0.634, D_loss_fake = 0.731, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 46/46, D_loss_real = 0.658, D_loss_fake = 0.675, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 1/46, D_loss_real = 0.679, D_loss_fake = 0.671, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 2/46, D_loss_real = 0.666, D_loss_fake = 0.692, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 3/46, D_loss_real = 0.623, D_loss_fake = 0.691, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 4/46, D_loss_real = 0.670, D_loss_fake = 0.739, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 5/46, D_loss_real = 0.726, D_loss_fake = 0.624, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 6/46, D_loss_real = 0.732, D_loss_fake = 0.634, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 7/46, D_loss_real = 0.724, D_loss_fake = 0.674, G_loss = 0.782
2/2 [=====] - 0s 4ms/step

Epoch 53, Iteration 8/46, D_loss_real = 0.722, D_loss_fake = 0.660, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.645, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 10/46, D_loss_real = 0.709, D_loss_fake = 0.663, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 11/46, D_loss_real = 0.686, D_loss_fake = 0.653, G_loss = 0.751
2/2 [=====] - 0s 5ms/step
Epoch 53, Iteration 12/46, D_loss_real = 0.675, D_loss_fake = 0.690, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 13/46, D_loss_real = 0.650, D_loss_fake = 0.741, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 14/46, D_loss_real = 0.665, D_loss_fake = 0.702, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 15/46, D_loss_real = 0.686, D_loss_fake = 0.715, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.722, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.657, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 18/46, D_loss_real = 0.710, D_loss_fake = 0.638, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.651, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 20/46, D_loss_real = 0.715, D_loss_fake = 0.654, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 21/46, D_loss_real = 0.665, D_loss_fake = 0.628, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 22/46, D_loss_real = 0.678, D_loss_fake = 0.674, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 23/46, D_loss_real = 0.666, D_loss_fake = 0.687, G_loss = 0.738
2/2 [=====] - 0s 4ms/step

Epoch 53, Iteration 24/46, D_loss_real = 0.680, D_loss_fake = 0.710, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 25/46, D_loss_real = 0.705, D_loss_fake = 0.674, G_loss = 0.781
2/2 [=====] - 0s 5ms/step
Epoch 53, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.680, G_loss = 0.817
2/2 [=====] - 0s 5ms/step
Epoch 53, Iteration 27/46, D_loss_real = 0.716, D_loss_fake = 0.672, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 28/46, D_loss_real = 0.674, D_loss_fake = 0.626, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 29/46, D_loss_real = 0.661, D_loss_fake = 0.614, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 30/46, D_loss_real = 0.732, D_loss_fake = 0.592, G_loss = 0.868
2/2 [=====] - 0s 5ms/step
Epoch 53, Iteration 31/46, D_loss_real = 0.687, D_loss_fake = 0.640, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 32/46, D_loss_real = 0.669, D_loss_fake = 0.681, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 33/46, D_loss_real = 0.694, D_loss_fake = 0.705, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 34/46, D_loss_real = 0.728, D_loss_fake = 0.716, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 35/46, D_loss_real = 0.726, D_loss_fake = 0.749, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 36/46, D_loss_real = 0.721, D_loss_fake = 0.721, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.681, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 38/46, D_loss_real = 0.736, D_loss_fake = 0.641, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 39/46, D_loss_real = 0.750, D_loss_fake = 0.631, G_loss = 0.803
2/2 [=====] - 0s 4ms/step

Epoch 53, Iteration 40/46, D_loss_real = 0.657, D_loss_fake = 0.648, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 41/46, D_loss_real = 0.713, D_loss_fake = 0.646, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.650, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 43/46, D_loss_real = 0.698, D_loss_fake = 0.669, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 44/46, D_loss_real = 0.670, D_loss_fake = 0.669, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 45/46, D_loss_real = 0.668, D_loss_fake = 0.683, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 46/46, D_loss_real = 0.720, D_loss_fake = 0.687, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 1/46, D_loss_real = 0.706, D_loss_fake = 0.722, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 2/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 3/46, D_loss_real = 0.735, D_loss_fake = 0.673, G_loss = 0.759
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 4/46, D_loss_real = 0.713, D_loss_fake = 0.659, G_loss = 0.763
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 5/46, D_loss_real = 0.728, D_loss_fake = 0.643, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 6/46, D_loss_real = 0.710, D_loss_fake = 0.656, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 7/46, D_loss_real = 0.713, D_loss_fake = 0.660, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 8/46, D_loss_real = 0.700, D_loss_fake = 0.677, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 9/46, D_loss_real = 0.682, D_loss_fake = 0.687, G_loss = 0.734
2/2 [=====] - 0s 4ms/step

Epoch 54, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.675, G_loss = 0.727
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 11/46, D_loss_real = 0.708, D_loss_fake = 0.693, G_loss = 0.722
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 12/46, D_loss_real = 0.675, D_loss_fake = 0.698, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 13/46, D_loss_real = 0.643, D_loss_fake = 0.698, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 14/46, D_loss_real = 0.658, D_loss_fake = 0.699, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 15/46, D_loss_real = 0.641, D_loss_fake = 0.663, G_loss = 0.738
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 16/46, D_loss_real = 0.626, D_loss_fake = 0.705, G_loss = 0.715
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 17/46, D_loss_real = 0.627, D_loss_fake = 0.699, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 18/46, D_loss_real = 0.617, D_loss_fake = 0.712, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 19/46, D_loss_real = 0.571, D_loss_fake = 0.726, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 20/46, D_loss_real = 0.593, D_loss_fake = 0.769, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 21/46, D_loss_real = 0.602, D_loss_fake = 0.842, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 22/46, D_loss_real = 0.645, D_loss_fake = 0.730, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 23/46, D_loss_real = 0.685, D_loss_fake = 0.723, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.685, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 25/46, D_loss_real = 0.722, D_loss_fake = 0.676, G_loss = 0.749
2/2 [=====] - 0s 3ms/step

Epoch 54, Iteration 26/46, D_loss_real = 0.735, D_loss_fake = 0.640, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 27/46, D_loss_real = 0.729, D_loss_fake = 0.626, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 28/46, D_loss_real = 0.754, D_loss_fake = 0.602, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 29/46, D_loss_real = 0.755, D_loss_fake = 0.599, G_loss = 0.860
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 30/46, D_loss_real = 0.794, D_loss_fake = 0.594, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 31/46, D_loss_real = 0.763, D_loss_fake = 0.598, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 32/46, D_loss_real = 0.753, D_loss_fake = 0.625, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 33/46, D_loss_real = 0.724, D_loss_fake = 0.635, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 34/46, D_loss_real = 0.729, D_loss_fake = 0.637, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 35/46, D_loss_real = 0.687, D_loss_fake = 0.650, G_loss = 0.753
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 36/46, D_loss_real = 0.684, D_loss_fake = 0.652, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 37/46, D_loss_real = 0.672, D_loss_fake = 0.696, G_loss = 0.731
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 38/46, D_loss_real = 0.649, D_loss_fake = 0.668, G_loss = 0.719
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 39/46, D_loss_real = 0.590, D_loss_fake = 0.745, G_loss = 0.723
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 40/46, D_loss_real = 0.639, D_loss_fake = 0.717, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 41/46, D_loss_real = 0.630, D_loss_fake = 0.748, G_loss = 0.739
2/2 [=====] - 0s 4ms/step

Epoch 54, Iteration 42/46, D_loss_real = 0.671, D_loss_fake = 0.696, G_loss = 0.707
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 43/46, D_loss_real = 0.676, D_loss_fake = 0.739, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.707, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 45/46, D_loss_real = 0.675, D_loss_fake = 0.672, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 46/46, D_loss_real = 0.735, D_loss_fake = 0.712, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 1/46, D_loss_real = 0.715, D_loss_fake = 0.673, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 2/46, D_loss_real = 0.719, D_loss_fake = 0.623, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 3/46, D_loss_real = 0.719, D_loss_fake = 0.641, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 4/46, D_loss_real = 0.701, D_loss_fake = 0.656, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.647, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.663, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 7/46, D_loss_real = 0.665, D_loss_fake = 0.711, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 8/46, D_loss_real = 0.668, D_loss_fake = 0.743, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 9/46, D_loss_real = 0.703, D_loss_fake = 0.686, G_loss = 0.728
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 10/46, D_loss_real = 0.708, D_loss_fake = 0.709, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 11/46, D_loss_real = 0.716, D_loss_fake = 0.683, G_loss = 0.753
2/2 [=====] - 0s 3ms/step

Epoch 55, Iteration 12/46, D_loss_real = 0.734, D_loss_fake = 0.672, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 13/46, D_loss_real = 0.704, D_loss_fake = 0.694, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.653, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.693, G_loss = 0.741
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 16/46, D_loss_real = 0.669, D_loss_fake = 0.714, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 17/46, D_loss_real = 0.651, D_loss_fake = 0.667, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 18/46, D_loss_real = 0.694, D_loss_fake = 0.663, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 19/46, D_loss_real = 0.682, D_loss_fake = 0.656, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 20/46, D_loss_real = 0.663, D_loss_fake = 0.684, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 21/46, D_loss_real = 0.637, D_loss_fake = 0.714, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 22/46, D_loss_real = 0.649, D_loss_fake = 0.741, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 23/46, D_loss_real = 0.652, D_loss_fake = 0.723, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.753, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.714, G_loss = 0.738
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 26/46, D_loss_real = 0.697, D_loss_fake = 0.708, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 27/46, D_loss_real = 0.697, D_loss_fake = 0.687, G_loss = 0.769
2/2 [=====] - 0s 3ms/step

Epoch 55, Iteration 28/46, D_loss_real = 0.714, D_loss_fake = 0.686, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 29/46, D_loss_real = 0.709, D_loss_fake = 0.680, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 30/46, D_loss_real = 0.698, D_loss_fake = 0.650, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 31/46, D_loss_real = 0.705, D_loss_fake = 0.638, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 32/46, D_loss_real = 0.683, D_loss_fake = 0.627, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 33/46, D_loss_real = 0.642, D_loss_fake = 0.651, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 34/46, D_loss_real = 0.653, D_loss_fake = 0.639, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 35/46, D_loss_real = 0.621, D_loss_fake = 0.691, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 36/46, D_loss_real = 0.605, D_loss_fake = 0.756, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 37/46, D_loss_real = 0.653, D_loss_fake = 0.767, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 38/46, D_loss_real = 0.674, D_loss_fake = 0.680, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 39/46, D_loss_real = 0.714, D_loss_fake = 0.678, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 40/46, D_loss_real = 0.713, D_loss_fake = 0.725, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.670, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 42/46, D_loss_real = 0.737, D_loss_fake = 0.588, G_loss = 0.845
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 43/46, D_loss_real = 0.756, D_loss_fake = 0.655, G_loss = 0.814
2/2 [=====] - 0s 4ms/step

Epoch 55, Iteration 44/46, D_loss_real = 0.746, D_loss_fake = 0.619, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 45/46, D_loss_real = 0.691, D_loss_fake = 0.613, G_loss = 0.789
2/2 [=====] - 0s 5ms/step
Epoch 55, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.647, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 1/46, D_loss_real = 0.655, D_loss_fake = 0.664, G_loss = 0.739
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 2/46, D_loss_real = 0.634, D_loss_fake = 0.713, G_loss = 0.734
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 3/46, D_loss_real = 0.619, D_loss_fake = 0.692, G_loss = 0.760
2/2 [=====] - 0s 5ms/step
Epoch 56, Iteration 4/46, D_loss_real = 0.607, D_loss_fake = 0.722, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 5/46, D_loss_real = 0.562, D_loss_fake = 0.701, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 6/46, D_loss_real = 0.573, D_loss_fake = 0.763, G_loss = 0.721
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 7/46, D_loss_real = 0.539, D_loss_fake = 0.792, G_loss = 0.716
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 8/46, D_loss_real = 0.593, D_loss_fake = 0.763, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 9/46, D_loss_real = 0.635, D_loss_fake = 0.729, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 10/46, D_loss_real = 0.672, D_loss_fake = 0.702, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 11/46, D_loss_real = 0.703, D_loss_fake = 0.617, G_loss = 0.806
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 12/46, D_loss_real = 0.690, D_loss_fake = 0.676, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 13/46, D_loss_real = 0.711, D_loss_fake = 0.647, G_loss = 0.795
2/2 [=====] - 0s 4ms/step

Epoch 56, Iteration 14/46, D_loss_real = 0.693, D_loss_fake = 0.652, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 15/46, D_loss_real = 0.709, D_loss_fake = 0.631, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 16/46, D_loss_real = 0.675, D_loss_fake = 0.638, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 17/46, D_loss_real = 0.713, D_loss_fake = 0.650, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.643, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.706, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 20/46, D_loss_real = 0.655, D_loss_fake = 0.647, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 21/46, D_loss_real = 0.699, D_loss_fake = 0.676, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 22/46, D_loss_real = 0.704, D_loss_fake = 0.667, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 23/46, D_loss_real = 0.707, D_loss_fake = 0.620, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 24/46, D_loss_real = 0.745, D_loss_fake = 0.669, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 25/46, D_loss_real = 0.751, D_loss_fake = 0.665, G_loss = 0.771
2/2 [=====] - 0s 5ms/step
Epoch 56, Iteration 26/46, D_loss_real = 0.739, D_loss_fake = 0.682, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 27/46, D_loss_real = 0.731, D_loss_fake = 0.645, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 28/46, D_loss_real = 0.715, D_loss_fake = 0.647, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 29/46, D_loss_real = 0.722, D_loss_fake = 0.652, G_loss = 0.756
2/2 [=====] - 0s 3ms/step

Epoch 56, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.658, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.661, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 32/46, D_loss_real = 0.640, D_loss_fake = 0.670, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 33/46, D_loss_real = 0.640, D_loss_fake = 0.653, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 34/46, D_loss_real = 0.680, D_loss_fake = 0.702, G_loss = 0.712
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 35/46, D_loss_real = 0.612, D_loss_fake = 0.682, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 36/46, D_loss_real = 0.631, D_loss_fake = 0.758, G_loss = 0.703
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 37/46, D_loss_real = 0.629, D_loss_fake = 0.731, G_loss = 0.703
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 38/46, D_loss_real = 0.648, D_loss_fake = 0.737, G_loss = 0.691
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 39/46, D_loss_real = 0.589, D_loss_fake = 0.789, G_loss = 0.714
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 40/46, D_loss_real = 0.620, D_loss_fake = 0.759, G_loss = 0.725
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 41/46, D_loss_real = 0.626, D_loss_fake = 0.777, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 42/46, D_loss_real = 0.699, D_loss_fake = 0.654, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.649, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 44/46, D_loss_real = 0.718, D_loss_fake = 0.615, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 45/46, D_loss_real = 0.682, D_loss_fake = 0.685, G_loss = 0.779
2/2 [=====] - 0s 4ms/step

Epoch 56, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.722, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 1/46, D_loss_real = 0.713, D_loss_fake = 0.702, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 2/46, D_loss_real = 0.732, D_loss_fake = 0.645, G_loss = 0.770
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 3/46, D_loss_real = 0.728, D_loss_fake = 0.662, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.685, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 5/46, D_loss_real = 0.702, D_loss_fake = 0.644, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 6/46, D_loss_real = 0.702, D_loss_fake = 0.638, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 7/46, D_loss_real = 0.646, D_loss_fake = 0.654, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 8/46, D_loss_real = 0.655, D_loss_fake = 0.646, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 9/46, D_loss_real = 0.685, D_loss_fake = 0.673, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.674, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 11/46, D_loss_real = 0.678, D_loss_fake = 0.685, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 12/46, D_loss_real = 0.666, D_loss_fake = 0.701, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.671, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 14/46, D_loss_real = 0.673, D_loss_fake = 0.666, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 15/46, D_loss_real = 0.716, D_loss_fake = 0.678, G_loss = 0.774
2/2 [=====] - 0s 4ms/step

Epoch 57, Iteration 16/46, D_loss_real = 0.709, D_loss_fake = 0.685, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 17/46, D_loss_real = 0.739, D_loss_fake = 0.653, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 18/46, D_loss_real = 0.717, D_loss_fake = 0.631, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 19/46, D_loss_real = 0.699, D_loss_fake = 0.668, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 20/46, D_loss_real = 0.723, D_loss_fake = 0.653, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 21/46, D_loss_real = 0.690, D_loss_fake = 0.671, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 22/46, D_loss_real = 0.721, D_loss_fake = 0.703, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 23/46, D_loss_real = 0.701, D_loss_fake = 0.687, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 24/46, D_loss_real = 0.662, D_loss_fake = 0.661, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 25/46, D_loss_real = 0.683, D_loss_fake = 0.671, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 26/46, D_loss_real = 0.689, D_loss_fake = 0.665, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 27/46, D_loss_real = 0.639, D_loss_fake = 0.697, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 28/46, D_loss_real = 0.665, D_loss_fake = 0.696, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 29/46, D_loss_real = 0.627, D_loss_fake = 0.697, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 30/46, D_loss_real = 0.656, D_loss_fake = 0.739, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 31/46, D_loss_real = 0.673, D_loss_fake = 0.710, G_loss = 0.784
2/2 [=====] - 0s 3ms/step

Epoch 57, Iteration 32/46, D_loss_real = 0.717, D_loss_fake = 0.648, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 33/46, D_loss_real = 0.692, D_loss_fake = 0.648, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 34/46, D_loss_real = 0.678, D_loss_fake = 0.640, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 35/46, D_loss_real = 0.705, D_loss_fake = 0.646, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 36/46, D_loss_real = 0.708, D_loss_fake = 0.626, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 37/46, D_loss_real = 0.693, D_loss_fake = 0.626, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 38/46, D_loss_real = 0.679, D_loss_fake = 0.692, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 39/46, D_loss_real = 0.673, D_loss_fake = 0.674, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 40/46, D_loss_real = 0.677, D_loss_fake = 0.655, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 41/46, D_loss_real = 0.654, D_loss_fake = 0.713, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 42/46, D_loss_real = 0.612, D_loss_fake = 0.677, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 43/46, D_loss_real = 0.637, D_loss_fake = 0.698, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 44/46, D_loss_real = 0.650, D_loss_fake = 0.632, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 45/46, D_loss_real = 0.662, D_loss_fake = 0.680, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 46/46, D_loss_real = 0.646, D_loss_fake = 0.723, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 1/46, D_loss_real = 0.641, D_loss_fake = 0.744, G_loss = 0.753
2/2 [=====] - 0s 3ms/step

Epoch 58, Iteration 2/46, D_loss_real = 0.657, D_loss_fake = 0.708, G_loss = 0.739
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.709, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 4/46, D_loss_real = 0.640, D_loss_fake = 0.723, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 5/46, D_loss_real = 0.682, D_loss_fake = 0.685, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 6/46, D_loss_real = 0.656, D_loss_fake = 0.708, G_loss = 0.713
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.689, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 8/46, D_loss_real = 0.681, D_loss_fake = 0.689, G_loss = 0.717
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 9/46, D_loss_real = 0.695, D_loss_fake = 0.719, G_loss = 0.730
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 10/46, D_loss_real = 0.717, D_loss_fake = 0.706, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.720, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 12/46, D_loss_real = 0.694, D_loss_fake = 0.686, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 13/46, D_loss_real = 0.692, D_loss_fake = 0.668, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.663, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.656, G_loss = 0.795
2/2 [=====] - 0s 5ms/step
Epoch 58, Iteration 16/46, D_loss_real = 0.704, D_loss_fake = 0.640, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 17/46, D_loss_real = 0.692, D_loss_fake = 0.657, G_loss = 0.805
2/2 [=====] - 0s 3ms/step

Epoch 58, Iteration 18/46, D_loss_real = 0.707, D_loss_fake = 0.662, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 19/46, D_loss_real = 0.663, D_loss_fake = 0.641, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 20/46, D_loss_real = 0.673, D_loss_fake = 0.665, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 21/46, D_loss_real = 0.687, D_loss_fake = 0.708, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.663, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 23/46, D_loss_real = 0.700, D_loss_fake = 0.675, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 24/46, D_loss_real = 0.695, D_loss_fake = 0.679, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 25/46, D_loss_real = 0.708, D_loss_fake = 0.662, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.654, G_loss = 0.744
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 27/46, D_loss_real = 0.689, D_loss_fake = 0.684, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.654, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.656, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.658, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 31/46, D_loss_real = 0.713, D_loss_fake = 0.655, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 32/46, D_loss_real = 0.705, D_loss_fake = 0.637, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 33/46, D_loss_real = 0.715, D_loss_fake = 0.647, G_loss = 0.777
2/2 [=====] - 0s 4ms/step

Epoch 58, Iteration 34/46, D_loss_real = 0.707, D_loss_fake = 0.655, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 35/46, D_loss_real = 0.693, D_loss_fake = 0.652, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 36/46, D_loss_real = 0.673, D_loss_fake = 0.662, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.664, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 38/46, D_loss_real = 0.681, D_loss_fake = 0.661, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 39/46, D_loss_real = 0.649, D_loss_fake = 0.671, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 40/46, D_loss_real = 0.705, D_loss_fake = 0.697, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 41/46, D_loss_real = 0.658, D_loss_fake = 0.641, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 42/46, D_loss_real = 0.677, D_loss_fake = 0.701, G_loss = 0.715
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 43/46, D_loss_real = 0.656, D_loss_fake = 0.699, G_loss = 0.736
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 44/46, D_loss_real = 0.630, D_loss_fake = 0.696, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 45/46, D_loss_real = 0.673, D_loss_fake = 0.693, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 46/46, D_loss_real = 0.656, D_loss_fake = 0.687, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 1/46, D_loss_real = 0.657, D_loss_fake = 0.689, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 2/46, D_loss_real = 0.643, D_loss_fake = 0.670, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 3/46, D_loss_real = 0.677, D_loss_fake = 0.697, G_loss = 0.752
2/2 [=====] - 0s 4ms/step

Epoch 59, Iteration 4/46, D_loss_real = 0.646, D_loss_fake = 0.686, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 5/46, D_loss_real = 0.653, D_loss_fake = 0.688, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 6/46, D_loss_real = 0.630, D_loss_fake = 0.658, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 7/46, D_loss_real = 0.625, D_loss_fake = 0.671, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 8/46, D_loss_real = 0.642, D_loss_fake = 0.722, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 9/46, D_loss_real = 0.658, D_loss_fake = 0.700, G_loss = 0.744
2/2 [=====] - 0s 5ms/step
Epoch 59, Iteration 10/46, D_loss_real = 0.638, D_loss_fake = 0.689, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 11/46, D_loss_real = 0.645, D_loss_fake = 0.656, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 12/46, D_loss_real = 0.689, D_loss_fake = 0.688, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 13/46, D_loss_real = 0.686, D_loss_fake = 0.674, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 14/46, D_loss_real = 0.653, D_loss_fake = 0.688, G_loss = 0.776
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.672, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 16/46, D_loss_real = 0.682, D_loss_fake = 0.679, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 17/46, D_loss_real = 0.634, D_loss_fake = 0.666, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 18/46, D_loss_real = 0.643, D_loss_fake = 0.723, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 19/46, D_loss_real = 0.653, D_loss_fake = 0.716, G_loss = 0.778
2/2 [=====] - 0s 4ms/step

Epoch 59, Iteration 20/46, D_loss_real = 0.648, D_loss_fake = 0.663, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 21/46, D_loss_real = 0.608, D_loss_fake = 0.701, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 22/46, D_loss_real = 0.631, D_loss_fake = 0.714, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 23/46, D_loss_real = 0.622, D_loss_fake = 0.785, G_loss = 0.724
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 24/46, D_loss_real = 0.667, D_loss_fake = 0.866, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 25/46, D_loss_real = 0.731, D_loss_fake = 0.649, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 26/46, D_loss_real = 0.699, D_loss_fake = 0.649, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.662, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 28/46, D_loss_real = 0.670, D_loss_fake = 0.669, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 29/46, D_loss_real = 0.640, D_loss_fake = 0.674, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 30/46, D_loss_real = 0.636, D_loss_fake = 0.669, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 31/46, D_loss_real = 0.585, D_loss_fake = 0.686, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 32/46, D_loss_real = 0.586, D_loss_fake = 0.752, G_loss = 0.733
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 33/46, D_loss_real = 0.631, D_loss_fake = 0.780, G_loss = 0.735
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 34/46, D_loss_real = 0.647, D_loss_fake = 0.740, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 35/46, D_loss_real = 0.717, D_loss_fake = 0.697, G_loss = 0.782
2/2 [=====] - 0s 4ms/step

Epoch 59, Iteration 36/46, D_loss_real = 0.718, D_loss_fake = 0.682, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 37/46, D_loss_real = 0.724, D_loss_fake = 0.658, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 38/46, D_loss_real = 0.733, D_loss_fake = 0.639, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 39/46, D_loss_real = 0.748, D_loss_fake = 0.627, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 40/46, D_loss_real = 0.712, D_loss_fake = 0.636, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 41/46, D_loss_real = 0.706, D_loss_fake = 0.635, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 42/46, D_loss_real = 0.733, D_loss_fake = 0.646, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 43/46, D_loss_real = 0.708, D_loss_fake = 0.640, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 44/46, D_loss_real = 0.704, D_loss_fake = 0.645, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 45/46, D_loss_real = 0.719, D_loss_fake = 0.632, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.658, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 1/46, D_loss_real = 0.684, D_loss_fake = 0.640, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 2/46, D_loss_real = 0.715, D_loss_fake = 0.665, G_loss = 0.786
2/2 [=====] - 0s 5ms/step
Epoch 60, Iteration 3/46, D_loss_real = 0.688, D_loss_fake = 0.656, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 4/46, D_loss_real = 0.646, D_loss_fake = 0.657, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 5/46, D_loss_real = 0.656, D_loss_fake = 0.675, G_loss = 0.769
2/2 [=====] - 0s 3ms/step

Epoch 60, Iteration 6/46, D_loss_real = 0.671, D_loss_fake = 0.661, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 7/46, D_loss_real = 0.676, D_loss_fake = 0.672, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 8/46, D_loss_real = 0.674, D_loss_fake = 0.687, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.641, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 10/46, D_loss_real = 0.619, D_loss_fake = 0.716, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 11/46, D_loss_real = 0.679, D_loss_fake = 0.659, G_loss = 0.761
2/2 [=====] - 0s 5ms/step
Epoch 60, Iteration 12/46, D_loss_real = 0.667, D_loss_fake = 0.689, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 13/46, D_loss_real = 0.642, D_loss_fake = 0.711, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 14/46, D_loss_real = 0.585, D_loss_fake = 0.758, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 15/46, D_loss_real = 0.617, D_loss_fake = 0.744, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 16/46, D_loss_real = 0.657, D_loss_fake = 0.647, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 17/46, D_loss_real = 0.608, D_loss_fake = 0.685, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 18/46, D_loss_real = 0.643, D_loss_fake = 0.634, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 19/46, D_loss_real = 0.664, D_loss_fake = 0.705, G_loss = 0.773
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 20/46, D_loss_real = 0.638, D_loss_fake = 0.766, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 21/46, D_loss_real = 0.650, D_loss_fake = 0.703, G_loss = 0.810
2/2 [=====] - 0s 4ms/step

Epoch 60, Iteration 22/46, D_loss_real = 0.690, D_loss_fake = 0.751, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 23/46, D_loss_real = 0.699, D_loss_fake = 0.666, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 24/46, D_loss_real = 0.735, D_loss_fake = 0.662, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 25/46, D_loss_real = 0.779, D_loss_fake = 0.644, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 26/46, D_loss_real = 0.728, D_loss_fake = 0.664, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 27/46, D_loss_real = 0.719, D_loss_fake = 0.655, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 28/46, D_loss_real = 0.676, D_loss_fake = 0.652, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 29/46, D_loss_real = 0.690, D_loss_fake = 0.678, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 30/46, D_loss_real = 0.690, D_loss_fake = 0.706, G_loss = 0.755
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 31/46, D_loss_real = 0.650, D_loss_fake = 0.686, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 32/46, D_loss_real = 0.644, D_loss_fake = 0.700, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 33/46, D_loss_real = 0.714, D_loss_fake = 0.674, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 34/46, D_loss_real = 0.658, D_loss_fake = 0.667, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 35/46, D_loss_real = 0.678, D_loss_fake = 0.623, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 36/46, D_loss_real = 0.675, D_loss_fake = 0.656, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.677, G_loss = 0.762
2/2 [=====] - 0s 4ms/step

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Epoch 60, Iteration 38/46, D_loss_real = 0.671, D_loss_fake = 0.650, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 39/46, D_loss_real = 0.665, D_loss_fake = 0.652, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.623, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 41/46, D_loss_real = 0.675, D_loss_fake = 0.660, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 42/46, D_loss_real = 0.660, D_loss_fake = 0.659, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 43/46, D_loss_real = 0.652, D_loss_fake = 0.660, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 44/46, D_loss_real = 0.679, D_loss_fake = 0.678, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 45/46, D_loss_real = 0.653, D_loss_fake = 0.769, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.633, G_loss = 0.823
2/2 [=====] - 0s 168ms/step

```

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

Kết quả truyền trực tuyến bị cắt bất đến 5000 dòng cuối.

```

2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 32/46, D_loss_real = 0.640, D_loss_fake = 0.740, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 33/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 34/46, D_loss_real = 0.685, D_loss_fake = 0.643, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 35/46, D_loss_real = 0.628, D_loss_fake = 0.745, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 36/46, D_loss_real = 0.668, D_loss_fake = 0.714, G_loss = 0.776
2/2 [=====] - 0s 4ms/step

```

Epoch 6, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.700, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 38/46, D_loss_real = 0.684, D_loss_fake = 0.686, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 39/46, D_loss_real = 0.692, D_loss_fake = 0.668, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 40/46, D_loss_real = 0.689, D_loss_fake = 0.630, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 41/46, D_loss_real = 0.665, D_loss_fake = 0.657, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 42/46, D_loss_real = 0.651, D_loss_fake = 0.649, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 43/46, D_loss_real = 0.649, D_loss_fake = 0.639, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 6, Iteration 44/46, D_loss_real = 0.647, D_loss_fake = 0.657, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 45/46, D_loss_real = 0.631, D_loss_fake = 0.739, G_loss = 0.777
2/2 [=====] - 0s 3ms/step
Epoch 6, Iteration 46/46, D_loss_real = 0.614, D_loss_fake = 0.732, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 1/46, D_loss_real = 0.666, D_loss_fake = 0.701, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 2/46, D_loss_real = 0.627, D_loss_fake = 0.699, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 3/46, D_loss_real = 0.619, D_loss_fake = 0.718, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 4/46, D_loss_real = 0.657, D_loss_fake = 0.724, G_loss = 0.778
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 5/46, D_loss_real = 0.652, D_loss_fake = 0.655, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.708, G_loss = 0.750
2/2 [=====] - 0s 4ms/step

Epoch 7, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.724, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 8/46, D_loss_real = 0.639, D_loss_fake = 0.684, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 9/46, D_loss_real = 0.680, D_loss_fake = 0.704, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 10/46, D_loss_real = 0.730, D_loss_fake = 0.654, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 11/46, D_loss_real = 0.750, D_loss_fake = 0.636, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 12/46, D_loss_real = 0.759, D_loss_fake = 0.649, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 13/46, D_loss_real = 0.736, D_loss_fake = 0.672, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 14/46, D_loss_real = 0.759, D_loss_fake = 0.642, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 15/46, D_loss_real = 0.722, D_loss_fake = 0.637, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 16/46, D_loss_real = 0.708, D_loss_fake = 0.645, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 17/46, D_loss_real = 0.735, D_loss_fake = 0.656, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 18/46, D_loss_real = 0.692, D_loss_fake = 0.634, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 19/46, D_loss_real = 0.698, D_loss_fake = 0.637, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 20/46, D_loss_real = 0.675, D_loss_fake = 0.659, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 21/46, D_loss_real = 0.692, D_loss_fake = 0.667, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.631, G_loss = 0.788
2/2 [=====] - 0s 4ms/step

Epoch 7, Iteration 23/46, D_loss_real = 0.673, D_loss_fake = 0.650, G_loss = 0.758
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 24/46, D_loss_real = 0.687, D_loss_fake = 0.718, G_loss = 0.745
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 25/46, D_loss_real = 0.656, D_loss_fake = 0.703, G_loss = 0.719
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 26/46, D_loss_real = 0.632, D_loss_fake = 0.724, G_loss = 0.725
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 27/46, D_loss_real = 0.653, D_loss_fake = 0.754, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 28/46, D_loss_real = 0.656, D_loss_fake = 0.696, G_loss = 0.726
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 29/46, D_loss_real = 0.678, D_loss_fake = 0.686, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 30/46, D_loss_real = 0.671, D_loss_fake = 0.666, G_loss = 0.754
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.700, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.679, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 33/46, D_loss_real = 0.700, D_loss_fake = 0.664, G_loss = 0.774
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 34/46, D_loss_real = 0.698, D_loss_fake = 0.666, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 35/46, D_loss_real = 0.677, D_loss_fake = 0.662, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 36/46, D_loss_real = 0.661, D_loss_fake = 0.672, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 37/46, D_loss_real = 0.654, D_loss_fake = 0.702, G_loss = 0.737
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 38/46, D_loss_real = 0.700, D_loss_fake = 0.687, G_loss = 0.766
2/2 [=====] - 0s 4ms/step

Epoch 7, Iteration 39/46, D_loss_real = 0.626, D_loss_fake = 0.674, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 40/46, D_loss_real = 0.640, D_loss_fake = 0.699, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 7, Iteration 41/46, D_loss_real = 0.638, D_loss_fake = 0.675, G_loss = 0.751
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 42/46, D_loss_real = 0.672, D_loss_fake = 0.658, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 43/46, D_loss_real = 0.668, D_loss_fake = 0.692, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 44/46, D_loss_real = 0.694, D_loss_fake = 0.665, G_loss = 0.736
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 45/46, D_loss_real = 0.646, D_loss_fake = 0.675, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 7, Iteration 46/46, D_loss_real = 0.670, D_loss_fake = 0.677, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 1/46, D_loss_real = 0.661, D_loss_fake = 0.677, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 2/46, D_loss_real = 0.640, D_loss_fake = 0.685, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 3/46, D_loss_real = 0.661, D_loss_fake = 0.614, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 4/46, D_loss_real = 0.674, D_loss_fake = 0.679, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 5/46, D_loss_real = 0.666, D_loss_fake = 0.725, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 6/46, D_loss_real = 0.683, D_loss_fake = 0.717, G_loss = 0.742
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 7/46, D_loss_real = 0.692, D_loss_fake = 0.684, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 8/46, D_loss_real = 0.680, D_loss_fake = 0.679, G_loss = 0.769
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 9/46, D_loss_real = 0.673, D_loss_fake = 0.672, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 10/46, D_loss_real = 0.682, D_loss_fake = 0.637, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 11/46, D_loss_real = 0.642, D_loss_fake = 0.675, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.674, G_loss = 0.746
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 13/46, D_loss_real = 0.626, D_loss_fake = 0.727, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 14/46, D_loss_real = 0.647, D_loss_fake = 0.725, G_loss = 0.730
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 15/46, D_loss_real = 0.643, D_loss_fake = 0.716, G_loss = 0.749
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 16/46, D_loss_real = 0.651, D_loss_fake = 0.729, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 17/46, D_loss_real = 0.679, D_loss_fake = 0.662, G_loss = 0.757
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 18/46, D_loss_real = 0.717, D_loss_fake = 0.647, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 19/46, D_loss_real = 0.672, D_loss_fake = 0.649, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.646, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 21/46, D_loss_real = 0.653, D_loss_fake = 0.665, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 22/46, D_loss_real = 0.635, D_loss_fake = 0.696, G_loss = 0.767
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.697, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 24/46, D_loss_real = 0.635, D_loss_fake = 0.700, G_loss = 0.715
2/2 [=====] - 0s 4ms/step

Epoch 8, Iteration 25/46, D_loss_real = 0.632, D_loss_fake = 0.716, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 26/46, D_loss_real = 0.632, D_loss_fake = 0.720, G_loss = 0.709
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 27/46, D_loss_real = 0.676, D_loss_fake = 0.743, G_loss = 0.704
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 28/46, D_loss_real = 0.659, D_loss_fake = 0.762, G_loss = 0.697
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 29/46, D_loss_real = 0.677, D_loss_fake = 0.765, G_loss = 0.711
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 30/46, D_loss_real = 0.666, D_loss_fake = 0.685, G_loss = 0.772
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 31/46, D_loss_real = 0.710, D_loss_fake = 0.618, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 32/46, D_loss_real = 0.687, D_loss_fake = 0.651, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 33/46, D_loss_real = 0.704, D_loss_fake = 0.607, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 34/46, D_loss_real = 0.682, D_loss_fake = 0.591, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 35/46, D_loss_real = 0.704, D_loss_fake = 0.604, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 36/46, D_loss_real = 0.720, D_loss_fake = 0.637, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.658, G_loss = 0.746
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 38/46, D_loss_real = 0.704, D_loss_fake = 0.705, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 39/46, D_loss_real = 0.697, D_loss_fake = 0.677, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 8, Iteration 40/46, D_loss_real = 0.708, D_loss_fake = 0.651, G_loss = 0.826
2/2 [=====] - 0s 3ms/step

Epoch 8, Iteration 41/46, D_loss_real = 0.744, D_loss_fake = 0.653, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 42/46, D_loss_real = 0.703, D_loss_fake = 0.649, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 43/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 44/46, D_loss_real = 0.653, D_loss_fake = 0.655, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 45/46, D_loss_real = 0.664, D_loss_fake = 0.670, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 8, Iteration 46/46, D_loss_real = 0.658, D_loss_fake = 0.667, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 1/46, D_loss_real = 0.653, D_loss_fake = 0.738, G_loss = 0.785
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 2/46, D_loss_real = 0.645, D_loss_fake = 0.698, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 3/46, D_loss_real = 0.642, D_loss_fake = 0.683, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 4/46, D_loss_real = 0.718, D_loss_fake = 0.699, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 5/46, D_loss_real = 0.738, D_loss_fake = 0.648, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 6/46, D_loss_real = 0.716, D_loss_fake = 0.635, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 7/46, D_loss_real = 0.690, D_loss_fake = 0.656, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 8/46, D_loss_real = 0.673, D_loss_fake = 0.641, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 9/46, D_loss_real = 0.644, D_loss_fake = 0.641, G_loss = 0.780
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 10/46, D_loss_real = 0.651, D_loss_fake = 0.678, G_loss = 0.757
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 11/46, D_loss_real = 0.605, D_loss_fake = 0.769, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 12/46, D_loss_real = 0.619, D_loss_fake = 0.689, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 13/46, D_loss_real = 0.672, D_loss_fake = 0.746, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.730, G_loss = 0.743
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 15/46, D_loss_real = 0.685, D_loss_fake = 0.718, G_loss = 0.731
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.685, G_loss = 0.718
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 17/46, D_loss_real = 0.703, D_loss_fake = 0.743, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 18/46, D_loss_real = 0.686, D_loss_fake = 0.681, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 19/46, D_loss_real = 0.710, D_loss_fake = 0.668, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.650, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 21/46, D_loss_real = 0.711, D_loss_fake = 0.632, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 22/46, D_loss_real = 0.706, D_loss_fake = 0.651, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 23/46, D_loss_real = 0.715, D_loss_fake = 0.659, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 24/46, D_loss_real = 0.680, D_loss_fake = 0.660, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 25/46, D_loss_real = 0.652, D_loss_fake = 0.676, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.653, G_loss = 0.760
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 27/46, D_loss_real = 0.681, D_loss_fake = 0.716, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 28/46, D_loss_real = 0.696, D_loss_fake = 0.654, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 29/46, D_loss_real = 0.706, D_loss_fake = 0.625, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 30/46, D_loss_real = 0.708, D_loss_fake = 0.632, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 31/46, D_loss_real = 0.702, D_loss_fake = 0.665, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 32/46, D_loss_real = 0.707, D_loss_fake = 0.646, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 33/46, D_loss_real = 0.674, D_loss_fake = 0.642, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 34/46, D_loss_real = 0.696, D_loss_fake = 0.682, G_loss = 0.740
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 35/46, D_loss_real = 0.659, D_loss_fake = 0.675, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 36/46, D_loss_real = 0.647, D_loss_fake = 0.718, G_loss = 0.744
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 37/46, D_loss_real = 0.621, D_loss_fake = 0.678, G_loss = 0.720
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 38/46, D_loss_real = 0.633, D_loss_fake = 0.690, G_loss = 0.732
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 39/46, D_loss_real = 0.651, D_loss_fake = 0.702, G_loss = 0.714
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 40/46, D_loss_real = 0.608, D_loss_fake = 0.727, G_loss = 0.704
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 41/46, D_loss_real = 0.653, D_loss_fake = 0.716, G_loss = 0.707
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 42/46, D_loss_real = 0.586, D_loss_fake = 0.725, G_loss = 0.726
2/2 [=====] - 0s 4ms/step

Epoch 9, Iteration 43/46, D_loss_real = 0.587, D_loss_fake = 0.730, G_loss = 0.694
2/2 [=====] - 0s 3ms/step
Epoch 9, Iteration 44/46, D_loss_real = 0.624, D_loss_fake = 0.744, G_loss = 0.767
2/2 [=====] - 0s 5ms/step
Epoch 9, Iteration 45/46, D_loss_real = 0.686, D_loss_fake = 0.688, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 9, Iteration 46/46, D_loss_real = 0.602, D_loss_fake = 0.709, G_loss = 0.771
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 1/46, D_loss_real = 0.614, D_loss_fake = 0.718, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 2/46, D_loss_real = 0.653, D_loss_fake = 0.699, G_loss = 0.756
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 3/46, D_loss_real = 0.626, D_loss_fake = 0.747, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 4/46, D_loss_real = 0.655, D_loss_fake = 0.627, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 5/46, D_loss_real = 0.673, D_loss_fake = 0.663, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 6/46, D_loss_real = 0.690, D_loss_fake = 0.663, G_loss = 0.766
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 7/46, D_loss_real = 0.706, D_loss_fake = 0.681, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 8/46, D_loss_real = 0.632, D_loss_fake = 0.662, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 9/46, D_loss_real = 0.675, D_loss_fake = 0.641, G_loss = 0.728
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 10/46, D_loss_real = 0.677, D_loss_fake = 0.745, G_loss = 0.761
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 11/46, D_loss_real = 0.646, D_loss_fake = 0.703, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 12/46, D_loss_real = 0.670, D_loss_fake = 0.726, G_loss = 0.794
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 13/46, D_loss_real = 0.674, D_loss_fake = 0.661, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 14/46, D_loss_real = 0.692, D_loss_fake = 0.642, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 15/46, D_loss_real = 0.706, D_loss_fake = 0.670, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 16/46, D_loss_real = 0.725, D_loss_fake = 0.654, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.666, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.681, G_loss = 0.742
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 19/46, D_loss_real = 0.685, D_loss_fake = 0.693, G_loss = 0.754
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 20/46, D_loss_real = 0.661, D_loss_fake = 0.659, G_loss = 0.761
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 21/46, D_loss_real = 0.628, D_loss_fake = 0.672, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 22/46, D_loss_real = 0.585, D_loss_fake = 0.673, G_loss = 0.753
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 23/46, D_loss_real = 0.555, D_loss_fake = 0.691, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 24/46, D_loss_real = 0.586, D_loss_fake = 0.813, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 25/46, D_loss_real = 0.552, D_loss_fake = 0.723, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 26/46, D_loss_real = 0.620, D_loss_fake = 0.709, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 27/46, D_loss_real = 0.683, D_loss_fake = 0.619, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 28/46, D_loss_real = 0.707, D_loss_fake = 0.594, G_loss = 0.893
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 29/46, D_loss_real = 0.661, D_loss_fake = 0.738, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 30/46, D_loss_real = 0.704, D_loss_fake = 0.680, G_loss = 0.741
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 31/46, D_loss_real = 0.676, D_loss_fake = 0.756, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 32/46, D_loss_real = 0.711, D_loss_fake = 0.679, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 33/46, D_loss_real = 0.667, D_loss_fake = 0.648, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 34/46, D_loss_real = 0.690, D_loss_fake = 0.636, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 35/46, D_loss_real = 0.673, D_loss_fake = 0.634, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 36/46, D_loss_real = 0.638, D_loss_fake = 0.667, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 37/46, D_loss_real = 0.648, D_loss_fake = 0.683, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 38/46, D_loss_real = 0.632, D_loss_fake = 0.629, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 39/46, D_loss_real = 0.623, D_loss_fake = 0.657, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 40/46, D_loss_real = 0.660, D_loss_fake = 0.692, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 41/46, D_loss_real = 0.656, D_loss_fake = 0.723, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 42/46, D_loss_real = 0.646, D_loss_fake = 0.732, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 10, Iteration 43/46, D_loss_real = 0.633, D_loss_fake = 0.736, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.657, G_loss = 0.788
2/2 [=====] - 0s 4ms/step

Epoch 10, Iteration 45/46, D_loss_real = 0.648, D_loss_fake = 0.657, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 10, Iteration 46/46, D_loss_real = 0.608, D_loss_fake = 0.659, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 1/46, D_loss_real = 0.658, D_loss_fake = 0.731, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 2/46, D_loss_real = 0.688, D_loss_fake = 0.744, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 3/46, D_loss_real = 0.695, D_loss_fake = 0.661, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 4/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 5/46, D_loss_real = 0.675, D_loss_fake = 0.680, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 6/46, D_loss_real = 0.711, D_loss_fake = 0.658, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 7/46, D_loss_real = 0.704, D_loss_fake = 0.654, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 8/46, D_loss_real = 0.648, D_loss_fake = 0.695, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.648, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 10/46, D_loss_real = 0.674, D_loss_fake = 0.607, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 11/46, D_loss_real = 0.678, D_loss_fake = 0.605, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 12/46, D_loss_real = 0.627, D_loss_fake = 0.612, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 13/46, D_loss_real = 0.614, D_loss_fake = 0.647, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 14/46, D_loss_real = 0.683, D_loss_fake = 0.687, G_loss = 0.762
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 15/46, D_loss_real = 0.662, D_loss_fake = 0.683, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 16/46, D_loss_real = 0.658, D_loss_fake = 0.713, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 17/46, D_loss_real = 0.661, D_loss_fake = 0.709, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 18/46, D_loss_real = 0.616, D_loss_fake = 0.727, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.717, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 20/46, D_loss_real = 0.694, D_loss_fake = 0.667, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 21/46, D_loss_real = 0.703, D_loss_fake = 0.634, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 22/46, D_loss_real = 0.678, D_loss_fake = 0.651, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 23/46, D_loss_real = 0.703, D_loss_fake = 0.666, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 24/46, D_loss_real = 0.681, D_loss_fake = 0.653, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 25/46, D_loss_real = 0.668, D_loss_fake = 0.666, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 26/46, D_loss_real = 0.665, D_loss_fake = 0.690, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 27/46, D_loss_real = 0.645, D_loss_fake = 0.622, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 28/46, D_loss_real = 0.645, D_loss_fake = 0.654, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 29/46, D_loss_real = 0.676, D_loss_fake = 0.629, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 30/46, D_loss_real = 0.648, D_loss_fake = 0.642, G_loss = 0.807
2/2 [=====] - 0s 3ms/step

Epoch 11, Iteration 31/46, D_loss_real = 0.664, D_loss_fake = 0.741, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 32/46, D_loss_real = 0.591, D_loss_fake = 0.688, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 33/46, D_loss_real = 0.616, D_loss_fake = 0.696, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 34/46, D_loss_real = 0.697, D_loss_fake = 0.641, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 35/46, D_loss_real = 0.687, D_loss_fake = 0.640, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 36/46, D_loss_real = 0.720, D_loss_fake = 0.653, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 37/46, D_loss_real = 0.739, D_loss_fake = 0.634, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 38/46, D_loss_real = 0.722, D_loss_fake = 0.649, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 39/46, D_loss_real = 0.689, D_loss_fake = 0.651, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 40/46, D_loss_real = 0.697, D_loss_fake = 0.601, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 41/46, D_loss_real = 0.661, D_loss_fake = 0.649, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 42/46, D_loss_real = 0.686, D_loss_fake = 0.676, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 43/46, D_loss_real = 0.732, D_loss_fake = 0.649, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.652, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 11, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.651, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 11, Iteration 46/46, D_loss_real = 0.706, D_loss_fake = 0.632, G_loss = 0.823
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 1/46, D_loss_real = 0.758, D_loss_fake = 0.665, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 2/46, D_loss_real = 0.717, D_loss_fake = 0.673, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 3/46, D_loss_real = 0.653, D_loss_fake = 0.644, G_loss = 0.760
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 4/46, D_loss_real = 0.624, D_loss_fake = 0.690, G_loss = 0.762
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 5/46, D_loss_real = 0.626, D_loss_fake = 0.651, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 6/46, D_loss_real = 0.573, D_loss_fake = 0.663, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 7/46, D_loss_real = 0.628, D_loss_fake = 0.710, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 8/46, D_loss_real = 0.633, D_loss_fake = 0.670, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 9/46, D_loss_real = 0.559, D_loss_fake = 0.692, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 10/46, D_loss_real = 0.565, D_loss_fake = 0.691, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 11/46, D_loss_real = 0.619, D_loss_fake = 0.733, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 12/46, D_loss_real = 0.650, D_loss_fake = 0.687, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 13/46, D_loss_real = 0.674, D_loss_fake = 0.695, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 14/46, D_loss_real = 0.672, D_loss_fake = 0.735, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 15/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.633, G_loss = 0.824
2/2 [=====] - 0s 3ms/step

Epoch 12, Iteration 17/46, D_loss_real = 0.676, D_loss_fake = 0.663, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 18/46, D_loss_real = 0.705, D_loss_fake = 0.636, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 19/46, D_loss_real = 0.676, D_loss_fake = 0.648, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 20/46, D_loss_real = 0.688, D_loss_fake = 0.650, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 21/46, D_loss_real = 0.662, D_loss_fake = 0.632, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 22/46, D_loss_real = 0.666, D_loss_fake = 0.653, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 23/46, D_loss_real = 0.594, D_loss_fake = 0.702, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.644, G_loss = 0.775
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 25/46, D_loss_real = 0.612, D_loss_fake = 0.720, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 26/46, D_loss_real = 0.601, D_loss_fake = 0.675, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 27/46, D_loss_real = 0.614, D_loss_fake = 0.687, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 28/46, D_loss_real = 0.628, D_loss_fake = 0.660, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 29/46, D_loss_real = 0.622, D_loss_fake = 0.694, G_loss = 0.760
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 30/46, D_loss_real = 0.583, D_loss_fake = 0.807, G_loss = 0.758
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 31/46, D_loss_real = 0.649, D_loss_fake = 0.717, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 32/46, D_loss_real = 0.670, D_loss_fake = 0.682, G_loss = 0.765
2/2 [=====] - 0s 4ms/step

Epoch 12, Iteration 33/46, D_loss_real = 0.669, D_loss_fake = 0.681, G_loss = 0.937
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 34/46, D_loss_real = 0.693, D_loss_fake = 0.690, G_loss = 0.766
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.668, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 36/46, D_loss_real = 0.685, D_loss_fake = 0.712, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 37/46, D_loss_real = 0.713, D_loss_fake = 0.656, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 38/46, D_loss_real = 0.699, D_loss_fake = 0.652, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 39/46, D_loss_real = 0.681, D_loss_fake = 0.659, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 40/46, D_loss_real = 0.693, D_loss_fake = 0.615, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 41/46, D_loss_real = 0.717, D_loss_fake = 0.633, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 42/46, D_loss_real = 0.677, D_loss_fake = 0.672, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 43/46, D_loss_real = 0.665, D_loss_fake = 0.653, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 44/46, D_loss_real = 0.712, D_loss_fake = 0.618, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 12, Iteration 45/46, D_loss_real = 0.726, D_loss_fake = 0.621, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 12, Iteration 46/46, D_loss_real = 0.718, D_loss_fake = 0.624, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 1/46, D_loss_real = 0.713, D_loss_fake = 0.669, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 2/46, D_loss_real = 0.719, D_loss_fake = 0.628, G_loss = 0.828
2/2 [=====] - 0s 3ms/step

Epoch 13, Iteration 3/46, D_loss_real = 0.703, D_loss_fake = 0.631, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 4/46, D_loss_real = 0.648, D_loss_fake = 0.634, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 5/46, D_loss_real = 0.687, D_loss_fake = 0.642, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 6/46, D_loss_real = 0.677, D_loss_fake = 0.659, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 7/46, D_loss_real = 0.639, D_loss_fake = 0.655, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 8/46, D_loss_real = 0.684, D_loss_fake = 0.676, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 9/46, D_loss_real = 0.674, D_loss_fake = 0.645, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 10/46, D_loss_real = 0.659, D_loss_fake = 0.628, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 11/46, D_loss_real = 0.627, D_loss_fake = 0.746, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 12/46, D_loss_real = 0.733, D_loss_fake = 0.636, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 13/46, D_loss_real = 0.737, D_loss_fake = 0.656, G_loss = 0.885
2/2 [=====] - 0s 7ms/step
Epoch 13, Iteration 14/46, D_loss_real = 0.676, D_loss_fake = 0.616, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 15/46, D_loss_real = 0.726, D_loss_fake = 0.594, G_loss = 0.918
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 16/46, D_loss_real = 0.733, D_loss_fake = 0.584, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 17/46, D_loss_real = 0.765, D_loss_fake = 0.604, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 18/46, D_loss_real = 0.713, D_loss_fake = 0.621, G_loss = 0.868
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 19/46, D_loss_real = 0.727, D_loss_fake = 0.592, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.607, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 21/46, D_loss_real = 0.660, D_loss_fake = 0.606, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 22/46, D_loss_real = 0.618, D_loss_fake = 0.706, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 23/46, D_loss_real = 0.667, D_loss_fake = 0.614, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 24/46, D_loss_real = 0.626, D_loss_fake = 0.637, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 25/46, D_loss_real = 0.602, D_loss_fake = 0.667, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 26/46, D_loss_real = 0.630, D_loss_fake = 0.716, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 27/46, D_loss_real = 0.618, D_loss_fake = 0.677, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 28/46, D_loss_real = 0.690, D_loss_fake = 0.686, G_loss = 0.812
2/2 [=====] - 0s 5ms/step
Epoch 13, Iteration 29/46, D_loss_real = 0.701, D_loss_fake = 0.690, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 30/46, D_loss_real = 0.723, D_loss_fake = 0.632, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 31/46, D_loss_real = 0.681, D_loss_fake = 0.616, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 32/46, D_loss_real = 0.680, D_loss_fake = 0.689, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 33/46, D_loss_real = 0.617, D_loss_fake = 0.659, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 34/46, D_loss_real = 0.649, D_loss_fake = 0.666, G_loss = 0.797
2/2 [=====] - 0s 4ms/step

Epoch 13, Iteration 35/46, D_loss_real = 0.683, D_loss_fake = 0.718, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.661, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 37/46, D_loss_real = 0.677, D_loss_fake = 0.641, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 38/46, D_loss_real = 0.695, D_loss_fake = 0.664, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 39/46, D_loss_real = 0.720, D_loss_fake = 0.604, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 40/46, D_loss_real = 0.715, D_loss_fake = 0.661, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 41/46, D_loss_real = 0.682, D_loss_fake = 0.612, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 42/46, D_loss_real = 0.714, D_loss_fake = 0.670, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 13, Iteration 43/46, D_loss_real = 0.673, D_loss_fake = 0.637, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 44/46, D_loss_real = 0.685, D_loss_fake = 0.633, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.646, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 13, Iteration 46/46, D_loss_real = 0.713, D_loss_fake = 0.627, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 1/46, D_loss_real = 0.772, D_loss_fake = 0.642, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 2/46, D_loss_real = 0.756, D_loss_fake = 0.614, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 3/46, D_loss_real = 0.750, D_loss_fake = 0.623, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 4/46, D_loss_real = 0.725, D_loss_fake = 0.623, G_loss = 0.847
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 5/46, D_loss_real = 0.697, D_loss_fake = 0.671, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 6/46, D_loss_real = 0.678, D_loss_fake = 0.620, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.650, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 8/46, D_loss_real = 0.642, D_loss_fake = 0.629, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.686, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 10/46, D_loss_real = 0.652, D_loss_fake = 0.693, G_loss = 0.775
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 11/46, D_loss_real = 0.577, D_loss_fake = 0.697, G_loss = 0.748
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 12/46, D_loss_real = 0.584, D_loss_fake = 0.711, G_loss = 0.780
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 13/46, D_loss_real = 0.581, D_loss_fake = 0.681, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 14/46, D_loss_real = 0.695, D_loss_fake = 0.635, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 15/46, D_loss_real = 0.652, D_loss_fake = 0.628, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 16/46, D_loss_real = 0.628, D_loss_fake = 0.622, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 17/46, D_loss_real = 0.683, D_loss_fake = 0.677, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 18/46, D_loss_real = 0.670, D_loss_fake = 0.671, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 19/46, D_loss_real = 0.666, D_loss_fake = 0.674, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 20/46, D_loss_real = 0.612, D_loss_fake = 0.663, G_loss = 0.812
2/2 [=====] - 0s 3ms/step

Epoch 14, Iteration 21/46, D_loss_real = 0.651, D_loss_fake = 0.652, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 22/46, D_loss_real = 0.660, D_loss_fake = 0.644, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 23/46, D_loss_real = 0.630, D_loss_fake = 0.668, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 24/46, D_loss_real = 0.618, D_loss_fake = 0.662, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 25/46, D_loss_real = 0.639, D_loss_fake = 0.670, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 26/46, D_loss_real = 0.677, D_loss_fake = 0.715, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 27/46, D_loss_real = 0.656, D_loss_fake = 0.689, G_loss = 0.769
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 28/46, D_loss_real = 0.611, D_loss_fake = 0.708, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 29/46, D_loss_real = 0.617, D_loss_fake = 0.709, G_loss = 0.767
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 30/46, D_loss_real = 0.650, D_loss_fake = 0.744, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 31/46, D_loss_real = 0.654, D_loss_fake = 0.732, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.626, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 33/46, D_loss_real = 0.687, D_loss_fake = 0.657, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 34/46, D_loss_real = 0.645, D_loss_fake = 0.660, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 35/46, D_loss_real = 0.698, D_loss_fake = 0.636, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 36/46, D_loss_real = 0.627, D_loss_fake = 0.664, G_loss = 0.839
2/2 [=====] - 0s 4ms/step

Epoch 14, Iteration 37/46, D_loss_real = 0.634, D_loss_fake = 0.624, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 38/46, D_loss_real = 0.712, D_loss_fake = 0.686, G_loss = 0.874
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 39/46, D_loss_real = 0.678, D_loss_fake = 0.609, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 40/46, D_loss_real = 0.725, D_loss_fake = 0.655, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 41/46, D_loss_real = 0.721, D_loss_fake = 0.641, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 42/46, D_loss_real = 0.771, D_loss_fake = 0.631, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 14, Iteration 43/46, D_loss_real = 0.705, D_loss_fake = 0.684, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 44/46, D_loss_real = 0.658, D_loss_fake = 0.684, G_loss = 0.763
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 45/46, D_loss_real = 0.679, D_loss_fake = 0.728, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 14, Iteration 46/46, D_loss_real = 0.684, D_loss_fake = 0.655, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 1/46, D_loss_real = 0.665, D_loss_fake = 0.623, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 2/46, D_loss_real = 0.636, D_loss_fake = 0.646, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 3/46, D_loss_real = 0.658, D_loss_fake = 0.634, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 4/46, D_loss_real = 0.626, D_loss_fake = 0.742, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 5/46, D_loss_real = 0.645, D_loss_fake = 0.658, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 6/46, D_loss_real = 0.695, D_loss_fake = 0.634, G_loss = 0.833
2/2 [=====] - 0s 7ms/step

Epoch 15, Iteration 7/46, D_loss_real = 0.634, D_loss_fake = 0.673, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 8/46, D_loss_real = 0.695, D_loss_fake = 0.602, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 9/46, D_loss_real = 0.684, D_loss_fake = 0.693, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 10/46, D_loss_real = 0.654, D_loss_fake = 0.676, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 11/46, D_loss_real = 0.702, D_loss_fake = 0.651, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 12/46, D_loss_real = 0.716, D_loss_fake = 0.601, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 13/46, D_loss_real = 0.673, D_loss_fake = 0.635, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 14/46, D_loss_real = 0.643, D_loss_fake = 0.624, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 15/46, D_loss_real = 0.643, D_loss_fake = 0.679, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 16/46, D_loss_real = 0.574, D_loss_fake = 0.710, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 17/46, D_loss_real = 0.630, D_loss_fake = 0.691, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 18/46, D_loss_real = 0.615, D_loss_fake = 0.670, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 19/46, D_loss_real = 0.634, D_loss_fake = 0.616, G_loss = 0.892
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 20/46, D_loss_real = 0.683, D_loss_fake = 0.670, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 21/46, D_loss_real = 0.736, D_loss_fake = 0.685, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 22/46, D_loss_real = 0.641, D_loss_fake = 0.690, G_loss = 0.810
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 23/46, D_loss_real = 0.716, D_loss_fake = 0.666, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 24/46, D_loss_real = 0.706, D_loss_fake = 0.648, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 25/46, D_loss_real = 0.646, D_loss_fake = 0.650, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 26/46, D_loss_real = 0.649, D_loss_fake = 0.599, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 27/46, D_loss_real = 0.658, D_loss_fake = 0.610, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 28/46, D_loss_real = 0.673, D_loss_fake = 0.659, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 29/46, D_loss_real = 0.667, D_loss_fake = 0.751, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 30/46, D_loss_real = 0.707, D_loss_fake = 0.754, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 31/46, D_loss_real = 0.700, D_loss_fake = 0.655, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 32/46, D_loss_real = 0.655, D_loss_fake = 0.647, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 33/46, D_loss_real = 0.676, D_loss_fake = 0.650, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.607, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 35/46, D_loss_real = 0.729, D_loss_fake = 0.655, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 36/46, D_loss_real = 0.626, D_loss_fake = 0.702, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 37/46, D_loss_real = 0.642, D_loss_fake = 0.648, G_loss = 0.778
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 38/46, D_loss_real = 0.633, D_loss_fake = 0.681, G_loss = 0.786
2/2 [=====] - 0s 4ms/step

Epoch 15, Iteration 39/46, D_loss_real = 0.644, D_loss_fake = 0.666, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 40/46, D_loss_real = 0.622, D_loss_fake = 0.604, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 41/46, D_loss_real = 0.660, D_loss_fake = 0.634, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 42/46, D_loss_real = 0.635, D_loss_fake = 0.635, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 43/46, D_loss_real = 0.502, D_loss_fake = 0.632, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 44/46, D_loss_real = 0.435, D_loss_fake = 0.671, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 15, Iteration 45/46, D_loss_real = 0.481, D_loss_fake = 0.821, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 15, Iteration 46/46, D_loss_real = 0.541, D_loss_fake = 0.743, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 1/46, D_loss_real = 0.626, D_loss_fake = 0.688, G_loss = 0.982
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 2/46, D_loss_real = 0.660, D_loss_fake = 0.667, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 3/46, D_loss_real = 0.724, D_loss_fake = 0.677, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 4/46, D_loss_real = 0.703, D_loss_fake = 0.633, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 5/46, D_loss_real = 0.730, D_loss_fake = 0.652, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 6/46, D_loss_real = 0.719, D_loss_fake = 0.658, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 7/46, D_loss_real = 0.723, D_loss_fake = 0.668, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 8/46, D_loss_real = 0.741, D_loss_fake = 0.665, G_loss = 0.864
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 9/46, D_loss_real = 0.734, D_loss_fake = 0.625, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 10/46, D_loss_real = 0.727, D_loss_fake = 0.620, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 11/46, D_loss_real = 0.748, D_loss_fake = 0.618, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 12/46, D_loss_real = 0.668, D_loss_fake = 0.670, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 13/46, D_loss_real = 0.747, D_loss_fake = 0.635, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 14/46, D_loss_real = 0.663, D_loss_fake = 0.622, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 15/46, D_loss_real = 0.649, D_loss_fake = 0.623, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 16/46, D_loss_real = 0.624, D_loss_fake = 0.723, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 17/46, D_loss_real = 0.627, D_loss_fake = 0.644, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 18/46, D_loss_real = 0.636, D_loss_fake = 0.675, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 19/46, D_loss_real = 0.664, D_loss_fake = 0.658, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 20/46, D_loss_real = 0.684, D_loss_fake = 0.625, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 21/46, D_loss_real = 0.730, D_loss_fake = 0.635, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 22/46, D_loss_real = 0.658, D_loss_fake = 0.610, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 23/46, D_loss_real = 0.678, D_loss_fake = 0.633, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 24/46, D_loss_real = 0.649, D_loss_fake = 0.637, G_loss = 0.797
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 25/46, D_loss_real = 0.603, D_loss_fake = 0.715, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 26/46, D_loss_real = 0.621, D_loss_fake = 0.738, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 27/46, D_loss_real = 0.678, D_loss_fake = 0.640, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 28/46, D_loss_real = 0.656, D_loss_fake = 0.610, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 29/46, D_loss_real = 0.612, D_loss_fake = 0.674, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 30/46, D_loss_real = 0.627, D_loss_fake = 0.713, G_loss = 0.815
2/2 [=====] - 0s 5ms/step
Epoch 16, Iteration 31/46, D_loss_real = 0.688, D_loss_fake = 0.681, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 32/46, D_loss_real = 0.679, D_loss_fake = 0.666, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 16, Iteration 33/46, D_loss_real = 0.711, D_loss_fake = 0.638, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 34/46, D_loss_real = 0.628, D_loss_fake = 0.647, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 35/46, D_loss_real = 0.670, D_loss_fake = 0.663, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 36/46, D_loss_real = 0.613, D_loss_fake = 0.664, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 37/46, D_loss_real = 0.624, D_loss_fake = 0.706, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 38/46, D_loss_real = 0.602, D_loss_fake = 0.665, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 39/46, D_loss_real = 0.615, D_loss_fake = 0.769, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 40/46, D_loss_real = 0.679, D_loss_fake = 0.662, G_loss = 0.846
2/2 [=====] - 0s 4ms/step

Epoch 16, Iteration 41/46, D_loss_real = 0.697, D_loss_fake = 0.658, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 42/46, D_loss_real = 0.681, D_loss_fake = 0.630, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 43/46, D_loss_real = 0.633, D_loss_fake = 0.718, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 44/46, D_loss_real = 0.625, D_loss_fake = 0.744, G_loss = 0.750
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 45/46, D_loss_real = 0.635, D_loss_fake = 0.730, G_loss = 0.785
2/2 [=====] - 0s 4ms/step
Epoch 16, Iteration 46/46, D_loss_real = 0.671, D_loss_fake = 0.688, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 1/46, D_loss_real = 0.676, D_loss_fake = 0.632, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 2/46, D_loss_real = 0.669, D_loss_fake = 0.658, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 3/46, D_loss_real = 0.641, D_loss_fake = 0.700, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 4/46, D_loss_real = 0.698, D_loss_fake = 0.696, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 5/46, D_loss_real = 0.652, D_loss_fake = 0.687, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 6/46, D_loss_real = 0.701, D_loss_fake = 0.602, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 7/46, D_loss_real = 0.729, D_loss_fake = 0.635, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 8/46, D_loss_real = 0.736, D_loss_fake = 0.641, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 9/46, D_loss_real = 0.772, D_loss_fake = 0.579, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 10/46, D_loss_real = 0.729, D_loss_fake = 0.644, G_loss = 0.841
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 11/46, D_loss_real = 0.673, D_loss_fake = 0.661, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 12/46, D_loss_real = 0.741, D_loss_fake = 0.633, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 13/46, D_loss_real = 0.718, D_loss_fake = 0.611, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 14/46, D_loss_real = 0.746, D_loss_fake = 0.642, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 15/46, D_loss_real = 0.714, D_loss_fake = 0.622, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 16/46, D_loss_real = 0.723, D_loss_fake = 0.602, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.630, G_loss = 0.895
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 18/46, D_loss_real = 0.696, D_loss_fake = 0.599, G_loss = 0.871
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 19/46, D_loss_real = 0.666, D_loss_fake = 0.652, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 20/46, D_loss_real = 0.722, D_loss_fake = 0.645, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 21/46, D_loss_real = 0.636, D_loss_fake = 0.612, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 22/46, D_loss_real = 0.623, D_loss_fake = 0.621, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 23/46, D_loss_real = 0.669, D_loss_fake = 0.699, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 24/46, D_loss_real = 0.633, D_loss_fake = 0.650, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 25/46, D_loss_real = 0.689, D_loss_fake = 0.613, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 26/46, D_loss_real = 0.631, D_loss_fake = 0.647, G_loss = 0.870
2/2 [=====] - 0s 4ms/step

Epoch 17, Iteration 27/46, D_loss_real = 0.600, D_loss_fake = 0.658, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 28/46, D_loss_real = 0.601, D_loss_fake = 0.721, G_loss = 0.773
2/2 [=====] - 0s 5ms/step
Epoch 17, Iteration 29/46, D_loss_real = 0.540, D_loss_fake = 0.777, G_loss = 0.746
2/2 [=====] - 0s 5ms/step
Epoch 17, Iteration 30/46, D_loss_real = 0.654, D_loss_fake = 0.710, G_loss = 0.756
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 31/46, D_loss_real = 0.631, D_loss_fake = 0.716, G_loss = 0.768
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 32/46, D_loss_real = 0.624, D_loss_fake = 0.771, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 33/46, D_loss_real = 0.650, D_loss_fake = 0.718, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 34/46, D_loss_real = 0.746, D_loss_fake = 0.635, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 35/46, D_loss_real = 0.736, D_loss_fake = 0.638, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 36/46, D_loss_real = 0.764, D_loss_fake = 0.624, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 37/46, D_loss_real = 0.745, D_loss_fake = 0.584, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 38/46, D_loss_real = 0.673, D_loss_fake = 0.649, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 39/46, D_loss_real = 0.708, D_loss_fake = 0.660, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 40/46, D_loss_real = 0.672, D_loss_fake = 0.707, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 41/46, D_loss_real = 0.642, D_loss_fake = 0.654, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 42/46, D_loss_real = 0.657, D_loss_fake = 0.629, G_loss = 0.804
2/2 [=====] - 0s 3ms/step

Epoch 17, Iteration 43/46, D_loss_real = 0.655, D_loss_fake = 0.668, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 17, Iteration 44/46, D_loss_real = 0.661, D_loss_fake = 0.633, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 45/46, D_loss_real = 0.611, D_loss_fake = 0.724, G_loss = 0.765
2/2 [=====] - 0s 4ms/step
Epoch 17, Iteration 46/46, D_loss_real = 0.659, D_loss_fake = 0.625, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 1/46, D_loss_real = 0.644, D_loss_fake = 0.660, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 2/46, D_loss_real = 0.682, D_loss_fake = 0.637, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 3/46, D_loss_real = 0.655, D_loss_fake = 0.653, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 4/46, D_loss_real = 0.668, D_loss_fake = 0.641, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 5/46, D_loss_real = 0.562, D_loss_fake = 0.682, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 6/46, D_loss_real = 0.616, D_loss_fake = 0.690, G_loss = 0.777
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 7/46, D_loss_real = 0.646, D_loss_fake = 0.691, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 8/46, D_loss_real = 0.635, D_loss_fake = 0.724, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 9/46, D_loss_real = 0.604, D_loss_fake = 0.652, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 10/46, D_loss_real = 0.634, D_loss_fake = 0.701, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.663, G_loss = 0.807
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 12/46, D_loss_real = 0.720, D_loss_fake = 0.675, G_loss = 0.843
2/2 [=====] - 0s 4ms/step

Epoch 18, Iteration 13/46, D_loss_real = 0.712, D_loss_fake = 0.652, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 14/46, D_loss_real = 0.674, D_loss_fake = 0.638, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 15/46, D_loss_real = 0.680, D_loss_fake = 0.642, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 16/46, D_loss_real = 0.663, D_loss_fake = 0.662, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 17/46, D_loss_real = 0.691, D_loss_fake = 0.677, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.649, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 19/46, D_loss_real = 0.733, D_loss_fake = 0.656, G_loss = 0.800
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 20/46, D_loss_real = 0.643, D_loss_fake = 0.661, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 21/46, D_loss_real = 0.703, D_loss_fake = 0.657, G_loss = 0.783
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 22/46, D_loss_real = 0.649, D_loss_fake = 0.672, G_loss = 0.762
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 23/46, D_loss_real = 0.616, D_loss_fake = 0.761, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 24/46, D_loss_real = 0.645, D_loss_fake = 0.666, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 25/46, D_loss_real = 0.707, D_loss_fake = 0.606, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 26/46, D_loss_real = 0.733, D_loss_fake = 0.626, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 27/46, D_loss_real = 0.718, D_loss_fake = 0.643, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 28/46, D_loss_real = 0.661, D_loss_fake = 0.608, G_loss = 0.782
2/2 [=====] - 0s 5ms/step

Epoch 18, Iteration 29/46, D_loss_real = 0.651, D_loss_fake = 0.644, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 30/46, D_loss_real = 0.657, D_loss_fake = 0.663, G_loss = 0.779
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 31/46, D_loss_real = 0.631, D_loss_fake = 0.690, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 32/46, D_loss_real = 0.692, D_loss_fake = 0.677, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 33/46, D_loss_real = 0.651, D_loss_fake = 0.646, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 34/46, D_loss_real = 0.648, D_loss_fake = 0.653, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.637, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 36/46, D_loss_real = 0.717, D_loss_fake = 0.632, G_loss = 0.788
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.634, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 38/46, D_loss_real = 0.719, D_loss_fake = 0.656, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 39/46, D_loss_real = 0.713, D_loss_fake = 0.703, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.635, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 41/46, D_loss_real = 0.692, D_loss_fake = 0.665, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 42/46, D_loss_real = 0.646, D_loss_fake = 0.662, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 18, Iteration 43/46, D_loss_real = 0.606, D_loss_fake = 0.658, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 44/46, D_loss_real = 0.654, D_loss_fake = 0.697, G_loss = 0.772
2/2 [=====] - 0s 3ms/step

Epoch 18, Iteration 45/46, D_loss_real = 0.635, D_loss_fake = 0.678, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 18, Iteration 46/46, D_loss_real = 0.649, D_loss_fake = 0.666, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 1/46, D_loss_real = 0.630, D_loss_fake = 0.660, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 2/46, D_loss_real = 0.600, D_loss_fake = 0.690, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 3/46, D_loss_real = 0.690, D_loss_fake = 0.658, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 4/46, D_loss_real = 0.691, D_loss_fake = 0.655, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 5/46, D_loss_real = 0.678, D_loss_fake = 0.713, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 6/46, D_loss_real = 0.661, D_loss_fake = 0.625, G_loss = 0.894
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 7/46, D_loss_real = 0.715, D_loss_fake = 0.597, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 8/46, D_loss_real = 0.734, D_loss_fake = 0.602, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 9/46, D_loss_real = 0.683, D_loss_fake = 0.634, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 10/46, D_loss_real = 0.713, D_loss_fake = 0.598, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 11/46, D_loss_real = 0.724, D_loss_fake = 0.620, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 12/46, D_loss_real = 0.687, D_loss_fake = 0.649, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 13/46, D_loss_real = 0.700, D_loss_fake = 0.635, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 14/46, D_loss_real = 0.653, D_loss_fake = 0.600, G_loss = 0.839
2/2 [=====] - 0s 4ms/step

Epoch 19, Iteration 15/46, D_loss_real = 0.675, D_loss_fake = 0.665, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 16/46, D_loss_real = 0.632, D_loss_fake = 0.654, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 17/46, D_loss_real = 0.650, D_loss_fake = 0.685, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 18/46, D_loss_real = 0.666, D_loss_fake = 0.641, G_loss = 0.773
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 19/46, D_loss_real = 0.642, D_loss_fake = 0.680, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 20/46, D_loss_real = 0.674, D_loss_fake = 0.672, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 21/46, D_loss_real = 0.721, D_loss_fake = 0.617, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 22/46, D_loss_real = 0.696, D_loss_fake = 0.633, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 23/46, D_loss_real = 0.721, D_loss_fake = 0.603, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 24/46, D_loss_real = 0.630, D_loss_fake = 0.621, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 25/46, D_loss_real = 0.658, D_loss_fake = 0.634, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 26/46, D_loss_real = 0.646, D_loss_fake = 0.685, G_loss = 0.779
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 27/46, D_loss_real = 0.642, D_loss_fake = 0.706, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 28/46, D_loss_real = 0.620, D_loss_fake = 0.678, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 29/46, D_loss_real = 0.604, D_loss_fake = 0.657, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 30/46, D_loss_real = 0.674, D_loss_fake = 0.678, G_loss = 0.804
2/2 [=====] - 0s 3ms/step

Epoch 19, Iteration 31/46, D_loss_real = 0.584, D_loss_fake = 0.695, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 32/46, D_loss_real = 0.647, D_loss_fake = 0.686, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 33/46, D_loss_real = 0.587, D_loss_fake = 0.663, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 34/46, D_loss_real = 0.628, D_loss_fake = 0.675, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 35/46, D_loss_real = 0.587, D_loss_fake = 0.686, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 36/46, D_loss_real = 0.626, D_loss_fake = 0.634, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 37/46, D_loss_real = 0.665, D_loss_fake = 0.711, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 38/46, D_loss_real = 0.651, D_loss_fake = 0.661, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 39/46, D_loss_real = 0.609, D_loss_fake = 0.706, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 40/46, D_loss_real = 0.637, D_loss_fake = 0.643, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 41/46, D_loss_real = 0.600, D_loss_fake = 0.691, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 42/46, D_loss_real = 0.617, D_loss_fake = 0.632, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 43/46, D_loss_real = 0.627, D_loss_fake = 0.722, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 44/46, D_loss_real = 0.627, D_loss_fake = 0.606, G_loss = 0.845
2/2 [=====] - 0s 3ms/step
Epoch 19, Iteration 45/46, D_loss_real = 0.559, D_loss_fake = 0.708, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 19, Iteration 46/46, D_loss_real = 0.569, D_loss_fake = 0.661, G_loss = 0.833
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 1/46, D_loss_real = 0.570, D_loss_fake = 0.782, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 2/46, D_loss_real = 0.621, D_loss_fake = 0.703, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 3/46, D_loss_real = 0.679, D_loss_fake = 0.759, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 4/46, D_loss_real = 0.733, D_loss_fake = 0.640, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 5/46, D_loss_real = 0.684, D_loss_fake = 0.623, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 6/46, D_loss_real = 0.749, D_loss_fake = 0.623, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 7/46, D_loss_real = 0.764, D_loss_fake = 0.685, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 8/46, D_loss_real = 0.759, D_loss_fake = 0.585, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 9/46, D_loss_real = 0.750, D_loss_fake = 0.594, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 10/46, D_loss_real = 0.694, D_loss_fake = 0.602, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 11/46, D_loss_real = 0.692, D_loss_fake = 0.638, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 12/46, D_loss_real = 0.684, D_loss_fake = 0.623, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 13/46, D_loss_real = 0.649, D_loss_fake = 0.660, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 14/46, D_loss_real = 0.679, D_loss_fake = 0.612, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 15/46, D_loss_real = 0.601, D_loss_fake = 0.665, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 16/46, D_loss_real = 0.634, D_loss_fake = 0.659, G_loss = 0.871
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 17/46, D_loss_real = 0.667, D_loss_fake = 0.660, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.711, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 19/46, D_loss_real = 0.652, D_loss_fake = 0.660, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 20/46, D_loss_real = 0.596, D_loss_fake = 0.675, G_loss = 0.830
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 21/46, D_loss_real = 0.572, D_loss_fake = 0.689, G_loss = 0.872
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 22/46, D_loss_real = 0.702, D_loss_fake = 0.658, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 23/46, D_loss_real = 0.678, D_loss_fake = 0.625, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 24/46, D_loss_real = 0.741, D_loss_fake = 0.638, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 25/46, D_loss_real = 0.726, D_loss_fake = 0.681, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 26/46, D_loss_real = 0.661, D_loss_fake = 0.668, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 27/46, D_loss_real = 0.648, D_loss_fake = 0.649, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 28/46, D_loss_real = 0.685, D_loss_fake = 0.704, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 29/46, D_loss_real = 0.707, D_loss_fake = 0.610, G_loss = 0.894
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 30/46, D_loss_real = 0.676, D_loss_fake = 0.596, G_loss = 0.920
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 31/46, D_loss_real = 0.715, D_loss_fake = 0.559, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 32/46, D_loss_real = 0.724, D_loss_fake = 0.639, G_loss = 0.854
2/2 [=====] - 0s 4ms/step

Epoch 20, Iteration 33/46, D_loss_real = 0.698, D_loss_fake = 0.659, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 34/46, D_loss_real = 0.633, D_loss_fake = 0.609, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 35/46, D_loss_real = 0.584, D_loss_fake = 0.755, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 36/46, D_loss_real = 0.627, D_loss_fake = 0.683, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 37/46, D_loss_real = 0.634, D_loss_fake = 0.698, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 38/46, D_loss_real = 0.608, D_loss_fake = 0.676, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 39/46, D_loss_real = 0.669, D_loss_fake = 0.614, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 40/46, D_loss_real = 0.681, D_loss_fake = 0.718, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 41/46, D_loss_real = 0.681, D_loss_fake = 0.640, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 42/46, D_loss_real = 0.609, D_loss_fake = 0.599, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 43/46, D_loss_real = 0.629, D_loss_fake = 0.677, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 20, Iteration 44/46, D_loss_real = 0.723, D_loss_fake = 0.771, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 45/46, D_loss_real = 0.631, D_loss_fake = 0.711, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 20, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.656, G_loss = 0.797
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 1/46, D_loss_real = 0.641, D_loss_fake = 0.734, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 2/46, D_loss_real = 0.663, D_loss_fake = 0.658, G_loss = 0.846
2/2 [=====] - 0s 4ms/step

Epoch 21, Iteration 3/46, D_loss_real = 0.689, D_loss_fake = 0.639, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 4/46, D_loss_real = 0.632, D_loss_fake = 0.658, G_loss = 0.765
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 5/46, D_loss_real = 0.625, D_loss_fake = 0.706, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 6/46, D_loss_real = 0.721, D_loss_fake = 0.633, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 7/46, D_loss_real = 0.696, D_loss_fake = 0.655, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 8/46, D_loss_real = 0.663, D_loss_fake = 0.645, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 9/46, D_loss_real = 0.643, D_loss_fake = 0.660, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 10/46, D_loss_real = 0.643, D_loss_fake = 0.644, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 11/46, D_loss_real = 0.670, D_loss_fake = 0.596, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 12/46, D_loss_real = 0.717, D_loss_fake = 0.698, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 13/46, D_loss_real = 0.652, D_loss_fake = 0.660, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 14/46, D_loss_real = 0.664, D_loss_fake = 0.711, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 15/46, D_loss_real = 0.715, D_loss_fake = 0.598, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 16/46, D_loss_real = 0.680, D_loss_fake = 0.675, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 17/46, D_loss_real = 0.702, D_loss_fake = 0.622, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 18/46, D_loss_real = 0.674, D_loss_fake = 0.659, G_loss = 0.807
2/2 [=====] - 0s 3ms/step

Epoch 21, Iteration 19/46, D_loss_real = 0.607, D_loss_fake = 0.634, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 20/46, D_loss_real = 0.634, D_loss_fake = 0.631, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 21/46, D_loss_real = 0.617, D_loss_fake = 0.659, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 22/46, D_loss_real = 0.672, D_loss_fake = 0.683, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 23/46, D_loss_real = 0.713, D_loss_fake = 0.698, G_loss = 0.787
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 24/46, D_loss_real = 0.600, D_loss_fake = 0.699, G_loss = 0.787
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 25/46, D_loss_real = 0.647, D_loss_fake = 0.685, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 26/46, D_loss_real = 0.651, D_loss_fake = 0.705, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 27/46, D_loss_real = 0.715, D_loss_fake = 0.688, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 28/46, D_loss_real = 0.732, D_loss_fake = 0.616, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 29/46, D_loss_real = 0.713, D_loss_fake = 0.639, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 30/46, D_loss_real = 0.701, D_loss_fake = 0.647, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 31/46, D_loss_real = 0.754, D_loss_fake = 0.614, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 32/46, D_loss_real = 0.715, D_loss_fake = 0.602, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 33/46, D_loss_real = 0.669, D_loss_fake = 0.609, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 34/46, D_loss_real = 0.646, D_loss_fake = 0.670, G_loss = 0.837
2/2 [=====] - 0s 4ms/step

Epoch 21, Iteration 35/46, D_loss_real = 0.663, D_loss_fake = 0.631, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 36/46, D_loss_real = 0.753, D_loss_fake = 0.631, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 37/46, D_loss_real = 0.684, D_loss_fake = 0.568, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.609, G_loss = 0.781
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 39/46, D_loss_real = 0.675, D_loss_fake = 0.653, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 40/46, D_loss_real = 0.662, D_loss_fake = 0.676, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 41/46, D_loss_real = 0.627, D_loss_fake = 0.677, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 42/46, D_loss_real = 0.570, D_loss_fake = 0.680, G_loss = 0.790
2/2 [=====] - 0s 3ms/step
Epoch 21, Iteration 43/46, D_loss_real = 0.610, D_loss_fake = 0.686, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 44/46, D_loss_real = 0.601, D_loss_fake = 0.655, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 45/46, D_loss_real = 0.634, D_loss_fake = 0.672, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 21, Iteration 46/46, D_loss_real = 0.597, D_loss_fake = 0.698, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 1/46, D_loss_real = 0.584, D_loss_fake = 0.709, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 2/46, D_loss_real = 0.648, D_loss_fake = 0.692, G_loss = 0.796
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 3/46, D_loss_real = 0.663, D_loss_fake = 0.613, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 4/46, D_loss_real = 0.656, D_loss_fake = 0.659, G_loss = 0.802
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 5/46, D_loss_real = 0.641, D_loss_fake = 0.674, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 6/46, D_loss_real = 0.662, D_loss_fake = 0.723, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 7/46, D_loss_real = 0.680, D_loss_fake = 0.546, G_loss = 0.924
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.653, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 9/46, D_loss_real = 0.709, D_loss_fake = 0.644, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 10/46, D_loss_real = 0.815, D_loss_fake = 0.639, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 11/46, D_loss_real = 0.699, D_loss_fake = 0.700, G_loss = 0.748
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 12/46, D_loss_real = 0.667, D_loss_fake = 0.722, G_loss = 0.797
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.674, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 14/46, D_loss_real = 0.696, D_loss_fake = 0.700, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.660, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 16/46, D_loss_real = 0.675, D_loss_fake = 0.642, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 17/46, D_loss_real = 0.621, D_loss_fake = 0.668, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 18/46, D_loss_real = 0.691, D_loss_fake = 0.639, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 19/46, D_loss_real = 0.667, D_loss_fake = 0.637, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 20/46, D_loss_real = 0.625, D_loss_fake = 0.652, G_loss = 0.862
2/2 [=====] - 0s 3ms/step

Epoch 22, Iteration 21/46, D_loss_real = 0.705, D_loss_fake = 0.666, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 22/46, D_loss_real = 0.616, D_loss_fake = 0.673, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 23/46, D_loss_real = 0.651, D_loss_fake = 0.627, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 24/46, D_loss_real = 0.605, D_loss_fake = 0.676, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 25/46, D_loss_real = 0.569, D_loss_fake = 0.668, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 26/46, D_loss_real = 0.600, D_loss_fake = 0.694, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 27/46, D_loss_real = 0.617, D_loss_fake = 0.722, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 28/46, D_loss_real = 0.612, D_loss_fake = 0.737, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 29/46, D_loss_real = 0.626, D_loss_fake = 0.675, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 30/46, D_loss_real = 0.586, D_loss_fake = 0.597, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 31/46, D_loss_real = 0.606, D_loss_fake = 0.596, G_loss = 0.956
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 32/46, D_loss_real = 0.613, D_loss_fake = 0.604, G_loss = 0.895
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 33/46, D_loss_real = 0.583, D_loss_fake = 0.686, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 34/46, D_loss_real = 0.627, D_loss_fake = 0.621, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 35/46, D_loss_real = 0.661, D_loss_fake = 0.650, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 36/46, D_loss_real = 0.551, D_loss_fake = 0.620, G_loss = 0.877
2/2 [=====] - 0s 4ms/step

Epoch 22, Iteration 37/46, D_loss_real = 0.522, D_loss_fake = 0.674, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 38/46, D_loss_real = 0.473, D_loss_fake = 0.698, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 39/46, D_loss_real = 0.628, D_loss_fake = 0.678, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 40/46, D_loss_real = 0.645, D_loss_fake = 0.677, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 41/46, D_loss_real = 0.676, D_loss_fake = 0.640, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 42/46, D_loss_real = 0.729, D_loss_fake = 0.687, G_loss = 0.894
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 43/46, D_loss_real = 0.676, D_loss_fake = 0.710, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 22, Iteration 44/46, D_loss_real = 0.718, D_loss_fake = 0.653, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 45/46, D_loss_real = 0.728, D_loss_fake = 0.655, G_loss = 0.896
2/2 [=====] - 0s 3ms/step
Epoch 22, Iteration 46/46, D_loss_real = 0.707, D_loss_fake = 0.623, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 1/46, D_loss_real = 0.733, D_loss_fake = 0.621, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 2/46, D_loss_real = 0.749, D_loss_fake = 0.642, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 3/46, D_loss_real = 0.716, D_loss_fake = 0.678, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.614, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.589, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 6/46, D_loss_real = 0.676, D_loss_fake = 0.595, G_loss = 0.899
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 7/46, D_loss_real = 0.726, D_loss_fake = 0.617, G_loss = 0.891
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 8/46, D_loss_real = 0.613, D_loss_fake = 0.582, G_loss = 0.897
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 9/46, D_loss_real = 0.629, D_loss_fake = 0.611, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 10/46, D_loss_real = 0.601, D_loss_fake = 0.659, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 11/46, D_loss_real = 0.646, D_loss_fake = 0.700, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 12/46, D_loss_real = 0.633, D_loss_fake = 0.733, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 13/46, D_loss_real = 0.571, D_loss_fake = 0.662, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 14/46, D_loss_real = 0.598, D_loss_fake = 0.712, G_loss = 0.772
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 15/46, D_loss_real = 0.654, D_loss_fake = 0.728, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 16/46, D_loss_real = 0.659, D_loss_fake = 0.660, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 17/46, D_loss_real = 0.642, D_loss_fake = 0.675, G_loss = 0.841
2/2 [=====] - 0s 6ms/step
Epoch 23, Iteration 18/46, D_loss_real = 0.673, D_loss_fake = 0.610, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 19/46, D_loss_real = 0.628, D_loss_fake = 0.652, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 20/46, D_loss_real = 0.657, D_loss_fake = 0.668, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 21/46, D_loss_real = 0.659, D_loss_fake = 0.650, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 22/46, D_loss_real = 0.657, D_loss_fake = 0.594, G_loss = 0.890
2/2 [=====] - 0s 3ms/step

Epoch 23, Iteration 23/46, D_loss_real = 0.650, D_loss_fake = 0.604, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 24/46, D_loss_real = 0.648, D_loss_fake = 0.597, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 25/46, D_loss_real = 0.617, D_loss_fake = 0.679, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 26/46, D_loss_real = 0.662, D_loss_fake = 0.591, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 27/46, D_loss_real = 0.558, D_loss_fake = 0.726, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 28/46, D_loss_real = 0.663, D_loss_fake = 0.763, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 29/46, D_loss_real = 0.631, D_loss_fake = 0.692, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 30/46, D_loss_real = 0.713, D_loss_fake = 0.710, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 31/46, D_loss_real = 0.680, D_loss_fake = 0.718, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.700, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 33/46, D_loss_real = 0.709, D_loss_fake = 0.630, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 34/46, D_loss_real = 0.660, D_loss_fake = 0.627, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 35/46, D_loss_real = 0.699, D_loss_fake = 0.690, G_loss = 0.966
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 36/46, D_loss_real = 0.744, D_loss_fake = 0.559, G_loss = 0.918
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 37/46, D_loss_real = 0.746, D_loss_fake = 0.595, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.624, G_loss = 0.839
2/2 [=====] - 0s 4ms/step

Epoch 23, Iteration 39/46, D_loss_real = 0.682, D_loss_fake = 0.644, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 40/46, D_loss_real = 0.669, D_loss_fake = 0.667, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.622, G_loss = 0.812
2/2 [=====] - 0s 5ms/step
Epoch 23, Iteration 42/46, D_loss_real = 0.626, D_loss_fake = 0.645, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 23, Iteration 43/46, D_loss_real = 0.661, D_loss_fake = 0.645, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 44/46, D_loss_real = 0.622, D_loss_fake = 0.684, G_loss = 0.757
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 45/46, D_loss_real = 0.553, D_loss_fake = 0.728, G_loss = 0.750
2/2 [=====] - 0s 3ms/step
Epoch 23, Iteration 46/46, D_loss_real = 0.549, D_loss_fake = 0.692, G_loss = 0.759
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 1/46, D_loss_real = 0.591, D_loss_fake = 0.707, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 2/46, D_loss_real = 0.599, D_loss_fake = 0.716, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 3/46, D_loss_real = 0.706, D_loss_fake = 0.663, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 4/46, D_loss_real = 0.695, D_loss_fake = 0.646, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 5/46, D_loss_real = 0.710, D_loss_fake = 0.632, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 6/46, D_loss_real = 0.653, D_loss_fake = 0.586, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 7/46, D_loss_real = 0.678, D_loss_fake = 0.668, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 8/46, D_loss_real = 0.647, D_loss_fake = 0.671, G_loss = 0.833
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 9/46, D_loss_real = 0.598, D_loss_fake = 0.661, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 10/46, D_loss_real = 0.657, D_loss_fake = 0.719, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 11/46, D_loss_real = 0.669, D_loss_fake = 0.652, G_loss = 0.808
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 12/46, D_loss_real = 0.597, D_loss_fake = 0.627, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 13/46, D_loss_real = 0.585, D_loss_fake = 0.685, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 14/46, D_loss_real = 0.613, D_loss_fake = 0.677, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 15/46, D_loss_real = 0.627, D_loss_fake = 0.672, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 16/46, D_loss_real = 0.686, D_loss_fake = 0.717, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 17/46, D_loss_real = 0.654, D_loss_fake = 0.678, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 18/46, D_loss_real = 0.705, D_loss_fake = 0.605, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 19/46, D_loss_real = 0.662, D_loss_fake = 0.650, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 20/46, D_loss_real = 0.623, D_loss_fake = 0.597, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 21/46, D_loss_real = 0.621, D_loss_fake = 0.708, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 22/46, D_loss_real = 0.678, D_loss_fake = 0.689, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 23/46, D_loss_real = 0.636, D_loss_fake = 0.709, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 24/46, D_loss_real = 0.652, D_loss_fake = 0.686, G_loss = 0.833
2/2 [=====] - 0s 3ms/step

Epoch 24, Iteration 25/46, D_loss_real = 0.666, D_loss_fake = 0.644, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 26/46, D_loss_real = 0.715, D_loss_fake = 0.608, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 27/46, D_loss_real = 0.707, D_loss_fake = 0.600, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 28/46, D_loss_real = 0.760, D_loss_fake = 0.596, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 29/46, D_loss_real = 0.653, D_loss_fake = 0.638, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 30/46, D_loss_real = 0.676, D_loss_fake = 0.658, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.658, G_loss = 0.860
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 32/46, D_loss_real = 0.682, D_loss_fake = 0.630, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 33/46, D_loss_real = 0.645, D_loss_fake = 0.611, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 34/46, D_loss_real = 0.641, D_loss_fake = 0.612, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 35/46, D_loss_real = 0.624, D_loss_fake = 0.702, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 36/46, D_loss_real = 0.590, D_loss_fake = 0.685, G_loss = 0.791
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 37/46, D_loss_real = 0.634, D_loss_fake = 0.762, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 38/46, D_loss_real = 0.649, D_loss_fake = 0.695, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 39/46, D_loss_real = 0.666, D_loss_fake = 0.694, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 40/46, D_loss_real = 0.690, D_loss_fake = 0.640, G_loss = 0.848
2/2 [=====] - 0s 4ms/step

Epoch 24, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.619, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 42/46, D_loss_real = 0.634, D_loss_fake = 0.649, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 43/46, D_loss_real = 0.616, D_loss_fake = 0.668, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.667, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 24, Iteration 45/46, D_loss_real = 0.670, D_loss_fake = 0.690, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 24, Iteration 46/46, D_loss_real = 0.717, D_loss_fake = 0.651, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 1/46, D_loss_real = 0.686, D_loss_fake = 0.690, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 2/46, D_loss_real = 0.755, D_loss_fake = 0.594, G_loss = 0.896
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 3/46, D_loss_real = 0.747, D_loss_fake = 0.604, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 4/46, D_loss_real = 0.663, D_loss_fake = 0.634, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 5/46, D_loss_real = 0.644, D_loss_fake = 0.646, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 6/46, D_loss_real = 0.650, D_loss_fake = 0.682, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.672, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 8/46, D_loss_real = 0.608, D_loss_fake = 0.602, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 9/46, D_loss_real = 0.648, D_loss_fake = 0.654, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 10/46, D_loss_real = 0.654, D_loss_fake = 0.651, G_loss = 0.832
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 11/46, D_loss_real = 0.624, D_loss_fake = 0.631, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 12/46, D_loss_real = 0.664, D_loss_fake = 0.695, G_loss = 0.892
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 13/46, D_loss_real = 0.747, D_loss_fake = 0.675, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 14/46, D_loss_real = 0.736, D_loss_fake = 0.617, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 15/46, D_loss_real = 0.696, D_loss_fake = 0.635, G_loss = 0.920
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 16/46, D_loss_real = 0.681, D_loss_fake = 0.670, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 17/46, D_loss_real = 0.676, D_loss_fake = 0.629, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 18/46, D_loss_real = 0.657, D_loss_fake = 0.644, G_loss = 0.784
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 19/46, D_loss_real = 0.677, D_loss_fake = 0.629, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 20/46, D_loss_real = 0.700, D_loss_fake = 0.667, G_loss = 0.793
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 21/46, D_loss_real = 0.688, D_loss_fake = 0.636, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 22/46, D_loss_real = 0.691, D_loss_fake = 0.636, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 23/46, D_loss_real = 0.742, D_loss_fake = 0.653, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 24/46, D_loss_real = 0.701, D_loss_fake = 0.643, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 25/46, D_loss_real = 0.738, D_loss_fake = 0.633, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 26/46, D_loss_real = 0.686, D_loss_fake = 0.654, G_loss = 0.829
2/2 [=====] - 0s 3ms/step

Epoch 25, Iteration 27/46, D_loss_real = 0.636, D_loss_fake = 0.638, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 28/46, D_loss_real = 0.753, D_loss_fake = 0.631, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 29/46, D_loss_real = 0.635, D_loss_fake = 0.684, G_loss = 0.788
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 30/46, D_loss_real = 0.630, D_loss_fake = 0.662, G_loss = 0.785
2/2 [=====] - 0s 5ms/step
Epoch 25, Iteration 31/46, D_loss_real = 0.647, D_loss_fake = 0.697, G_loss = 0.782
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.703, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 33/46, D_loss_real = 0.636, D_loss_fake = 0.668, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 34/46, D_loss_real = 0.687, D_loss_fake = 0.678, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 35/46, D_loss_real = 0.650, D_loss_fake = 0.612, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 36/46, D_loss_real = 0.661, D_loss_fake = 0.725, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 37/46, D_loss_real = 0.683, D_loss_fake = 0.643, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 38/46, D_loss_real = 0.705, D_loss_fake = 0.604, G_loss = 0.835
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 39/46, D_loss_real = 0.662, D_loss_fake = 0.625, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 40/46, D_loss_real = 0.674, D_loss_fake = 0.661, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 41/46, D_loss_real = 0.673, D_loss_fake = 0.630, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 25, Iteration 42/46, D_loss_real = 0.706, D_loss_fake = 0.622, G_loss = 0.832
2/2 [=====] - 0s 4ms/step

Epoch 25, Iteration 43/46, D_loss_real = 0.716, D_loss_fake = 0.631, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 44/46, D_loss_real = 0.663, D_loss_fake = 0.652, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 45/46, D_loss_real = 0.696, D_loss_fake = 0.638, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 25, Iteration 46/46, D_loss_real = 0.662, D_loss_fake = 0.655, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.643, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 2/46, D_loss_real = 0.683, D_loss_fake = 0.611, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 3/46, D_loss_real = 0.685, D_loss_fake = 0.630, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 4/46, D_loss_real = 0.679, D_loss_fake = 0.667, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 5/46, D_loss_real = 0.700, D_loss_fake = 0.632, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 6/46, D_loss_real = 0.685, D_loss_fake = 0.710, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 7/46, D_loss_real = 0.684, D_loss_fake = 0.653, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 8/46, D_loss_real = 0.705, D_loss_fake = 0.687, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.623, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 10/46, D_loss_real = 0.673, D_loss_fake = 0.653, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 11/46, D_loss_real = 0.641, D_loss_fake = 0.654, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 12/46, D_loss_real = 0.651, D_loss_fake = 0.581, G_loss = 0.837
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 13/46, D_loss_real = 0.677, D_loss_fake = 0.634, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 14/46, D_loss_real = 0.650, D_loss_fake = 0.720, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 15/46, D_loss_real = 0.654, D_loss_fake = 0.648, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 16/46, D_loss_real = 0.630, D_loss_fake = 0.647, G_loss = 0.771
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 17/46, D_loss_real = 0.597, D_loss_fake = 0.699, G_loss = 0.764
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 18/46, D_loss_real = 0.634, D_loss_fake = 0.711, G_loss = 0.813
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 19/46, D_loss_real = 0.586, D_loss_fake = 0.663, G_loss = 0.770
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 20/46, D_loss_real = 0.571, D_loss_fake = 0.753, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 21/46, D_loss_real = 0.714, D_loss_fake = 0.625, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 22/46, D_loss_real = 0.647, D_loss_fake = 0.689, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 23/46, D_loss_real = 0.636, D_loss_fake = 0.697, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 24/46, D_loss_real = 0.663, D_loss_fake = 0.572, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 25/46, D_loss_real = 0.672, D_loss_fake = 0.620, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 26/46, D_loss_real = 0.746, D_loss_fake = 0.581, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 27/46, D_loss_real = 0.712, D_loss_fake = 0.638, G_loss = 0.896
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.593, G_loss = 0.845
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 29/46, D_loss_real = 0.699, D_loss_fake = 0.603, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 30/46, D_loss_real = 0.710, D_loss_fake = 0.661, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 31/46, D_loss_real = 0.756, D_loss_fake = 0.683, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 32/46, D_loss_real = 0.680, D_loss_fake = 0.634, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 33/46, D_loss_real = 0.666, D_loss_fake = 0.647, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 34/46, D_loss_real = 0.639, D_loss_fake = 0.645, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 35/46, D_loss_real = 0.606, D_loss_fake = 0.685, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 36/46, D_loss_real = 0.604, D_loss_fake = 0.647, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 37/46, D_loss_real = 0.643, D_loss_fake = 0.602, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 38/46, D_loss_real = 0.611, D_loss_fake = 0.634, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 39/46, D_loss_real = 0.602, D_loss_fake = 0.682, G_loss = 0.747
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 40/46, D_loss_real = 0.581, D_loss_fake = 0.813, G_loss = 0.752
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 41/46, D_loss_real = 0.656, D_loss_fake = 0.766, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 26, Iteration 42/46, D_loss_real = 0.703, D_loss_fake = 0.632, G_loss = 0.896
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 43/46, D_loss_real = 0.697, D_loss_fake = 0.623, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 44/46, D_loss_real = 0.637, D_loss_fake = 0.661, G_loss = 0.847
2/2 [=====] - 0s 3ms/step

Epoch 26, Iteration 45/46, D_loss_real = 0.641, D_loss_fake = 0.638, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 26, Iteration 46/46, D_loss_real = 0.604, D_loss_fake = 0.725, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 1/46, D_loss_real = 0.610, D_loss_fake = 0.795, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 2/46, D_loss_real = 0.645, D_loss_fake = 0.738, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 3/46, D_loss_real = 0.682, D_loss_fake = 0.691, G_loss = 0.786
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 4/46, D_loss_real = 0.655, D_loss_fake = 0.669, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 5/46, D_loss_real = 0.729, D_loss_fake = 0.615, G_loss = 0.962
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 6/46, D_loss_real = 0.675, D_loss_fake = 0.609, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 7/46, D_loss_real = 0.691, D_loss_fake = 0.595, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 8/46, D_loss_real = 0.726, D_loss_fake = 0.611, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 9/46, D_loss_real = 0.643, D_loss_fake = 0.619, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 10/46, D_loss_real = 0.649, D_loss_fake = 0.639, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 11/46, D_loss_real = 0.654, D_loss_fake = 0.666, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 12/46, D_loss_real = 0.709, D_loss_fake = 0.671, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 13/46, D_loss_real = 0.661, D_loss_fake = 0.629, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 14/46, D_loss_real = 0.608, D_loss_fake = 0.642, G_loss = 0.835
2/2 [=====] - 0s 3ms/step

Epoch 27, Iteration 15/46, D_loss_real = 0.650, D_loss_fake = 0.645, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 16/46, D_loss_real = 0.558, D_loss_fake = 0.619, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 17/46, D_loss_real = 0.527, D_loss_fake = 0.725, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 18/46, D_loss_real = 0.596, D_loss_fake = 0.643, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 19/46, D_loss_real = 0.636, D_loss_fake = 0.779, G_loss = 0.776
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 20/46, D_loss_real = 0.586, D_loss_fake = 0.738, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 21/46, D_loss_real = 0.651, D_loss_fake = 0.700, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 22/46, D_loss_real = 0.716, D_loss_fake = 0.633, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 23/46, D_loss_real = 0.734, D_loss_fake = 0.517, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 24/46, D_loss_real = 0.656, D_loss_fake = 0.612, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 25/46, D_loss_real = 0.641, D_loss_fake = 0.640, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 26/46, D_loss_real = 0.662, D_loss_fake = 0.670, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 27/46, D_loss_real = 0.708, D_loss_fake = 0.619, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 28/46, D_loss_real = 0.733, D_loss_fake = 0.611, G_loss = 0.895
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 29/46, D_loss_real = 0.773, D_loss_fake = 0.620, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 30/46, D_loss_real = 0.670, D_loss_fake = 0.602, G_loss = 0.849
2/2 [=====] - 0s 4ms/step

Epoch 27, Iteration 31/46, D_loss_real = 0.712, D_loss_fake = 0.619, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 32/46, D_loss_real = 0.698, D_loss_fake = 0.613, G_loss = 0.800
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 33/46, D_loss_real = 0.614, D_loss_fake = 0.666, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 34/46, D_loss_real = 0.675, D_loss_fake = 0.689, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 35/46, D_loss_real = 0.663, D_loss_fake = 0.630, G_loss = 0.823
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 36/46, D_loss_real = 0.627, D_loss_fake = 0.631, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 37/46, D_loss_real = 0.641, D_loss_fake = 0.691, G_loss = 0.768
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 38/46, D_loss_real = 0.576, D_loss_fake = 0.747, G_loss = 0.793
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 39/46, D_loss_real = 0.639, D_loss_fake = 0.652, G_loss = 0.749
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 40/46, D_loss_real = 0.561, D_loss_fake = 0.694, G_loss = 0.774
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 41/46, D_loss_real = 0.583, D_loss_fake = 0.650, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 42/46, D_loss_real = 0.640, D_loss_fake = 0.704, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 43/46, D_loss_real = 0.606, D_loss_fake = 0.620, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 44/46, D_loss_real = 0.646, D_loss_fake = 0.663, G_loss = 0.860
2/2 [=====] - 0s 4ms/step
Epoch 27, Iteration 45/46, D_loss_real = 0.626, D_loss_fake = 0.669, G_loss = 0.799
2/2 [=====] - 0s 3ms/step
Epoch 27, Iteration 46/46, D_loss_real = 0.592, D_loss_fake = 0.682, G_loss = 0.797
2/2 [=====] - 0s 3ms/step

Epoch 28, Iteration 1/46, D_loss_real = 0.578, D_loss_fake = 0.727, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 2/46, D_loss_real = 0.645, D_loss_fake = 0.647, G_loss = 0.822
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 3/46, D_loss_real = 0.658, D_loss_fake = 0.658, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 4/46, D_loss_real = 0.629, D_loss_fake = 0.667, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 5/46, D_loss_real = 0.625, D_loss_fake = 0.617, G_loss = 0.846
2/2 [=====] - 0s 5ms/step
Epoch 28, Iteration 6/46, D_loss_real = 0.665, D_loss_fake = 0.653, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 7/46, D_loss_real = 0.607, D_loss_fake = 0.672, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 8/46, D_loss_real = 0.709, D_loss_fake = 0.702, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 9/46, D_loss_real = 0.648, D_loss_fake = 0.629, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 10/46, D_loss_real = 0.679, D_loss_fake = 0.737, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 11/46, D_loss_real = 0.670, D_loss_fake = 0.736, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 12/46, D_loss_real = 0.671, D_loss_fake = 0.710, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 13/46, D_loss_real = 0.710, D_loss_fake = 0.608, G_loss = 0.940
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 14/46, D_loss_real = 0.681, D_loss_fake = 0.565, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 15/46, D_loss_real = 0.695, D_loss_fake = 0.591, G_loss = 0.964
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 16/46, D_loss_real = 0.657, D_loss_fake = 0.622, G_loss = 0.948
2/2 [=====] - 0s 4ms/step

Epoch 28, Iteration 17/46, D_loss_real = 0.753, D_loss_fake = 0.588, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 18/46, D_loss_real = 0.647, D_loss_fake = 0.622, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 19/46, D_loss_real = 0.684, D_loss_fake = 0.644, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.639, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 21/46, D_loss_real = 0.633, D_loss_fake = 0.634, G_loss = 0.782
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 22/46, D_loss_real = 0.596, D_loss_fake = 0.677, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.616, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 24/46, D_loss_real = 0.621, D_loss_fake = 0.631, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 25/46, D_loss_real = 0.628, D_loss_fake = 0.607, G_loss = 0.937
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 26/46, D_loss_real = 0.608, D_loss_fake = 0.699, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.587, G_loss = 0.975
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 28/46, D_loss_real = 0.763, D_loss_fake = 0.616, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.671, G_loss = 0.783
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 30/46, D_loss_real = 0.636, D_loss_fake = 0.705, G_loss = 0.751
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 31/46, D_loss_real = 0.651, D_loss_fake = 0.733, G_loss = 0.798
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 32/46, D_loss_real = 0.606, D_loss_fake = 0.706, G_loss = 0.789
2/2 [=====] - 0s 4ms/step

Epoch 28, Iteration 33/46, D_loss_real = 0.618, D_loss_fake = 0.654, G_loss = 0.855
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 34/46, D_loss_real = 0.651, D_loss_fake = 0.708, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 35/46, D_loss_real = 0.689, D_loss_fake = 0.666, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 36/46, D_loss_real = 0.699, D_loss_fake = 0.617, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 37/46, D_loss_real = 0.651, D_loss_fake = 0.583, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 38/46, D_loss_real = 0.672, D_loss_fake = 0.635, G_loss = 0.871
2/2 [=====] - 0s 3ms/step
Epoch 28, Iteration 39/46, D_loss_real = 0.699, D_loss_fake = 0.633, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 40/46, D_loss_real = 0.706, D_loss_fake = 0.616, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 41/46, D_loss_real = 0.702, D_loss_fake = 0.666, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 42/46, D_loss_real = 0.720, D_loss_fake = 0.699, G_loss = 0.802
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 43/46, D_loss_real = 0.703, D_loss_fake = 0.606, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 44/46, D_loss_real = 0.715, D_loss_fake = 0.669, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 45/46, D_loss_real = 0.665, D_loss_fake = 0.681, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 28, Iteration 46/46, D_loss_real = 0.645, D_loss_fake = 0.678, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 1/46, D_loss_real = 0.655, D_loss_fake = 0.711, G_loss = 0.812
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 2/46, D_loss_real = 0.666, D_loss_fake = 0.669, G_loss = 0.867
2/2 [=====] - 0s 5ms/step

Epoch 29, Iteration 3/46, D_loss_real = 0.633, D_loss_fake = 0.595, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 4/46, D_loss_real = 0.659, D_loss_fake = 0.665, G_loss = 0.839
2/2 [=====] - 0s 5ms/step
Epoch 29, Iteration 5/46, D_loss_real = 0.639, D_loss_fake = 0.638, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 6/46, D_loss_real = 0.614, D_loss_fake = 0.666, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 7/46, D_loss_real = 0.639, D_loss_fake = 0.655, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 8/46, D_loss_real = 0.615, D_loss_fake = 0.640, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 9/46, D_loss_real = 0.613, D_loss_fake = 0.637, G_loss = 0.809
2/2 [=====] - 0s 5ms/step
Epoch 29, Iteration 10/46, D_loss_real = 0.607, D_loss_fake = 0.724, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 11/46, D_loss_real = 0.627, D_loss_fake = 0.712, G_loss = 0.822
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 12/46, D_loss_real = 0.602, D_loss_fake = 0.734, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 13/46, D_loss_real = 0.624, D_loss_fake = 0.657, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 14/46, D_loss_real = 0.614, D_loss_fake = 0.658, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 15/46, D_loss_real = 0.670, D_loss_fake = 0.733, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 16/46, D_loss_real = 0.648, D_loss_fake = 0.667, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 17/46, D_loss_real = 0.679, D_loss_fake = 0.614, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 18/46, D_loss_real = 0.625, D_loss_fake = 0.630, G_loss = 0.871
2/2 [=====] - 0s 4ms/step

Epoch 29, Iteration 19/46, D_loss_real = 0.710, D_loss_fake = 0.636, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 20/46, D_loss_real = 0.667, D_loss_fake = 0.645, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 21/46, D_loss_real = 0.634, D_loss_fake = 0.649, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 22/46, D_loss_real = 0.657, D_loss_fake = 0.704, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 23/46, D_loss_real = 0.650, D_loss_fake = 0.653, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 24/46, D_loss_real = 0.633, D_loss_fake = 0.752, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 25/46, D_loss_real = 0.671, D_loss_fake = 0.697, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 26/46, D_loss_real = 0.669, D_loss_fake = 0.667, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 27/46, D_loss_real = 0.634, D_loss_fake = 0.589, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 28/46, D_loss_real = 0.682, D_loss_fake = 0.666, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 29/46, D_loss_real = 0.715, D_loss_fake = 0.670, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.709, G_loss = 0.803
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.652, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 32/46, D_loss_real = 0.690, D_loss_fake = 0.717, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 33/46, D_loss_real = 0.721, D_loss_fake = 0.628, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 34/46, D_loss_real = 0.733, D_loss_fake = 0.595, G_loss = 0.875
2/2 [=====] - 0s 3ms/step

Epoch 29, Iteration 35/46, D_loss_real = 0.700, D_loss_fake = 0.594, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 36/46, D_loss_real = 0.678, D_loss_fake = 0.630, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 37/46, D_loss_real = 0.640, D_loss_fake = 0.627, G_loss = 0.815
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 38/46, D_loss_real = 0.650, D_loss_fake = 0.692, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 39/46, D_loss_real = 0.616, D_loss_fake = 0.646, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 40/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 41/46, D_loss_real = 0.716, D_loss_fake = 0.674, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 42/46, D_loss_real = 0.609, D_loss_fake = 0.647, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 43/46, D_loss_real = 0.646, D_loss_fake = 0.648, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 29, Iteration 44/46, D_loss_real = 0.623, D_loss_fake = 0.710, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 45/46, D_loss_real = 0.629, D_loss_fake = 0.638, G_loss = 0.920
2/2 [=====] - 0s 3ms/step
Epoch 29, Iteration 46/46, D_loss_real = 0.608, D_loss_fake = 0.665, G_loss = 0.918
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 1/46, D_loss_real = 0.675, D_loss_fake = 0.581, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 2/46, D_loss_real = 0.669, D_loss_fake = 0.629, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 3/46, D_loss_real = 0.694, D_loss_fake = 0.681, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 4/46, D_loss_real = 0.716, D_loss_fake = 0.549, G_loss = 0.881
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 5/46, D_loss_real = 0.670, D_loss_fake = 0.600, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 6/46, D_loss_real = 0.605, D_loss_fake = 0.616, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 7/46, D_loss_real = 0.655, D_loss_fake = 0.663, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 8/46, D_loss_real = 0.721, D_loss_fake = 0.610, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 9/46, D_loss_real = 0.647, D_loss_fake = 0.640, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 10/46, D_loss_real = 0.683, D_loss_fake = 0.627, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 11/46, D_loss_real = 0.654, D_loss_fake = 0.700, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 12/46, D_loss_real = 0.628, D_loss_fake = 0.620, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 13/46, D_loss_real = 0.640, D_loss_fake = 0.645, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 14/46, D_loss_real = 0.615, D_loss_fake = 0.693, G_loss = 0.824
2/2 [=====] - 0s 5ms/step
Epoch 30, Iteration 15/46, D_loss_real = 0.567, D_loss_fake = 0.734, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 16/46, D_loss_real = 0.636, D_loss_fake = 0.660, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 17/46, D_loss_real = 0.640, D_loss_fake = 0.661, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 18/46, D_loss_real = 0.652, D_loss_fake = 0.620, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 19/46, D_loss_real = 0.723, D_loss_fake = 0.662, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 20/46, D_loss_real = 0.657, D_loss_fake = 0.647, G_loss = 0.897
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 21/46, D_loss_real = 0.691, D_loss_fake = 0.621, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 22/46, D_loss_real = 0.716, D_loss_fake = 0.604, G_loss = 0.935
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 23/46, D_loss_real = 0.686, D_loss_fake = 0.604, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 24/46, D_loss_real = 0.689, D_loss_fake = 0.628, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 25/46, D_loss_real = 0.704, D_loss_fake = 0.626, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 26/46, D_loss_real = 0.670, D_loss_fake = 0.628, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 27/46, D_loss_real = 0.711, D_loss_fake = 0.655, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 28/46, D_loss_real = 0.708, D_loss_fake = 0.609, G_loss = 0.939
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 29/46, D_loss_real = 0.758, D_loss_fake = 0.598, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 30/46, D_loss_real = 0.660, D_loss_fake = 0.629, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 31/46, D_loss_real = 0.653, D_loss_fake = 0.584, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 32/46, D_loss_real = 0.711, D_loss_fake = 0.646, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 33/46, D_loss_real = 0.646, D_loss_fake = 0.616, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 34/46, D_loss_real = 0.681, D_loss_fake = 0.628, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 35/46, D_loss_real = 0.669, D_loss_fake = 0.630, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.663, G_loss = 0.844
2/2 [=====] - 0s 3ms/step

Epoch 30, Iteration 37/46, D_loss_real = 0.679, D_loss_fake = 0.644, G_loss = 0.829
2/2 [=====] - 0s 7ms/step
Epoch 30, Iteration 38/46, D_loss_real = 0.642, D_loss_fake = 0.599, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 39/46, D_loss_real = 0.650, D_loss_fake = 0.638, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 40/46, D_loss_real = 0.633, D_loss_fake = 0.656, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 41/46, D_loss_real = 0.634, D_loss_fake = 0.701, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 42/46, D_loss_real = 0.633, D_loss_fake = 0.691, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 43/46, D_loss_real = 0.592, D_loss_fake = 0.680, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 44/46, D_loss_real = 0.535, D_loss_fake = 0.666, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 30, Iteration 45/46, D_loss_real = 0.531, D_loss_fake = 0.754, G_loss = 0.745
2/2 [=====] - 0s 4ms/step
Epoch 30, Iteration 46/46, D_loss_real = 0.557, D_loss_fake = 0.822, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 1/46, D_loss_real = 0.604, D_loss_fake = 0.646, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 2/46, D_loss_real = 0.678, D_loss_fake = 0.637, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 3/46, D_loss_real = 0.675, D_loss_fake = 0.598, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 4/46, D_loss_real = 0.685, D_loss_fake = 0.609, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 5/46, D_loss_real = 0.655, D_loss_fake = 0.624, G_loss = 0.791
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 6/46, D_loss_real = 0.664, D_loss_fake = 0.684, G_loss = 0.824
2/2 [=====] - 0s 4ms/step

Epoch 31, Iteration 7/46, D_loss_real = 0.662, D_loss_fake = 0.662, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 8/46, D_loss_real = 0.646, D_loss_fake = 0.619, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 9/46, D_loss_real = 0.573, D_loss_fake = 0.638, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 10/46, D_loss_real = 0.655, D_loss_fake = 0.715, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.614, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 12/46, D_loss_real = 0.688, D_loss_fake = 0.616, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 13/46, D_loss_real = 0.599, D_loss_fake = 0.597, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 14/46, D_loss_real = 0.714, D_loss_fake = 0.618, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 15/46, D_loss_real = 0.706, D_loss_fake = 0.670, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 16/46, D_loss_real = 0.673, D_loss_fake = 0.729, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 17/46, D_loss_real = 0.674, D_loss_fake = 0.640, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 18/46, D_loss_real = 0.717, D_loss_fake = 0.636, G_loss = 0.896
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 19/46, D_loss_real = 0.659, D_loss_fake = 0.590, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 20/46, D_loss_real = 0.664, D_loss_fake = 0.579, G_loss = 0.970
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 21/46, D_loss_real = 0.616, D_loss_fake = 0.559, G_loss = 0.941
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 22/46, D_loss_real = 0.669, D_loss_fake = 0.625, G_loss = 0.914
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 23/46, D_loss_real = 0.687, D_loss_fake = 0.629, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 24/46, D_loss_real = 0.676, D_loss_fake = 0.651, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 25/46, D_loss_real = 0.661, D_loss_fake = 0.682, G_loss = 0.809
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 26/46, D_loss_real = 0.649, D_loss_fake = 0.707, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 27/46, D_loss_real = 0.662, D_loss_fake = 0.750, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 28/46, D_loss_real = 0.677, D_loss_fake = 0.682, G_loss = 0.903
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 29/46, D_loss_real = 0.642, D_loss_fake = 0.609, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 30/46, D_loss_real = 0.732, D_loss_fake = 0.627, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 31/46, D_loss_real = 0.691, D_loss_fake = 0.623, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.638, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 33/46, D_loss_real = 0.681, D_loss_fake = 0.597, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 34/46, D_loss_real = 0.663, D_loss_fake = 0.625, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 35/46, D_loss_real = 0.615, D_loss_fake = 0.624, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 36/46, D_loss_real = 0.645, D_loss_fake = 0.657, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 37/46, D_loss_real = 0.699, D_loss_fake = 0.669, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 38/46, D_loss_real = 0.675, D_loss_fake = 0.666, G_loss = 0.885
2/2 [=====] - 0s 3ms/step

Epoch 31, Iteration 39/46, D_loss_real = 0.604, D_loss_fake = 0.672, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 40/46, D_loss_real = 0.580, D_loss_fake = 0.632, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 41/46, D_loss_real = 0.690, D_loss_fake = 0.669, G_loss = 0.895
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 42/46, D_loss_real = 0.733, D_loss_fake = 0.687, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.644, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.599, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 31, Iteration 45/46, D_loss_real = 0.740, D_loss_fake = 0.622, G_loss = 0.896
2/2 [=====] - 0s 3ms/step
Epoch 31, Iteration 46/46, D_loss_real = 0.698, D_loss_fake = 0.620, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 1/46, D_loss_real = 0.663, D_loss_fake = 0.683, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 2/46, D_loss_real = 0.637, D_loss_fake = 0.597, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 3/46, D_loss_real = 0.635, D_loss_fake = 0.662, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 4/46, D_loss_real = 0.609, D_loss_fake = 0.686, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 5/46, D_loss_real = 0.637, D_loss_fake = 0.690, G_loss = 0.810
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 6/46, D_loss_real = 0.644, D_loss_fake = 0.632, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 7/46, D_loss_real = 0.589, D_loss_fake = 0.689, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 8/46, D_loss_real = 0.619, D_loss_fake = 0.700, G_loss = 0.791
2/2 [=====] - 0s 3ms/step

Epoch 32, Iteration 9/46, D_loss_real = 0.690, D_loss_fake = 0.677, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 10/46, D_loss_real = 0.693, D_loss_fake = 0.647, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 11/46, D_loss_real = 0.722, D_loss_fake = 0.631, G_loss = 1.052
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 12/46, D_loss_real = 0.660, D_loss_fake = 0.570, G_loss = 1.056
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 13/46, D_loss_real = 0.732, D_loss_fake = 0.614, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 14/46, D_loss_real = 0.678, D_loss_fake = 0.658, G_loss = 0.811
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 15/46, D_loss_real = 0.648, D_loss_fake = 0.639, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 16/46, D_loss_real = 0.628, D_loss_fake = 0.689, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 17/46, D_loss_real = 0.621, D_loss_fake = 0.742, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 18/46, D_loss_real = 0.719, D_loss_fake = 0.592, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 19/46, D_loss_real = 0.706, D_loss_fake = 0.689, G_loss = 0.786
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 20/46, D_loss_real = 0.660, D_loss_fake = 0.743, G_loss = 0.769
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 21/46, D_loss_real = 0.611, D_loss_fake = 0.679, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 22/46, D_loss_real = 0.627, D_loss_fake = 0.633, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 23/46, D_loss_real = 0.568, D_loss_fake = 0.642, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 24/46, D_loss_real = 0.595, D_loss_fake = 0.669, G_loss = 0.861
2/2 [=====] - 0s 3ms/step

Epoch 32, Iteration 25/46, D_loss_real = 0.614, D_loss_fake = 0.680, G_loss = 0.952
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 26/46, D_loss_real = 0.567, D_loss_fake = 0.671, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 27/46, D_loss_real = 0.705, D_loss_fake = 0.667, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 28/46, D_loss_real = 0.620, D_loss_fake = 0.648, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 29/46, D_loss_real = 0.703, D_loss_fake = 0.615, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 30/46, D_loss_real = 0.689, D_loss_fake = 0.641, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 31/46, D_loss_real = 0.670, D_loss_fake = 0.594, G_loss = 0.892
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 32/46, D_loss_real = 0.667, D_loss_fake = 0.603, G_loss = 0.919
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 33/46, D_loss_real = 0.699, D_loss_fake = 0.602, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 34/46, D_loss_real = 0.619, D_loss_fake = 0.615, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 35/46, D_loss_real = 0.703, D_loss_fake = 0.636, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 36/46, D_loss_real = 0.651, D_loss_fake = 0.645, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 37/46, D_loss_real = 0.647, D_loss_fake = 0.688, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 38/46, D_loss_real = 0.656, D_loss_fake = 0.579, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 39/46, D_loss_real = 0.630, D_loss_fake = 0.637, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 40/46, D_loss_real = 0.586, D_loss_fake = 0.752, G_loss = 0.784
2/2 [=====] - 0s 3ms/step

Epoch 32, Iteration 41/46, D_loss_real = 0.573, D_loss_fake = 0.828, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 42/46, D_loss_real = 0.626, D_loss_fake = 0.674, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 43/46, D_loss_real = 0.604, D_loss_fake = 0.696, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 44/46, D_loss_real = 0.663, D_loss_fake = 0.669, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 32, Iteration 45/46, D_loss_real = 0.704, D_loss_fake = 0.627, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 32, Iteration 46/46, D_loss_real = 0.693, D_loss_fake = 0.655, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 1/46, D_loss_real = 0.714, D_loss_fake = 0.624, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 2/46, D_loss_real = 0.642, D_loss_fake = 0.671, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 3/46, D_loss_real = 0.620, D_loss_fake = 0.631, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 4/46, D_loss_real = 0.618, D_loss_fake = 0.656, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 5/46, D_loss_real = 0.616, D_loss_fake = 0.625, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 6/46, D_loss_real = 0.681, D_loss_fake = 0.682, G_loss = 0.817
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 7/46, D_loss_real = 0.651, D_loss_fake = 0.728, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 8/46, D_loss_real = 0.704, D_loss_fake = 0.698, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 9/46, D_loss_real = 0.672, D_loss_fake = 0.606, G_loss = 0.927
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 10/46, D_loss_real = 0.701, D_loss_fake = 0.583, G_loss = 0.914
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 11/46, D_loss_real = 0.684, D_loss_fake = 0.624, G_loss = 0.889
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 12/46, D_loss_real = 0.692, D_loss_fake = 0.600, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 13/46, D_loss_real = 0.651, D_loss_fake = 0.606, G_loss = 0.916
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 14/46, D_loss_real = 0.639, D_loss_fake = 0.533, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 15/46, D_loss_real = 0.644, D_loss_fake = 0.656, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 16/46, D_loss_real = 0.638, D_loss_fake = 0.630, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 17/46, D_loss_real = 0.623, D_loss_fake = 0.665, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 18/46, D_loss_real = 0.655, D_loss_fake = 0.682, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 19/46, D_loss_real = 0.624, D_loss_fake = 0.645, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 20/46, D_loss_real = 0.628, D_loss_fake = 0.698, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 21/46, D_loss_real = 0.663, D_loss_fake = 0.688, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 22/46, D_loss_real = 0.657, D_loss_fake = 0.619, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 23/46, D_loss_real = 0.671, D_loss_fake = 0.643, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 24/46, D_loss_real = 0.615, D_loss_fake = 0.637, G_loss = 0.801
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 25/46, D_loss_real = 0.598, D_loss_fake = 0.683, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 26/46, D_loss_real = 0.623, D_loss_fake = 0.666, G_loss = 0.851
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 27/46, D_loss_real = 0.606, D_loss_fake = 0.668, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 28/46, D_loss_real = 0.590, D_loss_fake = 0.622, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 29/46, D_loss_real = 0.661, D_loss_fake = 0.685, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 30/46, D_loss_real = 0.629, D_loss_fake = 0.678, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 31/46, D_loss_real = 0.663, D_loss_fake = 0.651, G_loss = 0.872
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 32/46, D_loss_real = 0.647, D_loss_fake = 0.682, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 33/46, D_loss_real = 0.695, D_loss_fake = 0.620, G_loss = 0.960
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 34/46, D_loss_real = 0.709, D_loss_fake = 0.602, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 35/46, D_loss_real = 0.686, D_loss_fake = 0.676, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.688, G_loss = 0.806
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 37/46, D_loss_real = 0.701, D_loss_fake = 0.664, G_loss = 0.792
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 38/46, D_loss_real = 0.668, D_loss_fake = 0.638, G_loss = 0.784
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 39/46, D_loss_real = 0.663, D_loss_fake = 0.710, G_loss = 0.813
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 40/46, D_loss_real = 0.691, D_loss_fake = 0.681, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 41/46, D_loss_real = 0.639, D_loss_fake = 0.693, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 42/46, D_loss_real = 0.647, D_loss_fake = 0.622, G_loss = 0.838
2/2 [=====] - 0s 3ms/step

Epoch 33, Iteration 43/46, D_loss_real = 0.628, D_loss_fake = 0.609, G_loss = 0.832
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 44/46, D_loss_real = 0.658, D_loss_fake = 0.623, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 33, Iteration 45/46, D_loss_real = 0.559, D_loss_fake = 0.689, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 33, Iteration 46/46, D_loss_real = 0.587, D_loss_fake = 0.646, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 1/46, D_loss_real = 0.582, D_loss_fake = 0.751, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 2/46, D_loss_real = 0.623, D_loss_fake = 0.681, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 3/46, D_loss_real = 0.667, D_loss_fake = 0.632, G_loss = 0.906
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 4/46, D_loss_real = 0.689, D_loss_fake = 0.634, G_loss = 0.979
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 5/46, D_loss_real = 0.693, D_loss_fake = 0.599, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 6/46, D_loss_real = 0.691, D_loss_fake = 0.728, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 7/46, D_loss_real = 0.697, D_loss_fake = 0.624, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 8/46, D_loss_real = 0.636, D_loss_fake = 0.619, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 9/46, D_loss_real = 0.645, D_loss_fake = 0.637, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 10/46, D_loss_real = 0.653, D_loss_fake = 0.659, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 11/46, D_loss_real = 0.634, D_loss_fake = 0.596, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 12/46, D_loss_real = 0.615, D_loss_fake = 0.655, G_loss = 0.847
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 13/46, D_loss_real = 0.679, D_loss_fake = 0.645, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 14/46, D_loss_real = 0.632, D_loss_fake = 0.610, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 15/46, D_loss_real = 0.647, D_loss_fake = 0.650, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 16/46, D_loss_real = 0.659, D_loss_fake = 0.601, G_loss = 0.934
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 17/46, D_loss_real = 0.661, D_loss_fake = 0.583, G_loss = 0.955
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.599, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 19/46, D_loss_real = 0.662, D_loss_fake = 0.612, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 20/46, D_loss_real = 0.699, D_loss_fake = 0.649, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 21/46, D_loss_real = 0.654, D_loss_fake = 0.597, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 22/46, D_loss_real = 0.635, D_loss_fake = 0.650, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 23/46, D_loss_real = 0.619, D_loss_fake = 0.634, G_loss = 0.861
2/2 [=====] - 0s 5ms/step
Epoch 34, Iteration 24/46, D_loss_real = 0.694, D_loss_fake = 0.649, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 25/46, D_loss_real = 0.726, D_loss_fake = 0.592, G_loss = 0.846
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 26/46, D_loss_real = 0.629, D_loss_fake = 0.640, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 27/46, D_loss_real = 0.607, D_loss_fake = 0.619, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 28/46, D_loss_real = 0.562, D_loss_fake = 0.716, G_loss = 0.802
2/2 [=====] - 0s 3ms/step

Epoch 34, Iteration 29/46, D_loss_real = 0.577, D_loss_fake = 0.719, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 30/46, D_loss_real = 0.660, D_loss_fake = 0.687, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 31/46, D_loss_real = 0.601, D_loss_fake = 0.613, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 32/46, D_loss_real = 0.561, D_loss_fake = 0.751, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 33/46, D_loss_real = 0.630, D_loss_fake = 0.694, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 34/46, D_loss_real = 0.606, D_loss_fake = 0.618, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 35/46, D_loss_real = 0.646, D_loss_fake = 0.624, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 36/46, D_loss_real = 0.649, D_loss_fake = 0.693, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 37/46, D_loss_real = 0.641, D_loss_fake = 0.667, G_loss = 0.940
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 38/46, D_loss_real = 0.656, D_loss_fake = 0.610, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 39/46, D_loss_real = 0.691, D_loss_fake = 0.608, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 40/46, D_loss_real = 0.632, D_loss_fake = 0.620, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 41/46, D_loss_real = 0.585, D_loss_fake = 0.650, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 42/46, D_loss_real = 0.639, D_loss_fake = 0.619, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 34, Iteration 43/46, D_loss_real = 0.599, D_loss_fake = 0.633, G_loss = 0.790
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 44/46, D_loss_real = 0.546, D_loss_fake = 0.735, G_loss = 0.885
2/2 [=====] - 0s 4ms/step

Epoch 34, Iteration 45/46, D_loss_real = 0.644, D_loss_fake = 0.641, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 34, Iteration 46/46, D_loss_real = 0.651, D_loss_fake = 0.680, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 1/46, D_loss_real = 0.659, D_loss_fake = 0.603, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 2/46, D_loss_real = 0.751, D_loss_fake = 0.653, G_loss = 0.925
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 3/46, D_loss_real = 0.652, D_loss_fake = 0.665, G_loss = 0.905
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 4/46, D_loss_real = 0.740, D_loss_fake = 0.655, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 5/46, D_loss_real = 0.743, D_loss_fake = 0.573, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 6/46, D_loss_real = 0.682, D_loss_fake = 0.635, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 7/46, D_loss_real = 0.689, D_loss_fake = 0.646, G_loss = 0.821
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 8/46, D_loss_real = 0.660, D_loss_fake = 0.690, G_loss = 0.805
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 9/46, D_loss_real = 0.643, D_loss_fake = 0.667, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 10/46, D_loss_real = 0.656, D_loss_fake = 0.712, G_loss = 0.794
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 11/46, D_loss_real = 0.619, D_loss_fake = 0.713, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 12/46, D_loss_real = 0.646, D_loss_fake = 0.616, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 13/46, D_loss_real = 0.649, D_loss_fake = 0.592, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 14/46, D_loss_real = 0.646, D_loss_fake = 0.675, G_loss = 0.909
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 15/46, D_loss_real = 0.617, D_loss_fake = 0.584, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.636, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 17/46, D_loss_real = 0.634, D_loss_fake = 0.593, G_loss = 0.824
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 18/46, D_loss_real = 0.599, D_loss_fake = 0.700, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 19/46, D_loss_real = 0.668, D_loss_fake = 0.709, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 20/46, D_loss_real = 0.643, D_loss_fake = 0.648, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 21/46, D_loss_real = 0.640, D_loss_fake = 0.592, G_loss = 0.935
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 22/46, D_loss_real = 0.621, D_loss_fake = 0.601, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 23/46, D_loss_real = 0.677, D_loss_fake = 0.641, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 24/46, D_loss_real = 0.653, D_loss_fake = 0.623, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 25/46, D_loss_real = 0.648, D_loss_fake = 0.608, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 26/46, D_loss_real = 0.666, D_loss_fake = 0.659, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 27/46, D_loss_real = 0.588, D_loss_fake = 0.707, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 28/46, D_loss_real = 0.602, D_loss_fake = 0.631, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 29/46, D_loss_real = 0.666, D_loss_fake = 0.638, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 30/46, D_loss_real = 0.619, D_loss_fake = 0.600, G_loss = 0.859
2/2 [=====] - 0s 4ms/step

Epoch 35, Iteration 31/46, D_loss_real = 0.625, D_loss_fake = 0.640, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.773, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 33/46, D_loss_real = 0.649, D_loss_fake = 0.720, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 34/46, D_loss_real = 0.652, D_loss_fake = 0.692, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 35/46, D_loss_real = 0.635, D_loss_fake = 0.658, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 36/46, D_loss_real = 0.664, D_loss_fake = 0.666, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 37/46, D_loss_real = 0.660, D_loss_fake = 0.654, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 38/46, D_loss_real = 0.683, D_loss_fake = 0.635, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 39/46, D_loss_real = 0.668, D_loss_fake = 0.615, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 40/46, D_loss_real = 0.681, D_loss_fake = 0.621, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 41/46, D_loss_real = 0.644, D_loss_fake = 0.657, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 42/46, D_loss_real = 0.642, D_loss_fake = 0.642, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 43/46, D_loss_real = 0.669, D_loss_fake = 0.682, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 44/46, D_loss_real = 0.619, D_loss_fake = 0.667, G_loss = 0.794
2/2 [=====] - 0s 4ms/step
Epoch 35, Iteration 45/46, D_loss_real = 0.641, D_loss_fake = 0.644, G_loss = 0.789
2/2 [=====] - 0s 3ms/step
Epoch 35, Iteration 46/46, D_loss_real = 0.614, D_loss_fake = 0.707, G_loss = 0.801
2/2 [=====] - 0s 3ms/step

Epoch 36, Iteration 1/46, D_loss_real = 0.615, D_loss_fake = 0.676, G_loss = 0.747
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 2/46, D_loss_real = 0.617, D_loss_fake = 0.633, G_loss = 0.792
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 3/46, D_loss_real = 0.592, D_loss_fake = 0.725, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 4/46, D_loss_real = 0.599, D_loss_fake = 0.679, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 5/46, D_loss_real = 0.657, D_loss_fake = 0.580, G_loss = 0.869
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 6/46, D_loss_real = 0.612, D_loss_fake = 0.648, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 7/46, D_loss_real = 0.605, D_loss_fake = 0.621, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 8/46, D_loss_real = 0.608, D_loss_fake = 0.649, G_loss = 0.798
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 9/46, D_loss_real = 0.617, D_loss_fake = 0.624, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 10/46, D_loss_real = 0.658, D_loss_fake = 0.640, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 11/46, D_loss_real = 0.644, D_loss_fake = 0.629, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.665, G_loss = 0.873
2/2 [=====] - 0s 5ms/step
Epoch 36, Iteration 13/46, D_loss_real = 0.680, D_loss_fake = 0.642, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 14/46, D_loss_real = 0.635, D_loss_fake = 0.660, G_loss = 0.807
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 15/46, D_loss_real = 0.580, D_loss_fake = 0.667, G_loss = 0.799
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 16/46, D_loss_real = 0.615, D_loss_fake = 0.661, G_loss = 0.800
2/2 [=====] - 0s 5ms/step

Epoch 36, Iteration 17/46, D_loss_real = 0.583, D_loss_fake = 0.719, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 18/46, D_loss_real = 0.650, D_loss_fake = 0.714, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.664, G_loss = 0.918
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 20/46, D_loss_real = 0.739, D_loss_fake = 0.635, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 21/46, D_loss_real = 0.756, D_loss_fake = 0.569, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 22/46, D_loss_real = 0.705, D_loss_fake = 0.624, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 23/46, D_loss_real = 0.684, D_loss_fake = 0.591, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 24/46, D_loss_real = 0.703, D_loss_fake = 0.614, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 25/46, D_loss_real = 0.729, D_loss_fake = 0.599, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 26/46, D_loss_real = 0.722, D_loss_fake = 0.655, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 27/46, D_loss_real = 0.710, D_loss_fake = 0.672, G_loss = 0.845
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 28/46, D_loss_real = 0.686, D_loss_fake = 0.649, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 29/46, D_loss_real = 0.689, D_loss_fake = 0.656, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 30/46, D_loss_real = 0.635, D_loss_fake = 0.639, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 31/46, D_loss_real = 0.641, D_loss_fake = 0.672, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 32/46, D_loss_real = 0.635, D_loss_fake = 0.621, G_loss = 0.852
2/2 [=====] - 0s 4ms/step

Epoch 36, Iteration 33/46, D_loss_real = 0.623, D_loss_fake = 0.654, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 34/46, D_loss_real = 0.621, D_loss_fake = 0.637, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 35/46, D_loss_real = 0.640, D_loss_fake = 0.697, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 36/46, D_loss_real = 0.620, D_loss_fake = 0.671, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 37/46, D_loss_real = 0.666, D_loss_fake = 0.608, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 38/46, D_loss_real = 0.701, D_loss_fake = 0.615, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 39/46, D_loss_real = 0.684, D_loss_fake = 0.631, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 40/46, D_loss_real = 0.696, D_loss_fake = 0.627, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 41/46, D_loss_real = 0.614, D_loss_fake = 0.652, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 42/46, D_loss_real = 0.686, D_loss_fake = 0.621, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 36, Iteration 43/46, D_loss_real = 0.690, D_loss_fake = 0.626, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 44/46, D_loss_real = 0.671, D_loss_fake = 0.647, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 45/46, D_loss_real = 0.661, D_loss_fake = 0.652, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 36, Iteration 46/46, D_loss_real = 0.657, D_loss_fake = 0.580, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 1/46, D_loss_real = 0.623, D_loss_fake = 0.595, G_loss = 0.855
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 2/46, D_loss_real = 0.560, D_loss_fake = 0.634, G_loss = 0.814
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 3/46, D_loss_real = 0.558, D_loss_fake = 0.658, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 4/46, D_loss_real = 0.630, D_loss_fake = 0.743, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 5/46, D_loss_real = 0.562, D_loss_fake = 0.671, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 6/46, D_loss_real = 0.602, D_loss_fake = 0.657, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 7/46, D_loss_real = 0.671, D_loss_fake = 0.657, G_loss = 0.989
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 8/46, D_loss_real = 0.784, D_loss_fake = 0.600, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 9/46, D_loss_real = 0.699, D_loss_fake = 0.654, G_loss = 0.811
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 10/46, D_loss_real = 0.690, D_loss_fake = 0.680, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 11/46, D_loss_real = 0.655, D_loss_fake = 0.610, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 12/46, D_loss_real = 0.662, D_loss_fake = 0.677, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 13/46, D_loss_real = 0.639, D_loss_fake = 0.575, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 14/46, D_loss_real = 0.694, D_loss_fake = 0.592, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 15/46, D_loss_real = 0.636, D_loss_fake = 0.653, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 16/46, D_loss_real = 0.643, D_loss_fake = 0.640, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 17/46, D_loss_real = 0.617, D_loss_fake = 0.706, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 18/46, D_loss_real = 0.650, D_loss_fake = 0.631, G_loss = 0.880
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 19/46, D_loss_real = 0.705, D_loss_fake = 0.663, G_loss = 0.816
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 20/46, D_loss_real = 0.628, D_loss_fake = 0.720, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 21/46, D_loss_real = 0.658, D_loss_fake = 0.678, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 22/46, D_loss_real = 0.709, D_loss_fake = 0.622, G_loss = 0.832
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 23/46, D_loss_real = 0.596, D_loss_fake = 0.608, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 24/46, D_loss_real = 0.548, D_loss_fake = 0.729, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 25/46, D_loss_real = 0.614, D_loss_fake = 0.650, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 26/46, D_loss_real = 0.610, D_loss_fake = 0.661, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 27/46, D_loss_real = 0.579, D_loss_fake = 0.702, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 28/46, D_loss_real = 0.606, D_loss_fake = 0.661, G_loss = 0.796
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 29/46, D_loss_real = 0.585, D_loss_fake = 0.683, G_loss = 0.805
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 30/46, D_loss_real = 0.572, D_loss_fake = 0.672, G_loss = 0.827
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 31/46, D_loss_real = 0.617, D_loss_fake = 0.734, G_loss = 0.855
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 32/46, D_loss_real = 0.633, D_loss_fake = 0.650, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 33/46, D_loss_real = 0.647, D_loss_fake = 0.567, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 34/46, D_loss_real = 0.641, D_loss_fake = 0.595, G_loss = 0.903
2/2 [=====] - 0s 3ms/step

Epoch 37, Iteration 35/46, D_loss_real = 0.677, D_loss_fake = 0.634, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 36/46, D_loss_real = 0.706, D_loss_fake = 0.686, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 37/46, D_loss_real = 0.613, D_loss_fake = 0.602, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 38/46, D_loss_real = 0.636, D_loss_fake = 0.653, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 39/46, D_loss_real = 0.584, D_loss_fake = 0.650, G_loss = 0.941
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 40/46, D_loss_real = 0.662, D_loss_fake = 0.625, G_loss = 0.969
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 41/46, D_loss_real = 0.707, D_loss_fake = 0.604, G_loss = 0.962
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 42/46, D_loss_real = 0.661, D_loss_fake = 0.668, G_loss = 0.948
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 43/46, D_loss_real = 0.688, D_loss_fake = 0.614, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 44/46, D_loss_real = 0.739, D_loss_fake = 0.602, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 37, Iteration 45/46, D_loss_real = 0.692, D_loss_fake = 0.640, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 37, Iteration 46/46, D_loss_real = 0.652, D_loss_fake = 0.662, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 1/46, D_loss_real = 0.663, D_loss_fake = 0.644, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 2/46, D_loss_real = 0.659, D_loss_fake = 0.635, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 3/46, D_loss_real = 0.572, D_loss_fake = 0.741, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 4/46, D_loss_real = 0.575, D_loss_fake = 0.678, G_loss = 0.848
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 5/46, D_loss_real = 0.605, D_loss_fake = 0.781, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 6/46, D_loss_real = 0.610, D_loss_fake = 0.655, G_loss = 0.919
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 7/46, D_loss_real = 0.661, D_loss_fake = 0.600, G_loss = 0.941
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 8/46, D_loss_real = 0.658, D_loss_fake = 0.526, G_loss = 0.929
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.560, G_loss = 0.970
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 10/46, D_loss_real = 0.640, D_loss_fake = 0.598, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 11/46, D_loss_real = 0.613, D_loss_fake = 0.824, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 12/46, D_loss_real = 0.631, D_loss_fake = 0.722, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 13/46, D_loss_real = 0.644, D_loss_fake = 0.709, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 14/46, D_loss_real = 0.706, D_loss_fake = 0.597, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 15/46, D_loss_real = 0.664, D_loss_fake = 0.682, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 16/46, D_loss_real = 0.690, D_loss_fake = 0.635, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 17/46, D_loss_real = 0.640, D_loss_fake = 0.656, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 18/46, D_loss_real = 0.690, D_loss_fake = 0.655, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 19/46, D_loss_real = 0.654, D_loss_fake = 0.707, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 20/46, D_loss_real = 0.634, D_loss_fake = 0.672, G_loss = 0.868
2/2 [=====] - 0s 4ms/step

Epoch 38, Iteration 21/46, D_loss_real = 0.617, D_loss_fake = 0.649, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 22/46, D_loss_real = 0.646, D_loss_fake = 0.634, G_loss = 0.809
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 23/46, D_loss_real = 0.659, D_loss_fake = 0.641, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 24/46, D_loss_real = 0.662, D_loss_fake = 0.659, G_loss = 0.885
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 25/46, D_loss_real = 0.690, D_loss_fake = 0.620, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 26/46, D_loss_real = 0.667, D_loss_fake = 0.608, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 27/46, D_loss_real = 0.713, D_loss_fake = 0.610, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 28/46, D_loss_real = 0.688, D_loss_fake = 0.683, G_loss = 0.789
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 29/46, D_loss_real = 0.685, D_loss_fake = 0.688, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 30/46, D_loss_real = 0.599, D_loss_fake = 0.716, G_loss = 0.795
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 31/46, D_loss_real = 0.618, D_loss_fake = 0.822, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 32/46, D_loss_real = 0.686, D_loss_fake = 0.586, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 33/46, D_loss_real = 0.737, D_loss_fake = 0.589, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 34/46, D_loss_real = 0.711, D_loss_fake = 0.631, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 35/46, D_loss_real = 0.585, D_loss_fake = 0.660, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 36/46, D_loss_real = 0.641, D_loss_fake = 0.684, G_loss = 0.835
2/2 [=====] - 0s 3ms/step

Epoch 38, Iteration 37/46, D_loss_real = 0.589, D_loss_fake = 0.661, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 38/46, D_loss_real = 0.638, D_loss_fake = 0.682, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 39/46, D_loss_real = 0.653, D_loss_fake = 0.629, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 40/46, D_loss_real = 0.597, D_loss_fake = 0.607, G_loss = 0.889
2/2 [=====] - 0s 3ms/step
Epoch 38, Iteration 41/46, D_loss_real = 0.591, D_loss_fake = 0.608, G_loss = 0.922
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 42/46, D_loss_real = 0.622, D_loss_fake = 0.663, G_loss = 0.932
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 43/46, D_loss_real = 0.701, D_loss_fake = 0.605, G_loss = 0.952
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 44/46, D_loss_real = 0.654, D_loss_fake = 0.697, G_loss = 0.938
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 45/46, D_loss_real = 0.681, D_loss_fake = 0.633, G_loss = 1.003
2/2 [=====] - 0s 4ms/step
Epoch 38, Iteration 46/46, D_loss_real = 0.643, D_loss_fake = 0.621, G_loss = 0.938
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 1/46, D_loss_real = 0.733, D_loss_fake = 0.605, G_loss = 1.028
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 2/46, D_loss_real = 0.708, D_loss_fake = 0.551, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 3/46, D_loss_real = 0.696, D_loss_fake = 0.596, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 4/46, D_loss_real = 0.658, D_loss_fake = 0.669, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 5/46, D_loss_real = 0.723, D_loss_fake = 0.590, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 6/46, D_loss_real = 0.561, D_loss_fake = 0.651, G_loss = 0.806
2/2 [=====] - 0s 3ms/step

Epoch 39, Iteration 7/46, D_loss_real = 0.644, D_loss_fake = 0.678, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 8/46, D_loss_real = 0.650, D_loss_fake = 0.651, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 9/46, D_loss_real = 0.654, D_loss_fake = 0.661, G_loss = 0.837
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 10/46, D_loss_real = 0.600, D_loss_fake = 0.671, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 11/46, D_loss_real = 0.530, D_loss_fake = 0.582, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 12/46, D_loss_real = 0.607, D_loss_fake = 0.665, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 13/46, D_loss_real = 0.605, D_loss_fake = 0.665, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 14/46, D_loss_real = 0.688, D_loss_fake = 0.594, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 15/46, D_loss_real = 0.665, D_loss_fake = 0.612, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 16/46, D_loss_real = 0.630, D_loss_fake = 0.673, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 17/46, D_loss_real = 0.699, D_loss_fake = 0.705, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 18/46, D_loss_real = 0.697, D_loss_fake = 0.669, G_loss = 0.835
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 19/46, D_loss_real = 0.721, D_loss_fake = 0.645, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 20/46, D_loss_real = 0.755, D_loss_fake = 0.622, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 21/46, D_loss_real = 0.709, D_loss_fake = 0.624, G_loss = 0.802
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 22/46, D_loss_real = 0.714, D_loss_fake = 0.640, G_loss = 0.777
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 23/46, D_loss_real = 0.620, D_loss_fake = 0.687, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 24/46, D_loss_real = 0.606, D_loss_fake = 0.714, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 25/46, D_loss_real = 0.648, D_loss_fake = 0.627, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 26/46, D_loss_real = 0.654, D_loss_fake = 0.643, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 27/46, D_loss_real = 0.577, D_loss_fake = 0.666, G_loss = 0.829
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 28/46, D_loss_real = 0.585, D_loss_fake = 0.667, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 29/46, D_loss_real = 0.598, D_loss_fake = 0.745, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 30/46, D_loss_real = 0.627, D_loss_fake = 0.719, G_loss = 0.815
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 31/46, D_loss_real = 0.625, D_loss_fake = 0.690, G_loss = 0.817
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 32/46, D_loss_real = 0.714, D_loss_fake = 0.664, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 33/46, D_loss_real = 0.662, D_loss_fake = 0.676, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 34/46, D_loss_real = 0.650, D_loss_fake = 0.608, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 35/46, D_loss_real = 0.647, D_loss_fake = 0.632, G_loss = 0.906
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 36/46, D_loss_real = 0.622, D_loss_fake = 0.600, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 37/46, D_loss_real = 0.695, D_loss_fake = 0.581, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 38/46, D_loss_real = 0.575, D_loss_fake = 0.651, G_loss = 0.855
2/2 [=====] - 0s 4ms/step

Epoch 39, Iteration 39/46, D_loss_real = 0.509, D_loss_fake = 0.721, G_loss = 0.840
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 40/46, D_loss_real = 0.516, D_loss_fake = 0.706, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 41/46, D_loss_real = 0.598, D_loss_fake = 0.712, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 39, Iteration 42/46, D_loss_real = 0.678, D_loss_fake = 0.737, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 43/46, D_loss_real = 0.605, D_loss_fake = 0.779, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 44/46, D_loss_real = 0.606, D_loss_fake = 0.651, G_loss = 0.897
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 45/46, D_loss_real = 0.587, D_loss_fake = 0.602, G_loss = 0.904
2/2 [=====] - 0s 4ms/step
Epoch 39, Iteration 46/46, D_loss_real = 0.601, D_loss_fake = 0.610, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 1/46, D_loss_real = 0.591, D_loss_fake = 0.654, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 2/46, D_loss_real = 0.674, D_loss_fake = 0.685, G_loss = 0.976
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 3/46, D_loss_real = 0.702, D_loss_fake = 0.656, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 4/46, D_loss_real = 0.698, D_loss_fake = 0.591, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 5/46, D_loss_real = 0.671, D_loss_fake = 0.619, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 6/46, D_loss_real = 0.660, D_loss_fake = 0.649, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 7/46, D_loss_real = 0.664, D_loss_fake = 0.656, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 8/46, D_loss_real = 0.699, D_loss_fake = 0.642, G_loss = 0.908
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 9/46, D_loss_real = 0.696, D_loss_fake = 0.605, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 10/46, D_loss_real = 0.648, D_loss_fake = 0.576, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 11/46, D_loss_real = 0.640, D_loss_fake = 0.594, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 12/46, D_loss_real = 0.629, D_loss_fake = 0.648, G_loss = 0.825
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 13/46, D_loss_real = 0.661, D_loss_fake = 0.687, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 14/46, D_loss_real = 0.617, D_loss_fake = 0.649, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 15/46, D_loss_real = 0.644, D_loss_fake = 0.563, G_loss = 0.951
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 16/46, D_loss_real = 0.684, D_loss_fake = 0.593, G_loss = 1.022
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 17/46, D_loss_real = 0.740, D_loss_fake = 0.619, G_loss = 0.974
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 18/46, D_loss_real = 0.732, D_loss_fake = 0.577, G_loss = 0.939
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 19/46, D_loss_real = 0.770, D_loss_fake = 0.594, G_loss = 0.947
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 20/46, D_loss_real = 0.665, D_loss_fake = 0.603, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 21/46, D_loss_real = 0.712, D_loss_fake = 0.617, G_loss = 0.957
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 22/46, D_loss_real = 0.611, D_loss_fake = 0.589, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 23/46, D_loss_real = 0.695, D_loss_fake = 0.607, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 24/46, D_loss_real = 0.610, D_loss_fake = 0.621, G_loss = 0.842
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 25/46, D_loss_real = 0.646, D_loss_fake = 0.634, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 26/46, D_loss_real = 0.572, D_loss_fake = 0.668, G_loss = 0.812
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 27/46, D_loss_real = 0.585, D_loss_fake = 0.639, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 28/46, D_loss_real = 0.606, D_loss_fake = 0.697, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 29/46, D_loss_real = 0.617, D_loss_fake = 0.619, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 30/46, D_loss_real = 0.640, D_loss_fake = 0.583, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 31/46, D_loss_real = 0.643, D_loss_fake = 0.584, G_loss = 0.924
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 32/46, D_loss_real = 0.609, D_loss_fake = 0.716, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 33/46, D_loss_real = 0.599, D_loss_fake = 0.625, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 34/46, D_loss_real = 0.724, D_loss_fake = 0.620, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 35/46, D_loss_real = 0.704, D_loss_fake = 0.636, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 36/46, D_loss_real = 0.695, D_loss_fake = 0.653, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 40, Iteration 37/46, D_loss_real = 0.664, D_loss_fake = 0.653, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 38/46, D_loss_real = 0.665, D_loss_fake = 0.685, G_loss = 0.900
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 39/46, D_loss_real = 0.639, D_loss_fake = 0.613, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 40/46, D_loss_real = 0.587, D_loss_fake = 0.593, G_loss = 0.884
2/2 [=====] - 0s 4ms/step

Epoch 40, Iteration 41/46, D_loss_real = 0.588, D_loss_fake = 0.669, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 42/46, D_loss_real = 0.651, D_loss_fake = 0.617, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 43/46, D_loss_real = 0.659, D_loss_fake = 0.672, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 44/46, D_loss_real = 0.604, D_loss_fake = 0.673, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 45/46, D_loss_real = 0.575, D_loss_fake = 0.687, G_loss = 0.825
2/2 [=====] - 0s 4ms/step
Epoch 40, Iteration 46/46, D_loss_real = 0.556, D_loss_fake = 0.629, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 1/46, D_loss_real = 0.577, D_loss_fake = 0.631, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 2/46, D_loss_real = 0.653, D_loss_fake = 0.698, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 3/46, D_loss_real = 0.677, D_loss_fake = 0.632, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 4/46, D_loss_real = 0.663, D_loss_fake = 0.683, G_loss = 0.937
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 5/46, D_loss_real = 0.642, D_loss_fake = 0.629, G_loss = 0.906
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 6/46, D_loss_real = 0.585, D_loss_fake = 0.601, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 7/46, D_loss_real = 0.638, D_loss_fake = 0.649, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 8/46, D_loss_real = 0.500, D_loss_fake = 0.759, G_loss = 0.819
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 9/46, D_loss_real = 0.563, D_loss_fake = 0.697, G_loss = 0.801
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 10/46, D_loss_real = 0.603, D_loss_fake = 0.846, G_loss = 0.949
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 11/46, D_loss_real = 0.666, D_loss_fake = 0.612, G_loss = 0.904
2/2 [=====] - 0s 6ms/step
Epoch 41, Iteration 12/46, D_loss_real = 0.734, D_loss_fake = 0.532, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 13/46, D_loss_real = 0.676, D_loss_fake = 0.566, G_loss = 0.954
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 14/46, D_loss_real = 0.627, D_loss_fake = 0.652, G_loss = 0.923
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 15/46, D_loss_real = 0.704, D_loss_fake = 0.587, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 16/46, D_loss_real = 0.667, D_loss_fake = 0.605, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 17/46, D_loss_real = 0.651, D_loss_fake = 0.635, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 18/46, D_loss_real = 0.656, D_loss_fake = 0.741, G_loss = 0.887
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 19/46, D_loss_real = 0.645, D_loss_fake = 0.628, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 20/46, D_loss_real = 0.626, D_loss_fake = 0.687, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 21/46, D_loss_real = 0.717, D_loss_fake = 0.632, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 22/46, D_loss_real = 0.684, D_loss_fake = 0.607, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 23/46, D_loss_real = 0.647, D_loss_fake = 0.662, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 24/46, D_loss_real = 0.591, D_loss_fake = 0.608, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 25/46, D_loss_real = 0.607, D_loss_fake = 0.604, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 26/46, D_loss_real = 0.562, D_loss_fake = 0.720, G_loss = 0.788
2/2 [=====] - 0s 4ms/step

Epoch 41, Iteration 27/46, D_loss_real = 0.665, D_loss_fake = 0.651, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 28/46, D_loss_real = 0.522, D_loss_fake = 0.705, G_loss = 0.994
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 29/46, D_loss_real = 0.645, D_loss_fake = 0.562, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 30/46, D_loss_real = 0.691, D_loss_fake = 0.625, G_loss = 0.934
2/2 [=====] - 0s 5ms/step
Epoch 41, Iteration 31/46, D_loss_real = 0.584, D_loss_fake = 0.552, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 32/46, D_loss_real = 0.594, D_loss_fake = 0.727, G_loss = 0.908
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 33/46, D_loss_real = 0.588, D_loss_fake = 0.660, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 34/46, D_loss_real = 0.653, D_loss_fake = 0.656, G_loss = 0.918
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 35/46, D_loss_real = 0.638, D_loss_fake = 0.781, G_loss = 0.996
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 36/46, D_loss_real = 0.759, D_loss_fake = 0.576, G_loss = 0.972
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 37/46, D_loss_real = 0.774, D_loss_fake = 0.595, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 38/46, D_loss_real = 0.687, D_loss_fake = 0.608, G_loss = 0.855
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 39/46, D_loss_real = 0.701, D_loss_fake = 0.640, G_loss = 0.831
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 40/46, D_loss_real = 0.720, D_loss_fake = 0.709, G_loss = 0.843
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 41/46, D_loss_real = 0.743, D_loss_fake = 0.604, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 42/46, D_loss_real = 0.695, D_loss_fake = 0.595, G_loss = 0.874
2/2 [=====] - 0s 3ms/step

Epoch 41, Iteration 43/46, D_loss_real = 0.717, D_loss_fake = 0.606, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 44/46, D_loss_real = 0.653, D_loss_fake = 0.602, G_loss = 0.933
2/2 [=====] - 0s 4ms/step
Epoch 41, Iteration 45/46, D_loss_real = 0.664, D_loss_fake = 0.661, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 41, Iteration 46/46, D_loss_real = 0.639, D_loss_fake = 0.580, G_loss = 0.952
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 1/46, D_loss_real = 0.682, D_loss_fake = 0.602, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 2/46, D_loss_real = 0.656, D_loss_fake = 0.625, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 3/46, D_loss_real = 0.593, D_loss_fake = 0.549, G_loss = 0.937
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 4/46, D_loss_real = 0.643, D_loss_fake = 0.647, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 5/46, D_loss_real = 0.638, D_loss_fake = 0.685, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 6/46, D_loss_real = 0.640, D_loss_fake = 0.673, G_loss = 0.837
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 7/46, D_loss_real = 0.610, D_loss_fake = 0.641, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 8/46, D_loss_real = 0.523, D_loss_fake = 0.707, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 9/46, D_loss_real = 0.634, D_loss_fake = 0.664, G_loss = 0.820
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 10/46, D_loss_real = 0.642, D_loss_fake = 0.691, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 11/46, D_loss_real = 0.570, D_loss_fake = 0.621, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 12/46, D_loss_real = 0.661, D_loss_fake = 0.637, G_loss = 0.888
2/2 [=====] - 0s 4ms/step

Epoch 42, Iteration 13/46, D_loss_real = 0.643, D_loss_fake = 0.698, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 14/46, D_loss_real = 0.702, D_loss_fake = 0.639, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 15/46, D_loss_real = 0.700, D_loss_fake = 0.599, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 16/46, D_loss_real = 0.659, D_loss_fake = 0.615, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 17/46, D_loss_real = 0.717, D_loss_fake = 0.635, G_loss = 0.929
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 18/46, D_loss_real = 0.742, D_loss_fake = 0.607, G_loss = 0.844
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 19/46, D_loss_real = 0.653, D_loss_fake = 0.633, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 20/46, D_loss_real = 0.711, D_loss_fake = 0.630, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 21/46, D_loss_real = 0.698, D_loss_fake = 0.682, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 22/46, D_loss_real = 0.650, D_loss_fake = 0.671, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 23/46, D_loss_real = 0.641, D_loss_fake = 0.610, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 24/46, D_loss_real = 0.722, D_loss_fake = 0.587, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 25/46, D_loss_real = 0.639, D_loss_fake = 0.649, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 26/46, D_loss_real = 0.733, D_loss_fake = 0.640, G_loss = 0.925
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 27/46, D_loss_real = 0.712, D_loss_fake = 0.642, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 28/46, D_loss_real = 0.758, D_loss_fake = 0.555, G_loss = 0.926
2/2 [=====] - 0s 3ms/step

Epoch 42, Iteration 29/46, D_loss_real = 0.739, D_loss_fake = 0.597, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 30/46, D_loss_real = 0.677, D_loss_fake = 0.591, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.586, G_loss = 0.887
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 32/46, D_loss_real = 0.646, D_loss_fake = 0.648, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 33/46, D_loss_real = 0.682, D_loss_fake = 0.651, G_loss = 0.858
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 34/46, D_loss_real = 0.668, D_loss_fake = 0.636, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 35/46, D_loss_real = 0.676, D_loss_fake = 0.597, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 36/46, D_loss_real = 0.643, D_loss_fake = 0.629, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 37/46, D_loss_real = 0.637, D_loss_fake = 0.612, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 38/46, D_loss_real = 0.656, D_loss_fake = 0.699, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 39/46, D_loss_real = 0.639, D_loss_fake = 0.613, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 40/46, D_loss_real = 0.717, D_loss_fake = 0.601, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 41/46, D_loss_real = 0.718, D_loss_fake = 0.606, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 42/46, D_loss_real = 0.648, D_loss_fake = 0.658, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 43/46, D_loss_real = 0.607, D_loss_fake = 0.702, G_loss = 0.818
2/2 [=====] - 0s 4ms/step
Epoch 42, Iteration 44/46, D_loss_real = 0.678, D_loss_fake = 0.702, G_loss = 0.835
2/2 [=====] - 0s 3ms/step

Epoch 42, Iteration 45/46, D_loss_real = 0.589, D_loss_fake = 0.634, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 42, Iteration 46/46, D_loss_real = 0.655, D_loss_fake = 0.668, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 1/46, D_loss_real = 0.649, D_loss_fake = 0.645, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 2/46, D_loss_real = 0.620, D_loss_fake = 0.622, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 3/46, D_loss_real = 0.694, D_loss_fake = 0.614, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 4/46, D_loss_real = 0.682, D_loss_fake = 0.643, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 5/46, D_loss_real = 0.747, D_loss_fake = 0.601, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 6/46, D_loss_real = 0.652, D_loss_fake = 0.653, G_loss = 0.816
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 7/46, D_loss_real = 0.618, D_loss_fake = 0.679, G_loss = 0.871
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 8/46, D_loss_real = 0.636, D_loss_fake = 0.617, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 9/46, D_loss_real = 0.650, D_loss_fake = 0.612, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 10/46, D_loss_real = 0.648, D_loss_fake = 0.662, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 11/46, D_loss_real = 0.658, D_loss_fake = 0.633, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 12/46, D_loss_real = 0.723, D_loss_fake = 0.710, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 13/46, D_loss_real = 0.563, D_loss_fake = 0.634, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 14/46, D_loss_real = 0.625, D_loss_fake = 0.602, G_loss = 0.902
2/2 [=====] - 0s 3ms/step

Epoch 43, Iteration 15/46, D_loss_real = 0.617, D_loss_fake = 0.676, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 16/46, D_loss_real = 0.584, D_loss_fake = 0.605, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 17/46, D_loss_real = 0.688, D_loss_fake = 0.588, G_loss = 0.845
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 18/46, D_loss_real = 0.627, D_loss_fake = 0.642, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 19/46, D_loss_real = 0.629, D_loss_fake = 0.630, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 20/46, D_loss_real = 0.607, D_loss_fake = 0.622, G_loss = 0.922
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 21/46, D_loss_real = 0.687, D_loss_fake = 0.662, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 22/46, D_loss_real = 0.663, D_loss_fake = 0.616, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 23/46, D_loss_real = 0.721, D_loss_fake = 0.604, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 24/46, D_loss_real = 0.671, D_loss_fake = 0.650, G_loss = 0.824
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 25/46, D_loss_real = 0.675, D_loss_fake = 0.694, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 26/46, D_loss_real = 0.656, D_loss_fake = 0.592, G_loss = 0.834
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 27/46, D_loss_real = 0.687, D_loss_fake = 0.695, G_loss = 0.841
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 28/46, D_loss_real = 0.643, D_loss_fake = 0.650, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 29/46, D_loss_real = 0.692, D_loss_fake = 0.618, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 30/46, D_loss_real = 0.692, D_loss_fake = 0.604, G_loss = 0.854
2/2 [=====] - 0s 4ms/step

Epoch 43, Iteration 31/46, D_loss_real = 0.646, D_loss_fake = 0.659, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 32/46, D_loss_real = 0.666, D_loss_fake = 0.627, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 33/46, D_loss_real = 0.669, D_loss_fake = 0.603, G_loss = 0.855
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 34/46, D_loss_real = 0.620, D_loss_fake = 0.597, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 35/46, D_loss_real = 0.590, D_loss_fake = 0.714, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 36/46, D_loss_real = 0.611, D_loss_fake = 0.684, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 37/46, D_loss_real = 0.690, D_loss_fake = 0.630, G_loss = 0.864
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 38/46, D_loss_real = 0.629, D_loss_fake = 0.664, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 39/46, D_loss_real = 0.582, D_loss_fake = 0.668, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 40/46, D_loss_real = 0.580, D_loss_fake = 0.667, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 41/46, D_loss_real = 0.533, D_loss_fake = 0.689, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 42/46, D_loss_real = 0.611, D_loss_fake = 0.620, G_loss = 0.946
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 43/46, D_loss_real = 0.586, D_loss_fake = 0.611, G_loss = 0.904
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 44/46, D_loss_real = 0.703, D_loss_fake = 0.711, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 43, Iteration 45/46, D_loss_real = 0.702, D_loss_fake = 0.628, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 43, Iteration 46/46, D_loss_real = 0.676, D_loss_fake = 0.655, G_loss = 0.902
2/2 [=====] - 0s 4ms/step

Epoch 44, Iteration 1/46, D_loss_real = 0.670, D_loss_fake = 0.632, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 2/46, D_loss_real = 0.673, D_loss_fake = 0.605, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 3/46, D_loss_real = 0.600, D_loss_fake = 0.621, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 4/46, D_loss_real = 0.734, D_loss_fake = 0.628, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 5/46, D_loss_real = 0.660, D_loss_fake = 0.624, G_loss = 0.821
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 6/46, D_loss_real = 0.586, D_loss_fake = 0.656, G_loss = 0.838
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 7/46, D_loss_real = 0.616, D_loss_fake = 0.662, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 8/46, D_loss_real = 0.553, D_loss_fake = 0.710, G_loss = 0.828
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 9/46, D_loss_real = 0.557, D_loss_fake = 0.686, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 10/46, D_loss_real = 0.550, D_loss_fake = 0.671, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 11/46, D_loss_real = 0.546, D_loss_fake = 0.658, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 12/46, D_loss_real = 0.594, D_loss_fake = 0.626, G_loss = 0.919
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 13/46, D_loss_real = 0.605, D_loss_fake = 0.594, G_loss = 0.951
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 14/46, D_loss_real = 0.669, D_loss_fake = 0.718, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 15/46, D_loss_real = 0.668, D_loss_fake = 0.696, G_loss = 0.921
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 16/46, D_loss_real = 0.719, D_loss_fake = 0.690, G_loss = 0.924
2/2 [=====] - 0s 3ms/step

Epoch 44, Iteration 17/46, D_loss_real = 0.715, D_loss_fake = 0.580, G_loss = 0.982
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 18/46, D_loss_real = 0.693, D_loss_fake = 0.669, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 19/46, D_loss_real = 0.629, D_loss_fake = 0.714, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 20/46, D_loss_real = 0.679, D_loss_fake = 0.661, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 21/46, D_loss_real = 0.620, D_loss_fake = 0.562, G_loss = 0.927
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 22/46, D_loss_real = 0.626, D_loss_fake = 0.618, G_loss = 0.923
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 23/46, D_loss_real = 0.639, D_loss_fake = 0.587, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 24/46, D_loss_real = 0.522, D_loss_fake = 0.664, G_loss = 0.803
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 25/46, D_loss_real = 0.549, D_loss_fake = 0.803, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 26/46, D_loss_real = 0.539, D_loss_fake = 0.645, G_loss = 0.876
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 27/46, D_loss_real = 0.652, D_loss_fake = 0.688, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 28/46, D_loss_real = 0.606, D_loss_fake = 0.679, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 29/46, D_loss_real = 0.619, D_loss_fake = 0.674, G_loss = 0.820
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 30/46, D_loss_real = 0.605, D_loss_fake = 0.676, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 31/46, D_loss_real = 0.593, D_loss_fake = 0.720, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 32/46, D_loss_real = 0.671, D_loss_fake = 0.616, G_loss = 0.923
2/2 [=====] - 0s 4ms/step

Epoch 44, Iteration 33/46, D_loss_real = 0.708, D_loss_fake = 0.620, G_loss = 0.949
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.568, G_loss = 0.912
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 35/46, D_loss_real = 0.604, D_loss_fake = 0.624, G_loss = 0.933
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 36/46, D_loss_real = 0.630, D_loss_fake = 0.623, G_loss = 0.971
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 37/46, D_loss_real = 0.659, D_loss_fake = 0.648, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 38/46, D_loss_real = 0.668, D_loss_fake = 0.601, G_loss = 0.942
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 39/46, D_loss_real = 0.635, D_loss_fake = 0.607, G_loss = 0.901
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 40/46, D_loss_real = 0.720, D_loss_fake = 0.575, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 41/46, D_loss_real = 0.641, D_loss_fake = 0.649, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 42/46, D_loss_real = 0.618, D_loss_fake = 0.682, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.638, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 44/46, D_loss_real = 0.657, D_loss_fake = 0.618, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 44, Iteration 45/46, D_loss_real = 0.683, D_loss_fake = 0.680, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 44, Iteration 46/46, D_loss_real = 0.661, D_loss_fake = 0.650, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 1/46, D_loss_real = 0.646, D_loss_fake = 0.657, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 2/46, D_loss_real = 0.654, D_loss_fake = 0.706, G_loss = 0.852
2/2 [=====] - 0s 4ms/step

Epoch 45, Iteration 3/46, D_loss_real = 0.622, D_loss_fake = 0.622, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 4/46, D_loss_real = 0.665, D_loss_fake = 0.672, G_loss = 0.865
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 5/46, D_loss_real = 0.713, D_loss_fake = 0.622, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 6/46, D_loss_real = 0.692, D_loss_fake = 0.614, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 7/46, D_loss_real = 0.702, D_loss_fake = 0.658, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 8/46, D_loss_real = 0.677, D_loss_fake = 0.622, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 9/46, D_loss_real = 0.699, D_loss_fake = 0.621, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 10/46, D_loss_real = 0.592, D_loss_fake = 0.628, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 11/46, D_loss_real = 0.538, D_loss_fake = 0.610, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 12/46, D_loss_real = 0.611, D_loss_fake = 0.678, G_loss = 0.858
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 13/46, D_loss_real = 0.584, D_loss_fake = 0.669, G_loss = 0.795
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 14/46, D_loss_real = 0.595, D_loss_fake = 0.686, G_loss = 0.841
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 15/46, D_loss_real = 0.574, D_loss_fake = 0.694, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 16/46, D_loss_real = 0.569, D_loss_fake = 0.794, G_loss = 0.892
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 17/46, D_loss_real = 0.679, D_loss_fake = 0.627, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 18/46, D_loss_real = 0.679, D_loss_fake = 0.626, G_loss = 0.858
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 19/46, D_loss_real = 0.643, D_loss_fake = 0.658, G_loss = 0.814
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 20/46, D_loss_real = 0.647, D_loss_fake = 0.672, G_loss = 0.833
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 21/46, D_loss_real = 0.684, D_loss_fake = 0.653, G_loss = 0.859
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 22/46, D_loss_real = 0.689, D_loss_fake = 0.668, G_loss = 0.887
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 23/46, D_loss_real = 0.689, D_loss_fake = 0.692, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 24/46, D_loss_real = 0.759, D_loss_fake = 0.603, G_loss = 0.945
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 25/46, D_loss_real = 0.753, D_loss_fake = 0.589, G_loss = 0.952
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 26/46, D_loss_real = 0.712, D_loss_fake = 0.591, G_loss = 0.955
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 27/46, D_loss_real = 0.668, D_loss_fake = 0.634, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 28/46, D_loss_real = 0.712, D_loss_fake = 0.604, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 29/46, D_loss_real = 0.705, D_loss_fake = 0.606, G_loss = 0.819
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 30/46, D_loss_real = 0.592, D_loss_fake = 0.684, G_loss = 0.935
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 31/46, D_loss_real = 0.680, D_loss_fake = 0.571, G_loss = 0.981
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 32/46, D_loss_real = 0.678, D_loss_fake = 0.601, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 33/46, D_loss_real = 0.652, D_loss_fake = 0.600, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 34/46, D_loss_real = 0.663, D_loss_fake = 0.683, G_loss = 0.881
2/2 [=====] - 0s 3ms/step

Epoch 45, Iteration 35/46, D_loss_real = 0.669, D_loss_fake = 0.651, G_loss = 0.842
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 36/46, D_loss_real = 0.609, D_loss_fake = 0.690, G_loss = 0.823
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 37/46, D_loss_real = 0.596, D_loss_fake = 0.698, G_loss = 0.810
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 38/46, D_loss_real = 0.600, D_loss_fake = 0.767, G_loss = 0.919
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 39/46, D_loss_real = 0.614, D_loss_fake = 0.638, G_loss = 0.919
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 40/46, D_loss_real = 0.647, D_loss_fake = 0.577, G_loss = 0.960
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 41/46, D_loss_real = 0.680, D_loss_fake = 0.579, G_loss = 0.950
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 42/46, D_loss_real = 0.608, D_loss_fake = 0.621, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 43/46, D_loss_real = 0.580, D_loss_fake = 0.644, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 45, Iteration 44/46, D_loss_real = 0.644, D_loss_fake = 0.655, G_loss = 0.916
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 45/46, D_loss_real = 0.634, D_loss_fake = 0.681, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 45, Iteration 46/46, D_loss_real = 0.675, D_loss_fake = 0.659, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 1/46, D_loss_real = 0.618, D_loss_fake = 0.599, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 2/46, D_loss_real = 0.629, D_loss_fake = 0.647, G_loss = 0.828
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 3/46, D_loss_real = 0.657, D_loss_fake = 0.663, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 4/46, D_loss_real = 0.660, D_loss_fake = 0.761, G_loss = 0.828
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 5/46, D_loss_real = 0.588, D_loss_fake = 0.671, G_loss = 0.826
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 6/46, D_loss_real = 0.677, D_loss_fake = 0.666, G_loss = 0.831
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 7/46, D_loss_real = 0.637, D_loss_fake = 0.701, G_loss = 0.859
2/2 [=====] - 0s 5ms/step
Epoch 46, Iteration 8/46, D_loss_real = 0.656, D_loss_fake = 0.648, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 9/46, D_loss_real = 0.639, D_loss_fake = 0.638, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 10/46, D_loss_real = 0.643, D_loss_fake = 0.657, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 11/46, D_loss_real = 0.647, D_loss_fake = 0.671, G_loss = 0.911
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 12/46, D_loss_real = 0.660, D_loss_fake = 0.566, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 13/46, D_loss_real = 0.670, D_loss_fake = 0.613, G_loss = 0.883
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 14/46, D_loss_real = 0.637, D_loss_fake = 0.704, G_loss = 0.938
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 15/46, D_loss_real = 0.671, D_loss_fake = 0.657, G_loss = 0.948
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 16/46, D_loss_real = 0.700, D_loss_fake = 0.613, G_loss = 0.954
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 17/46, D_loss_real = 0.736, D_loss_fake = 0.584, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 18/46, D_loss_real = 0.680, D_loss_fake = 0.558, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 19/46, D_loss_real = 0.623, D_loss_fake = 0.591, G_loss = 0.860
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 20/46, D_loss_real = 0.635, D_loss_fake = 0.651, G_loss = 0.854
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 21/46, D_loss_real = 0.560, D_loss_fake = 0.618, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 22/46, D_loss_real = 0.572, D_loss_fake = 0.639, G_loss = 0.851
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 23/46, D_loss_real = 0.639, D_loss_fake = 0.657, G_loss = 0.839
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 24/46, D_loss_real = 0.522, D_loss_fake = 0.651, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 25/46, D_loss_real = 0.580, D_loss_fake = 0.689, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 26/46, D_loss_real = 0.637, D_loss_fake = 0.613, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 27/46, D_loss_real = 0.601, D_loss_fake = 0.604, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 28/46, D_loss_real = 0.575, D_loss_fake = 0.613, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 29/46, D_loss_real = 0.596, D_loss_fake = 0.675, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 30/46, D_loss_real = 0.616, D_loss_fake = 0.692, G_loss = 0.814
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 31/46, D_loss_real = 0.543, D_loss_fake = 0.671, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 32/46, D_loss_real = 0.563, D_loss_fake = 0.662, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 33/46, D_loss_real = 0.596, D_loss_fake = 0.713, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 34/46, D_loss_real = 0.598, D_loss_fake = 0.655, G_loss = 0.972
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 35/46, D_loss_real = 0.636, D_loss_fake = 0.552, G_loss = 0.947
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 36/46, D_loss_real = 0.672, D_loss_fake = 0.642, G_loss = 0.927
2/2 [=====] - 0s 3ms/step

Epoch 46, Iteration 37/46, D_loss_real = 0.592, D_loss_fake = 0.619, G_loss = 0.896
2/2 [=====] - 0s 4ms/step
Epoch 46, Iteration 38/46, D_loss_real = 0.648, D_loss_fake = 0.653, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 39/46, D_loss_real = 0.681, D_loss_fake = 0.614, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 40/46, D_loss_real = 0.635, D_loss_fake = 0.666, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 41/46, D_loss_real = 0.647, D_loss_fake = 0.608, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 42/46, D_loss_real = 0.642, D_loss_fake = 0.635, G_loss = 0.860
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 43/46, D_loss_real = 0.641, D_loss_fake = 0.678, G_loss = 0.918
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 44/46, D_loss_real = 0.672, D_loss_fake = 0.660, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 46, Iteration 45/46, D_loss_real = 0.675, D_loss_fake = 0.608, G_loss = 0.917
2/2 [=====] - 0s 5ms/step
Epoch 46, Iteration 46/46, D_loss_real = 0.620, D_loss_fake = 0.647, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 1/46, D_loss_real = 0.648, D_loss_fake = 0.613, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 2/46, D_loss_real = 0.592, D_loss_fake = 0.700, G_loss = 0.894
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 3/46, D_loss_real = 0.661, D_loss_fake = 0.719, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 4/46, D_loss_real = 0.622, D_loss_fake = 0.647, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 5/46, D_loss_real = 0.681, D_loss_fake = 0.624, G_loss = 0.873
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 6/46, D_loss_real = 0.697, D_loss_fake = 0.628, G_loss = 0.906
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 7/46, D_loss_real = 0.647, D_loss_fake = 0.625, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 8/46, D_loss_real = 0.690, D_loss_fake = 0.553, G_loss = 0.987
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 9/46, D_loss_real = 0.662, D_loss_fake = 0.551, G_loss = 1.011
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 10/46, D_loss_real = 0.660, D_loss_fake = 0.534, G_loss = 0.952
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 11/46, D_loss_real = 0.629, D_loss_fake = 0.628, G_loss = 0.949
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 12/46, D_loss_real = 0.669, D_loss_fake = 0.637, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 13/46, D_loss_real = 0.655, D_loss_fake = 0.593, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 14/46, D_loss_real = 0.615, D_loss_fake = 0.667, G_loss = 0.970
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 15/46, D_loss_real = 0.663, D_loss_fake = 0.639, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 16/46, D_loss_real = 0.699, D_loss_fake = 0.670, G_loss = 0.874
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 17/46, D_loss_real = 0.729, D_loss_fake = 0.628, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 18/46, D_loss_real = 0.653, D_loss_fake = 0.581, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 19/46, D_loss_real = 0.654, D_loss_fake = 0.607, G_loss = 1.006
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.549, G_loss = 0.903
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 21/46, D_loss_real = 0.625, D_loss_fake = 0.669, G_loss = 0.906
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 22/46, D_loss_real = 0.650, D_loss_fake = 0.589, G_loss = 0.896
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 23/46, D_loss_real = 0.566, D_loss_fake = 0.619, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 24/46, D_loss_real = 0.592, D_loss_fake = 0.730, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 25/46, D_loss_real = 0.609, D_loss_fake = 0.610, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 26/46, D_loss_real = 0.607, D_loss_fake = 0.645, G_loss = 0.851
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 27/46, D_loss_real = 0.609, D_loss_fake = 0.765, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 28/46, D_loss_real = 0.547, D_loss_fake = 0.742, G_loss = 0.804
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 29/46, D_loss_real = 0.634, D_loss_fake = 0.666, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 30/46, D_loss_real = 0.653, D_loss_fake = 0.649, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 31/46, D_loss_real = 0.690, D_loss_fake = 0.660, G_loss = 0.970
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 32/46, D_loss_real = 0.699, D_loss_fake = 0.579, G_loss = 0.975
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 33/46, D_loss_real = 0.693, D_loss_fake = 0.630, G_loss = 0.942
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 34/46, D_loss_real = 0.638, D_loss_fake = 0.649, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 35/46, D_loss_real = 0.662, D_loss_fake = 0.653, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 36/46, D_loss_real = 0.666, D_loss_fake = 0.691, G_loss = 0.866
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 37/46, D_loss_real = 0.682, D_loss_fake = 0.677, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 38/46, D_loss_real = 0.731, D_loss_fake = 0.662, G_loss = 0.923
2/2 [=====] - 0s 4ms/step

Epoch 47, Iteration 39/46, D_loss_real = 0.738, D_loss_fake = 0.587, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 40/46, D_loss_real = 0.714, D_loss_fake = 0.548, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 41/46, D_loss_real = 0.710, D_loss_fake = 0.565, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 42/46, D_loss_real = 0.598, D_loss_fake = 0.602, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 43/46, D_loss_real = 0.636, D_loss_fake = 0.562, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 44/46, D_loss_real = 0.626, D_loss_fake = 0.596, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 47, Iteration 45/46, D_loss_real = 0.529, D_loss_fake = 0.590, G_loss = 0.827
2/2 [=====] - 0s 3ms/step
Epoch 47, Iteration 46/46, D_loss_real = 0.538, D_loss_fake = 0.789, G_loss = 0.847
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 1/46, D_loss_real = 0.521, D_loss_fake = 0.675, G_loss = 0.925
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 2/46, D_loss_real = 0.567, D_loss_fake = 0.647, G_loss = 0.989
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 3/46, D_loss_real = 0.648, D_loss_fake = 0.547, G_loss = 1.086
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 4/46, D_loss_real = 0.659, D_loss_fake = 0.580, G_loss = 0.940
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 5/46, D_loss_real = 0.680, D_loss_fake = 0.581, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 6/46, D_loss_real = 0.650, D_loss_fake = 0.628, G_loss = 0.844
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 7/46, D_loss_real = 0.607, D_loss_fake = 0.747, G_loss = 0.840
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 8/46, D_loss_real = 0.650, D_loss_fake = 0.671, G_loss = 0.891
2/2 [=====] - 0s 4ms/step

Epoch 48, Iteration 9/46, D_loss_real = 0.669, D_loss_fake = 0.662, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 10/46, D_loss_real = 0.626, D_loss_fake = 0.648, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 11/46, D_loss_real = 0.647, D_loss_fake = 0.603, G_loss = 0.808
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 12/46, D_loss_real = 0.650, D_loss_fake = 0.637, G_loss = 0.804
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 13/46, D_loss_real = 0.587, D_loss_fake = 0.804, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 14/46, D_loss_real = 0.653, D_loss_fake = 0.621, G_loss = 0.994
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 15/46, D_loss_real = 0.739, D_loss_fake = 0.622, G_loss = 0.935
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 16/46, D_loss_real = 0.670, D_loss_fake = 0.591, G_loss = 0.940
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 17/46, D_loss_real = 0.700, D_loss_fake = 0.626, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 18/46, D_loss_real = 0.740, D_loss_fake = 0.618, G_loss = 0.849
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 19/46, D_loss_real = 0.605, D_loss_fake = 0.651, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 20/46, D_loss_real = 0.582, D_loss_fake = 0.607, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 21/46, D_loss_real = 0.588, D_loss_fake = 0.664, G_loss = 0.889
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 22/46, D_loss_real = 0.612, D_loss_fake = 0.629, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 23/46, D_loss_real = 0.673, D_loss_fake = 0.674, G_loss = 0.853
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 24/46, D_loss_real = 0.688, D_loss_fake = 0.709, G_loss = 0.865
2/2 [=====] - 0s 3ms/step

Epoch 48, Iteration 25/46, D_loss_real = 0.641, D_loss_fake = 0.699, G_loss = 0.850
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 26/46, D_loss_real = 0.598, D_loss_fake = 0.676, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 27/46, D_loss_real = 0.657, D_loss_fake = 0.615, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 28/46, D_loss_real = 0.691, D_loss_fake = 0.621, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 29/46, D_loss_real = 0.703, D_loss_fake = 0.665, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 30/46, D_loss_real = 0.720, D_loss_fake = 0.537, G_loss = 1.006
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 31/46, D_loss_real = 0.721, D_loss_fake = 0.565, G_loss = 1.012
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 32/46, D_loss_real = 0.739, D_loss_fake = 0.540, G_loss = 0.939
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 33/46, D_loss_real = 0.722, D_loss_fake = 0.595, G_loss = 0.937
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 34/46, D_loss_real = 0.710, D_loss_fake = 0.586, G_loss = 0.919
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 35/46, D_loss_real = 0.710, D_loss_fake = 0.567, G_loss = 0.935
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 36/46, D_loss_real = 0.692, D_loss_fake = 0.578, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 37/46, D_loss_real = 0.685, D_loss_fake = 0.571, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 38/46, D_loss_real = 0.672, D_loss_fake = 0.627, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 39/46, D_loss_real = 0.578, D_loss_fake = 0.588, G_loss = 0.836
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 40/46, D_loss_real = 0.641, D_loss_fake = 0.605, G_loss = 0.853
2/2 [=====] - 0s 3ms/step

Epoch 48, Iteration 41/46, D_loss_real = 0.722, D_loss_fake = 0.661, G_loss = 0.843
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 42/46, D_loss_real = 0.627, D_loss_fake = 0.659, G_loss = 0.829
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 43/46, D_loss_real = 0.581, D_loss_fake = 0.679, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 48, Iteration 44/46, D_loss_real = 0.640, D_loss_fake = 0.648, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 45/46, D_loss_real = 0.621, D_loss_fake = 0.628, G_loss = 0.854
2/2 [=====] - 0s 3ms/step
Epoch 48, Iteration 46/46, D_loss_real = 0.556, D_loss_fake = 0.701, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 1/46, D_loss_real = 0.659, D_loss_fake = 0.639, G_loss = 0.931
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 2/46, D_loss_real = 0.677, D_loss_fake = 0.589, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 3/46, D_loss_real = 0.638, D_loss_fake = 0.590, G_loss = 0.880
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 4/46, D_loss_real = 0.579, D_loss_fake = 0.614, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 5/46, D_loss_real = 0.654, D_loss_fake = 0.631, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 6/46, D_loss_real = 0.550, D_loss_fake = 0.595, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 7/46, D_loss_real = 0.604, D_loss_fake = 0.631, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 8/46, D_loss_real = 0.557, D_loss_fake = 0.670, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 9/46, D_loss_real = 0.656, D_loss_fake = 0.670, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 10/46, D_loss_real = 0.653, D_loss_fake = 0.740, G_loss = 0.886
2/2 [=====] - 0s 4ms/step

Epoch 49, Iteration 11/46, D_loss_real = 0.639, D_loss_fake = 0.657, G_loss = 0.859
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 12/46, D_loss_real = 0.582, D_loss_fake = 0.642, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 13/46, D_loss_real = 0.605, D_loss_fake = 0.704, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 14/46, D_loss_real = 0.662, D_loss_fake = 0.688, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 15/46, D_loss_real = 0.580, D_loss_fake = 0.711, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 16/46, D_loss_real = 0.635, D_loss_fake = 0.643, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 17/46, D_loss_real = 0.657, D_loss_fake = 0.600, G_loss = 0.931
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 18/46, D_loss_real = 0.605, D_loss_fake = 0.606, G_loss = 0.961
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 19/46, D_loss_real = 0.628, D_loss_fake = 0.566, G_loss = 0.980
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 20/46, D_loss_real = 0.671, D_loss_fake = 0.564, G_loss = 0.948
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 21/46, D_loss_real = 0.652, D_loss_fake = 0.660, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 22/46, D_loss_real = 0.661, D_loss_fake = 0.618, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 23/46, D_loss_real = 0.583, D_loss_fake = 0.632, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 24/46, D_loss_real = 0.650, D_loss_fake = 0.799, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 25/46, D_loss_real = 0.662, D_loss_fake = 0.712, G_loss = 0.938
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 26/46, D_loss_real = 0.687, D_loss_fake = 0.636, G_loss = 0.925
2/2 [=====] - 0s 3ms/step

Epoch 49, Iteration 27/46, D_loss_real = 0.702, D_loss_fake = 0.562, G_loss = 0.925
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 28/46, D_loss_real = 0.711, D_loss_fake = 0.594, G_loss = 0.972
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 29/46, D_loss_real = 0.680, D_loss_fake = 0.599, G_loss = 0.957
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 30/46, D_loss_real = 0.695, D_loss_fake = 0.627, G_loss = 0.967
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 31/46, D_loss_real = 0.677, D_loss_fake = 0.611, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 32/46, D_loss_real = 0.577, D_loss_fake = 0.645, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 33/46, D_loss_real = 0.632, D_loss_fake = 0.593, G_loss = 0.874
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 34/46, D_loss_real = 0.634, D_loss_fake = 0.627, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 35/46, D_loss_real = 0.716, D_loss_fake = 0.617, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 36/46, D_loss_real = 0.652, D_loss_fake = 0.658, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 37/46, D_loss_real = 0.696, D_loss_fake = 0.632, G_loss = 0.912
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 38/46, D_loss_real = 0.654, D_loss_fake = 0.590, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 39/46, D_loss_real = 0.698, D_loss_fake = 0.590, G_loss = 0.877
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 40/46, D_loss_real = 0.623, D_loss_fake = 0.640, G_loss = 0.868
2/2 [=====] - 0s 4ms/step
Epoch 49, Iteration 41/46, D_loss_real = 0.616, D_loss_fake = 0.695, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 42/46, D_loss_real = 0.610, D_loss_fake = 0.730, G_loss = 0.813
2/2 [=====] - 0s 3ms/step

Epoch 49, Iteration 43/46, D_loss_real = 0.608, D_loss_fake = 0.660, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 44/46, D_loss_real = 0.721, D_loss_fake = 0.678, G_loss = 0.955
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 45/46, D_loss_real = 0.661, D_loss_fake = 0.554, G_loss = 0.959
2/2 [=====] - 0s 3ms/step
Epoch 49, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.639, G_loss = 0.931
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 1/46, D_loss_real = 0.625, D_loss_fake = 0.655, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 2/46, D_loss_real = 0.670, D_loss_fake = 0.596, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 3/46, D_loss_real = 0.681, D_loss_fake = 0.581, G_loss = 0.948
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 4/46, D_loss_real = 0.716, D_loss_fake = 0.605, G_loss = 0.970
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 5/46, D_loss_real = 0.667, D_loss_fake = 0.572, G_loss = 0.962
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 6/46, D_loss_real = 0.659, D_loss_fake = 0.578, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 7/46, D_loss_real = 0.610, D_loss_fake = 0.558, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 8/46, D_loss_real = 0.716, D_loss_fake = 0.627, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 9/46, D_loss_real = 0.566, D_loss_fake = 0.643, G_loss = 0.818
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 10/46, D_loss_real = 0.538, D_loss_fake = 0.687, G_loss = 0.839
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 11/46, D_loss_real = 0.585, D_loss_fake = 0.695, G_loss = 0.848
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 12/46, D_loss_real = 0.619, D_loss_fake = 0.647, G_loss = 0.857
2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 13/46, D_loss_real = 0.638, D_loss_fake = 0.783, G_loss = 0.888
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 14/46, D_loss_real = 0.582, D_loss_fake = 0.628, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 15/46, D_loss_real = 0.557, D_loss_fake = 0.646, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 16/46, D_loss_real = 0.551, D_loss_fake = 0.671, G_loss = 0.878
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 17/46, D_loss_real = 0.537, D_loss_fake = 0.658, G_loss = 0.920
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 18/46, D_loss_real = 0.580, D_loss_fake = 0.700, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 19/46, D_loss_real = 0.598, D_loss_fake = 0.538, G_loss = 0.932
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 20/46, D_loss_real = 0.631, D_loss_fake = 0.649, G_loss = 0.950
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 21/46, D_loss_real = 0.734, D_loss_fake = 0.647, G_loss = 0.868
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 22/46, D_loss_real = 0.639, D_loss_fake = 0.661, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 23/46, D_loss_real = 0.661, D_loss_fake = 0.625, G_loss = 0.908
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 24/46, D_loss_real = 0.669, D_loss_fake = 0.602, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 25/46, D_loss_real = 0.674, D_loss_fake = 0.639, G_loss = 0.916
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 26/46, D_loss_real = 0.658, D_loss_fake = 0.608, G_loss = 0.903
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 27/46, D_loss_real = 0.625, D_loss_fake = 0.653, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 28/46, D_loss_real = 0.647, D_loss_fake = 0.717, G_loss = 0.956
2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 29/46, D_loss_real = 0.717, D_loss_fake = 0.625, G_loss = 0.952
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 30/46, D_loss_real = 0.694, D_loss_fake = 0.512, G_loss = 0.996
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 31/46, D_loss_real = 0.659, D_loss_fake = 0.596, G_loss = 0.983
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 32/46, D_loss_real = 0.641, D_loss_fake = 0.560, G_loss = 1.019
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 33/46, D_loss_real = 0.665, D_loss_fake = 0.550, G_loss = 0.920
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 34/46, D_loss_real = 0.551, D_loss_fake = 0.590, G_loss = 0.968
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 35/46, D_loss_real = 0.640, D_loss_fake = 0.605, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 36/46, D_loss_real = 0.579, D_loss_fake = 0.722, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 37/46, D_loss_real = 0.585, D_loss_fake = 0.725, G_loss = 0.931
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 38/46, D_loss_real = 0.561, D_loss_fake = 0.616, G_loss = 0.976
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 39/46, D_loss_real = 0.599, D_loss_fake = 0.557, G_loss = 0.961
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 40/46, D_loss_real = 0.629, D_loss_fake = 0.760, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 41/46, D_loss_real = 0.520, D_loss_fake = 0.805, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 42/46, D_loss_real = 0.604, D_loss_fake = 0.674, G_loss = 1.069
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 43/46, D_loss_real = 0.687, D_loss_fake = 0.527, G_loss = 1.053
2/2 [=====] - 0s 3ms/step
Epoch 50, Iteration 44/46, D_loss_real = 0.715, D_loss_fake = 0.577, G_loss = 0.963
2/2 [=====] - 0s 3ms/step

Epoch 50, Iteration 45/46, D_loss_real = 0.666, D_loss_fake = 0.634, G_loss = 0.956
2/2 [=====] - 0s 4ms/step
Epoch 50, Iteration 46/46, D_loss_real = 0.622, D_loss_fake = 0.558, G_loss = 0.931
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 1/46, D_loss_real = 0.665, D_loss_fake = 0.574, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 2/46, D_loss_real = 0.661, D_loss_fake = 0.605, G_loss = 0.951
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 3/46, D_loss_real = 0.671, D_loss_fake = 0.612, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 4/46, D_loss_real = 0.619, D_loss_fake = 0.606, G_loss = 0.935
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 5/46, D_loss_real = 0.596, D_loss_fake = 0.613, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 6/46, D_loss_real = 0.569, D_loss_fake = 0.612, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 7/46, D_loss_real = 0.609, D_loss_fake = 0.700, G_loss = 0.974
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 8/46, D_loss_real = 0.586, D_loss_fake = 0.582, G_loss = 1.069
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 9/46, D_loss_real = 0.660, D_loss_fake = 0.537, G_loss = 1.038
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 10/46, D_loss_real = 0.667, D_loss_fake = 0.541, G_loss = 1.007
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 11/46, D_loss_real = 0.749, D_loss_fake = 0.631, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 12/46, D_loss_real = 0.709, D_loss_fake = 0.645, G_loss = 0.888
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 13/46, D_loss_real = 0.747, D_loss_fake = 0.615, G_loss = 0.847
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 14/46, D_loss_real = 0.645, D_loss_fake = 0.669, G_loss = 0.828
2/2 [=====] - 0s 4ms/step

Epoch 51, Iteration 15/46, D_loss_real = 0.591, D_loss_fake = 0.726, G_loss = 0.764
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 16/46, D_loss_real = 0.604, D_loss_fake = 0.710, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 17/46, D_loss_real = 0.560, D_loss_fake = 0.674, G_loss = 0.964
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 18/46, D_loss_real = 0.640, D_loss_fake = 0.559, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 19/46, D_loss_real = 0.662, D_loss_fake = 0.563, G_loss = 0.903
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 20/46, D_loss_real = 0.613, D_loss_fake = 0.638, G_loss = 0.931
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 21/46, D_loss_real = 0.594, D_loss_fake = 0.561, G_loss = 0.982
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 22/46, D_loss_real = 0.607, D_loss_fake = 0.605, G_loss = 0.978
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 23/46, D_loss_real = 0.603, D_loss_fake = 0.643, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 24/46, D_loss_real = 0.600, D_loss_fake = 0.621, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 25/46, D_loss_real = 0.615, D_loss_fake = 0.623, G_loss = 0.945
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 26/46, D_loss_real = 0.609, D_loss_fake = 0.627, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 27/46, D_loss_real = 0.700, D_loss_fake = 0.635, G_loss = 0.884
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 28/46, D_loss_real = 0.589, D_loss_fake = 0.597, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 29/46, D_loss_real = 0.616, D_loss_fake = 0.667, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 30/46, D_loss_real = 0.630, D_loss_fake = 0.716, G_loss = 0.955
2/2 [=====] - 0s 4ms/step

Epoch 51, Iteration 31/46, D_loss_real = 0.709, D_loss_fake = 0.693, G_loss = 0.980
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 32/46, D_loss_real = 0.806, D_loss_fake = 0.540, G_loss = 0.959
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 33/46, D_loss_real = 0.747, D_loss_fake = 0.543, G_loss = 0.964
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 34/46, D_loss_real = 0.742, D_loss_fake = 0.622, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 35/46, D_loss_real = 0.675, D_loss_fake = 0.599, G_loss = 0.942
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 36/46, D_loss_real = 0.584, D_loss_fake = 0.633, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 37/46, D_loss_real = 0.740, D_loss_fake = 0.612, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 38/46, D_loss_real = 0.638, D_loss_fake = 0.660, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 39/46, D_loss_real = 0.652, D_loss_fake = 0.700, G_loss = 0.906
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 40/46, D_loss_real = 0.599, D_loss_fake = 0.625, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 41/46, D_loss_real = 0.641, D_loss_fake = 0.613, G_loss = 0.920
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 42/46, D_loss_real = 0.613, D_loss_fake = 0.639, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 43/46, D_loss_real = 0.665, D_loss_fake = 0.620, G_loss = 0.921
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 44/46, D_loss_real = 0.633, D_loss_fake = 0.641, G_loss = 0.923
2/2 [=====] - 0s 4ms/step
Epoch 51, Iteration 45/46, D_loss_real = 0.685, D_loss_fake = 0.595, G_loss = 0.888
2/2 [=====] - 0s 3ms/step
Epoch 51, Iteration 46/46, D_loss_real = 0.647, D_loss_fake = 0.681, G_loss = 0.910
2/2 [=====] - 0s 3ms/step

Epoch 52, Iteration 1/46, D_loss_real = 0.663, D_loss_fake = 0.629, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 2/46, D_loss_real = 0.643, D_loss_fake = 0.624, G_loss = 0.932
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 3/46, D_loss_real = 0.655, D_loss_fake = 0.645, G_loss = 0.945
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 4/46, D_loss_real = 0.715, D_loss_fake = 0.643, G_loss = 0.950
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 5/46, D_loss_real = 0.600, D_loss_fake = 0.645, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 6/46, D_loss_real = 0.623, D_loss_fake = 0.649, G_loss = 0.940
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 7/46, D_loss_real = 0.613, D_loss_fake = 0.585, G_loss = 0.972
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 8/46, D_loss_real = 0.585, D_loss_fake = 0.578, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 9/46, D_loss_real = 0.616, D_loss_fake = 0.648, G_loss = 0.876
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 10/46, D_loss_real = 0.594, D_loss_fake = 0.674, G_loss = 0.852
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 11/46, D_loss_real = 0.551, D_loss_fake = 0.693, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 12/46, D_loss_real = 0.659, D_loss_fake = 0.715, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 13/46, D_loss_real = 0.670, D_loss_fake = 0.621, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 14/46, D_loss_real = 0.684, D_loss_fake = 0.508, G_loss = 0.978
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 15/46, D_loss_real = 0.636, D_loss_fake = 0.658, G_loss = 0.938
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 16/46, D_loss_real = 0.637, D_loss_fake = 0.722, G_loss = 0.874
2/2 [=====] - 0s 3ms/step

Epoch 52, Iteration 17/46, D_loss_real = 0.656, D_loss_fake = 0.672, G_loss = 0.918
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 18/46, D_loss_real = 0.631, D_loss_fake = 0.609, G_loss = 0.948
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 19/46, D_loss_real = 0.694, D_loss_fake = 0.640, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 20/46, D_loss_real = 0.658, D_loss_fake = 0.666, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 21/46, D_loss_real = 0.671, D_loss_fake = 0.673, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 22/46, D_loss_real = 0.611, D_loss_fake = 0.665, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 23/46, D_loss_real = 0.645, D_loss_fake = 0.610, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 24/46, D_loss_real = 0.610, D_loss_fake = 0.619, G_loss = 0.842
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 25/46, D_loss_real = 0.602, D_loss_fake = 0.663, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 26/46, D_loss_real = 0.619, D_loss_fake = 0.616, G_loss = 0.905
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 27/46, D_loss_real = 0.631, D_loss_fake = 0.606, G_loss = 0.966
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 28/46, D_loss_real = 0.635, D_loss_fake = 0.687, G_loss = 0.953
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 29/46, D_loss_real = 0.670, D_loss_fake = 0.672, G_loss = 0.960
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 30/46, D_loss_real = 0.681, D_loss_fake = 0.567, G_loss = 0.962
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 31/46, D_loss_real = 0.695, D_loss_fake = 0.598, G_loss = 0.946
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 32/46, D_loss_real = 0.533, D_loss_fake = 0.620, G_loss = 0.923
2/2 [=====] - 0s 3ms/step

Epoch 52, Iteration 33/46, D_loss_real = 0.628, D_loss_fake = 0.642, G_loss = 0.985
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 34/46, D_loss_real = 0.601, D_loss_fake = 0.628, G_loss = 1.077
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 35/46, D_loss_real = 0.763, D_loss_fake = 0.540, G_loss = 1.060
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 36/46, D_loss_real = 0.725, D_loss_fake = 0.634, G_loss = 0.998
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 37/46, D_loss_real = 0.723, D_loss_fake = 0.590, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 38/46, D_loss_real = 0.655, D_loss_fake = 0.550, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 39/46, D_loss_real = 0.626, D_loss_fake = 0.603, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 52, Iteration 40/46, D_loss_real = 0.557, D_loss_fake = 0.656, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 41/46, D_loss_real = 0.567, D_loss_fake = 0.704, G_loss = 0.830
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 42/46, D_loss_real = 0.579, D_loss_fake = 0.681, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.646, G_loss = 0.899
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 44/46, D_loss_real = 0.613, D_loss_fake = 0.667, G_loss = 0.928
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 45/46, D_loss_real = 0.693, D_loss_fake = 0.596, G_loss = 0.996
2/2 [=====] - 0s 3ms/step
Epoch 52, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.540, G_loss = 0.969
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 1/46, D_loss_real = 0.654, D_loss_fake = 0.612, G_loss = 0.934
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 2/46, D_loss_real = 0.695, D_loss_fake = 0.642, G_loss = 0.923
2/2 [=====] - 0s 3ms/step

Epoch 53, Iteration 3/46, D_loss_real = 0.630, D_loss_fake = 0.603, G_loss = 0.983
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 4/46, D_loss_real = 0.694, D_loss_fake = 0.559, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 5/46, D_loss_real = 0.631, D_loss_fake = 0.754, G_loss = 0.952
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 6/46, D_loss_real = 0.588, D_loss_fake = 0.564, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 7/46, D_loss_real = 0.577, D_loss_fake = 0.663, G_loss = 0.927
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 8/46, D_loss_real = 0.658, D_loss_fake = 0.600, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 9/46, D_loss_real = 0.631, D_loss_fake = 0.741, G_loss = 0.939
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 10/46, D_loss_real = 0.668, D_loss_fake = 0.610, G_loss = 0.923
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 11/46, D_loss_real = 0.653, D_loss_fake = 0.627, G_loss = 0.957
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 12/46, D_loss_real = 0.673, D_loss_fake = 0.607, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 13/46, D_loss_real = 0.736, D_loss_fake = 0.676, G_loss = 0.958
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 14/46, D_loss_real = 0.720, D_loss_fake = 0.600, G_loss = 0.974
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 15/46, D_loss_real = 0.702, D_loss_fake = 0.554, G_loss = 1.001
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 16/46, D_loss_real = 0.741, D_loss_fake = 0.546, G_loss = 0.957
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 17/46, D_loss_real = 0.684, D_loss_fake = 0.594, G_loss = 0.975
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 18/46, D_loss_real = 0.598, D_loss_fake = 0.573, G_loss = 0.907
2/2 [=====] - 0s 3ms/step

Epoch 53, Iteration 19/46, D_loss_real = 0.568, D_loss_fake = 0.637, G_loss = 0.921
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 20/46, D_loss_real = 0.644, D_loss_fake = 0.652, G_loss = 0.869
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 21/46, D_loss_real = 0.561, D_loss_fake = 0.648, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 22/46, D_loss_real = 0.533, D_loss_fake = 0.681, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 23/46, D_loss_real = 0.593, D_loss_fake = 0.724, G_loss = 0.826
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 24/46, D_loss_real = 0.630, D_loss_fake = 0.652, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 25/46, D_loss_real = 0.688, D_loss_fake = 0.671, G_loss = 0.855
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 26/46, D_loss_real = 0.622, D_loss_fake = 0.634, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 27/46, D_loss_real = 0.678, D_loss_fake = 0.634, G_loss = 0.907
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 28/46, D_loss_real = 0.594, D_loss_fake = 0.639, G_loss = 0.874
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 29/46, D_loss_real = 0.696, D_loss_fake = 0.619, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 30/46, D_loss_real = 0.599, D_loss_fake = 0.659, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 31/46, D_loss_real = 0.648, D_loss_fake = 0.640, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 32/46, D_loss_real = 0.624, D_loss_fake = 0.577, G_loss = 0.944
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 33/46, D_loss_real = 0.675, D_loss_fake = 0.599, G_loss = 0.969
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 34/46, D_loss_real = 0.704, D_loss_fake = 0.580, G_loss = 0.936
2/2 [=====] - 0s 3ms/step

Epoch 53, Iteration 35/46, D_loss_real = 0.692, D_loss_fake = 0.586, G_loss = 0.975
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 36/46, D_loss_real = 0.674, D_loss_fake = 0.600, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 37/46, D_loss_real = 0.675, D_loss_fake = 0.592, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 38/46, D_loss_real = 0.651, D_loss_fake = 0.611, G_loss = 0.974
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 39/46, D_loss_real = 0.539, D_loss_fake = 0.606, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 40/46, D_loss_real = 0.605, D_loss_fake = 0.570, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 41/46, D_loss_real = 0.649, D_loss_fake = 0.657, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 42/46, D_loss_real = 0.550, D_loss_fake = 0.602, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 43/46, D_loss_real = 0.644, D_loss_fake = 0.618, G_loss = 0.867
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 44/46, D_loss_real = 0.549, D_loss_fake = 0.717, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 53, Iteration 45/46, D_loss_real = 0.554, D_loss_fake = 0.655, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 53, Iteration 46/46, D_loss_real = 0.604, D_loss_fake = 0.644, G_loss = 0.833
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 1/46, D_loss_real = 0.615, D_loss_fake = 0.643, G_loss = 0.862
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 2/46, D_loss_real = 0.717, D_loss_fake = 0.623, G_loss = 0.836
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 3/46, D_loss_real = 0.628, D_loss_fake = 0.685, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 4/46, D_loss_real = 0.593, D_loss_fake = 0.643, G_loss = 0.941
2/2 [=====] - 0s 4ms/step

Epoch 54, Iteration 5/46, D_loss_real = 0.720, D_loss_fake = 0.688, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 6/46, D_loss_real = 0.676, D_loss_fake = 0.656, G_loss = 0.939
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 7/46, D_loss_real = 0.733, D_loss_fake = 0.673, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 8/46, D_loss_real = 0.671, D_loss_fake = 0.630, G_loss = 0.881
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 9/46, D_loss_real = 0.720, D_loss_fake = 0.632, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 10/46, D_loss_real = 0.652, D_loss_fake = 0.579, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 11/46, D_loss_real = 0.644, D_loss_fake = 0.587, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 12/46, D_loss_real = 0.652, D_loss_fake = 0.617, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 13/46, D_loss_real = 0.668, D_loss_fake = 0.596, G_loss = 0.933
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 14/46, D_loss_real = 0.660, D_loss_fake = 0.576, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 15/46, D_loss_real = 0.638, D_loss_fake = 0.593, G_loss = 0.945
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 16/46, D_loss_real = 0.611, D_loss_fake = 0.625, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 17/46, D_loss_real = 0.598, D_loss_fake = 0.585, G_loss = 0.958
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 18/46, D_loss_real = 0.675, D_loss_fake = 0.609, G_loss = 0.888
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 19/46, D_loss_real = 0.586, D_loss_fake = 0.625, G_loss = 0.896
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 20/46, D_loss_real = 0.666, D_loss_fake = 0.592, G_loss = 0.912
2/2 [=====] - 0s 3ms/step

Epoch 54, Iteration 21/46, D_loss_real = 0.632, D_loss_fake = 0.668, G_loss = 0.834
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 22/46, D_loss_real = 0.587, D_loss_fake = 0.611, G_loss = 0.838
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 23/46, D_loss_real = 0.620, D_loss_fake = 0.661, G_loss = 0.849
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 24/46, D_loss_real = 0.567, D_loss_fake = 0.636, G_loss = 0.861
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 25/46, D_loss_real = 0.577, D_loss_fake = 0.654, G_loss = 0.879
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 26/46, D_loss_real = 0.548, D_loss_fake = 0.616, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 27/46, D_loss_real = 0.611, D_loss_fake = 0.628, G_loss = 0.853
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 28/46, D_loss_real = 0.631, D_loss_fake = 0.622, G_loss = 0.885
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 29/46, D_loss_real = 0.595, D_loss_fake = 0.779, G_loss = 0.870
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 30/46, D_loss_real = 0.622, D_loss_fake = 0.632, G_loss = 1.007
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 31/46, D_loss_real = 0.720, D_loss_fake = 0.525, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 32/46, D_loss_real = 0.657, D_loss_fake = 0.606, G_loss = 0.867
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 33/46, D_loss_real = 0.629, D_loss_fake = 0.608, G_loss = 0.897
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 34/46, D_loss_real = 0.700, D_loss_fake = 0.784, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 35/46, D_loss_real = 0.685, D_loss_fake = 0.654, G_loss = 0.863
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 36/46, D_loss_real = 0.649, D_loss_fake = 0.618, G_loss = 0.878
2/2 [=====] - 0s 4ms/step

Epoch 54, Iteration 37/46, D_loss_real = 0.598, D_loss_fake = 0.669, G_loss = 0.887
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 38/46, D_loss_real = 0.655, D_loss_fake = 0.594, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 39/46, D_loss_real = 0.593, D_loss_fake = 0.699, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 40/46, D_loss_real = 0.593, D_loss_fake = 0.689, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 41/46, D_loss_real = 0.625, D_loss_fake = 0.634, G_loss = 0.856
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 42/46, D_loss_real = 0.697, D_loss_fake = 0.646, G_loss = 0.846
2/2 [=====] - 0s 3ms/step
Epoch 54, Iteration 43/46, D_loss_real = 0.682, D_loss_fake = 0.682, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 44/46, D_loss_real = 0.651, D_loss_fake = 0.621, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 45/46, D_loss_real = 0.709, D_loss_fake = 0.618, G_loss = 0.906
2/2 [=====] - 0s 4ms/step
Epoch 54, Iteration 46/46, D_loss_real = 0.606, D_loss_fake = 0.664, G_loss = 0.936
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 1/46, D_loss_real = 0.646, D_loss_fake = 0.619, G_loss = 0.901
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 2/46, D_loss_real = 0.609, D_loss_fake = 0.612, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 3/46, D_loss_real = 0.545, D_loss_fake = 0.653, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 4/46, D_loss_real = 0.711, D_loss_fake = 0.590, G_loss = 0.886
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 5/46, D_loss_real = 0.648, D_loss_fake = 0.618, G_loss = 0.931
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 6/46, D_loss_real = 0.743, D_loss_fake = 0.609, G_loss = 0.937
2/2 [=====] - 0s 4ms/step

Epoch 55, Iteration 7/46, D_loss_real = 0.687, D_loss_fake = 0.649, G_loss = 1.013
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 8/46, D_loss_real = 0.673, D_loss_fake = 0.624, G_loss = 0.958
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 9/46, D_loss_real = 0.692, D_loss_fake = 0.586, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 10/46, D_loss_real = 0.603, D_loss_fake = 0.618, G_loss = 0.930
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 11/46, D_loss_real = 0.606, D_loss_fake = 0.635, G_loss = 0.871
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 12/46, D_loss_real = 0.626, D_loss_fake = 0.617, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 13/46, D_loss_real = 0.581, D_loss_fake = 0.642, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 14/46, D_loss_real = 0.585, D_loss_fake = 0.711, G_loss = 0.974
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 15/46, D_loss_real = 0.558, D_loss_fake = 0.707, G_loss = 0.919
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 16/46, D_loss_real = 0.665, D_loss_fake = 0.581, G_loss = 0.935
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 17/46, D_loss_real = 0.687, D_loss_fake = 0.564, G_loss = 0.988
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 18/46, D_loss_real = 0.716, D_loss_fake = 0.626, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 19/46, D_loss_real = 0.641, D_loss_fake = 0.629, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 20/46, D_loss_real = 0.623, D_loss_fake = 0.657, G_loss = 0.898
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 21/46, D_loss_real = 0.565, D_loss_fake = 0.605, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 22/46, D_loss_real = 0.605, D_loss_fake = 0.644, G_loss = 0.886
2/2 [=====] - 0s 3ms/step

Epoch 55, Iteration 23/46, D_loss_real = 0.583, D_loss_fake = 0.731, G_loss = 0.988
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 24/46, D_loss_real = 0.567, D_loss_fake = 0.587, G_loss = 0.961
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 25/46, D_loss_real = 0.709, D_loss_fake = 0.598, G_loss = 0.964
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 26/46, D_loss_real = 0.643, D_loss_fake = 0.642, G_loss = 0.890
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 27/46, D_loss_real = 0.656, D_loss_fake = 0.640, G_loss = 0.870
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 28/46, D_loss_real = 0.602, D_loss_fake = 0.721, G_loss = 0.848
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 29/46, D_loss_real = 0.613, D_loss_fake = 0.736, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 30/46, D_loss_real = 0.616, D_loss_fake = 0.691, G_loss = 0.959
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 31/46, D_loss_real = 0.684, D_loss_fake = 0.586, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 32/46, D_loss_real = 0.684, D_loss_fake = 0.645, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 33/46, D_loss_real = 0.661, D_loss_fake = 0.651, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 34/46, D_loss_real = 0.592, D_loss_fake = 0.599, G_loss = 0.955
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 35/46, D_loss_real = 0.669, D_loss_fake = 0.576, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 36/46, D_loss_real = 0.648, D_loss_fake = 0.686, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 37/46, D_loss_real = 0.645, D_loss_fake = 0.670, G_loss = 0.927
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 38/46, D_loss_real = 0.664, D_loss_fake = 0.661, G_loss = 0.994
2/2 [=====] - 0s 4ms/step

Epoch 55, Iteration 39/46, D_loss_real = 0.662, D_loss_fake = 0.565, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 40/46, D_loss_real = 0.738, D_loss_fake = 0.638, G_loss = 0.857
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 41/46, D_loss_real = 0.695, D_loss_fake = 0.609, G_loss = 0.875
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 42/46, D_loss_real = 0.598, D_loss_fake = 0.553, G_loss = 0.941
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 43/46, D_loss_real = 0.634, D_loss_fake = 0.575, G_loss = 0.956
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 44/46, D_loss_real = 0.644, D_loss_fake = 0.605, G_loss = 0.936
2/2 [=====] - 0s 3ms/step
Epoch 55, Iteration 45/46, D_loss_real = 0.645, D_loss_fake = 0.626, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 55, Iteration 46/46, D_loss_real = 0.666, D_loss_fake = 0.612, G_loss = 0.950
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 1/46, D_loss_real = 0.664, D_loss_fake = 0.655, G_loss = 0.946
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 2/46, D_loss_real = 0.646, D_loss_fake = 0.568, G_loss = 0.875
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 3/46, D_loss_real = 0.606, D_loss_fake = 0.659, G_loss = 0.861
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 4/46, D_loss_real = 0.575, D_loss_fake = 0.634, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 5/46, D_loss_real = 0.652, D_loss_fake = 0.616, G_loss = 0.909
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 6/46, D_loss_real = 0.650, D_loss_fake = 0.655, G_loss = 0.904
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 7/46, D_loss_real = 0.646, D_loss_fake = 0.654, G_loss = 0.887
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 8/46, D_loss_real = 0.565, D_loss_fake = 0.592, G_loss = 0.883
2/2 [=====] - 0s 4ms/step

Epoch 56, Iteration 9/46, D_loss_real = 0.565, D_loss_fake = 0.594, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 10/46, D_loss_real = 0.560, D_loss_fake = 0.706, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 11/46, D_loss_real = 0.636, D_loss_fake = 0.712, G_loss = 0.921
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 12/46, D_loss_real = 0.636, D_loss_fake = 0.614, G_loss = 0.856
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 13/46, D_loss_real = 0.636, D_loss_fake = 0.626, G_loss = 0.869
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 14/46, D_loss_real = 0.569, D_loss_fake = 0.758, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 15/46, D_loss_real = 0.582, D_loss_fake = 0.688, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 16/46, D_loss_real = 0.702, D_loss_fake = 0.681, G_loss = 0.975
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 17/46, D_loss_real = 0.720, D_loss_fake = 0.639, G_loss = 0.980
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 18/46, D_loss_real = 0.726, D_loss_fake = 0.601, G_loss = 0.997
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 19/46, D_loss_real = 0.689, D_loss_fake = 0.565, G_loss = 0.953
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 20/46, D_loss_real = 0.680, D_loss_fake = 0.691, G_loss = 0.973
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 21/46, D_loss_real = 0.654, D_loss_fake = 0.577, G_loss = 0.984
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 22/46, D_loss_real = 0.643, D_loss_fake = 0.605, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 23/46, D_loss_real = 0.672, D_loss_fake = 0.625, G_loss = 0.951
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 24/46, D_loss_real = 0.586, D_loss_fake = 0.685, G_loss = 0.889
2/2 [=====] - 0s 4ms/step

Epoch 56, Iteration 25/46, D_loss_real = 0.635, D_loss_fake = 0.642, G_loss = 0.895
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 26/46, D_loss_real = 0.627, D_loss_fake = 0.669, G_loss = 0.904
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 27/46, D_loss_real = 0.629, D_loss_fake = 0.670, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 28/46, D_loss_real = 0.655, D_loss_fake = 0.668, G_loss = 0.916
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 29/46, D_loss_real = 0.708, D_loss_fake = 0.582, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 30/46, D_loss_real = 0.611, D_loss_fake = 0.621, G_loss = 0.917
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 31/46, D_loss_real = 0.597, D_loss_fake = 0.666, G_loss = 0.893
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 32/46, D_loss_real = 0.617, D_loss_fake = 0.626, G_loss = 0.857
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 33/46, D_loss_real = 0.616, D_loss_fake = 0.718, G_loss = 0.999
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 34/46, D_loss_real = 0.682, D_loss_fake = 0.540, G_loss = 0.999
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 35/46, D_loss_real = 0.736, D_loss_fake = 0.533, G_loss = 1.013
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 36/46, D_loss_real = 0.682, D_loss_fake = 0.622, G_loss = 0.937
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 37/46, D_loss_real = 0.612, D_loss_fake = 0.583, G_loss = 0.899
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 38/46, D_loss_real = 0.646, D_loss_fake = 0.614, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 39/46, D_loss_real = 0.656, D_loss_fake = 0.585, G_loss = 0.882
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 40/46, D_loss_real = 0.635, D_loss_fake = 0.669, G_loss = 0.941
2/2 [=====] - 0s 3ms/step

Epoch 56, Iteration 41/46, D_loss_real = 0.622, D_loss_fake = 0.596, G_loss = 0.956
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 42/46, D_loss_real = 0.605, D_loss_fake = 0.672, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 43/46, D_loss_real = 0.606, D_loss_fake = 0.668, G_loss = 0.954
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 44/46, D_loss_real = 0.613, D_loss_fake = 0.611, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 56, Iteration 45/46, D_loss_real = 0.645, D_loss_fake = 0.694, G_loss = 0.880
2/2 [=====] - 0s 4ms/step
Epoch 56, Iteration 46/46, D_loss_real = 0.505, D_loss_fake = 0.675, G_loss = 0.829
2/2 [=====] - 0s 5ms/step
Epoch 57, Iteration 1/46, D_loss_real = 0.613, D_loss_fake = 0.709, G_loss = 0.951
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 2/46, D_loss_real = 0.690, D_loss_fake = 0.567, G_loss = 1.010
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 3/46, D_loss_real = 0.629, D_loss_fake = 0.602, G_loss = 0.895
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 4/46, D_loss_real = 0.629, D_loss_fake = 0.628, G_loss = 0.898
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 5/46, D_loss_real = 0.600, D_loss_fake = 0.654, G_loss = 0.915
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 6/46, D_loss_real = 0.567, D_loss_fake = 0.603, G_loss = 0.928
2/2 [=====] - 0s 6ms/step
Epoch 57, Iteration 7/46, D_loss_real = 0.639, D_loss_fake = 0.687, G_loss = 0.884
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 8/46, D_loss_real = 0.555, D_loss_fake = 0.625, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 9/46, D_loss_real = 0.613, D_loss_fake = 0.693, G_loss = 0.890
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 10/46, D_loss_real = 0.581, D_loss_fake = 0.583, G_loss = 0.912
2/2 [=====] - 0s 4ms/step

Epoch 57, Iteration 11/46, D_loss_real = 0.652, D_loss_fake = 0.583, G_loss = 0.959
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 12/46, D_loss_real = 0.532, D_loss_fake = 0.561, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 13/46, D_loss_real = 0.640, D_loss_fake = 0.596, G_loss = 0.933
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 14/46, D_loss_real = 0.666, D_loss_fake = 0.641, G_loss = 0.913
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 15/46, D_loss_real = 0.630, D_loss_fake = 0.673, G_loss = 0.910
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 16/46, D_loss_real = 0.578, D_loss_fake = 0.667, G_loss = 0.882
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 17/46, D_loss_real = 0.578, D_loss_fake = 0.661, G_loss = 0.893
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 18/46, D_loss_real = 0.568, D_loss_fake = 0.676, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 19/46, D_loss_real = 0.686, D_loss_fake = 0.614, G_loss = 0.883
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 20/46, D_loss_real = 0.579, D_loss_fake = 0.648, G_loss = 0.892
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 21/46, D_loss_real = 0.520, D_loss_fake = 0.601, G_loss = 0.905
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 22/46, D_loss_real = 0.555, D_loss_fake = 0.603, G_loss = 0.911
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 23/46, D_loss_real = 0.585, D_loss_fake = 0.670, G_loss = 0.942
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 24/46, D_loss_real = 0.561, D_loss_fake = 0.689, G_loss = 0.920
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 25/46, D_loss_real = 0.686, D_loss_fake = 0.682, G_loss = 0.871
2/2 [=====] - 0s 5ms/step
Epoch 57, Iteration 26/46, D_loss_real = 0.641, D_loss_fake = 0.617, G_loss = 0.906
2/2 [=====] - 0s 4ms/step

Epoch 57, Iteration 27/46, D_loss_real = 0.713, D_loss_fake = 0.661, G_loss = 0.958
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 28/46, D_loss_real = 0.729, D_loss_fake = 0.566, G_loss = 0.970
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 29/46, D_loss_real = 0.699, D_loss_fake = 0.535, G_loss = 1.009
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 30/46, D_loss_real = 0.729, D_loss_fake = 0.614, G_loss = 1.007
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 31/46, D_loss_real = 0.701, D_loss_fake = 0.537, G_loss = 0.972
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 32/46, D_loss_real = 0.625, D_loss_fake = 0.744, G_loss = 0.966
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 33/46, D_loss_real = 0.670, D_loss_fake = 0.620, G_loss = 0.862
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 34/46, D_loss_real = 0.648, D_loss_fake = 0.669, G_loss = 0.900
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 35/46, D_loss_real = 0.679, D_loss_fake = 0.645, G_loss = 0.886
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 36/46, D_loss_real = 0.670, D_loss_fake = 0.601, G_loss = 0.854
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 37/46, D_loss_real = 0.573, D_loss_fake = 0.664, G_loss = 0.863
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 38/46, D_loss_real = 0.595, D_loss_fake = 0.551, G_loss = 0.922
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 39/46, D_loss_real = 0.678, D_loss_fake = 0.669, G_loss = 0.930
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 40/46, D_loss_real = 0.666, D_loss_fake = 0.658, G_loss = 0.935
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 41/46, D_loss_real = 0.678, D_loss_fake = 0.571, G_loss = 1.038
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 42/46, D_loss_real = 0.680, D_loss_fake = 0.552, G_loss = 0.910
2/2 [=====] - 0s 3ms/step

Epoch 57, Iteration 43/46, D_loss_real = 0.662, D_loss_fake = 0.605, G_loss = 0.908
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 44/46, D_loss_real = 0.629, D_loss_fake = 0.701, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 57, Iteration 45/46, D_loss_real = 0.622, D_loss_fake = 0.587, G_loss = 0.944
2/2 [=====] - 0s 3ms/step
Epoch 57, Iteration 46/46, D_loss_real = 0.626, D_loss_fake = 0.679, G_loss = 0.941
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 1/46, D_loss_real = 0.619, D_loss_fake = 0.601, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 2/46, D_loss_real = 0.705, D_loss_fake = 0.596, G_loss = 0.993
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 3/46, D_loss_real = 0.708, D_loss_fake = 0.612, G_loss = 0.989
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 4/46, D_loss_real = 0.669, D_loss_fake = 0.563, G_loss = 0.980
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 5/46, D_loss_real = 0.648, D_loss_fake = 0.596, G_loss = 0.999
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 6/46, D_loss_real = 0.620, D_loss_fake = 0.635, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 7/46, D_loss_real = 0.686, D_loss_fake = 0.622, G_loss = 0.891
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 8/46, D_loss_real = 0.605, D_loss_fake = 0.636, G_loss = 0.881
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 9/46, D_loss_real = 0.656, D_loss_fake = 0.655, G_loss = 0.879
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 10/46, D_loss_real = 0.607, D_loss_fake = 0.590, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 11/46, D_loss_real = 0.667, D_loss_fake = 0.610, G_loss = 0.919
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 12/46, D_loss_real = 0.620, D_loss_fake = 0.615, G_loss = 0.851
2/2 [=====] - 0s 3ms/step

Epoch 58, Iteration 13/46, D_loss_real = 0.556, D_loss_fake = 0.678, G_loss = 0.901
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 14/46, D_loss_real = 0.549, D_loss_fake = 0.646, G_loss = 0.897
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 15/46, D_loss_real = 0.603, D_loss_fake = 0.625, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 16/46, D_loss_real = 0.548, D_loss_fake = 0.617, G_loss = 0.889
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 17/46, D_loss_real = 0.564, D_loss_fake = 0.612, G_loss = 0.919
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 18/46, D_loss_real = 0.595, D_loss_fake = 0.548, G_loss = 0.924
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 19/46, D_loss_real = 0.626, D_loss_fake = 0.706, G_loss = 0.913
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 20/46, D_loss_real = 0.480, D_loss_fake = 0.634, G_loss = 0.921
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 21/46, D_loss_real = 0.577, D_loss_fake = 0.668, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 22/46, D_loss_real = 0.632, D_loss_fake = 0.667, G_loss = 0.900
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 23/46, D_loss_real = 0.633, D_loss_fake = 0.601, G_loss = 0.850
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 24/46, D_loss_real = 0.701, D_loss_fake = 0.728, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 25/46, D_loss_real = 0.656, D_loss_fake = 0.531, G_loss = 1.022
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 26/46, D_loss_real = 0.637, D_loss_fake = 0.595, G_loss = 0.972
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 27/46, D_loss_real = 0.599, D_loss_fake = 0.547, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 28/46, D_loss_real = 0.636, D_loss_fake = 0.667, G_loss = 0.934
2/2 [=====] - 0s 3ms/step

Epoch 58, Iteration 29/46, D_loss_real = 0.625, D_loss_fake = 0.580, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 30/46, D_loss_real = 0.625, D_loss_fake = 0.656, G_loss = 0.928
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 31/46, D_loss_real = 0.586, D_loss_fake = 0.597, G_loss = 0.877
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 32/46, D_loss_real = 0.633, D_loss_fake = 0.626, G_loss = 0.929
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 33/46, D_loss_real = 0.617, D_loss_fake = 0.620, G_loss = 0.976
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 34/46, D_loss_real = 0.573, D_loss_fake = 0.624, G_loss = 0.965
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 35/46, D_loss_real = 0.602, D_loss_fake = 0.598, G_loss = 0.929
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 36/46, D_loss_real = 0.551, D_loss_fake = 0.689, G_loss = 0.866
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 37/46, D_loss_real = 0.447, D_loss_fake = 0.658, G_loss = 0.944
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 38/46, D_loss_real = 0.574, D_loss_fake = 0.559, G_loss = 0.973
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 39/46, D_loss_real = 0.532, D_loss_fake = 0.677, G_loss = 0.905
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 40/46, D_loss_real = 0.565, D_loss_fake = 0.630, G_loss = 0.927
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 41/46, D_loss_real = 0.566, D_loss_fake = 0.905, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 42/46, D_loss_real = 0.585, D_loss_fake = 0.649, G_loss = 0.894
2/2 [=====] - 0s 4ms/step
Epoch 58, Iteration 43/46, D_loss_real = 0.627, D_loss_fake = 0.600, G_loss = 0.949
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 44/46, D_loss_real = 0.673, D_loss_fake = 0.540, G_loss = 0.901
2/2 [=====] - 0s 4ms/step

Epoch 58, Iteration 45/46, D_loss_real = 0.620, D_loss_fake = 0.639, G_loss = 0.872
2/2 [=====] - 0s 3ms/step
Epoch 58, Iteration 46/46, D_loss_real = 0.679, D_loss_fake = 0.694, G_loss = 0.953
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 1/46, D_loss_real = 0.608, D_loss_fake = 0.564, G_loss = 0.878
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 2/46, D_loss_real = 0.719, D_loss_fake = 0.651, G_loss = 0.902
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 3/46, D_loss_real = 0.666, D_loss_fake = 0.589, G_loss = 0.950
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 4/46, D_loss_real = 0.600, D_loss_fake = 0.569, G_loss = 0.852
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 5/46, D_loss_real = 0.573, D_loss_fake = 0.700, G_loss = 0.954
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 6/46, D_loss_real = 0.551, D_loss_fake = 0.656, G_loss = 0.992
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 7/46, D_loss_real = 0.622, D_loss_fake = 0.590, G_loss = 1.032
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 8/46, D_loss_real = 0.751, D_loss_fake = 0.591, G_loss = 0.984
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 9/46, D_loss_real = 0.681, D_loss_fake = 0.586, G_loss = 0.998
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 10/46, D_loss_real = 0.697, D_loss_fake = 0.566, G_loss = 0.912
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 11/46, D_loss_real = 0.593, D_loss_fake = 0.621, G_loss = 0.935
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 12/46, D_loss_real = 0.654, D_loss_fake = 0.630, G_loss = 0.965
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 13/46, D_loss_real = 0.681, D_loss_fake = 0.593, G_loss = 0.969
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 14/46, D_loss_real = 0.800, D_loss_fake = 0.580, G_loss = 0.930
2/2 [=====] - 0s 3ms/step

Epoch 59, Iteration 15/46, D_loss_real = 0.672, D_loss_fake = 0.578, G_loss = 0.936
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 16/46, D_loss_real = 0.628, D_loss_fake = 0.568, G_loss = 0.941
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 17/46, D_loss_real = 0.619, D_loss_fake = 0.687, G_loss = 0.864
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 18/46, D_loss_real = 0.615, D_loss_fake = 0.654, G_loss = 0.951
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 19/46, D_loss_real = 0.692, D_loss_fake = 0.612, G_loss = 0.912
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 20/46, D_loss_real = 0.629, D_loss_fake = 0.623, G_loss = 0.873
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 21/46, D_loss_real = 0.575, D_loss_fake = 0.626, G_loss = 0.909
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 22/46, D_loss_real = 0.607, D_loss_fake = 0.645, G_loss = 0.891
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 23/46, D_loss_real = 0.570, D_loss_fake = 0.598, G_loss = 0.927
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 24/46, D_loss_real = 0.626, D_loss_fake = 0.678, G_loss = 0.925
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 25/46, D_loss_real = 0.613, D_loss_fake = 0.628, G_loss = 0.971
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 26/46, D_loss_real = 0.668, D_loss_fake = 0.635, G_loss = 0.942
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 27/46, D_loss_real = 0.623, D_loss_fake = 0.649, G_loss = 0.955
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 28/46, D_loss_real = 0.589, D_loss_fake = 0.568, G_loss = 0.977
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 29/46, D_loss_real = 0.594, D_loss_fake = 0.579, G_loss = 0.956
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 30/46, D_loss_real = 0.610, D_loss_fake = 0.626, G_loss = 1.009
2/2 [=====] - 0s 4ms/step

Epoch 59, Iteration 31/46, D_loss_real = 0.595, D_loss_fake = 0.618, G_loss = 0.985
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 32/46, D_loss_real = 0.670, D_loss_fake = 0.624, G_loss = 0.949
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 33/46, D_loss_real = 0.653, D_loss_fake = 0.597, G_loss = 0.907
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 34/46, D_loss_real = 0.657, D_loss_fake = 0.664, G_loss = 0.939
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 35/46, D_loss_real = 0.690, D_loss_fake = 0.593, G_loss = 0.914
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 36/46, D_loss_real = 0.672, D_loss_fake = 0.600, G_loss = 0.940
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 37/46, D_loss_real = 0.596, D_loss_fake = 0.646, G_loss = 0.928
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 38/46, D_loss_real = 0.608, D_loss_fake = 0.545, G_loss = 0.926
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 39/46, D_loss_real = 0.557, D_loss_fake = 0.629, G_loss = 0.934
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 40/46, D_loss_real = 0.582, D_loss_fake = 0.579, G_loss = 0.942
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 41/46, D_loss_real = 0.638, D_loss_fake = 0.691, G_loss = 0.946
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 42/46, D_loss_real = 0.540, D_loss_fake = 0.663, G_loss = 0.910
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 43/46, D_loss_real = 0.605, D_loss_fake = 0.591, G_loss = 0.999
2/2 [=====] - 0s 4ms/step
Epoch 59, Iteration 44/46, D_loss_real = 0.588, D_loss_fake = 0.655, G_loss = 0.983
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 45/46, D_loss_real = 0.631, D_loss_fake = 0.677, G_loss = 1.009
2/2 [=====] - 0s 3ms/step
Epoch 59, Iteration 46/46, D_loss_real = 0.673, D_loss_fake = 0.588, G_loss = 0.942
2/2 [=====] - 0s 4ms/step

Epoch 60, Iteration 1/46, D_loss_real = 0.693, D_loss_fake = 0.601, G_loss = 0.983
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 2/46, D_loss_real = 0.671, D_loss_fake = 0.609, G_loss = 0.961
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 3/46, D_loss_real = 0.630, D_loss_fake = 0.592, G_loss = 0.985
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 4/46, D_loss_real = 0.656, D_loss_fake = 0.553, G_loss = 0.982
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 5/46, D_loss_real = 0.656, D_loss_fake = 0.545, G_loss = 0.922
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 6/46, D_loss_real = 0.577, D_loss_fake = 0.658, G_loss = 0.996
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 7/46, D_loss_real = 0.642, D_loss_fake = 0.585, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 8/46, D_loss_real = 0.564, D_loss_fake = 0.648, G_loss = 0.995
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 9/46, D_loss_real = 0.604, D_loss_fake = 0.607, G_loss = 0.997
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 10/46, D_loss_real = 0.588, D_loss_fake = 0.560, G_loss = 0.987
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 11/46, D_loss_real = 0.566, D_loss_fake = 0.565, G_loss = 0.969
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 12/46, D_loss_real = 0.665, D_loss_fake = 0.582, G_loss = 0.956
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 13/46, D_loss_real = 0.602, D_loss_fake = 0.614, G_loss = 1.020
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 14/46, D_loss_real = 0.682, D_loss_fake = 0.584, G_loss = 1.008
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 15/46, D_loss_real = 0.593, D_loss_fake = 0.579, G_loss = 0.983
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 16/46, D_loss_real = 0.562, D_loss_fake = 0.579, G_loss = 0.978
2/2 [=====] - 0s 3ms/step

Epoch 60, Iteration 17/46, D_loss_real = 0.522, D_loss_fake = 0.617, G_loss = 0.938
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 18/46, D_loss_real = 0.685, D_loss_fake = 0.623, G_loss = 1.096
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 19/46, D_loss_real = 0.610, D_loss_fake = 0.531, G_loss = 1.214
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 20/46, D_loss_real = 0.738, D_loss_fake = 0.540, G_loss = 1.013
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 21/46, D_loss_real = 0.718, D_loss_fake = 0.608, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 22/46, D_loss_real = 0.598, D_loss_fake = 0.738, G_loss = 0.865
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 23/46, D_loss_real = 0.585, D_loss_fake = 0.675, G_loss = 0.915
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 24/46, D_loss_real = 0.624, D_loss_fake = 0.606, G_loss = 0.977
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 25/46, D_loss_real = 0.495, D_loss_fake = 0.587, G_loss = 0.902
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 26/46, D_loss_real = 0.644, D_loss_fake = 0.652, G_loss = 0.957
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 27/46, D_loss_real = 0.638, D_loss_fake = 0.606, G_loss = 0.943
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 28/46, D_loss_real = 0.610, D_loss_fake = 0.595, G_loss = 0.914
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 29/46, D_loss_real = 0.633, D_loss_fake = 0.666, G_loss = 0.917
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 30/46, D_loss_real = 0.599, D_loss_fake = 0.648, G_loss = 0.932
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 31/46, D_loss_real = 0.572, D_loss_fake = 0.657, G_loss = 0.980
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 32/46, D_loss_real = 0.622, D_loss_fake = 0.577, G_loss = 1.094
2/2 [=====] - 0s 3ms/step


```

Epoch 60, Iteration 33/46, D_loss_real = 0.781, D_loss_fake = 0.633, G_loss =
1.008
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 34/46, D_loss_real = 0.735, D_loss_fake = 0.524, G_loss =
0.929
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 35/46, D_loss_real = 0.682, D_loss_fake = 0.594, G_loss =
0.862
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 36/46, D_loss_real = 0.599, D_loss_fake = 0.782, G_loss =
0.847
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 37/46, D_loss_real = 0.705, D_loss_fake = 0.664, G_loss =
0.890
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 38/46, D_loss_real = 0.626, D_loss_fake = 0.648, G_loss =
0.913
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 39/46, D_loss_real = 0.710, D_loss_fake = 0.651, G_loss =
0.921
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 40/46, D_loss_real = 0.652, D_loss_fake = 0.650, G_loss =
0.867
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 41/46, D_loss_real = 0.657, D_loss_fake = 0.595, G_loss =
0.930
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 42/46, D_loss_real = 0.637, D_loss_fake = 0.584, G_loss =
0.916
2/2 [=====] - 0s 4ms/step
Epoch 60, Iteration 43/46, D_loss_real = 0.559, D_loss_fake = 0.616, G_loss =
0.864
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 44/46, D_loss_real = 0.622, D_loss_fake = 0.643, G_loss =
0.892
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 45/46, D_loss_real = 0.527, D_loss_fake = 0.719, G_loss =
0.922
2/2 [=====] - 0s 3ms/step
Epoch 60, Iteration 46/46, D_loss_real = 0.589, D_loss_fake = 0.705, G_loss =
0.973
2/2 [=====] - 0s 4ms/step

```

WARNING:tensorflow:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty until you train or evaluate the model.

```

>Saved: plot_032x032-tuned.png, g_model_032x032-tuned.sav,
d_model_032x032-tuned.sav and gan_model_032x032-tuned.sav

```

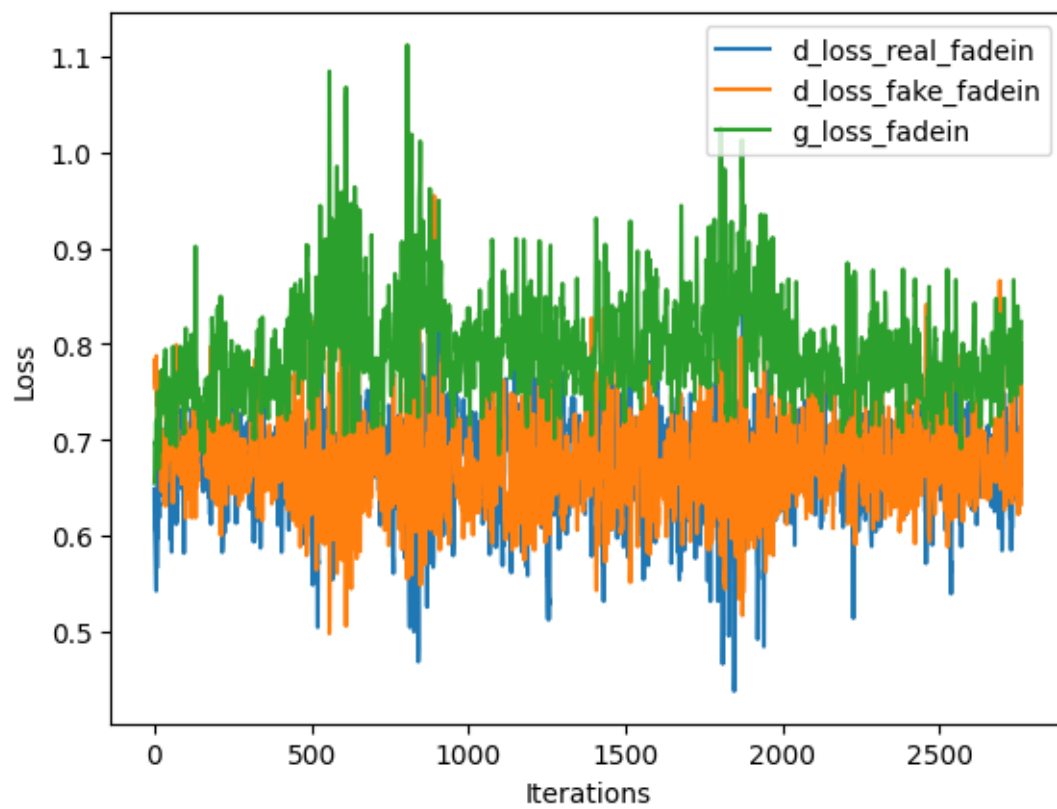
```
[19]: ### Plot the training loss of the Generator and the Discriminator of the last
      ↳ level of growth (32x32 resolution) over iterations

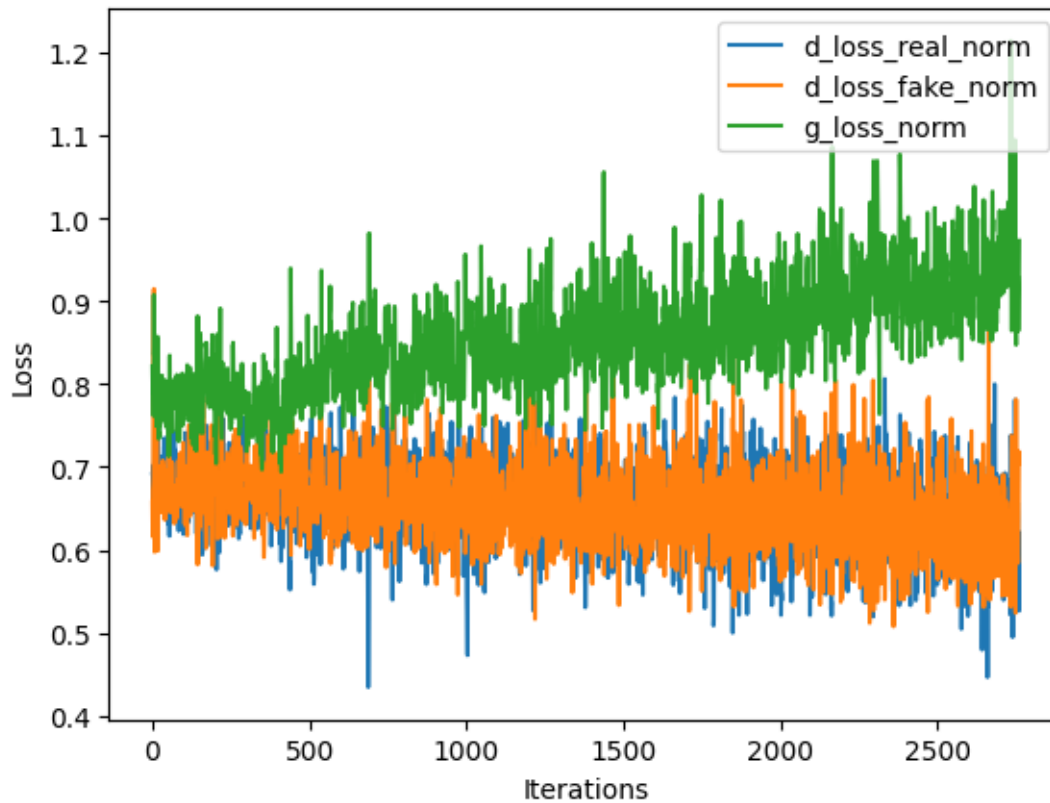
      # Initialize values of x-axis (iterations)
      gx_fadein = np.linspace(0, len(g_losses_fadein), len(g_losses_fadein))
      dx_real_fadein = np.linspace(0, len(d_losses_real_fadein),
      ↳ len(d_losses_real_fadein))
      dx_fake_fadein = np.linspace(0, len(d_losses_fake_fadein),
      ↳ len(d_losses_fake_fadein))

      gx_norm = np.linspace(0, len(g_losses_norm), len(g_losses_norm))
      dx_real_norm = np.linspace(0, len(d_losses_real_norm), len(d_losses_real_norm))
      dx_fake_norm = np.linspace(0, len(d_losses_fake_norm), len(d_losses_fake_norm))

      # Plot the training loss of G and D
      plt.plot(dx_real_fadein, d_losses_real_fadein)
      plt.plot(dx_fake_fadein, d_losses_fake_fadein)
      plt.plot(gx_fadein, g_losses_fadein)
      plt.ylabel('Loss')
      plt.xlabel('Iterations')
      plt.legend(['d_loss_real_fadein', 'd_loss_fake_fadein', 'g_loss_fadein'],
      ↳ loc='upper right')
      plt.show()

      plt.plot(dx_real_norm, d_losses_real_norm)
      plt.plot(dx_fake_norm, d_losses_fake_norm)
      plt.plot(gx_norm, g_losses_norm)
      plt.ylabel('Loss')
      plt.xlabel('Iterations')
      plt.legend(['d_loss_real_norm', 'd_loss_fake_norm', 'g_loss_norm'], loc='upper
      ↳ right')
      plt.show()
```





10 Load the models

```
[20]: # Load the Generator from the final saved model
g_model = keras.models.load_model("g_model_032x032-tuned.sav",
    ↪ custom_objects={"WeightedSum": WeightedSum})

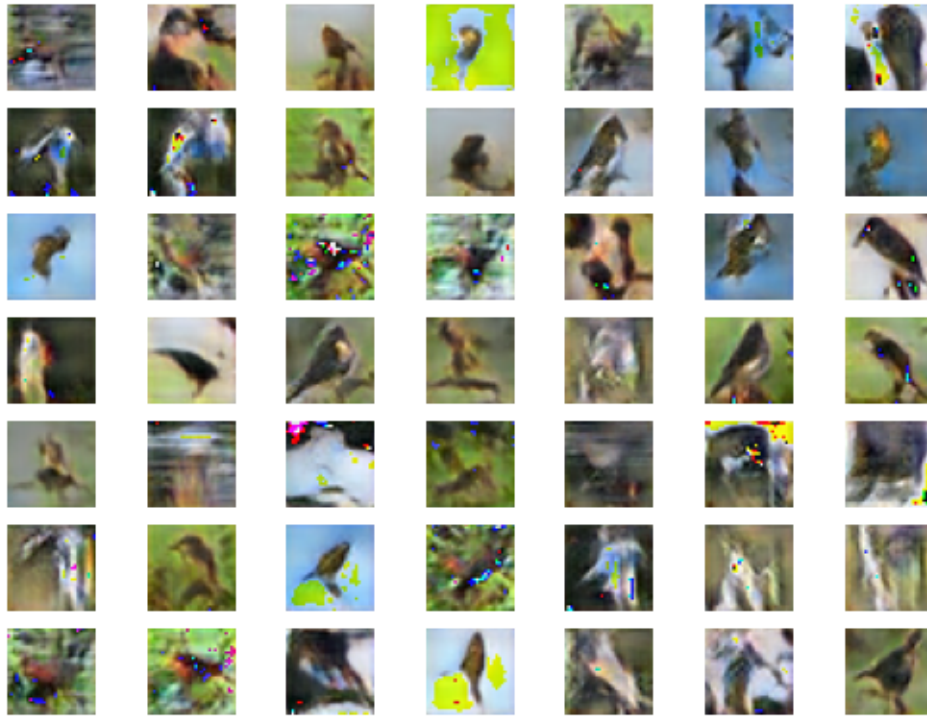
# Generate 49 fake images by passing 256-dimensional random vectors through the
    ↪ Generator
X_fake, _ = generate_fake_samples(g_model, 256, 49)
```

WARNING:tensorflow:No training configuration found in save file, so the model was *not* compiled. Compile it manually.

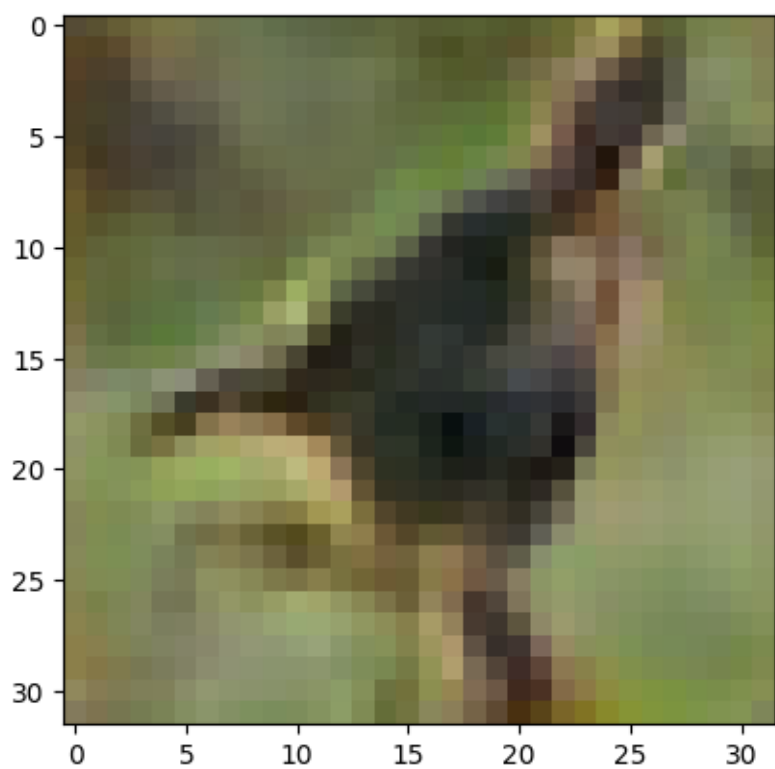
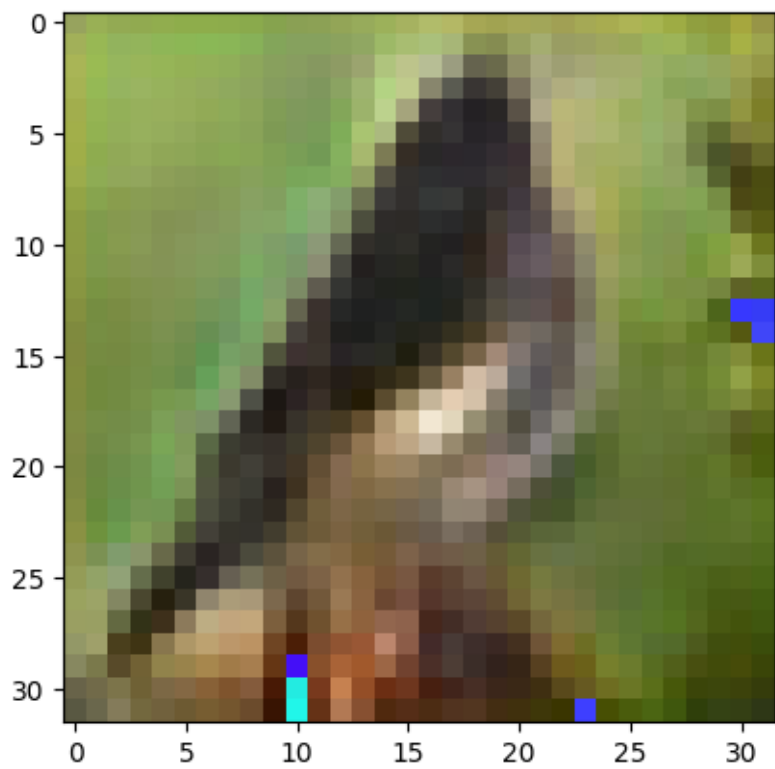
2/2 [=====] - 0s 4ms/step

```
[21]: # Visualize all generated images
for i in range(49):
    plt.subplot(7, 7, 1 + i)
    plt.axis('off')
    plt.imshow(((X_fake[i] * 127.5) + 127.5).astype(np.uint8)) # Convert pixel
    ↪ values from range [-1,1] into range [0,255]
```

```
plt.show()
```



```
[31]: # Visualize some individual generated images
imgplot = plt.imshow(((X_fake[26] * 127.5) + 127.5).astype(np.uint8))
plt.show()
imgplot = plt.imshow(((X_fake[48] * 127.5) + 127.5).astype(np.uint8))
plt.show()
```



```
[23]: # Compress output folders for downloading
!zip -r /content/g_model_032x032-tuned.zip /content/g_model_032x032-tuned.sav
!zip -r /content/d_model_032x032-tuned.zip /content/d_model_032x032-tuned.sav
!zip -r /content/gan_model_032x032-tuned.zip /content/gan_model_032x032-tuned.
↪sav
```

```
adding: content/g_model_032x032-tuned.sav/ (stored 0%)
adding: content/g_model_032x032-tuned.sav/assets/ (stored 0%)
adding: content/g_model_032x032-tuned.sav/fingerprint.pb (stored 0%)
adding: content/g_model_032x032-tuned.sav/keras_metadata.pb (deflated 94%)
adding: content/g_model_032x032-tuned.sav/saved_model.pb (deflated 90%)
adding: content/g_model_032x032-tuned.sav/variables/ (stored 0%)
adding: content/g_model_032x032-tuned.sav/variables/variables.index (deflated
65%)
adding:
content/g_model_032x032-tuned.sav/variables/variables.data-00000-of-00001
(deflated 8%)
adding: content/d_model_032x032-tuned.sav/ (stored 0%)
adding: content/d_model_032x032-tuned.sav/assets/ (stored 0%)
adding: content/d_model_032x032-tuned.sav/fingerprint.pb (stored 0%)
adding: content/d_model_032x032-tuned.sav/keras_metadata.pb (deflated 94%)
adding: content/d_model_032x032-tuned.sav/saved_model.pb (deflated 89%)
adding: content/d_model_032x032-tuned.sav/variables/ (stored 0%)
adding: content/d_model_032x032-tuned.sav/variables/variables.index (deflated
73%)
adding:
content/d_model_032x032-tuned.sav/variables/variables.data-00000-of-00001
(deflated 8%)
adding: content/gan_model_032x032-tuned.sav/ (stored 0%)
adding: content/gan_model_032x032-tuned.sav/assets/ (stored 0%)
adding: content/gan_model_032x032-tuned.sav/fingerprint.pb (stored 0%)
adding: content/gan_model_032x032-tuned.sav/keras_metadata.pb (deflated 95%)
adding: content/gan_model_032x032-tuned.sav/saved_model.pb (deflated 90%)
adding: content/gan_model_032x032-tuned.sav/variables/ (stored 0%)
adding: content/gan_model_032x032-tuned.sav/variables/variables.index
(deflated 76%)
adding:
content/gan_model_032x032-tuned.sav/variables/variables.data-00000-of-00001
(deflated 9%)
```

```
[33]: # Download the final Generator, Discriminator and composite straight-through
↪model
files.download("/content/g_model_032x032-tuned.zip")
files.download("/content/d_model_032x032-tuned.zip")
files.download("/content/gan_model_032x032-tuned.zip")
```

<IPython.core.display.Javascript object>
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